

Retained fibres, nonstandard Riemann hypotheses, and geometric complexity

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1 Scope and the directed construction

The starting objects are the original polynomial map and its retained inverse-root cubic, followed by its two-sheet conductor and arithmetic operator extension. The geometric-complexity route intended here is Mulmuley's nonstandard Riemann-hypothesis programme over finite fields. Its source-stated proposed steps are recorded at their actual scope. All the algebraic maps proved in this reader are explicit. No off-critical zero of the classical Riemann zeta function and no equality or separation of Boolean complexity classes is established here.

The following complete calculations retain the constants in the original map, including the coordinate relation $a = 2s = -r^2$. The term *normalization ring* below denotes a specified integral-closure construction together with its injection, conductor and cokernel; the original singular ring remains in every diagram.

2 The nonstandard Riemann-hypothesis bridge in the original GCT sources

The bridge named by the user is explicitly present in Mulmuley's *On P vs. NP, Geometric Complexity Theory, and the Riemann Hypothesis*, arXiv:0908.1936v2 [11]. Its final section, headed "How to prove PH1 and why should it hold?", proposes extending the geometric ingredients surrounding the Riemann hypothesis over finite fields to nonstandard quantum groups. The stated destination is a characteristic-zero complexity separation. This is an actual research programme in the source. In that source the nonstandard Riemann hypotheses themselves are described as still unknown; no nonstandard zeta function, Frobenius space, weight bound, or numbered nonstandard-RH conjecture is specified there. Consequently the reconstruction below retains the source's proved constructions and its conjectured arrows with their original status. It does not replace the proposed finite-field bridge by the classical hypothesis for the complex zeros of $\zeta_{\mathbb{Q}}(s)$.

2.1 Objects and arrows before any complexity claim

In GCT VII, arXiv:0709.0749v2 [9], Section 1, H is a complex connected classical reductive group, μ is a highest weight, $X = V_{\mu}(H)$, $G = GL(X)$, and

$$\rho : H \longrightarrow G$$

is the representation homomorphism. The plethysm multiplicity is

$$a_{\lambda, \mu}^{\pi} = \dim_{\mathbb{C}} \operatorname{Hom}_H(V_{\pi}(H), \operatorname{Res}_{\rho} V_{\lambda}(G)).$$

Here λ is a highest weight of G and π a highest weight of H . This notation is that of GCT VII; the survey uses a different order of indices in its plethysm discussion, which is not silently substituted in these formulas.

The quantum parameter q is independent of any prime-power cardinality. GCT VII's Section 6, **srefined**, explicitly warns that a splitting extension $K/\mathbb{Q}(q)$ may be needed. Its general reciprocity statement is not asserted over $\mathbb{Q}(q)$ without that extension, and Section 7.1 gives a failure over $\mathbb{Q}(q)$. The following description therefore retains the chosen coefficient field K , rather than erasing it.

Let $X_q = V_{q, \mu}(H_q)$ and let $\widehat{R}_{X, X}^H$ be the braiding operator attached to the standard quantum group H_q . GCT VII, Section 2, splits $X_q \otimes_K X_q$ according to the signs of the eigenvalues $\pm q^{a/2}$ and writes

$$I = P_{X, X}^{+, H} + P_{X, X}^{-, H}.$$

With u_j^i the entries of a variable $\dim(X) \times \dim(X)$ matrix, its matrix bialgebra is the explicit quotient

$$\mathcal{O}(M_q^H(X)) = K\langle u_j^i \rangle / \langle P_{X,X}^{+,H}(\mathbf{u} \otimes \mathbf{u}) - (\mathbf{u} \otimes \mathbf{u})P_{X,X}^{+,H} \rangle. \quad (1)$$

Replacing $P^{+,H}$ by $P^{-,H} = I - P^{+,H}$ gives the same ideal because the corresponding matrix of relations changes by the scalar -1 . The source defines braided symmetric and exterior algebras with the relations $P^{-,H}\mathbf{x}_1\mathbf{x}_2 = 0$ and $P^{+,H}\mathbf{x}_1\mathbf{x}_2 = 0$, respectively. Nonstandard minors are the matrix coefficients of the resulting exterior-algebra coactions. The top exterior power can vanish. Thus the determinant localization and an antipode cannot be presumed to exist in the general plethysm case. The source’s term “quantum group” includes this possible semigroup case.

Theorems labelled `tnonstd` and `tnonstdformal` in GCT VII state the construction of the quantum map $H_q \rightarrow G_q^H$, its matrix coordinate bialgebra map, cosemisimplicity, and a Peter–Weyl decomposition. The coordinate-ring direction is

$$\mathcal{O}(M_q^H(X)) \longrightarrow \mathcal{O}(M_q(V)) \quad (H = GL(V)),$$

which is opposite to the displayed direction $H_q \rightarrow G_q^H$ of virtual quantum spaces. We use the concrete coordinate map, whose elementary quotient construction is proved below, without importing an unproved canonical basis from the existence of the bialgebra.

GCT VII’s conjecture `cduality` proposes a decomposition

$$X_q^{\otimes r} \cong \bigoplus_{\alpha} W_{q,\alpha} \otimes_K T_{q,\alpha} \quad (2)$$

for the commuting nonstandard quantum-group and $\mathcal{B}_r^H(q)$ actions. Its conjecture `creciprocity`, refined by `crecirefined`, proposes

$$V_{q,\lambda}^H = \bigoplus_{\alpha} (W_{q,\alpha})^{\oplus m_{\lambda}^{\alpha}}, \quad V_{q,\lambda}^H|_{q=1} \cong \text{Res}_{\rho} V_{\lambda}(G). \quad (3)$$

The multiplicity m_{λ}^{α} counts the factors isomorphic to the Specht module S_{λ} in a composition series of $T_{q,\alpha}(1)$ as a $\mathcal{B}_r^H(1)$ -module. A semisimple generic algebra does not license replacing that composition series by an assumed semisimple special fibre. The source itself makes this point in Section 6. If n_{π}^{α} denotes the multiplicity of $V_{q,\pi}$ in $W_{q,\alpha}$, the proposed reciprocity yields the source’s equation

$$a_{\lambda,\mu}^{\pi} = \sum_{\alpha} m_{\lambda}^{\alpha} n_{\pi}^{\alpha}. \quad (4)$$

This displayed equality is reported here as the consequence asserted by that reciprocity conjecture, not as a newly proved general-plethysm result. The elementary dimension calculation involved is explicit: on a direct sum as in (3), an H_q -linear map from $V_{q,\pi}$ is exactly one such map to each summand, and projection and inclusion supply inverse linear maps between the corresponding Hom spaces. Each of the m_{λ}^{α} copies contributes n_{π}^{α} dimensions. What this calculation does not establish is the asserted specialization isomorphism in (3).

2.2 The positivity and effectivity steps retained separately

GCT VIII, arXiv:0709.0751v2 [10], presents algorithms whose correctness and required properties are conjectured. Its source labels include `clocalcrystalb` (local crystal-basis existence), `cuniquemindegleft` and `cunimindegright` (minimum degree), `ccelldecompleft` and `ccelldecompright` (cell decomposition), and `cposkroneckerleft`, `cposkroneckerright` (positivity). The cells must recover the irreducible modules, and the subcells must recover the restricted H_q -modules. This is structural data beyond a list of positive numbers.

In the Kronecker setting

$$H = GL(V) \times GL(W), \quad X = V \otimes W,$$

the left positivity conjecture in Section 2.4.2 does not assert that every raw multiplication constant is a nonnegative polynomial. It allows the exact form

$$\pm(q - q^{-1})^a f(q), \quad a \in \mathbb{Z}_{\geq 0}, \quad (5)$$

where f is invariant under $q \leftrightarrow q^{-1}$ and has nonnegative rational coefficients, together with the stated unimodality requirement. The factor $(q - q^{-1})^a$ and its sign are part of the conjectured statement and cannot be suppressed at $q = 1$.

The general-plethysm proposal is weaker and more complicated. The coefficients can be algebraic over $\mathbb{Q}(q)$, and Section 2.4.3 discusses coefficients of their minimal polynomials. The conjecture `cnonstdposleft0` uses positivity of a nonzero residual polynomial at $q = 1$ after explicitly retaining signs and powers of $q - q^{-1}$. The conjecture `cnonstdposleft1` proposes a decomposition $c(q) = c^0(q) + c^1(q)$ with a dominant positive unimodal term and a small correction. The source includes asymptotic language and a polynomial bitlength bound; it does not provide a fully specified Frobenius complex whose weights prove that decomposition.

Section 6 of GCT VIII then adds computational requirements on the labels of the basis and its cells. In the standard setting, efficiently computable tableau/path labels and crystal operations yield counting rules even when constructing all basis vectors is expensive. The nonstandard analogues are proposed there as additional properties whose full specification is deferred. Thus coefficient positivity alone does not provide a uniform polynomial-time algorithm or a $\#P$ formula. For example, from a predicate $R(x, y)$ and a bound $|y| \leq p(|x|)$, both explicitly supplied, one obtains a $\#P$ count by nondeterministically choosing a padded witness and checking R . Without a witness encoding, polynomial bound, and polynomial-time checking procedure, the mere nonnegative integer value of a multiplicity supplies no such machine.

The survey's final section next discusses Plethysm PH0, the existence of compatible positive bases for $\rho : H \rightarrow GL(V_\mu(H))$, and Plethysm PH/PH1, positive polyhedral formulas and their effective versions. The intended further step is to quantize the triple

$$\overline{H}_{\widehat{h}} \longrightarrow \overline{H} = SL(\overline{X}) \longrightarrow L = GL(\overline{W})$$

associated with a class variety. It is not a proved automatic passage from a single plethysm pair to all class-variety multiplicities. The survey's general PH1, labelled `hph1restatement`, requires bounds polynomial in $\langle \lambda \rangle, n, \langle m \rangle$, including the bitlength of m , for the padded polynomial $\overline{h}(\overline{X}) = z^{m-n} h(X)$.

Finally the displayed source theorem `tgct6refined` also requires an obstruction hypothesis OH. It takes $m = 2^{\log^a n}$, with $a > 1$ fixed, and asks for labels λ with $P_{\lambda, n, m}(k) \neq \emptyset$ for all sufficiently large k but $Q_\lambda = \emptyset$. This separates a positive formula from the additional proof that the required obstruction actually exists. Its displayed target is the general $\#P$ versus NC problem in characteristic zero; the source routes the analogous P versus NP formulation to GCT VI. This record does not substitute a Boolean uniform P versus NP theorem for that stated algebraic target.

The primary-source chain therefore has the following exact status:

1. The standard geometric positivity story uses the finite-field purity results surrounding Deligne's work and the canonical bases of standard quantum groups. This is the source's established model.
2. The matrix bialgebra quantization is explicitly constructed.

3. General nonstandard canonical bases, their required cells, positivity, reciprocity and effective labels are proposed additional statements. They are not all consequences of the existence of the matrix bialgebra.
4. The nonstandard finite-field purity/RH theory is a proposed way to prove those statements; the cited works do not state it as a complete zeta-function hypothesis.
5. Effective positive formulas must still be connected to the class varieties and to actual obstruction existence, with the declared size parameters, before the intended complexity separation follows.

These labels describe the literature's proposed programme, and introduce no conditional theorem as a replacement for a missing proof in the present work.

2.3 A fully explicit FRT coordinate morphism

Here is the elementary algebraic construction that underlies (1). It is included with a full proof so that the quantum-space arrow has a concrete coordinate-ring meaning. Let K be a field, $E = K^d$ with its ordered basis $e_1, \dots, e_d$, and $P \in \text{End}_K(E \otimes E)$. Put

$$T = K\langle u_{ij} : 1 \leq i, j \leq d \rangle, \quad W_{(i,k),(j,l)} = u_{ij}u_{kl}.$$

$$R = PW - WP, \quad A_P = T/(R_{ab} : a, b \in \{1, \dots, d\}^2).$$

Here (R_{ab}) denotes the two-sided ideal generated by the displayed entries; no commutation between different u_{ij} is presumed. On the free algebra define

$$\Delta(u_{ij}) = \sum_{h=1}^d u_{ih} \otimes u_{hj}, \quad \epsilon(u_{ij}) = \delta_{ij}.$$

Both extend uniquely as algebra homomorphisms. Their coassociativity and counit identities hold on a generator because both iterated coproducts equal $\sum_{h,k} u_{ih} \otimes u_{hk} \otimes u_{kj}$, and contracting either factor with ϵ gives u_{ij} . The same identities then hold on each word by multiplicativity and on all of T by linearity.

For indices a, b, c in $\{1, \dots, d\}^2$,

$$\Delta(W_{ab}) = \sum_c W_{ac} \otimes W_{cb}.$$

Consequently, with each occurrence of the scalar matrix P acting on these pair-indices,

$$\Delta(R_{ab}) = \sum_c R_{ac} \otimes W_{cb} + \sum_c W_{ac} \otimes R_{cb}, \quad \epsilon(R_{ab}) = 0.$$

The two middle terms cancel entry by entry. The ideal of relations is therefore a coideal and lies in the kernel of the counit. It follows that these formulas define a bialgebra structure on A_P .

Now let B be a bialgebra with a right comodule E written

$$\delta_B(e_j) = \sum_i e_i \otimes y_{ij}.$$

The comodule identities are exactly $\Delta_B(y_{ij}) = \sum_h y_{ih} \otimes y_{hj}$ and $\epsilon_B(y_{ij}) = \delta_{ij}$. When P is a comodule endomorphism of $E \otimes E$, applying $\delta_B P = (P \otimes 1)\delta_B$ to every basis vector gives $P(y \otimes y) = (y \otimes y)P$ entry by entry. Thus the explicit map

$$\Phi_P : A_P \longrightarrow B, \quad [u_{ij}] \longmapsto y_{ij} \tag{6}$$

is well-defined. It preserves coproduct and counit on generators and hence everywhere. It is the unique bialgebra map with those specified generator images. Its image is the subbialgebra generated by the y_{ij} ; its exact kernel is

$$\{[F(u)] \in A_P : F(y) = 0 \text{ in } B\}.$$

This does not claim that the map is injective, nor that its image is all of B .

For comparison at the classical symmetric projector, assume $\text{char}(K) \neq 2$ and take $P = (I + \tau)/2$, where $\tau(e_i \otimes e_k) = e_k \otimes e_i$. Then

$$2R_{(i,k),(j,l)} = u_{kj}u_{il} - u_{il}u_{kj}.$$

As the four indices vary these are exactly all pairwise commutators of the matrix variables. Sending $[u_{ij}]$ to the corresponding commuting variable defines an isomorphism

$$A_{(I+\tau)/2} \longrightarrow K[u_{ij} : 1 \leq i, j \leq d],$$

whose inverse sends a commuting variable to $[u_{ij}]$. The commutator relations prove that the inverse is defined, and the two compositions fix every generator. This proves the comparison map with its field, size and characteristic restriction retained.

2.4 Specialization and the information in its kernel

Let $D = \mathbb{Q}[q, q^{-1}]_{(q-1)}$, $t = q - 1$, L a finite free D -module, $J \subseteq L$ a submodule, and $A = L/J$. These are explicit algebraic data rather than a claim that a particular GCT conjecture holds. Let $A[t] = \{a \in A : ta = 0\}$. There is an exact sequence

$$0 \longrightarrow A[t] \xrightarrow{\eta} J/tJ \longrightarrow L/tL \longrightarrow A/tA \longrightarrow 0, \quad \eta([v]) = [tv]. \quad (7)$$

Indeed $ta = 0$ means that a lift $v \in L$ satisfies $tv \in J$. Changing the lift by $j \in J$ changes tv by tj , so η is well-defined. If $tv \in tJ$, then $tv = tj$ for some $j \in J$; multiplication by t is injective on the free module L , so $v = j$ and $[v] = 0$. Thus η is injective. An element $j \in J$ maps to zero in L/tL exactly when $j = tv$ for some $v \in L$, which is exactly the image of η . The kernel of $L/tL \rightarrow A/tA$ consists of the classes of $J + tL$, and the last map is surjective by lifting a representative from A to L . These observations prove every assertion of exactness.

Put

$$J^{\text{sat}} = \{v \in L : t^e v \in J \text{ for some } e \in \mathbb{Z}_{\geq 0}\}.$$

The map $A \rightarrow A \otimes_D \mathbb{Q}(q)$ has kernel J^{sat}/J . To prove this, localization sends the class of v to zero precisely when some nonzero element of D multiplies v into J . Every nonzero element of D is a power of t times a unit: factor its numerator by its exact order of vanishing at $q = 1$ and note that its denominator and remaining numerator are units in this local ring. Multiplying or dividing by that unit does not change membership in J . This proves the kernel formula. Passing to a generic fibre therefore removes an explicitly identified module, not an unspecified “small error”. Formula (7) records the first special-fibre layer that a proposed reciprocity construction must preserve in the data to which it is applied.

2.5 A source-specific coefficient and its retained double zero

The final GCT IV, by Blasiak, Mulmuley and Sohoni, arXiv:cs/0703110v4 [1], proves nonstandard Schur–Weyl duality in Section 12 (source label `t nonstandard schur-weyl duality`). Its displayed statement uses $\check{X}^{\otimes r}$, and the immediately following remark explicitly records

the dual convention needed by its proof. The source then constructs a global crystal-basis rule for the two-row problem, $\dim V = \dim W = 2$, with

$$g_{\lambda\mu\nu} = \#\{\text{highest-weight elements of } +\text{HNSTC}(\nu) \text{ of weight } (\lambda, \mu)\}.$$

This is a proved bounded case in the source, not a claim about all dimensions. Theorem **t positivity** gives sign-coherent coefficients for individual Chevalley generators. Example **ex not positive** displays a divided-power expression with mixed signs. The literal printed target tableau has a weight inconsistency. Section 3 proves that obstruction, constructs the corrected target Q from the actual action rules, and proves that its divided-power matrix entry is the displayed Laurent polynomial. The printed and corrected targets remain separately specified objects.

We now independently prove all algebraic identities used from that example. The example takes column-count shape $\nu = [0, n_3, n_2, 2]$ with $n_3 = 3, n_2 = 2$; in ordinary row-length notation this is the same partition $(7, 5, 3)$, since there are three columns of height three, two of height two and two of height one. Its size is $3 \cdot 3 + 2 \cdot 2 + 2 = 15$. We retain the original column-count shape. For $n \geq 1$ set

$$[n]_q = \sum_{j=0}^{n-1} q^{n-1-2j} \quad \text{in } \mathbb{Z}[q, q^{-1}].$$

Multiplication gives $(q - q^{-1})[n]_q = q^n - q^{-n}$ by cancellation of explicitly adjacent terms; the finite sum also defines its values at $q = 1$ and $q = -1$. The source displays the following rational expression and Laurent polynomial:

$$C(q) = \frac{[6]_q([7]_q - [3]_q) - [3]_q[8]_q}{[2]_q} = q^{10} - q^4 - q^{-4} + q^{-10}. \quad (8)$$

The first expression lives in $\mathbb{Q}(q)$ and the second is its Laurent-polynomial representative; multiplying the equality by $[2]_q = q + q^{-1}$ verifies it without imposing any evaluation at a zero of $[2]_q$.

The original Laurent polynomial, including all four signs, satisfies

$$\begin{aligned} C(q) &= (q^7 - q^{-7})(q^3 - q^{-3}) \\ &= (q - q^{-1})^2 [7]_q [3]_q. \end{aligned} \quad (9)$$

The first equality follows by multiplying its two binomials, giving the four exponents $10, 4, -4, -10$ with coefficients $1, -1, -1, 1$. The second follows from the preceding finite-sum identity for $n = 7, 3$. Moreover

$$[7]_q [3]_q = q^8 + 2q^6 + 3q^4 + 3q^2 + 3 + 3q^{-2} + 3q^{-4} + 2q^{-6} + q^{-8}.$$

The displayed coefficients count the ordered pairs of summands with each exponent. This proves nonnegativity of this retained factor, while C itself has the mixed signs displayed in (8). No assertion about every positivity requirement of GCT VIII follows from this single example.

Since

$$q - q^{-1} = \frac{(q-1)(q+1)}{q},$$

equation (9) gives the equality in D with all factors retained:

$$C(q) = (q-1)^2 \frac{(q+1)^2}{q^2} [7]_q [3]_q, \quad \left. \frac{(q+1)^2}{q^2} [7]_q [3]_q \right|_{q=1} = 4 \cdot 7 \cdot 3 = 84.$$

Thus $C(1) = C'(1) = 0$, $C''(1) = 168$, and its exact order at $q = 1$ is two. Specializing C to its value 0 would lose the nonzero retained second jet $84(q-1)^2$ in $D/(q-1)^3$.

There is also an explicit two-sheet presentation. Here p is a new trace coordinate, distinct from the source polynomial map's retained coordinate w . Define

$$B = \mathbb{Q}[p, Q]/(Q^2 - pQ + 1), \quad \theta : B \longrightarrow \mathbb{Q}[q, q^{-1}], \quad Q \longmapsto q, \quad p \longmapsto q + q^{-1}.$$

In B , $Q(p - Q) = 1$. Hence the inverse homomorphism sends q to Q and q^{-1} to $p - Q$. Both compositions fix the displayed generators, proving that θ is an isomorphism. The ring B is free of rank two over $\mathbb{Q}[p]$, with ordered basis $(1, Q)$: division by the monic relation produces a unique representative $a(p) + b(p)Q$, and a nonzero multiple of a monic degree-two polynomial cannot have smaller Q -degree. The deck involution sends Q to $p - Q$; its image under θ is $q \mapsto q^{-1}$.

Put $r_{\text{GCT}} = 2Q - p$ in B . Then

$$r_{\text{GCT}}^2 = p^2 - 4.$$

The associated conic presentation is the isomorphism

$$\mathbb{Q}[p, r_{\text{GCT}}]/(p^2 - r_{\text{GCT}}^2 - 4) \longrightarrow \mathbb{Q}[q, q^{-1}], \quad (p, r_{\text{GCT}}) \longmapsto (q + q^{-1}, q - q^{-1}),$$

whose inverse sends q to $(p + r_{\text{GCT}})/2$ and q^{-1} to $(p - r_{\text{GCT}})/2$. Their product is 1 by the conic equation. The use of $1/2$ is explicit; these displayed inverses are over $\mathbb{Q}$ and are not asserted in characteristic two.

Writing $p = q + q^{-1}$ and applying the recurrence $[n+1]_q = p[n]_q - [n-1]_q$ gives

$$[3]_q = p^2 - 1, \quad [7]_q = p^6 - 5p^4 + 6p^2 - 1.$$

The recurrence is verified directly by multiplying the two finite sums and cancelling the shared interior terms. Thus the same retained coefficient is

$$\begin{aligned} C(q) &= (p^2 - 4)(p^6 - 5p^4 + 6p^2 - 1)(p^2 - 1) \\ &= p^{10} - 10p^8 + 35p^6 - 51p^4 + 29p^2 - 4. \end{aligned}$$

At $p = 2$ the factor multiplying $p - 2$ has value $4 \cdot 7 \cdot 3 = 84$, whereas

$$p - 2 = \frac{(q - 1)^2}{q}.$$

This exact ramification equation explains why a first zero in the trace coordinate p is the retained second zero in the coordinate q . The discriminant of $Q^2 - pQ + 1$ is $p^2 - 4$; the two excluded branch values for the étale double cover are $p = 2$ and $p = -2$. Indeed the relative differential module is $\Omega_{B/\mathbb{Q}[p]} = B/(2Q - p)dQ$, by differentiating the single monic relation with p held fixed. Since $(2Q - p)^2 = p^2 - 4$, this module vanishes after inverting $p^2 - 4$. The algebra is already free of rank two over the base, so this localization is finite étale. At $p = 2$ the fibre is $\mathbb{Q}[Q]/(Q - 1)^2$, and at $p = -2$ it is $\mathbb{Q}[Q]/(Q + 1)^2$, proving that both exceptional fibres retain a nonzero square-zero element. Further, if $a(p) + b(p)Q$ is invariant under $Q \mapsto p - Q$, then $b(p)(2Q - p) = 0$. The proved embedding into the Laurent-polynomial domain gives $b = 0$, so the invariant subring is exactly $\mathbb{Q}[p]$. These identities give a concrete, fully calculated nonstandard-GCT coefficient and its rank-two coordinate algebra for any subsequent arithmetic transport to act on. They prove no general canonical-basis conjecture and no complexity separation.

3 A source-correct GCT-IV generator interval and its integral endpoint

Put $A = \mathbb{Z}[q, q^{-1}]$, $K = \mathbb{Q}(q)$ and $[m] = q^{m-1} + q^{m-3} + \cdots + q^{1-m}$ for $m \geq 1$, with $[0] = 0$. Write $p = [2] = q + q^{-1}$ and $r = q - q^{-1}$ throughout this section. These retain the source parameter (called u in its TeX), rather than specializing it in the generator calculation. A column is denoted by its entries in parentheses; concatenation means the source's projected upper based product, not an ordinary commutative monomial product. In particular (23) and (32) are distinct columns.

The source is the April 2013 version of GCT IV, *Geometric complexity theory IV: nonstandard quantum group for the Kronecker problem*, arXiv:cs/0703110v4. Its literal example labelled **ex not positive** and equations labelled **e aFV computation** must be read together with the column dictionary, the contextual nonintegral straightening rule, and the Chevalley action rule. The precise source bytes and line intervals are recorded in `source_receipts.json`; this proof retains the printed target as a separate object.

3.1 Objects, weights, and the printed endpoint

Let $X = \check{X}_{(7,5,3)}$ be the original source quotient $\check{Y}_{(3,3,3,2,2,1,1)}/\check{Y}_{\triangleright(3,3,3,2,2,1,1)}$. Its source basis is the set of positive honest nonstandard tabloid classes. Its integral form is the A -span of those classes. We use the following actual vectors, including the source's rescaling:

$$\begin{aligned} T &= (123)(123)(124)(12)(12)(1)(1), \\ T_1 &= (123)(123)(124)(23)(12)(1)(1), \\ T_2 &= -p^{-1}(123)(123)(123)(12)(12)(4)(1), \\ P_{\text{printed}} &= -p^{-1}(123)(123)(124)(23)(12)(4)(1), \\ Q &= -p^{-1}(123)(123)(124)(13)(12)(4)(1). \end{aligned}$$

Thus $\nu = [0, 3, 2, 2]$ in the source's column-count notation and $\nu = (7, 5, 3)$ in row-length notation. There are fifteen boxes. The alphabet dictionary is $1 = (1, 1)$, $2 = (1, 2)$, $3 = (2, 1)$, $4 = (2, 2)$ for the V, W letters. Counting each letter, without reordering columns, gives

	V -weight	W -weight
T	$(12, 3)$	$(9, 6)$
T_1, T_2	$(11, 4)$	$(9, 6)$
P_{printed}	$(10, 5)$	$(8, 7)$
Q	$(10, 5)$	$(9, 6)$

The Chevalley operator F_V changes the V -weight by $(-1, 1)$ and preserves both W -weight coordinates. Since the basis consists of weight vectors and q is an indeterminate, its distinct W -weight spaces form a direct sum. Therefore the coefficient of P_{printed} in $F_V^{(2)}T$ is zero. This proves a discrepancy in the literal displayed endpoint. It does not infer a corrected endpoint from the scalar formula. The direct calculation below constructs Q and its coefficient.

3.2 Contextual straightening and every two-step path

The selected column words in the source dictionary are

column	V	W	column	V	W
1	1	1	3	2	1
4	2	2	12	11	21
13	21	11	23	21	12
32	12	21	123	211	121
124	211	221	134	212	121
234	212	221			

These are diagram labels: they are not claims that the corresponding nonstandard column equals just one canonical tensor word. For example the column (23) includes the source's additional p^{-1} term. The source's sign convention takes $(-p^{-1})^{\deg}$ times an honest representative with no 3–2 arc, or with its 3–2 arc on the W side, as positive. The displayed T, T_1 have degree zero, while T_2, Q have degree one. The diagram of Q has its 3–2 arc on the W side. Thus the displayed rescalings have the source's positive orientation. The diagram classification places T, T_1, T_2, Q respectively in the pure 3–1 W , 3–2–1 W , no-external-arc, and pure 3–2 W cases after deleting invariants; these are honest classes in the source basis. Pair each 2 with the next available 1 to its right, recursively matching 21 pairs; equivalently scan left to right using a stack of unmatched 2 positions. The source's generator rule is

$$F_V D = \sum_{j=1}^{\varphi_V(D)} [j] \mathcal{F}_{(j)V}(D),$$

where φ_V is the number of unpaired V -ones. The j th term changes the column containing that one by its column Kashiwara operator, unless the changed word has an extra internal arc, in which case the term is zero. The changes needed below are $123 \rightarrow 134$, $124 \rightarrow 234$, $12 \rightarrow 32$, $1 \rightarrow 3$. For an unpaired (12) column, choosing the first of its two V -ones creates an extra internal arc and gives zero; choosing the second gives (32).

For clarity we specify the contextual nonintegral rule used here. For a consecutive pair of equal-height columns in one of the source nonintegral strings, write the entire based product as BCD . Let I be the corresponding equal-height invariant, and let ω be the unpaired word of C on the indicated side. The source's contextual projection identity, after taking the quotient, is

$$BCD = -p^{-1} \begin{cases} B'ID & \omega = 11, \\ BID & \omega = 12, \\ BID' & \omega = 22. \end{cases}$$

In the 11 instances below, B' is obtained by changing the unpaired 2 in B paired to the first local 1 to a 1; if there is no such 2, $B' = 0$. The 12 instance leaves both outside words unchanged. These identities follow by projecting the source formula $B \heartsuit_g \tilde{C} \heartsuit_g D = BCD + p^{-1}B'ID$ (and its 12 version) to the quotient in which the left-hand side is zero. This is the source corollary labelled **c projecting to heartproj basis Kronecker**; the distinction between the two based products in that formula is essential.

The pairs used here are (134)(123), with W -word 11 and height-three invariant (234)(123); (134)(124) with W -word 12 and the same invariant; (32)(12) with V -word 11 and height-two invariant (34)(12); and (3)(1) with W -word 11 and height-one invariant (4)(1). No isolated relation such as $(3)(1) = 0$ is substituted inside a longer based product.

The integral relations needed in the same quotient are

$$\begin{aligned} (234)(12) &= -(124)(23), & (134)(12) &= -(124)(13), \\ (32)(1) &= (23)(1), & (12)(23) &= (23)(12). \end{aligned}$$

They are the source's height-(3, 2), (2, 1) and (2, 2) degree-preserving relations. Their contextual products remain relations: for integral relations the source's two based products agree. Thus there is no suppressed nonintegral correction in these four replacements.

In T , the prefix before its last two columns has exactly one unpaired W -two, in column (124). It is paired with the first of the two suffix W -ones. The contextual rule therefore gives

$$(123)(123)(124)(12)(12)(3)(1) = -p^{-1}(123)(123)(123)(12)(12)(4)(1) = T_2.$$

For T_1 the analogous unpaired W -two is in column (23), and changing it to a 1 changes that column to (13). Hence

$$(123)(123)(124)(23)(12)(3)(1) = -p^{-1}(123)(123)(124)(13)(12)(4)(1) = Q.$$

This constructs the endpoint and its minus sign directly. The second calculation is source case (F.1-1.5): its hypothesis holds because T_1 has a 3-2-1 W -arc and no pure 3-1 W -arc.

We record how invariant counts exclude paths. An equal-height invariant pair has all of its letters paired internally or within the pair on both diagrams, with no exiting arc. An operator $\mathcal{F}_{(j)V}$ changes an unpaired letter, so it cannot change such a pair. Degree-preserving straightening moves preserve the invariant record. In a nonintegral correction, the added pair is an invariant and the contextual change is at an outside unpaired letter; it therefore leaves pre-existing pairs intact. Integral graded alternatives give zero. These facts prove, step by step, that nonzero generator coefficients cannot decrease any of the height-specific invariant counts. They also prove the direction of containment used here independently of the reversed containment sentence in source line 5007. In particular a vector with a height-three or height-two invariant cannot reach Q , whose only invariant is its final (4)(1).

For T the ordered unpaired V -one positions (column, word position) are

$$(1, 3), (2, 3), (3, 3), (4, 1), (4, 2), (5, 1), (5, 2), (6, 1), (7, 1).$$

Put

$$H = -p^{-1}(123)(234)(123)(12)(12)(1)(1), \quad R_0 = (123)(123)(124)(12)(12)(1)(3).$$

The following table lists all nine terms before multiplication by their quantum integer $[j]$:

j	$\mathcal{F}_{(j)V}T$	verification
1	0	(134)(123), $W : 11$, $B' = 0$
2	H	(134)(124), $W : 12$, context unchanged
3	$-T_1$	(234)(12) = -(124)(23)
4	0	extra internal arc in column 4
5	0	(32)(12), $V : 11$, $B' = 0$
6	0	extra internal arc in column 5
7	T_1	(32)(1) = (23)(1), then commute 12, 23
8	T_2	the first contextual suffix calculation above
9	R_0	last unpaired one, the crystal term

All displayed zero-prefix assertions follow by the stack pairing: for $j = 1$ the prefix is empty; for $j = 5$ the prefix has no unpaired V -two. The vector H has a height-three invariant, so none of its nonzero successors can equal Q . This proves $F_V T = ([7] - [3])T_1 + [8]T_2 + [2]H + [9]R_0$, and only the last term besides T_1, T_2 still requires a possible-return check.

There are eight unpaired V -ones in R_0 , the first eight positions in the preceding list; its last 3 supplies an unpaired V -two. Its W -diagram equals that of T . Put

$$H_R = -p^{-1}(123)(234)(123)(12)(12)(1)(3), \quad R_1 = (123)(123)(124)(23)(12)(1)(3),$$

and $R_2 = (123)(123)(124)(12)(12)(3)(3)$. Every term is

$$\begin{array}{c|cccccccc} j & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \hline \mathcal{F}_{(j)V}R_0 & 0 & H_R & -R_1 & 0 & 0 & 0 & R_1 & R_2. \end{array}$$

The same local relations and zero-prefix calculations apply since the last column lies outside them. The class H_R has a height-three invariant. The honest class R_1 has no invariant and one unpaired V -two; R_2 has no invariant and two unpaired V -twos. The source diagram classification identifies these honest classes, so neither is Q , which has invariant record $(0, 0, 0, 1)$ and no unpaired V -two. This direct enumeration proves that the coefficient of Q in $F_V R_0$ is zero; no general monotonicity assertion about unpaired twos is needed.

The ordered positions for T_1 are $(1, 3), (2, 3), (3, 3), (5, 1), (5, 2), (6, 1), (7, 1)$. Its seven terms, with the source cases made explicit, are

j	outcome and reason it can or cannot equal Q
1	0 : $(134)(123)$, $W : 11$, empty prefix
2	a height-three invariant, case (F.3-3.1)
3	a height-two invariant, case (F.3-2)
4	0 : extra internal arc in column 5
5	a height-two invariant, case (F.2-1)
6	Q : the second contextual suffix calculation above
7	R_1 : the crystal term, invariant count $i_1 = 0$.

The stated invariant cases can also be checked with the finite relations: for $j = 2$, apply the $W : 12$ replacement $(134)(124) \mapsto -p^{-1}(234)(123)$. For $j = 3$, first use $(234)(23) = -(234)(32)$ and then the contextual $V : 11$ replacement of $(32)(12)$, whose preceding unpaired V -two is in (234) ; it raises that column to (124) and creates $(34)(12)$. For $j = 5$, use $(32)(1) = (23)(1)$ and then the $W : 12$ replacement $(23)(23) \mapsto -p^{-1}(24)(13)$, which creates a height-two invariant. The extra integral relation here is the source's $(3, 2)$ bottom-weight relation, and the latter contextual relation is its height-two nonintegral string. These computations account for all indices and prove $a_{Q, T_1}^{F_V} = [6]$.

For T_2 the seven positions are $(1, 3), (2, 3), (3, 3), (4, 1), (4, 2), (5, 1), (5, 2)$; its existing $(4)(1)$ invariant contributes none. Terms $j = 1, 2$ vanish because their $(134)(123)$ $W : 11$ replacements have no preceding unpaired W -two. Terms $j = 4, 6$ vanish by extra internal arcs. Term $j = 5$ vanishes by the $V : 11$ $(32)(12)$ relation with no preceding unpaired V -two. Term $j = 7$ is the crystal vector $-p^{-1}(123)(123)(123)(12)(32)(4)(1)$, with one unpaired V -two; after removing its invariant its diagram has no external arc, whereas Q has a pure $3-2$ W -arc and no unpaired V -two. Finally the exact $j = 3$ calculation is

$$-p^{-1}(123)(123)(134)(12)(12)(4)(1) = p^{-1}(123)(123)(124)(13)(12)(4)(1) = -Q.$$

Thus $a_{Q, T_2}^{F_V} = -[3]$. Every possible first step and every remaining second step has now been accounted for. Therefore precisely T_1 and T_2 contribute to the two-step path from T to Q , with four edge coefficients $[7] - [3]$, $[8]$, $[6]$, and $-[3]$.

3.3 An actual negative-Borel subquotient

Here B_K^- denotes the algebra generated by F_V, F_W , invertible K_V, K_W , with commuting Cartan generators, $K_V F_V K_V^{-1} = q^{-2} F_V$, $K_W F_W K_W^{-1} = q^{-2} F_W$, commuting F_V, F_W , and trivial cross Cartan commutators. These are the two $\mathfrak{sl}_2$ Cartan ratios in the original $U_q(\mathfrak{gl}_2 \oplus \mathfrak{gl}_2)$; the two central total-weight generators act by q^{15} .

Take the finite directed support graph on the actual source basis of X : an arrow $b \rightarrow b'$ of colour V or W means the corresponding coefficient in $F_V b$ or $F_W b$ is nonzero in K . Let $\mathcal{D}$ be

all descendants of T , including T , and let $\mathcal{J}$ be the descendants which cannot reach Q , with length-zero paths allowed. Both sets are closed under outgoing arrows. Indeed a descendant's successor is a descendant; and a successor which could reach Q would give a path to Q from its predecessor. Consequently

$$X_{\mathcal{D}} = \bigoplus_{b \in \mathcal{D}} Kb, \quad X_{\mathcal{J}} = \bigoplus_{b \in \mathcal{J}} Kb$$

are B_K^- -submodules, since Cartan acts diagonally. Their quotient $M = X_{\mathcal{D}}/X_{\mathcal{J}}$ is an actual source subquotient. This construction uses the span of support descendants; it makes no claim that the span equals the cyclic submodule generated by T .

Every path from T to Q has two V arrows and no W arrows: each W arrow lowers the first W coordinate by one, which cannot be restored, and each V arrow lowers the first V coordinate by one. The exhaustive calculation above therefore proves $\mathcal{D} \setminus \mathcal{J} = \{T, T_1, T_2, Q\}$. In this ordered quotient basis, denote the vectors by e_0, e_1, e_2, e_3 . Set

$$\alpha = [7] - [3], \quad \beta = [8], \quad d = [3], \quad z = q^3 + q^{-3}, \quad \gamma = [6] = dz, \quad \delta = -d.$$

Then every entry of the induced operators is

$$N = [F_V] = \begin{pmatrix} 0 & 0 & 0 & 0 \\ \alpha & 0 & 0 & 0 \\ \beta & 0 & 0 & 0 \\ 0 & \gamma & \delta & 0 \end{pmatrix}, \quad [F_W] = 0, \quad [K_V] = \text{diag}(q^9, q^7, q^7, q^5), \quad [K_W] = q^3 I_4.$$

Direct multiplication verifies every negative-Borel relation, including $K_V N = q^{-2} N K_V$ and the cross relations. In particular the four vectors are not asserted to span an invariant subspace inside X : the discarded vectors in the first-step table are genuinely present before taking this specified quotient.

The Laurent identity calculated from the two paths is

$$C = \frac{[6]([7] - [3]) - [3][8]}{[2]} = q^{10} - q^4 - q^{-4} + q^{-10} = r^2 [7][3]. \quad (10)$$

For a direct verification, $[6] = [3](q^3 + q^{-3})$ and $(q^3 + q^{-3})([7] - [3]) - [8] = [2]r^2[7]$ follow by multiplying their displayed finite Laurent sums; substituting gives the last equality, and expanding gives the four displayed terms. Thus

$$N^2 = pD, \quad D = CE_{30}, \quad N^3 = ND = DN = D^2 = 0.$$

Here $E_{30}e_0 = e_3$ and it kills the other three vectors. This is the divided square $F_V^{(2)} = F_V^2/[2]$, with its denominator retained until the Laurent divisibility has been proved.

Define $M_A = \bigoplus_{i=0}^3 Ae_i$. It is precisely the quotient of the A -spans of $\mathcal{D}$ and $\mathcal{J}$ in the source integral form: the source Chevalley coefficients are in A , so both spans are stable and their quotient has this basis. The displayed matrices preserve M_A . The induced divided powers preserve it as well: $F_V^{(0)} = I$, $F_V^{(1)} = N$, $F_V^{(2)} = D$, and higher V divided powers and all positive W divided powers vanish. The only product identity with a nonzero product of positive divided powers is $F_V^{(1)} F_V^{(1)} = [2]F_V^{(2)}$, already verified. All other divided-power product identities have both sides zero. This specifies the divided-power integral negative-Borel action on this subquotient.

There is no extension of these four matrices to the full original quantum group. Such an extension would require an endomorphism E_V satisfying $E_V N - N E_V = (K_V - K_V^{-1})/(q - q^{-1})$. The left side has trace zero because $\text{tr}(AB) = \text{tr}(BA)$, whereas the right side has trace

$[9] + 2[7] + [5]$, a nonzero Laurent polynomial (its q^8 coefficient is one). This proves the exact obstruction to that extension. The full source module X , with its additional weights and all four Chevalley generators, is constructed explicitly in Section 4. Its original 1260 basis vectors, all divided powers, and the 420-dimensional domain with 416-dimensional kernel give the exact quotient morphism to M . The commutator obstruction on M itself remains the one just proved.

3.4 The retained integral kernel, cokernel, and specialization

We first retain the original base $A = \mathbb{Z}[q, q^{-1}]$ before inverting 2. Define

$$a_* = q^5 + q^{-5}, \quad b_* = q^6 + q^2 + q^{-2} + q^{-6}, \quad I = (a_*, b_*) \subset A.$$

Then $I = (4, q^2 + 1) = (4, p)$. Here is an explicit proof of the ideal equality. Put $x = q^2$, $a_x = x^5 + 1 = q^5 a_*$ and $b_x = x^6 + x^4 + x^2 + 1 = q^6 b_*$. Direct multiplication gives

$$(x^4 + x^3 + x^2 + x + 1)a_x - x^2(x + 1)b_x = x + 1.$$

The polynomials displayed below in the identity $u_* a_* + v_* b_* = 1$ have denominators only 4, so multiplication by 4 gives $4 \in I$ already over A . Conversely a_x is divisible by $x + 1$, and division of b_x by $x + 1$ has remainder 4 because $b_x(-1) = 4$. This proves both containments. Since q is invertible and $q^2 + 1 = qp$, the second equality follows. In particular $A/I \simeq (\mathbb{Z}/4)[q]/(q^2 + 1)$, with inverse of q equal to $-q$; the monic relation makes this free of rank two over $\mathbb{Z}/4$.

The equation $Nx = 0$ gives, over the domain A ,

$$\ker_A N = A(e_1 + ze_2) \oplus Ae_3.$$

For the cokernel let $P = A^2/pA(a_*, b_*)$. Since the first nonzero column of N is $p(a_*, b_*)$ and the last row generates dAe_3 ,

$$\text{coker}_A N = Ae_0 \oplus P \oplus A/(d).$$

There is the exact sequence

$$0 \longrightarrow A/(p) \xrightarrow{\bar{f} \mapsto [f(a_*, b_*)]} P \xrightarrow{[(x_1, x_2)] \mapsto -b_* x_1 + a_* x_2} I \longrightarrow 0.$$

Well-definedness and surjectivity of the last map follow from its formula. The two elements a_*, b_* have no common nonunit divisor: a common divisor would divide both 4 and $q^2 + 1$, which is impossible in the unique factorization domain A . Thus $a_* x_2 = b_* x_1$ implies $x_1 = a_* f$, $x_2 = b_* f$ in A by prime-factor divisibility. Quotienting this kernel by $pA(a_*, b_*)$ gives exactly $A/(p)$, proving injectivity of the first map and exactness in the middle. This records the original 2-primary ideal defect which is removed by the base change made next.

At $q = 1$ over $\mathbb{Z}$, the image in the middle two coordinates is generated by $(4, 8)$ and the bottom-coordinate image is $3\mathbb{Z}$. The integral basis $(e_1 + 2e_2, e_2)$ therefore proves $\text{coker}_{\mathbb{Z}} N_0 \simeq \mathbb{Z}^2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/3$.

For the node comparison make the explicit base change $A \rightarrow R = \mathbb{Z}[1/2][q, q^{-1}]$, and put $O = \mathbb{Z}[1/2]$. No quantum integer, 3, 7, or $q - 1$ is inverted. The finite-sum identities give $\alpha = pa_*$, $\beta = pb_*$ and $za_* - b_* = r^2[7]$. For an explicit Bezout identity put

$$u_* = \frac{q^5}{4}(-q^{10} + q^8 + 2q^6 + 2q^4 + q^2 + 3),$$

$$v_* = \frac{q^6}{4}(q^8 - q^6 - 3q^4 - q^2 + 1).$$

Multiplication gives $u_*a_* + v_*b_* = 1$. Hence $f = a_*e_1 + b_*e_2$ and $g = -v_*e_1 + u_*e_2$ are an R -basis of $Re_1 \oplus Re_2$: the determinant of their two columns is one and their inverse sends $x_1 + ye_2$ to $(u_*x + v_*y)f + (-b_*x + a_*y)g$.

For $x = \sum x_i e_i$, the equation $Nx = 0$ first gives $x_0 = 0$, because α is nonzero in the domain R . The last coordinate then gives $d(zx_1 - x_2) = 0$, hence $x_2 = zx_1$. Therefore

$$\ker_R N = R(e_1 + ze_2) \oplus Re_3.$$

Also $Ne_0 = pf$, $Ne_1 = dze_3$, $Ne_2 = -de_3$, $Ne_3 = 0$, so $\text{im } N = Rpf \oplus Rde_3$ and the explicit quotient map is

$$\sum x_i e_i \mapsto (x_0, -b_*x_1 + a_*x_2, \overline{u_*x_1 + v_*x_2}, \overline{x_3}) \in R \oplus R \oplus R/(p) \oplus R/(d).$$

Its kernel is exactly the displayed image, by the inverse basis formula, and its surjectivity follows by lifting the four coordinates. Thus this is a complete cokernel description, including integral torsion.

Specialize by $R \rightarrow O$, $q \mapsto 1$, and denote reduction by a subscript zero. Since $\alpha_0 = 4$, $\beta_0 = 8$, $\gamma_0 = 6$, $\delta_0 = -3$,

$$N_0 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 0 \\ 8 & 0 & 0 & 0 \\ 0 & 6 & -3 & 0 \end{pmatrix}, \quad N_0^2 = 0, \quad \ker_O N_0 = O(e_1 + 2e_2) \oplus Oe_3, \quad \text{coker}_O N_0 \simeq O^2 \oplus O/(3).$$

The same coordinate proof establishes the kernel and cokernel after specialization; $p_0 = 2$ is a unit of O but $d_0 = 3$ is not. In particular the integral torsion cannot be inferred from rational rank alone. The kernel base-change map takes the two displayed generators to the two displayed generators and is an isomorphism.

Over K , $N^2 \neq 0$, $N^3 = 0$, and $\ker N$ has dimension two. For completeness the columns $e_0, Ne_0, N^2e_0, e_1 + ze_2$ have determinant $-p^2r^4[7]^2[3] \neq 0$. Thus they give the exact chain decomposition with Jordan block lengths $(3, 1)$. Over $\mathbb{Q}$ at $q = 1$, the columns e_0, N_0e_0, e_2, N_0e_2 have determinant $-12 \neq 0$, giving block lengths $(2, 2)$. These are proved changes of basis over their stated fields, not identifications of integral lattices.

The unrescaled tableau span has ordered basis

$$(e_0, e_1, -pe_2, -pe_3).$$

Its inclusion into M_R has matrix $\text{diag}(1, 1, -p, -p)$. Its inclusion cokernel is $(R/(p))^2$, and its kernel is zero because R is a domain. It is stable under N : the four images in that ordered basis are respectively $\alpha e_1 - (\beta/p)(-pe_2)$, $-(\gamma/p)(-pe_3)$, $\delta(-pe_3)$, and zero. Here $\beta/p = b_*$ and $\gamma/p = [3](q^2 - 1 + q^{-2})$ lie in R . It need not be stable under divided powers: $De_0 = -(C/p)(-pe_3)$, and p does not divide C . Indeed evaluation at $q = i$ over $\mathbb{C}$ gives $p(i) = 0$, $r(i)^2 = -4$, $[7](i) = -1$, $[3](i) = -1$, so $C(i) = -4 \neq 0$. This retains the distinction between the actual source-rescaled lattice and the raw tableau lattice.

3.5 An equivariant endpoint map and the original node

The flag $Re_3 \subset Re_1 \oplus Re_2 \oplus Re_3 \subset M_R$ is negative-Borel stable, as follows directly from the matrices. Let $L_{\text{top}} = M_R/(Re_1 \oplus Re_2 \oplus Re_3)$ and $L_{\text{bot}} = Re_3$. The first line has Cartan ratios (q^9, q^3) and the second (q^5, q^3) ; both lowering operators kill both lines. Let χ be the rank-one negative-Borel module with V -weight $(-2, 2)$ and W -weight $(0, 0)$, hence Cartan ratios $(q^{-4}, 1)$ and zero lowering operators. With the usual tensor-product action, define

$$\Phi : L_{\text{top}} \otimes \chi \longrightarrow L_{\text{bot}}, \quad \bar{e}_0 \otimes 1 \mapsto Ce_3.$$

This is the actual map induced by $F_V^{(2)}$ on M_R : it kills the submodule divided out, and its image is $CR\epsilon_3$ inside the displayed bottom line. Its Cartan actions agree after the specified twist, and both lowering actions vanish on source and target, proving equivariance on every generator. The map is injective because $C \neq 0$ in R , and its exact cokernel is $(R/(C))\epsilon_3$.

Here is the exact comparison with the original two-sheet node, including all coordinates. Over $O[a]$ set

$$B = O[a, r_1, r_2]/(r_1^2 + a, r_2^2 + a), \quad N_{\text{node}} = O[y] \oplus O[y], \quad a = (-y^2, -y^2).$$

Restriction sends $r_1 \mapsto (y, y)$ and $r_2 \mapsto (y, -y)$. In the ordered bases $(1, r_1, r_2, r_1 r_2)$ and $((1, 1), (y, y), (y, -y), (1, -1))$, respectively, it is $\text{diag}(1, 1, 1, -a)$: the last image is $(y^2, -y^2) = -a(1, -1)$. Monic division proves that the first is a basis; the even/odd decomposition and invertibility of 2 prove that the second is a basis. Substitute the original base map $a = -r^2$ by $O[a] \rightarrow R$. The resulting restriction map is

$$V : B_R \longrightarrow (N_{\text{node}})_R, \quad [V] = \text{diag}(1, 1, 1, r^2).$$

Write $h = [7][3]$. Stabilizing the actual endpoint map Φ by three identity maps gives a rank-four map $\hat{\Phi}$ with matrix $\text{diag}(1, 1, 1, C)$. Give both endpoint coordinates their already specified negative-Borel character (q^5, q^3) and the three added coordinates any fixed equal characters on source and target. Give the intermediate and node modules these same diagonal actions. These explicit module actions make the following maps equivariant as module maps; no action by algebra automorphisms of the node ring is asserted:

$$\hat{\Phi} = VH, \quad H = \text{diag}(1, 1, 1, h).$$

Both maps are injective, and, inside the common target, the image lattices satisfy $L_C = R^3 \oplus CR \subset L_r = R^3 \oplus r^2 R$. Consequently

$$0 \longrightarrow R/(h) \xrightarrow{\bar{f} \mapsto r^2 \bar{f}} R/(C) \xrightarrow{\bar{f} \mapsto \bar{f}} R/(r^2) \longrightarrow 0.$$

The first map is well defined because $r^2 h = C$; its kernel is zero by cancellation of r^2 . The kernel of the second map is precisely the classes of $r^2 f$, and surjectivity follows by lifting a representative. This proves the full sequence and connects the actual GCT generator endpoint to the retained conductor lattice.

Finally put $t = q - 1$, $u = (q + 1)/q$, so $r = tu$, $u(1) = 2$ and $h(1) = 21$. The first nonzero t -coefficient of $C = t^2 u^2 h$ is $84t^2$; equivalently $C(1) = C'(1) = 0$ and $C''(1) = 168$. The two rank-four maps V and $\hat{\Phi}$ specialize to $\text{diag}(1, 1, 1, 0)$, with kernel and cokernel O on the fourth coordinate. The induced kernel map from H is 21, and the cokernel map is the identity. To retain the full derived sequence, the free resolution of $O = R/(t)$ by multiplication by t gives

$$0 \longrightarrow O \xrightarrow{21} O \xrightarrow{\text{mod } 21} O/(21) \xrightarrow{0} O \xrightarrow{1} O \longrightarrow 0.$$

There is no omitted initial Tor term: if $tf = hg$ in R , reduction at $q = 1$ gives $21g(1) = 0$ in the domain O , hence $g = tg_1$. Cancellation of t then gives $f = hg_1$. Therefore $\text{Tor}_1^R(R/(h), O) = 0$. The resolution has length one, so all higher Tor groups vanish as well. Indeed the two Tor generators are C/t and r^2/t ; projection takes the first to h times the second and hence to 21 after reduction. Lifting r^2/t to $R/(C)$ and multiplying by t gives r^2 , the image of $1 \in R/(h)$, proving the positive sign of the connecting map. The next map is multiplication by r^2 and vanishes modulo t ; the final map is the identity. Multiplication by 21 is injective on O , with exactly the displayed cokernel. No passage to a field has discarded this residual module.

The finite interval, its exact source quotient maps, the divided-power endpoint, the two integral lattices, and this conductor correspondence are therefore constructed. They do not constitute the full generator representation of $\check{X}_{(7,5,3)}$ or a Frobenius realization of the non-standard Riemann-hypothesis programme.

4 The complete original GCT-IV module and its integral projection

Let $A = \mathbb{Z}[q, q^{-1}]$, $K = \mathbb{Q}(q)$, $p = q + q^{-1}$, and $[j] = q^{j-1} + q^{j-3} + \dots + q^{1-j}$ for $j > 0$. This section constructs the four original Chevalley matrices on $\check{X}_{(7,5,3)}$, rather than extending the four-dimensional negative-Borel quotient of Section 3. The column shape is exactly $\alpha = (3, 3, 3, 2, 2, 1, 1)$; the row shape is $(7, 5, 3, 0)$; the column-count shape is $[0, 3, 2, 2]$. The source parameter, denoted u in its TeX, remains the indeterminate q here.

The construction is finite and explicit. The file `agents/full_source_generators/full_generators.json` contains every entry of all four 1260×1260 matrices and the image of every original tabloid. Its coefficient convention is a list of pairs (k, a_k) denoting the Laurent polynomial $\sum_k a_k q^k$; its sparse matrix convention lists target indices in each source column. Indices start at zero. The file `source_basis/basis_1260.json` gives every original column label, scalar, invariant record, twelve-tuple, and weight. The definitions and recurrences below specify these files exactly; the attached Python programs implement those definitions with integer arithmetic and exhaustive, finite checks.

4.1 The original ambient basis and all adjacent relations

Write $Y = \check{Y}_\alpha$. Its original NST basis is indexed by the set $\mathcal{W}$ of all column tuples in

$$\{123, 124, 134, 234\}^3 \times \{12, 13, 23, 32, 24, 34\}^2 \times \{1, 2, 3, 4\}^2.$$

The product is the source based product, with the indicated column composition retained. Thus $|\mathcal{W}| = 4^3 6^2 4^2 = 36864$. These are column labels, not ordinary tensor monomials. In particular, the two columns 23 and 32 are distinct. The alphabet is $1 = (1, 1), 2 = (1, 2), 3 = (2, 1), 4 = (2, 2)$. For diagram calculations read each column from bottom to top, then extract its V and W letters:

c	V	W	c	V	W
1	1	1	2	1	2
3	2	1	4	2	2
12	11	21	13	21	11
23	21	12	32	12	21
24	21	22	34	22	21
123	211	121	124	211	221
134	221	211	234	221	212

Some selected canonical tensor summands of the height-three columns use other words. The displayed dictionary is the literal diagram reading rule; it avoids identifying a whole column with one summand of its canonical expansion.

Let R_K be the original sum of the six adjacent source submodules $\check{Y}_{\check{s}i_\alpha} \subset Y$. The following tables give all of its integral two-column basis relations. Each row is an equality of the images in Y/R_K ; the equality means that the difference belongs to the indicated local submodule before taking the quotient:

$(2, 1)$	$(23)(1) = (32)(1), \quad (23)(3) = (34)(1),$ $(24)(3) = (34)(2), \quad (24)(1) = (32)(2)$
$(3, 2)$	$(124)(13) = -(134)(12), \quad (234)(23) = -(234)(32),$ $(234)(13) = -(134)(32), \quad (124)(23) = -(234)(12)$
$(2, 2)$	$(32)(24) = (24)(32), \quad (12)(13) = (13)(12),$ $(34)(24) = (24)(34), \quad (12)(23) = (23)(12),$ $(32)(13) = (13)(32), \quad (34)(13) = (13)(34),$ $(34)(23) = (23)(34), \quad (12)(24) = (24)(12),$ $(32)(23) = (23)(32), \quad (24)(13) = (34)(12).$

The first nine $(2, 2)$ relations are its nine-dimensional degree-preserving component; the last is its one-dimensional invariant component. For an integral local relation the source based products for the submodule and ambient module agree, so these relations hold with every left and right NST context.

There are two further local strings at each equal height. The active side, its three words C_{11}, C_{12}, C_{22} , and its invariant I are as follows:

h	side	C_{11}	C_{12}	C_{22}	I
1	V	$(2)(1)$	$(2)(3)$	$(4)(3)$	$(4)(1)$
1	W	$(3)(1)$	$(3)(2)$	$(4)(2)$	$(4)(1)$
2	V	$(32)(12)$	$(32)(32)$	$(34)(32)$	$(34)(12)$
2	W	$(23)(13)$	$(23)(23)$	$(24)(23)$	$(24)(13)$
3	V	$(124)(123)$	$(124)(134)$	$(234)(134)$	$(234)(123)$
3	W	$(134)(123)$	$(134)(124)$	$(234)(124)$	$(234)(123)$

In the ambient two-column module their projected string vectors are $\tilde{C}_{11} = C_{11}$, $\tilde{C}_{12} = C_{12} + p^{-1}I$, $\tilde{C}_{22} = C_{22}$. Their contextual products in $\check{Y}_{\triangleright^i \alpha}$ give exactly

$$BC_{11}D = -p^{-1}B'ID, \quad BC_{12}D = -p^{-1}BID, \quad BC_{22}D = -p^{-1}BID'. \quad (11)$$

Here is the full meaning of B' and D' . Pair every 2 with the next available 1 to its right, using a stack of unmatched 2 positions. For 11, take the leftmost local unpaired 1. If it has no partner in B , set $B' = 0$; otherwise change its partner from 2 to 1 and apply the raising column Kashiwara operator at that column. For 22, take the rightmost local unpaired 2. If it has no partner in D , set $D' = 0$; otherwise change that partner from 1 to 2 and apply the lowering column Kashiwara operator there. In either case an extra internal arc makes the corrected word zero. The other context is unchanged. The middle equality changes no context. These are the source contextual projection formulas, with every outside letter retained. An isolated equality from this table cannot be substituted in a longer product while discarding B' or D' .

The integral table and these six strings exhaust the local submodule bases for the adjacent shapes occurring in α : $(3, 3)$ twice, $(3, 2)$ once, $(2, 2)$ once, $(2, 1)$ once and $(1, 1)$ once. Taking all contexts therefore spans exactly R_K , by its defining sum. This proves that the following relation construction presents the original $\check{X}_{(7,5,3)}$.

4.2 A finite exact quotient algorithm

Order $\mathcal{W}$ lexicographically using the column orders in its displayed Cartesian product. There are 22528 contextual instances of integral table rows and 47616 instances of (11). Each says $t_i = \epsilon p^k t_j$ with $\epsilon \in \{1, -1\}$ and $k \in \{0, -1\}$, or $t_i = 0$. Join the vertices by these equalities, keeping for each vertex an accumulated scalar ϵp^k relative to a chosen root. An equality within one component either agrees with its accumulated scalar, or yields $(1 - \epsilon p^k)t_{\text{root}} = 0$. In the latter case set that component to zero; do the same when a zero relation is present.

This is Gaussian elimination specialized to binomial relations over K . Indeed a spanning tree in each component expresses every vertex as its root times a nonzero scalar. Every remaining edge is then either $0 = 0$ or a nonzero scalar times that root. Consequently a component contributes exactly one free K coordinate if every cycle agrees and no zero relation is present, and contributes zero otherwise. For this original presentation the algorithm leaves exactly 1260 nonzero components; 18652 original words are zero and 18212 original words have nonzero image. The program checks every equation again through its computed projection.

This elimination takes place over K . Inverting a nonzero $1 - \epsilon p^k$ in a conflicting component does not prove that an integral presentation obtained merely by clearing denominators has no torsion. We make no such claim. The integral lattice and its exact kernel are constructed separately below inside the original field quotient.

4.3 The complete original honest basis and its signs

We specify the original basis at the original parameter. Put $K = \mathbb{Q}(q)$, $p = q + q^{-1}$, $\nu = (7, 5, 3, 0)$, and $\alpha = \nu' = (3, 3, 3, 2, 2, 1, 1)$. In this subsection a string such as 124 denotes a single column, read from top to bottom; parentheses list columns from left to right. In particular 23 and 32 are distinct nonstandard columns. The source column alphabets, in their source order, are

$$\mathcal{C}_3 = (123, 124, 134, 234), \quad \mathcal{C}_2 = (12, 13, 23, 32, 24, 34), \quad \mathcal{C}_1 = (1, 2, 3, 4).$$

All references to source lines below refer to the retained original `Apr13KroneckerGCT4.tex`, whose SHA-256 is

5c5b990306c3b817d822df9f5088fdd42dda146fc3449155a0b2f6202d2ed49e

The alphabets and their full canonical tensor expressions occur at lines 4630–4635; their order occurs at 6064.

The ambient source space is

$$\check{Y}_\alpha = \bigotimes_{j=1}^7 \check{\Lambda}^{\alpha_j} X, \quad \dim_K \check{Y}_\alpha = 4^3 6^2 4^2 = 36864.$$

Its NST basis uses the source projected canonical product: if $T = (C_1, \dots, C_7)$, its vector is $\check{p}_\alpha(C_1 \heartsuit \dots \heartsuit C_7)$. Here $\heartsuit$ concatenates the original upper canonical words in each of the V, W factors, and $\check{p}_\alpha$ is the source projector. This is the definition at 4751–4765 and agrees with the based tensor product by 4795–4805. Thus these words do not mean an unqualified pure tensor in the NSC basis. The original quotient is

$$\check{X}_\nu = \check{Y}_\alpha / \sum_{j=1}^6 \check{Y}_{>j\alpha}, \quad \check{Y}_{>j\alpha} = \check{Y}_{(\alpha_1, \dots, \alpha_{j-1})} \otimes \check{Y}_{>(\alpha_j, \alpha_{j+1})} \otimes \check{Y}_{(\alpha_{j+2}, \dots, \alpha_7)}.$$

These are exactly the objects in source 4918–4920, with the source local relation submodules. When written in the canonical NST product the nonintegral relations must retain their context, as in 5773–5785.

Here is an exhaustive finite index set for the honest basis. Choose $i_3, i_2, i_1 \in \{0, 1\}$ and nonnegative integers m_C for each column $C \in \mathcal{C}_3 \sqcup \mathcal{C}_2 \sqcup \mathcal{C}_1$, subject to

$$\sum_{C \in \mathcal{C}_3} m_C = 3 - 2i_3, \quad \sum_{C \in \mathcal{C}_2} m_C = 2 - 2i_2, \quad \sum_{C \in \mathcal{C}_1} m_C = 2 - 2i_1. \quad (12)$$

Retain precisely the tuples satisfying all the following source conditions:

$$\begin{aligned} m_2 m_3 &= 0, & m_1 > 0 &\implies m_{34} = m_{24} = m_{32} = 0, \\ m_2 > 0 &\implies m_{34} = 0, & m_3 > 1 &\implies m_{24} = 0, \\ m_{23} &\leq 1, & m_{32} &\leq 1, \\ m_{124} m_{134} &= 0, & m_{234} > 0 &\implies m_{12} = m_{13} = m_{32} = 0, \\ m_{134} > 0 &\implies m_{12} = 0, & m_{124} > 1 &\implies m_{13} = 0, \\ m_{234} m_1 &= 0. \end{aligned} \quad (13)$$

Within each height the invariant-free columns are in the displayed alphabet order. Equations (13) retain every condition of source 6114–6130; conditions with identical antecedents have been grouped without changing their content.

From the tuple form the exact source word

$$\begin{aligned}
T(m, i) = & 123^{m_{123}} 124^{m_{124}} 134^{m_{134}} (234, 123)^{i_3} 234^{m_{234}} \\
& 12^{m_{12}} 13^{m_{13}} 23^{m_{23}} 32^{m_{32}} (24, 13)^{i_2} 24^{m_{24}} 34^{m_{34}} \\
& 1^{m_1} 2^{m_2} 3^{m_3} (4, 1)^{i_1} 4^{m_4}.
\end{aligned} \tag{14}$$

Exponentiation of a column or pair means that many consecutive copies. The invariant insertion positions and orientations are those of source 6169. The invariant record is $(0, i_3, i_2, i_1)$ and the degree is $d = i_3 + i_2 + i_1$. Indeed the inserted pairs have two arcs connecting their columns and no arcs exiting the pair. Removing them gives the invariant-free word with multiplicities m_C . The source classification at 6263–6269 identifies these straightened words bijectively with the honest classes. The complete quotient computation in this chapter also checks, independently for this fixed shape, that the words in (14) project to distinct nonzero components and exhaust all nonzero quotient components. Thus both the representatives and the module are the original source objects.

For completeness we give an exact diagram algorithm. Read the columns bottom to top and then left to right and apply

$$1 \mapsto (1, 1), \quad 2 \mapsto (1, 2), \quad 3 \mapsto (2, 1), \quad 4 \mapsto (2, 2).$$

This produces the original length-15 V and W words (source 4723). For each word scan left to right, pushing every 2 on a stack; a 1 pairs with the last unmatched 2 if there is one. Otherwise it remains unpaired. This is exactly the source parenthesis rule 2448. Record the column containing each letter. An arc is external when its ends lie in different columns; two arcs between a pair of columns give an invariant pair (source 4938–4954). A two-edge component involving one column of each height 3, 2, 1 is a 3–2–1 arc. Its color is the common edge color, or the color of the longer edge if the colors differ, as in source 6320. All other remaining edges are pure. A height-2 column with no external edge is externally free; its type is V for 12, 32, 34 and W for 13, 23, 24 (source 6068–6073).

The exact twelve-entry label is

$$\zeta = (i_4, i_3, i_2, i_1, a_{21}, a_{32}, a_{31V}, a_{31W}, a_{321V}, a_{321W}, f_V, f_W). \tag{15}$$

Here a_{21}, a_{32} count pure arcs, the other a 's retain their specified source colors, and f_V, f_W count externally free columns. This is the original order in source 7233–7238. Together with the two unpaired words, necessarily of the form $1^a 2^b$, it determines the honest class by 6473–6481. Original torus weights count all letters, including paired letters: each factor's weight is $(\#1, \#2)$. This retains the determinant-weight contribution of every invariant.

A source sign ambiguity and its exact resolution. Source 5489–5490 specifies a positive class using the multiplier $(-1/p)^d$ and a representative with a W -colored 3–2 arc whenever the class has a 3–2 arc. It also asserts that having no 3–2 arc for every representative is equivalent to having none for any representative. That latter assertion fails; the 3–2–1 example at 6337–6388 already has equivalent representatives of both types. We preserve the original sign choice through its stated W -arc representative criterion and explicitly record the needed correction, instead of treating the inaccurate equivalence as a theorem.

Here is the exact integral witness relevant to this shape:

$$\begin{aligned}
(134, 13, 23, 1, x) &= (134, 13, 32, 1, x) \\
&= (134, 32, 13, 1, x) \\
&= -(234, 13, 13, 1, x), \quad x \in \{1, 2, 3, 4\}.
\end{aligned} \tag{16}$$

The first equality is the source integral $(23, 1) = (32, 1)$ relation at 5080. The second is the type- V /type- W commutation at 5139. The last is $(134, 32) = -(234, 13)$ at 5209. All are degree-preserving integral moves, so their insertion into the retained contexts is valid. The original left word has a 3–2–1 V -arc and no 3–2 edge; the right word has a W -colored 3–2 edge. Direct application of the pairing algorithm verifies both assertions for each of the four values of x . Consequently the positive vector represented by the left word is $-(-1/p)^d$ times that word.

Within the present shape, the additional words requiring this sign are exactly

$$\begin{aligned} &(123, 123, 134, 13, 23, 1, x), \quad (123, 134, 134, 13, 23, 1, x), \\ &(134, 134, 134, 13, 23, 1, x), \quad (134, 234, 123, 13, 23, 1, x), \\ &x \in \{1, 2, 3, 4\}. \end{aligned}$$

For the last family first use $(134, 234, 123) = (234, 123, 134)$, the invariant commutation at 5286–5287, then apply (16). The first three families use (16) directly in their height-3 prefixes. Exhaustiveness follows by evaluating (13) and the pairing algorithm on the finite index set; every index and its predicate values are computed by `enumerate_basis.py`, and independently by `audit/independent_audit.py`. All sixteen equalities are also checked against the full original quotient projection by `check_sign_witnesses.py`. In source lexicographic order their zero-based indices are 262–265, 478–481, 1000–1003, and 1108–1111.

Thus the final completely explicit scaling is

$$b_{m,i} = \epsilon(T)(-1/p)^d T, \quad \epsilon(T) = \begin{cases} -1, & T \text{ has a } V\text{-colored 3–2 edge,} \\ -1, & T \text{ has no 3–2 edge and has a 3–2–1 } V\text{-arc,} \\ 1, & \text{otherwise.} \end{cases} \quad (17)$$

The sixteen corrections affect neither the index set nor any original quotient relation. They give an explicitly specified diagonal sign change, with its exact inverse equal to itself. All four retained labels below have their original established scalings unchanged.

Exhaustive enumeration and a second exact bijection check. For each invariant record the numbers of words in (14) are as follows:

(i_3, i_2, i_1)	000	001	010	011	100	101	110	111
number	628	148	108	16	268	56	32	4.

Their sum is 1260; the degree counts are 628, 524, 104, 4 for $d = 0, 1, 2, 3$. All 1260 keys consisting of (15) and the two unpaired words are distinct. There are 48 distinct twelve-tuples. For each such tuple, if its two unpaired lengths are h_V, h_W , its vectors contain exactly all pairs

$$(1^{h_V-a} 2^a, 1^{h_W-b} 2^b), \quad 0 \leq a \leq h_V, \quad 0 \leq b \leq h_W.$$

The complete data are in `basis_1260.json` and `ssyt_bijection_check.json`.

We additionally verify the original map to ordinary semistandard tableaux, retaining every source case rather than using its cardinality alone. For clarity here is that map in multiplicity form. Let m'_C be the ordinary output multiplicities, initially equal to the invariant-free m_C ; set $m'_{32} = 0$. The columns 123, 234, 1, 4 keep their invariant-free multiplicities. The following tables specify all other multiplicities; entries not changed in a row keep their invariant-free

values. First, the right part (source 6195–6210):

condition	m'_2	m'_3	m'_{24}	m'_{34}
$m_1 > 0$	$i_1 + m_2$	$i_1 + m_3$	m_{24}	m_{34}
$m_1 = 0, m_{24} > 0, m_2 = 0$	0	$2i_1 + m_3$	m_{24}	m_{34}
$m_1 = 0, m_{24} > 0, m_2 > 0$	$i_1 + m_2$	i_1	m_{24}	0
$m_1 = m_{24} = 0, i_1 > 0$	$i_1 + m_2$	$i_1 + m_3$	m_{34}	0
$m_1 = m_{24} = i_1 = 0$	m_2	m_3	0	m_{34}

Next the middle part (source 6217–6232):

condition	m'_{14}	m'_{23}
$m_{234} = 0, m_1 > 0$	$2i_2 + m_{23}$	0
$m_{234} > 0, m_1 = 0$	0	$2i_2 + m_{23}$
$m_{234} = m_1 = m_{23} = 0$	$2i_2 + m_{32}$	0
$m_{234} = m_1 = 0, m_{23} > 0$	0	$2i_2 + m_{32} + m_{23}$

Finally the left part (source 6237–6250):

condition	m'_{124}	m'_{134}	m'_{12}	m'_{13}
$m_{234} > 0$	$i_3 + m_{124}$	$i_3 + m_{134}$	m_{12}	m_{13}
$m_{234} = 0, m_{13} > 0, m_{134} = 0$	$2i_3 + m_{124}$	0	m_{12}	m_{13}
$m_{234} = 0, m_{13} > 0, m_{134} > 0$	i_3	$i_3 + m_{134}$	0	m_{13}
$m_{234} = m_{13} = 0, i_3 > 0$	$i_3 + m_{124}$	$i_3 + m_{134}$	0	m_{12}
$m_{234} = m_{13} = i_3 = 0$	m_{124}	m_{134}	m_{12}	0

Conditions (13) imply that exactly one row of each table applies. The three tables write disjoint groups of output multiplicities and use only original m_C , so they do not interfere. Arrange the output columns in ordinary column lexicographic order.

The exact finite verification in `check_ssyt_bijection.py` enumerates every tuple in (12) by combinations with repetition, tests every predicate (13), and evaluates the three tables. Independently it enumerates every ordinary height-3 column multiset of size 3, height-2 column multiset of size 2, and height-1 column multiset of size 2. Each ordinary column is a strictly increasing sequence in $\{1, 2, 3, 4\}$; every common row coordinate of adjacent columns is tested for weak increase. These are exactly the defining semistandard conditions, and the enumeration has no other exclusion. The two sets are equal, have 1260 elements, and the map's 1260 outputs are distinct. The complete inverse is therefore given without an existence assumption by the reversed finite correspondence in `ssyt_correspondence.json`. As a further exact check the ordinary Weyl product is

$$\prod_{1 \leq i < j \leq 4} \frac{\nu_i - \nu_j + j - i}{j - i} = 3 \cdot 3 \cdot \frac{10}{3} \cdot 3 \cdot \frac{7}{2} \cdot 4 = 1260.$$

The numerical identity is a check on the explicitly constructed bijection, not a replacement for its construction.

Retained original labels. In the zero-based basis order defined above,

$$\begin{aligned} b_{126} &= T = (123, 123, 124, 12, 12, 1, 1), \\ b_{146} &= T^1 = (123, 123, 124, 23, 12, 1, 1), \\ b_8 &= T^2 = -p^{-1}(123, 123, 123, 12, 12, 4, 1), \\ b_{144} &= Q = -p^{-1}(123, 123, 124, 13, 12, 4, 1). \end{aligned}$$

For T^1 and Q , the lexicographic word has the 12 column before 23 and 13, respectively; their equality with the displayed original representatives is the integral type- V /type- W commutation. Their twelve-tuples and unpaired lengths are

$$\begin{array}{l|l|l} T & (0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 2, 0) & (9, 3) \\ T^1 & (0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0) & (7, 3) \\ T^2 & (0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 2, 0) & (7, 3) \\ Q & (0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0) & (5, 3). \end{array}$$

All these unpaired words consist entirely of 1's. The original weights (wt_V, wt_W) are respectively $((12, 3), (9, 6)), ((11, 4), (9, 6)), ((11, 4), (9, 6)),$ and $((10, 5), (9, 6)).$

The literal source target at 7333 is retained separately:

$$b_{154} = T'_{\text{printed}} = -p^{-1}(123, 123, 124, 23, 12, 4, 1).$$

It has the same twelve-tuple as Q but unpaired words (11111, 112) and original weights $((10, 5), (8, 7)).$ Since F_V commutes with the W torus, its coefficient in $F_V^2 T$ or $F_V^{(2)} T$ is zero. This is the precise weight obstruction to the printed target; the corrected Q retains the already proved contextual endpoint and is not silently identified with the printed object.

Original ambient generator algorithm. For $Z = V, W$, let $\varphi_Z(T)$ count its unpaired 1's. Source 4881–4908 gives

$$F_Z T = \sum_{j=1}^{\varphi_Z(T)} [j] F_{(j)Z} T.$$

The j th operation changes the column containing the j th unpaired 1, in left-to-right order, by its local Kashiwara arrow; it is zero if changing that selected letter creates an extra internal arc. For E_Z , replace the j th unpaired 2 counted from the right, with coefficient $[j]$, exactly as in source 2508–2517. The local F strings are

$$\begin{array}{l|l} V & 1 \rightarrow 3 \quad 2 \rightarrow 4 \quad 123 \rightarrow 134 \quad 124 \rightarrow 234 \quad 12 \rightarrow 32 \rightarrow 34 \\ W & 1 \rightarrow 2 \quad 3 \rightarrow 4 \quad 123 \rightarrow 124 \quad 134 \rightarrow 234 \quad 13 \rightarrow 23 \rightarrow 24. \end{array}$$

All unlisted arrows are zero, and the local E arrows reverse these strings. Choosing the first of the two unpaired 1's in a height-2 spin-one column creates an extra internal arc; the second gives its first F arrow. This zero test is performed for each summand before the contextual quotient reduction. Scaling the input by (17), reducing in the original quotient, and expressing each output in that same signed basis specifies every original Chevalley coefficient. The complete matrices, divided powers, and their exact checks are supplied by the full quotient construction in this chapter.

For each of the 1260 displayed representatives b_n , let $b_n = \sigma_n p^{-d_n} \bar{t}_{w_n}$, with its fully specified sign and degree. Each nonzero relation component contains exactly one such representative. This is checked by matching all 1260 roots; hence the vectors b_n are an actual basis of the original quotient. For every original word w the table records

$$\pi(t_w) = 0 \quad \text{or} \quad \pi(t_w) = \epsilon_w p^{k_w} b_{n_w}. \quad (18)$$

The algorithm produces only $k_w \in \{0, 1, 2, 3\}$. The numbers of nonzero words with these four exponents are 12544, 5100, 552, 16, respectively. No sign is suppressed in the table.

4.4 Every entry of all four original generators

For a side $Z \in \{V, W\}$ concatenate the displayed diagram words of an original tabloid w . Let its unpaired ones from left to right be $a_1, \dots, a_\varphi$, and its unpaired twos from left to right be $c_1, \dots, c_\varepsilon$. Define $\mathcal{F}_{jZ}(w)$ by changing the column containing a_j with its lowering column Kashiwara operator; define $\mathcal{E}_{jZ}(w)$ by raising the column containing $c_{\varepsilon-j+1}$. An extra internal arc gives zero. To specify the column operator without an implicit convention: the other side's weight and internal-arc number stay fixed, the chosen side's number of twos changes by one, and the chosen side's original internal-arc number stays fixed. Among the columns of that same height there is exactly one such column. If the intermediate changed diagram has more internal arcs than the old diagram, the result is zero instead.

The original ambient action is exactly

$$F_Z t_w = \sum_{j=1}^{\varphi_Z(w)} [j] t_{\mathcal{F}_{jZ}(w)}, \quad E_Z t_w = \sum_{j=1}^{\varepsilon_Z(w)} [j] t_{\mathcal{E}_{jZ}(w)}. \quad (19)$$

Terms whose changed word is zero are omitted. Apply (18) to every summand and multiply by $\sigma_n p^{-d_n}$ to obtain the column for $g b_n$. Equivalently, for $g = F_Z$ or E_Z its (m, n) entry is the finite sum

$$(G_g)_{mn} = \sum_{j: \pi(t_{g_j Z}(w_n)) = \epsilon_j p^{k_j} b_m} [j] \sigma_n \epsilon_j p^{k_j - d_n}. \quad (20)$$

This supplies every matrix entry, including all cancellations, signs, and possible intermediate denominators.

The nonzero entry counts are

g	F_V	E_V	F_W	E_W
$\# \text{supp}(G_g)$	1998	2083	1998	2083.

Every resulting entry lies in A : finite integral Laurent addition and division by p with remainder zero verify this directly. The original projection commutes with the action on every one of the 36864 ambient words for each of the four generators, including all 18652 words with zero image. These 147456 equalities prove descent throughout the original relation space, independently of the choice of representatives.

Let v_n and w'_n be the numbers of V and W ones in the fifteen-letter diagram of b_n . The four diagonal $\mathfrak{gl}_2$ weight generators have eigenvalues $q^{v_n}, q^{15-v_n}, q^{w'_n}, q^{15-w'_n}$. Thus the two Cartan ratios are $K_V b_n = q^{2v_n-15} b_n$ and $K_W b_n = q^{2w'_n-15} b_n$; their central products are q^{15} . The exact matrices satisfy

$$[E_Z, F_Z] = \frac{K_Z - K_Z^{-1}}{q - q^{-1}}, \quad [g_V, h_W] = 0 \quad (g, h \in \{E, F\}),$$

and $K_Z E_Z K_Z^{-1} = q^2 E_Z$, $K_Z F_Z K_Z^{-1} = q^{-2} F_Z$, with trivial cross Cartan commutators. These equalities are checked as equality of integer Laurent coefficient dictionaries in every matrix column. They also follow by descent from (19), the source graphical action on the original tensor product. Thus these are the full original $U_q(\mathfrak{gl}_2 \oplus \mathfrak{gl}_2)$ generators, with their exact quotient identification.

4.5 Integral lattices, divided powers, and specialization

Define $L_A = \bigoplus_{n=0}^{1259} A b_n \subset Y/R_K$. The exact Laurent integrality just proved makes this an integral generator lattice. For each $g \in \{E_V, F_V, E_W, F_W\}$ define the matrices recursively by

$$G_g^{(0)} = I, \quad G_g^{(m)} = G_g G_g^{(m-1)} / [m].$$

This retains the original divided-power denominator until the division is established. The certificate computes each coefficient by exact integral Laurent division and verifies zero remainder for every $m = 1, \dots, 10$. Both raising and lowering matrices have $G_g^{(9)} \neq 0$ and $G_g^{(10)} = 0$. Consequently all divided powers preserve L_A and every higher divided power is zero. The nonzero-entry counts are

m	0	1	2	3	4	5	6	7	8	9	10
F_V, F_W	1260	1998	2314	2100	1489	892	447	185	56	12	0
E_V, E_W	1260	2083	2425	2082	1486	907	452	186	56	12	0

Over K the recurrence is $G_g^m/[m]!$, by induction. The divided-power identities therefore hold over K ; both sides are integral matrices, and the injection $A \hookrightarrow K$ proves the same identities over A . This proves the full divided-power integral action, not only the four ordinary generator identities.

There is also an exact map from the original ambient integral word span. Put $Y_A = \bigoplus_{w \in \mathcal{W}} A t_w$ and use (18) to define $\pi_A : Y_A \rightarrow L_A$. Its exponents are nonnegative, so this map is defined over A . For each n choose the first word $w(n)$ in the fixed order whose projection is $\epsilon(n)b_n$, with exponent zero. The exhaustive table contains such a word for every n . Define $s_A(b_n) = \epsilon(n)t_{w(n)}$. Then $\pi_A s_A(b_n) = b_n$ for each basis vector, proving $\pi_A s_A = I$.

Here is the entire kernel, with no unspecified torsion. For a nonpivot word w , set

$$h_w = \begin{cases} t_w, & \pi_A(t_w) = 0, \\ t_w - \epsilon_w \epsilon(n_w) p^{k_w} t_{w(n_w)}, & \pi_A(t_w) = \epsilon_w p^{k_w} b_{n_w}. \end{cases}$$

The 35604 vectors h_w form an A -basis of $\ker \pi_A$. Their image is zero by the displayed formula. They are independent because in a relation among them the coefficient of each distinct nonpivot original word is its own coefficient. Subtracting $\sum_w a_w h_w$ from any kernel vector leaves only pivot words; their images are signed distinct basis vectors of L_A , so all remaining coefficients vanish. This proves spanning. Therefore

$$0 \longrightarrow A^{35604} \xrightarrow{h} Y_A \xrightarrow{\pi_A} L_A \longrightarrow 0 \quad (21)$$

is split exact as a sequence of A -modules, with cokernel zero. Its section is a specified linear section; no equivariance of that section is asserted. The projection itself is equivariant by the 147456 descent identities. The ambient graphical module has its original divided-power action, so its kernel is stable under that action as well. The section and all kernel vectors are listed in `raw_integral_projection.json`.

Upon $q = 1$ over $\mathbb{Z}$, $p = 2$, and the same formulas with $p^{k_w} = 2^{k_w}$ remain a split exact sequence of free abelian groups. In particular the natural kernel base-change map is an isomorphism. Every full generator and divided-power matrix specializes by replacing $\sum a_k q^k$ by $\sum a_k$; `specialized_generators.json` gives all these exact integer entries. Thus this specialization retains every coefficient and has rank 1260 over $\mathbb{Z}$. This is a statement about the actual image lattice inside the field quotient. It does not replace the unsaturated integral span of the original binomial relations by its saturation without identifying the map.

For clarity the corresponding saturation map can also be stated exactly. Let $P_A \subset Y_A$ be generated by the integral contextual rows and by every nonintegral row $pt_w + t_{w'}$ (or t_w when its correction is zero). Then $P_A \subset \ker \pi_A$ and $K \otimes_A P_A = K \otimes_A \ker \pi_A = R_K$. The induced map $Y_A/P_A \rightarrow L_A$ is onto and has exactly the torsion kernel $(\ker \pi_A)/P_A$. Here “torsion” follows because tensoring that kernel with K is zero: each of its elements is annihilated by some nonzero element of A . It need not be zero. A fully explicit presentation of this kernel is obtained by expressing each displayed generator of P_A in the basis h_w : its coefficient at h_w is

just its coefficient at that nonpivot word, as proved in the elimination above. This preserves the original relation presentation and states the exact map to its torsion-free image lattice.

There is a sharper exact answer after a specified localization. Put $B = A[p^{-1}]$, retaining 2 as a nonunit. In this ring every tree scalar in the original relation graph is a unit. Elimination along a spanning tree is therefore an invertible change of presentation coordinates over B . For each connected component it leaves one root and relations $(1 - \epsilon p^k)\text{root} = 0$ for its non-tree edges, plus $\text{root} = 0$ if there is an explicit zero relation. The exhaustive graph has 1260 components with no defect and no zero relation, 4147 with an explicit zero relation, and 40 further components whose entire nonzero cycle ideal is generated by $1 - (-1) = 2$. Consequently the actual localized presentation has the exact decomposition

$$\begin{aligned} B^{36864}/P_B &\simeq B^{1260} \oplus (B/(2))^{40}, \\ \ker(B^{36864}/P_B \longrightarrow B \otimes_A L_A) &\simeq (B/(2))^{40}. \end{aligned} \tag{22}$$

The map to the image lattice is the previously specified original-word projection; the displayed isomorphism is given by the spanning-tree coordinates. This identifies the kernel of the actual map, with every original component listed in `quotient_audit/relation_graph_certificate.json`.

One of the forty torsion generators has the explicit original representative

$$u = (123)(123)(234)(23)(23)(1)(1).$$

The three original integral rows give, in succession,

$$\begin{aligned} u &= -v, & v &= (123)(123)(234)(32)(23)(1)(1), \\ v &= w, & w &= (123)(123)(234)(23)(32)(1)(1), \\ w &= u. \end{aligned}$$

The first row uses the adjacent $(3, 2)$ relation, the second commutes $(32)(23)$, and the third uses $(32)(1) = (23)(1)$. Thus $2u = 0$. Its component has six original vertices, no explicit zero relation, and cycle ideal exactly (2) ; the spanning-tree proof shows that $u \neq 0$ and generates one $B/(2)$ summand. Indeed $B/(2)$ is the localization of $\mathbb{F}_2[q, q^{-1}]$ at its nonzero element $q + q^{-1}$, so it is not the zero ring. This proves nonvanishing, not merely the absence of a proof that the torsion class vanishes.

4.6 The original rational specialization bridge

We first specify the source's map s_X , which is a map of quotients. For any original NST t_w , let $d(w)$ be its number of invariant column pairs, including pairs of different heights. Such a pair is detected by two arcs between its two columns. Define $Y_A^S = \bigoplus_w A(-p^{-1})^{d(w)} t_w$ inside Y . Let R_A^S be the source's sum of the correspondingly scaled adjacent submodule lattices: scale each contextual submodule basis element $\tilde{t}$ by $(-p^{-1})^{\deg(\tilde{t})}$, with its ambient filtered degree retained. Thus the source has the canonical surjection

$$s_X : Y_A^S / R_A^S \longrightarrow \pi(Y_A^S), \quad [y] \longmapsto \pi(y).$$

The identification of its localized relation lattice with P_B requires more than equality of their K -spans. We give both containments explicitly. By the source definition at lines 5620–5624, a contextual basis vector is $g = B_0 \heartsuit_g \tilde{C} \heartsuit_g D_0$, with B_0, D_0 ranging over all original NST bases of the two contexts. Its actual lattice generator at lines 5881–5889 is $(-p^{-1})^{\deg(g)} g$. For an integral local $\tilde{C}$, the equality at lines 5680–5685 is exact in Y : g equals the ambient based product of the listed binomial with its two contexts. Thus it is the corresponding

integral graph row with its original coefficients $+1, -1$ or $+1, +1$. For a nonintegral local $\tilde{C}$, the equality at lines 5773–5785 is also exact in Y : $g = t_w + p^{-1}t_{w'}$, or $g = t_w$ when the contextual correction is zero. This uses the explicit contextual corollary, not the weaker congruence for nonintegral vectors in lines 5680–5685. The cleared graph row is respectively pg or g . Therefore each source lattice generator is $\pm p^j$ times its corresponding graph row for a specified integer j , and conversely each graph row is $\pm p^{-j}$ times that source lattice generator. Every such coefficient is a unit in B ; no factor 2 is inverted in either containment. The contexts and local rows are exhaustive in both definitions. It follows that $B \otimes_A R_A^S = P_B$ as actual submodules of $B \otimes_A Y_A^S = B^{36864}$. Hence the localization of this exact source presentation is the presentation in (22); its localized map s_X has precisely the same nonzero kernel $(B/(2))^{40}$. In particular the source remark at lines 5901–5908 asserting equality of its two integral forms cannot be used as an injectivity proof for these literal prescribed contextual lattices. The displayed original cycle gives the exact integral obstruction, while the rational specialization below constructs their isomorphism over its stated base. In this shape $\pi(Y_A^S) = L_A$: applying the explicit projection to each scaled word gives zero or $\epsilon(-1)^{d(w)}p^{k_w-d(w)}b_{n_w}$. All nonzero exponents are 0 or 1; their counts are 17728 and 484. Conversely the scaled honest representatives include every b_n . This proves both containments without assuming that s_X is injective over A .

Its rational specialization is injective, by the following additional argument. Every edge of the original relation graph has scalar ϵp^k , and every cycle product has that form. At $p = 2$ all edges remain nonzero, and $\epsilon 2^k = 1$ if and only if $\epsilon = 1$ and $k = 0$. These are exactly the conditions for $\epsilon p^k = 1$ in K . Hence the 1260 nonzero components remain nonzero, and every zero component remains zero. More specifically, the independently reconstructed graph has 4147 components with an explicit zero relation and 40 further components killed by a cycle of scalar -1 . The specialized relation space has rank $36864 - 1260 = 35604$ over $\mathbb{Q}$. At $q = 1$ the rescalings $(-p^{-1})^d = (-1/2)^d$ are nonzero rational numbers, so they change no rational span. Thus $\mathbb{Q} \otimes_A (Y_A^S/R_A^S)$ has dimension 1260. The specialization of s_X is a surjection onto $\mathbb{Q} \otimes_A L_A$, also of dimension 1260 by (21); hence it is an isomorphism. Its formula is the original quotient map, not a dimension comparison without a specified morphism.

The rational specialization of L_A is the restriction of the classical Schur module of row shape $(7, 5, 3, 0)$, through an explicit quotient map, as follows. Realize each original column at $q = 1$ in $(V \otimes W)^{\otimes h}$ by its source canonical column formula; take the ordinary exterior quotient $e_h : (V \otimes W)^{\otimes h} \rightarrow \bigwedge^h(V \otimes W)$ on each column factor. The restriction to the original nonstandard exterior subspace is a $\mathrm{GL}(V) \times \mathrm{GL}(W)$ isomorphism over $\mathbb{Q}$. To see injectivity, define $\mathrm{Alt}_h : \bigwedge^h X \rightarrow X^{\otimes h}$ by the unnormalized signed sum over all permutations. Directly from the alternating relations, $e_h \mathrm{Alt}_h = h!I$ on $\bigwedge^h X$ and $\mathrm{Alt}_h e_h = h!I$ on the alternating tensor subspace. The source subspace at $q = 1$ is exactly this alternating image, since the two tensor permutation actions on V, W combine to the permutation action on $X = V \otimes W$. Thus e_h has inverse $\mathrm{Alt}_h/h!$ on these subspaces. This proof retains the factor $h!$ and works over $\mathbb{Q}$, as stated.

Tensor these seven maps, keeping the original ordered column factors. Apply this tensor product linearly to the actual specialized based-product vector t_w and its tensor expansion. A based product at $q = 1$ need not be the ordinary tensor product of its column labels; replacing each label independently and concatenating would not specify this map. The source defines its two-column submodules as the exact U^τ summands removed by the classical two-column Schur quotient: this is its defining prescription at source lines 5015–5021, not an inference from the global dimension 1260. Here the ambient two-column module is itself multiplicity-free as a U^τ -module; the involution τ interchanges V, W and distinguishes otherwise repeated restrictions. We verify the small cases to justify uniqueness of the prescribed subspace. Write $D_V = \det V, D_W = \det W$. First $Y_{(1,1)}$ has the three distinct U^τ summands of dimensions

9, 6, 1, obtained from $(S^2V \oplus D_V) \otimes (S^2W \oplus D_W)$ by pairing the two cross terms. Next

$$\bigwedge^2 X = (S^2V \otimes D_W) \oplus (D_V \otimes S^2W).$$

Tensor with X and use $S^2V \otimes V = S^3V \oplus D_V V$ and its W version. The two cross outer terms pair to dimension 16; the two copies of $D_V V \otimes D_W W$ split into the two distinct τ signs, each of dimension 4. Thus $Y_{(2,1)}$ is multiplicity-free with dimensions 16, 4, 4, the two last summands having opposite signs. For $Y_{(2,2)}$ use $S^2V \otimes S^2V = S^4V \oplus D_V S^2V \oplus D_V^2$ and its W version in the previous exterior decomposition. The outer terms pair to dimensions 10 and 6; the two cross terms of dimension 9 split with opposite τ signs, and the two invariant lines also split with opposite signs. These six summands, of dimensions 10, 6, 9, 9, 1, 1, are pairwise inequivalent as U^τ -modules. The displayed tensor decompositions follow by the explicit symmetric product and alternating contraction maps for two-dimensional V, W : in each case the weights are the disjoint strings with respective highest weights $(a + b - j, j)$, for $0 \leq j \leq \min(a, b)$; their dimensions sum to $(a + 1)(b + 1)$, proving exhaustion. Finally $\bigwedge^3 X \simeq D_V D_W \otimes X$ with the corresponding overall τ sign. Hence $Y_{(3,2)}$ and $Y_{(3,3)}$ are character twists of $Y_{(1,2)}$ and $Y_{(1,1)}$ and remain multiplicity-free. This proves the ambient fact needed here. Selecting the source's specified classical quotient summands therefore uniquely determines its kernel as the sum of all remaining ambient summands. The local bases in the two tables are the explicit bases given for this prescription in the source's Figures **straightening11** through **straightening33**. Consequently the columnwise exterior maps take each such specialized local submodule to the classical adjacent straightening submodule. This is the exact local identification used in the source proof at lines 6013–6018; all its contexts are the unchanged tensor factors. Quotient by their sum on both sides. The relation graph argument above identifies the specialized relation space with the entire specialized kernel, so the resulting map is

$$s_{(7,5,3)} : \mathbb{Q} \otimes_A L_A \longrightarrow \left(\bigwedge^3 X \right)^{\otimes 3} \otimes \left(\bigwedge^2 X \right)^{\otimes 2} \otimes X^{\otimes 2} / \text{classical adjacent relations}.$$

Its formula takes any representative in Y_A to the class of its columnwise exterior images. Two representatives differ by the kernel, whose rational specialization was proved equal to the specialized relation space; hence the map is well defined and injective. It is onto because the columnwise exterior maps are onto. The target is the classical Schur module $S_{(7,5,3)}X$ by its exterior-column presentation, and each map is equivariant. This is the explicit rational specialization correspondence used for the source character calculation. No literal identification of a crystal-highest tabloid with an E -annihilated vector is used here.

4.7 The full source map to the retained interval

In the full basis the original vectors are exactly

$$T = b_{126}, \quad T_1 = b_{146}, \quad T_2 = b_8, \quad Q = b_{144}.$$

These equalities include their original signs and the factors $-p^{-1}$ in T_2, Q . The height-two columns in T_1, Q commute by an integral table row when their order differs from the representative table; no scalar correction occurs.

Form the directed support graph of the full F_V, F_W matrices. Let $\mathcal{D}$ be the descendants of 126, and let $\mathcal{J}$ be the descendants which cannot reach 144. Exhaustive traversal of these explicit matrices gives

$$|\mathcal{D}| = 420, \quad |\mathcal{J}| = 416, \quad \mathcal{D} \setminus \mathcal{J} = \{126, 146, 8, 144\}.$$

Both sets are closed under outgoing lowering edges: a successor remains a descendant, and a successor able to reach 144 would provide such a path from its predecessor. Their A -spans are therefore stable under the integral negative-Borel action and its divided powers. The exact projection $\bigoplus_{n \in \mathcal{D}} Ab_n \rightarrow M_A$ kills the 416 basis vectors in $\mathcal{J}$ and takes the four remaining vectors, in the displayed order, to e_0, e_1, e_2, e_3 . Its kernel is exactly the displayed 416-dimensional span and its cokernel is zero.

Projecting every column of the full matrices gives

$$F_V = \begin{pmatrix} 0 & 0 & 0 & 0 \\ [7] - [3] & 0 & 0 & 0 \\ [8] & 0 & 0 & 0 \\ 0 & [6] & -[3] & 0 \end{pmatrix}, \quad F_W = 0,$$

and the original Cartan ratios $\text{diag}(q^9, q^7, q^7, q^5)$ and $q^3 I$. The full divided square has the actual entry

$$(F_V^{(2)})_{144,126} = \frac{[6]([7] - [3]) - [3][8]}{[2]} = q^{10} - q^4 - q^{-4} + q^{-10}.$$

The relation with the earlier restricted raw tableau lattice is especially explicit. Its four raw words have images $(e_0, e_1, -pe_2, -pe_3)$, and therefore their inclusion matrix in M_A is $\text{diag}(1, 1, -p, -p)$, with the previously proved cokernel $(A/(p))^2$. Enlarging the raw domain to Y_A supplies other original words. The section chosen above gives the following four preimages, each with coefficient +1:

$$\begin{array}{l|l} T & (123)(123)(124)(12)(12)(1)(1) \\ T_1 & (123)(123)(124)(12)(23)(1)(1) \\ T_2 & (123)(123)(123)(12)(12)(2)(3) \\ Q & (123)(123)(124)(12)(13)(2)(3). \end{array}$$

For the last two rows the active V -string is $C_{12} = (2)(3)$; the middle formula in (11) gives $(2)(3) = -p^{-1}(4)(1)$ with these exact contexts. This proves their stated images directly. The T_1 and Q column-order adjustments use the integral $(2, 2)$ commutations already listed.

Put $Y_{\mathcal{D},A} = \pi_A^{-1}(\bigoplus_{n \in \mathcal{D}} Ab_n)$. Inclusion of the four old raw words into this module and the composite $Y_{\mathcal{D},A} \xrightarrow{\pi_A} \bigoplus_{n \in \mathcal{D}} Ab_n \twoheadrightarrow M_A$ give the commutative square

$$\begin{array}{ccc} A_{\text{old raw}}^4 & \longrightarrow & Y_{\mathcal{D},A} \\ \downarrow \text{diag}(1,1,-p,-p) & & \downarrow \pi_A \text{ followed by quotient} \\ M_A & \xrightarrow{I} & M_A. \end{array}$$

Its right arrow is onto by the four additional preimages. Its left arrow retains its exact nonzero cokernel; enlarging the domain does not erase that restricted-lattice defect. Thus the earlier four-vector construction is recovered by an exact morphism from the full original generator module. Its integral endpoint map, conductor factorization and specialization kernel/cokernel remain the maps already proved in Section 3. The extra 1256 basis vectors here are original source vectors with their original raising and lowering actions; they are not chosen to repair the trace of a hypothetical four-dimensional quantum-group representation.

5 The full original character and specialization-compatible projectors

The original row shape in this section is $\nu = (7, 5, 3)$, its column lengths are $(3, 3, 3, 2, 2, 1, 1)$, and its column-count shape is $[0, 3, 2, 2]$. None of these data is replaced by a smaller partition.

Set $A = \mathbb{Z}[q, q^{-1}]$, $K = \mathbb{Q}(q)$, and $X_A = \text{ApNSTC}(\nu)$ inside the source quotient

$$X = \check{X}_\nu = \check{Y}_{(3,3,3,2,2,1,1)} / \sum_{i=1}^6 \check{Y}_{\triangleright^i(3,3,3,2,2,1,1)}.$$

Write $[t] = (q^t - q^{-t})/(q - q^{-1})$ for every integer t , with $[0] = 0$ and $[-t] = -[t]$. The source parameter is retained.

We use the actual two-row theorem of the April 2013 GCT-IV source, whose TeX labels are **t main canonical basis**, **p highest weight straightened NST**, and **c basis correct size**. Its hypothesis is $\dim V = \dim W = 2$, with arbitrary ν of length at most four; it is not a theorem constructing all conjectured nonstandard tensor fat cells. At this scope the positive honest classes are an A -basis, the divided Chevalley generators preserve that lattice, and the crystal cells have the highest weights obtained by the source's straightened-class rule. These are facts about the quotient just displayed. Crystal-highest class representatives are not being identified with simultaneous Chevalley kernels in X .

5.1 An exhaustive count in the original column shape

An original straightened highest class is specified by its invariant record $(0, i_3, i_2, i_1)$ and invariant-free columns

$$(123)^A(124)^B(134)^C(12)^D(13)^E(23)^G(1)^f.$$

Here powers count consecutive columns, not commutative products. All signs and scalings remain those of the positive source class: an underlying tabloid of degree $h = i_3 + i_2 + i_1$ has the source factor $(-1/[2])^h$, followed by its prescribed orientation sign. Counting a class does not change this scalar. The inserted height-three invariant is $(234)(123)$, the prescribed height-two invariant is $(24)(13)$, and the height-one invariant is $(4)(1)$. They are positioned by the source's straightening convention; their respective two-sided weights are $(3, 3; 3, 3)$, $(2, 2; 2, 2)$, and $(1, 1; 1, 1)$. Thus the complete list of constraints in the retained shape is

$$i_3, i_2, i_1 \in \{0, 1\}, \quad A + B + C = 3 - 2i_3, \quad D + E + G = 2 - 2i_2, \quad f = 2 - 2i_1, \quad (23)$$

$$A, B, C, D, E, G, f \geq 0, \quad BC = 0, \quad D > 0 \implies C = 0, \quad E > 0 \implies B \leq 1, \\ G \leq 1, \quad G = 1 \implies f > 0, \quad B \leq f + E, \quad C \leq f + D. \quad (24)$$

The source's highest-class classification and invariant insertion give one positive honest highest class for each solution, and its unique straightened representative recovers these integers. Hence there is neither an omitted class nor multiplicity from choosing different representatives of one class.

Put $k = 3i_3 + 2i_2 + i_1$. The original alphabet $1 = (1, 1)$, $2 = (1, 2)$, $3 = (2, 1)$, $4 = (2, 2)$ gives, by counting each letter in these columns,

$$\begin{aligned} a &= 2A + 2B + C + 2D + E + G + f + k, \\ b &= 2A + B + 2C + D + 2E + G + f + k, \\ \lambda &= (a, 15 - a), \quad \mu = (b, 15 - b). \end{aligned} \quad (25)$$

The invariants add the displayed k to both coordinates of both weights. Consequently the total degree stays fifteen on each side. The inequalities ensure $a, b \geq 8$.

Here is the entire finite enumeration. Each row is one solution of (23)–(24); the last column is $(2a - 14)(2b - 14)$.

(i_3, i_2, i_1)	(A, B, C)	(D, E, G)	f	λ	μ	dimension
(0, 0, 0)	(1, 0, 2)	(0, 1, 1)	2	(8, 7)	(11, 4)	16
(0, 0, 0)	(1, 0, 2)	(0, 2, 0)	2	(8, 7)	(12, 3)	20
(0, 0, 0)	(1, 2, 0)	(1, 0, 1)	2	(11, 4)	(8, 7)	16
(0, 0, 0)	(1, 2, 0)	(2, 0, 0)	2	(12, 3)	(8, 7)	20
(0, 0, 0)	(2, 0, 1)	(0, 1, 1)	2	(9, 6)	(11, 4)	32
(0, 0, 0)	(2, 0, 1)	(0, 2, 0)	2	(9, 6)	(12, 3)	40
(0, 0, 0)	(2, 1, 0)	(0, 1, 1)	2	(10, 5)	(10, 5)	36
(0, 0, 0)	(2, 1, 0)	(0, 2, 0)	2	(10, 5)	(11, 4)	48
(0, 0, 0)	(2, 1, 0)	(1, 0, 1)	2	(11, 4)	(9, 6)	32
(0, 0, 0)	(2, 1, 0)	(1, 1, 0)	2	(11, 4)	(10, 5)	48
(0, 0, 0)	(2, 1, 0)	(2, 0, 0)	2	(12, 3)	(9, 6)	40
(0, 0, 0)	(3, 0, 0)	(0, 1, 1)	2	(10, 5)	(11, 4)	48
(0, 0, 0)	(3, 0, 0)	(0, 2, 0)	2	(10, 5)	(12, 3)	60
(0, 0, 0)	(3, 0, 0)	(1, 0, 1)	2	(11, 4)	(10, 5)	48
(0, 0, 0)	(3, 0, 0)	(1, 1, 0)	2	(11, 4)	(11, 4)	64
(0, 0, 0)	(3, 0, 0)	(2, 0, 0)	2	(12, 3)	(10, 5)	60
(0, 0, 1)	(2, 1, 0)	(0, 2, 0)	0	(9, 6)	(10, 5)	24
(0, 0, 1)	(2, 1, 0)	(1, 1, 0)	0	(10, 5)	(9, 6)	24
(0, 0, 1)	(3, 0, 0)	(0, 2, 0)	0	(9, 6)	(11, 4)	32
(0, 0, 1)	(3, 0, 0)	(1, 1, 0)	0	(10, 5)	(10, 5)	36
(0, 0, 1)	(3, 0, 0)	(2, 0, 0)	0	(11, 4)	(9, 6)	32
(0, 1, 0)	(1, 0, 2)	(0, 0, 0)	2	(8, 7)	(10, 5)	12
(0, 1, 0)	(1, 2, 0)	(0, 0, 0)	2	(10, 5)	(8, 7)	12
(0, 1, 0)	(2, 0, 1)	(0, 0, 0)	2	(9, 6)	(10, 5)	24

(i_3, i_2, i_1)	(A, B, C)	(D, E, G)	f	λ	μ	dimension
(0, 1, 0)	(2, 1, 0)	(0, 0, 0)	2	(10, 5)	(9, 6)	24
(0, 1, 0)	(3, 0, 0)	(0, 0, 0)	2	(10, 5)	(10, 5)	36
(0, 1, 1)	(3, 0, 0)	(0, 0, 0)	0	(9, 6)	(9, 6)	16
(1, 0, 0)	(0, 0, 1)	(0, 1, 1)	2	(8, 7)	(10, 5)	12
(1, 0, 0)	(0, 0, 1)	(0, 2, 0)	2	(8, 7)	(11, 4)	16
(1, 0, 0)	(0, 1, 0)	(0, 1, 1)	2	(9, 6)	(9, 6)	16
(1, 0, 0)	(0, 1, 0)	(0, 2, 0)	2	(9, 6)	(10, 5)	24
(1, 0, 0)	(0, 1, 0)	(1, 0, 1)	2	(10, 5)	(8, 7)	12
(1, 0, 0)	(0, 1, 0)	(1, 1, 0)	2	(10, 5)	(9, 6)	24
(1, 0, 0)	(0, 1, 0)	(2, 0, 0)	2	(11, 4)	(8, 7)	16
(1, 0, 0)	(1, 0, 0)	(0, 1, 1)	2	(9, 6)	(10, 5)	24
(1, 0, 0)	(1, 0, 0)	(0, 2, 0)	2	(9, 6)	(11, 4)	32
(1, 0, 0)	(1, 0, 0)	(1, 0, 1)	2	(10, 5)	(9, 6)	24
(1, 0, 0)	(1, 0, 0)	(1, 1, 0)	2	(10, 5)	(10, 5)	36
(1, 0, 0)	(1, 0, 0)	(2, 0, 0)	2	(11, 4)	(9, 6)	32
(1, 0, 1)	(0, 1, 0)	(0, 2, 0)	0	(8, 7)	(9, 6)	8
(1, 0, 1)	(0, 1, 0)	(1, 1, 0)	0	(9, 6)	(8, 7)	8
(1, 0, 1)	(1, 0, 0)	(0, 2, 0)	0	(8, 7)	(10, 5)	12
(1, 0, 1)	(1, 0, 0)	(1, 1, 0)	0	(9, 6)	(9, 6)	16
(1, 0, 1)	(1, 0, 0)	(2, 0, 0)	0	(10, 5)	(8, 7)	12
(1, 1, 0)	(0, 0, 1)	(0, 0, 0)	2	(8, 7)	(9, 6)	8
(1, 1, 0)	(0, 1, 0)	(0, 0, 0)	2	(9, 6)	(8, 7)	8
(1, 1, 0)	(1, 0, 0)	(0, 0, 0)	2	(9, 6)	(9, 6)	16
(1, 1, 1)	(1, 0, 0)	(0, 0, 0)	0	(8, 7)	(8, 7)	4

For a fully specified exhaustive procedure, choose the eight invariant records, then $A = 0, \dots, 3 - 2i_3$, $B = 0, \dots, 3 - 2i_3 - A$, and set $C = 3 - 2i_3 - A - B$. Choose $D = 0, \dots, 2 - 2i_2$ and $E = 0, \dots, 2 - 2i_2 - D$, and set $G = 2 - 2i_2 - D - E$. Reject exactly the failed inequalities

above, set $f = 2 - 2i_1$, and evaluate (25). Every nonnegative solution is visited exactly once because all its free coordinates are visited exactly once. This proves the enumeration without a numerical interpolation or a guess about saturation.

Let g_{ab} be the number of these solutions with highest first coordinates (a, b) . Rows and columns in the following matrix have the order 12, 11, 10, 9, 8:

$$(g_{ab}) = \begin{pmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 2 \\ 1 & 2 & 4 & 4 & 3 \\ 1 & 3 & 4 & 4 & 2 \\ 1 & 2 & 3 & 2 & 1 \end{pmatrix}. \quad (26)$$

All other highest weights have multiplicity zero. The sum of the entries is 48, and

$$\dim_K X = \sum_{a,b=8}^{12} g_{ab}(2a-14)(2b-14) = 1260. \quad (27)$$

5.2 The full character and the classical specialization map

Let x_1, x_2 and y_1, y_2 record the two original torus weights. The complete character, retaining every coordinate, is

$$\text{ch } X = \sum_{a,b=8}^{12} g_{ab} \left(\sum_{i=0}^{2a-15} x_1^{a-i} x_2^{15-a+i} \right) \left(\sum_{j=0}^{2b-15} y_1^{b-j} y_2^{15-b+j} \right). \quad (28)$$

Each crystal component is a rectangle with one basis vector at every listed weight, so the source highest-class count proves this formula for the original quotient, not for an unrelated space of the same dimension.

For reference, the dimensions w_{uv} of the dominant weight spaces $(u, 15-u; v, 15-v)$, with rows and columns again in order 12, 11, 10, 9, 8, are

$$(w_{uv}) = \begin{pmatrix} 0 & 0 & 1 & 2 & 3 \\ 0 & 1 & 4 & 8 & 11 \\ 1 & 4 & 11 & 19 & 25 \\ 2 & 8 & 19 & 31 & 39 \\ 3 & 11 & 25 & 39 & 48 \end{pmatrix}. \quad (29)$$

The other weights satisfy $w_{uv} = w_{15-u,v} = w_{u,15-v}$, and vanish outside $3 \leq u, v \leq 12$. Formula (28) proves these reflection identities, since each of its two intervals is invariant under the corresponding reflection. On the dominant chamber the exact inverse to rectangle summation is

$$g_{ab} = w_{ab} - w_{a+1,b} - w_{a,b+1} + w_{a+1,b+1}.$$

Indeed $w_{ab} = \sum_{c \geq a, d \geq b} g_{cd}$ there, and both successive finite differences cancel precisely the terms outside $(c, d) = (a, b)$.

The source specialization is a diagram of actual morphisms

$$X \longleftarrow X_A \longrightarrow X_A/(q-1)X_A \longrightarrow \mathbb{Q} \otimes_{\mathbb{Z}} (X_A/(q-1)X_A) \xrightarrow{\sigma} S_{(7,5,3)}(V_{\mathbb{Q}} \otimes W_{\mathbb{Q}}).$$

The first arrow is inclusion followed by scalar extension; the second is the quotient. We make explicit the source's two integral presentations. With Y_A^S its rescaled tabloid lattice and M_A^S the sum of its six specified integral relation lattices, the source first forms $X'_A = Y_A^S/M_A^S$. The inclusion $M_A^S \subseteq (KM_A^S) \cap Y_A^S$ gives the actual surjection

$$s_X : X'_A \longrightarrow Y_A^S/((KM_A^S) \cap Y_A^S) = X_A, \quad [y] \longmapsto [y].$$

The source’s specialization theorem is stated first for X'_A . Its remark at TeX lines 5901–5908 also asserts an integral identification, but that stronger assertion is not used here: an integral relation presentation can retain torsion that the honest image lattice does not retain. We now prove exactly the rational specialization map required for the displayed diagram. Let $\mathbb{Q}_1$ denote $\mathbb{Q}$ as an A -algebra through $q \mapsto 1$. Right exactness gives a surjective module map

$$\bar{s}_X = \mathbb{Q}_1 \otimes_A s_X : \mathbb{Q}_1 \otimes_A X'_A \rightarrow \mathbb{Q}_1 \otimes_A X_A.$$

The original contextual relation graph has 36864 tabloid vertices; its specialization at $q = 1$, $p = [2] = 2$, has relation rank 35604, so its quotient has dimension 1260 over $\mathbb{Q}$. All original rescaling factors are signed powers of p , hence units over $\mathbb{Q}_1$. They do not change that rational row span. This is the explicit source relation specialization computation, recorded in `specialization_bridge.json` with its graph hash and the original scaled-image check. Equivalently it is the rational dimension statement in the source’s two-row theorem. Thus the domain of $\bar{s}_X$ has dimension 1260. Its codomain has dimension 1260 because the original honest lattice has its proved 1260-element A -basis.

The inverse can be given directly, without an inverse to s_X over A . For each original positive honest basis vector b , choose its displayed positive rescaled tabloid representative $\tilde{b} \in Y_A^S$. Define

$$j : \mathbb{Q}_1 \otimes_A X_A \longrightarrow \mathbb{Q}_1 \otimes_A X'_A, \quad 1 \otimes b \longmapsto 1 \otimes [\tilde{b}].$$

This is a well-defined linear map on a basis, and $\bar{s}_X j = 1$ by the quotient definitions. These 1260 lifted vectors are independent, since their images are the honest basis; the dimension calculation proves that they are also a basis of the domain. It follows that $j \bar{s}_X = 1$. As the inverse of the surjective module map $\bar{s}_X$, j intertwines the specialized original actions. In particular its value is independent of the choice of representative. This proves the exact rational isomorphism and makes no injectivity assertion about s_X over A .

The last arrow in the displayed diagram is $\sigma = \tau j$, where τ is the source theorem’s rational Schur-quotient map, described intrinsically as follows. The nonstandard exterior algebra is the quotient of the tensor algebra on the four original letters by its nonstandard symmetric-square relations. At $q = 1$ these relations are the ordinary symmetric-square relations. The identity on the four letters therefore induces an isomorphism of the specialized exterior algebra with the ordinary exterior algebra. Apply its degree maps in the original seven tensor factors. The source’s adjacent-column relation spaces specialize to the adjacent Schur relation spaces. Thus the tensor map carries the sum of the six relation spaces onto the sum of the six classical relation spaces, and it induces τ on their rational quotients. Applying the inverse tensor map gives an inverse on these quotients. Naturality on the four-letter tensor algebra shows that these maps intertwine the $\mathfrak{gl}_2 \oplus \mathfrak{gl}_2$ actions and their polynomial $\mathrm{GL}_2 \times \mathrm{GL}_2$ actions over $\mathbb{Q}$. This is the relation proved in the source’s specialization theorem; it does not assert specialization of arbitrary elements of $K \otimes_A X_A$ that have poles at $q = 1$.

In particular

$$\mathrm{ch} X = s_{(7,5,3)}(x_1 y_1, x_1 y_2, x_2 y_1, x_2 y_2).$$

There is also a separate exact verification of the entire polynomial. Enumerate all weakly increasing rows of lengths 7, 5, 3 on $\{1, 2, 3, 4\}$, retaining a triple of rows exactly when its columns strictly increase. This gives all 1260 semistandard tableaux exactly once. If their contents are (c_1, c_2, c_3, c_4) , their monomial is

$$x_1^{c_1+c_2} x_2^{c_3+c_4} y_1^{c_1+c_3} y_2^{c_2+c_4}.$$

The resulting weight multiplicities equal (28) at every weight. The separate checker `independent_check/check_character.py` additionally evaluates the Jacobi–Trudi determinant in exact integer polynomials. The source’s set bijection between honest classes and

ordinary semistandard tableaux is not being used as a weight-preserving linear map; σ is the module morphism that establishes the classical connection.

5.3 Explicit intertwiners into the actual source module

We prove the needed splitting and then give maps whose input operators are the full original source generators. For $U_q(\mathfrak{sl}_2)$ use $KEK^{-1} = q^2E$, $KFK^{-1} = q^{-2}F$ and $EF - FE = (K - K^{-1})/(q - q^{-1})$. An integer h denotes a weight on which K acts by q^h .

Lemma 5.1. *Every finite-dimensional type-one integrable K -module for this algebra is the direct sum of its highest-weight strings. A string from a vector z with $Ez = 0$ and $Kz = q^n z$ has $n \geq 0$, basis $F^{(i)}z$, $0 \leq i \leq n$, and actions*

$$FF^{(i)}z = [i+1]F^{(i+1)}z, \quad EF^{(i)}z = [n-i+1]F^{(i-1)}z.$$

Proof. Repeatedly applying the displayed commutator gives $EF^jz = [j][n-j+1]F^{j-1}z$. This follows by induction: moving E past one additional F adds the next Cartan term, and the finite Laurent identity $[j+1][n-j] - [j][n-j+1] = [n-2j]$ supplies precisely that term. Let d be the last nonzero power of F on z ; it exists by integrability. Applying E to $F^{d+1}z = 0$ gives $[d+1][n-d]F^d z = 0$. Over K , $[t] = 0$ for an integer t only when $t = 0$, so $d = n \geq 0$. Distinct string vectors have distinct weights and are independent. Dividing by $[i]!$ yields the claimed formulas.

For completeness, splitting is proved by induction on dimension. Choose the largest weight n occurring in a nonzero module. Its entire weight space consists of highest vectors. The strings from a basis of that space are independent: a relation at weight $n - 2i$ can be raised by E^i , which multiplies each top vector by the same nonzero scalar. They span a submodule N , a sum of copies of the length- n string. The quotient has smaller dimension and no weight n , so by induction it is a sum of strings with highest weights $0 \leq m < n$. Lift each quotient highest vector to a vector z of weight m . Then $Ez \in N_{m+2}$. On every copy of the length- n string, $E : N_m \rightarrow N_{m+2}$ is surjective whenever the target occurs: its coefficient in the formulas above is nonzero. Subtract a preimage in N_m to obtain a highest lift. Its length- m string maps isomorphically onto the chosen quotient string by the preceding formulas. All these lifted strings together map isomorphically to the quotient, so their sum is disjoint from N and splits the original module. This completes the induction. $\square$

For $a, b \in \{8, 9, 10, 11, 12\}$ put $n = 2a - 15$, $m = 2b - 15$ and define the actual source subspace

$$H_{ab} = X_{(a,15-a;b,15-b)} \cap \ker E_V \cap \ker E_W.$$

Let $L_{(a,15-a)}$ be the standard quantum $\mathfrak{gl}_2$ module with basis e_i , $0 \leq i \leq n$, of weight $(a-i, 15-a+i)$, with $Fe_i = [i+1]e_{i+1}$ and $Ee_i = [n-i+1]e_{i-1}$; the out-of-range terms are zero. Define $L_{(b,15-b)}$ in the same way, with basis f_j . In the next tensor product H_{ab} is used as a multiplicity vector space with trivial quantum-group action, not with an action inherited from its embedding in X . Then the map

$$\Phi : \bigoplus_{a,b=8}^{12} L_{(a,15-a)} \otimes L_{(b,15-b)} \otimes H_{ab} \longrightarrow X, \quad e_i \otimes f_j \otimes z \longmapsto F_V^{(i)} F_W^{(j)} z \quad (30)$$

is an isomorphism of the full quantum $\mathfrak{gl}_2 \oplus \mathfrak{gl}_2$ modules. To prove this, first apply Lemma 5.1 to the V generators. The commuting W generators act on each V highest subspace. Apply the lemma to that action. The two successive string maps are exactly (30). The lemma verifies both raising and lowering operators, and the weights verify all four original diagonal generators and both total degree-fifteen central characters. This proves the intertwiner and

its bijectivity. The character comparison now gives $\dim_K H_{ab} = g_{ab}$; no choice of purported highest source basis vectors was made.

Here is an explicit inverse requiring only the original operators. For either colour, put

$$C = (q - q^{-1})^2 FE + qK + q^{-1}K^{-1}.$$

Its commutators with K, E, F are zero. For example its commutator with E is

$$-(q - q^{-1})E(q^2K - q^{-2}K^{-1}) + q(q^2 - 1)EK + q^{-1}(q^{-2} - 1)EK^{-1} = 0;$$

the F calculation follows by the same defining relations. On a highest vector of weight n it acts by $c_n = q^{n+1} + q^{-n-1}$, hence by that scalar on the whole string. All source weights h are odd. Consequently the operator

$$B = FE + \sum_h [(h+1)/2]^2 \text{pr}_h = \frac{C - 2}{(q - q^{-1})^2} \quad (31)$$

preserves X_A . The equality holds because $q^{h+1} + q^{-h-1} - 2 = (q^{(h+1)/2} - q^{-(h+1)/2})^2$. Here pr_h is projection to the original weight subspace of the source weight basis, so it is A -linear. The apparent quotient in (31) has already been replaced by its exact integral operator formula; it is not a limit used to discard any term.

For $n \in \mathcal{N} = \{1, 3, 5, 7, 9\}$ set $\beta_n = [(n+1)/2]^2$ and

$$\Pi_n(B) = \prod_{t \in \mathcal{N} \setminus \{n\}} \frac{B - \beta_t}{\beta_n - \beta_t}. \quad (32)$$

The denominators are nonzero since at $q = 1$ the five eigenvalues are respectively 1, 4, 9, 16, 25. The string decomposition proves that these are orthogonal idempotents with sum one, acting as identity exactly on the strings of highest ratio weight n .

For an original vector x of weight $(u, 15 - u; v, 15 - v)$ set $i = a - u$, $j = b - v$. Keep a pair (a, b) exactly when $0 \leq i \leq n$ and $0 \leq j \leq m$. Its inverse component is

$$e_i \otimes f_j \otimes \frac{E_V^{(i)} E_W^{(j)} \Pi_n(B_V) \Pi_m(B_W) x}{\begin{bmatrix} n \\ i \end{bmatrix}_q \begin{bmatrix} m \\ j \end{bmatrix}_q}, \quad \begin{bmatrix} n \\ i \end{bmatrix}_q = \frac{[n]!}{[i]! [n-i]!}. \quad (33)$$

To check both compositions, apply this formula to $F_V^{(r)} F_W^{(s)} z$ from each string in (30). The projectors kill all different highest-weight pairs. Weight equality forces $i = r, j = s$ in the surviving pair, and repeated raising gives $E_V^{(i)} F_V^{(i)} z = \begin{bmatrix} n \\ i \end{bmatrix}_q z$ and its W analogue. Thus the formula recovers z and every component, proving both inverse identities. This is a formula on the actual source generators, with concrete finite products; it is not a character-only identification.

5.4 The precise integral localization and specialization

Let S be the multiplicative set generated by $[1], \dots, [9]$ and $\beta_n - \beta_t$ for distinct $n, t \in \mathcal{N}$, and write $R = S^{-1}A$. The exact Laurent identity

$$[r]^2 - [s]^2 = [r - s][r + s]$$

follows by expanding the four powers after multiplication by $(q - q^{-1})^2$, including its sign when $r < s$. For distinct $r, s \in \{1, 2, 3, 4, 5\}$ its two nonzero factors are signed members of $[1], \dots, [9]$. Thus this displayed localization equals $A([1] \cdots [9])^{-1}$; the Casimir denominators

were retained and have now been proved to be units. The lattice X_A remains the original lattice; this explicitly stated scalar extension is needed for the following splitting. Every operator in (30) and (33) is defined over R : the divided generators already preserve X_A , the integral B does also, and the remaining denominators are units in R . Put $H_{ab,R} = X_{R(a,15-a;b,15-b)} \cap \ker E_V \cap \ker E_W$. The formulas give mutually inverse R -linear maps

$$\bigoplus_{a,b} L_{(a,15-a),R} \otimes L_{(b,15-b),R} \otimes H_{ab,R} \cong X_R. \quad (34)$$

Indeed the numerator in the inverse formula lies in this highest kernel over K by the proved inverse, and its annihilation by both E holds over R because X_R is torsion-free. Both composition identities also descend from K by torsion-freeness. Moreover $H_{ab,R} = \Pi_n(B_V)\Pi_m(B_W)X_{R(a,15-a;b,15-b)}$: one inclusion follows from the inverse formula at $i = j = 0$, and a highest vector has the indicated two Casimir eigenvalues, which proves the other inclusion. It is therefore a finite projective R -module of rank g_{ab} .

Evaluation at $q = 1$ sends $[j]$ to j and the distinct Casimir differences to differences of 1, 4, 9, 16, 25. These nonzero differences have prime factors only in $\{2, 3, 5, 7\}$, and $[1], \dots, [9]$ already require all four primes. The universal property of localization therefore gives

$$R/(q-1)R = \mathbb{Z}[1/210].$$

Tensoring the two inverse maps in (34) with this ring keeps them inverse. In particular the specialized highest kernel is exactly the specialization of $H_{ab,R}$. One can check this also directly on the strings: the coefficient of every non-top raising step is one of $1, \dots, 9$, a unit in $\mathbb{Z}[1/210]$, so simultaneous raising kills exactly the two top positions. No additional highest vectors appear. The specialized B is $FE + \sum_h ((h+1)/2)^2 \text{pr}_h$ and retains its five distinct eigenvalues. Thus these projectors specialize without a pole, and the string intertwiners become the classical divided-power intertwiners. After tensoring with $\mathbb{Q}$, the source map σ above relates them to the classical Schur module. This does not claim that the splitting holds over unlocalized A , or identify an integral Schur lattice without its own lattice comparison.

5.5 Exact locations of the three retained weights

The weight $(12, 3; 9, 6)$ has dimension two. Its two components are $(a, b) = (12, 10)$ at positions $(i, j) = (0, 1)$ and $(12, 9)$ at positions $(0, 0)$, each with multiplicity one. The complete locations at the next two weights are as follows; “copies” is the dimension of the indicated actual H_{ab} .

(u, v)	a	b	(i, j)	copies
(11, 9)	12	10	(1, 1)	1
	12	9	(1, 0)	1
	11	11	(0, 2)	1
	11	10	(0, 1)	2
	11	9	(0, 0)	3
(10, 9)	12	10	(2, 1)	1
	12	9	(2, 0)	1
	11	11	(1, 2)	1
	11	10	(1, 1)	2
	11	9	(1, 0)	3
	10	12	(0, 3)	1
	10	11	(0, 2)	2
	10	10	(0, 1)	4
	10	9	(0, 0)	4

The corresponding dimensions are 8 and 19. Every location is obtained by $i = a - u$, $j = b - v$ from an actual source isotypic projector and the proved string isomorphism. A specific original honest vector such as T, T_1, T_2, Q is located inside these summands by applying (33) to that vector. Merely knowing its weight does not assign it to one summand. The original four-vector negative-Borel interval is still the specified descendant quotient of this full 1260-dimensional module; these character and projector formulas do not identify that interval with an invariant full quantum group module.

For these four particular vectors the full original source matrices now permit a stronger determination than their weights alone. In their original order and with their original scalars they are

$$\begin{aligned} T &= (123)(123)(124)(12)(12)(1)(1), \\ T_1 &= (123)(123)(124)(23)(12)(1)(1), \\ T_2 &= -[2]^{-1}(123)(123)(123)(12)(12)(4)(1), \\ Q &= -[2]^{-1}(123)(123)(124)(13)(12)(4)(1). \end{aligned}$$

The source's integral height-two commutation relation moves the columns (23)(12) of T_1 to (12)(23) and the columns (13)(12) of Q to (12)(13) with coefficient +1. Their resulting canonical positive representatives, including $-[2]^{-1}$ for T_2, Q , are basis indices 126, 146, 8, 144 of the full original matrix certificate. The quotient projection on the original ambient words proves these four equalities, not merely a match of their weights. The graphical raising rule pairs a 2 with a later 1 in the original V or W word and sums over unpaired 2's. For the four representatives its unpaired-word data are

	V	W
T	1^9	1^3
T_1	1^7	1^3
T_2	1^7	1^3
Q	1^5	1^3

Thus each raising sum is empty, proving $E_V T = E_W T = E_V T_1 = E_W T_1 = E_V T_2 = E_W T_2 = E_V Q = E_W Q = 0$ directly in the original module. Hence

$$T \in H_{12,9}, \quad T_1, T_2 \in H_{11,9}, \quad Q \in H_{10,9}.$$

The two middle vectors are independent because they are distinct members of the original honest basis. In particular they span two dimensions in the three-dimensional actual $H_{11,9}$. The exact projector computation on all four original vectors returns the vector itself for this listed pair and zero for every other pair; its raising and lowering inverse checks use the full original matrices over $\mathbb{Q}(q)$. For example $F_V^{(2)} T$ remains in the isotypic summand $(\lambda, \mu) = ((12, 3), (9, 6))$, whereas Q is in the different summand $((10, 5), (9, 6))$. The original nonzero coefficient of Q in the honest-basis expansion of $F_V^{(2)} T$ is therefore not an isotypic projection. Applying $\Pi_5(B_V)\Pi_3(B_W)$ to the complete expansion gives zero: the contribution of the Q basis term is canceled by the projections of the other terms. This identity follows because the projectors commute with F_V and kill T for that highest pair. It preserves the original coefficient and explains its exact relation to the full irreducible decomposition.

6 The retained inverse fibre as an explicit determinantal family

All rings in the algebraic statements are over a specified field k of characteristic zero. The bit statements use $k = \mathbb{Q}$ and signed binary integers. No operation changes the coefficients

of the original map. The third source coordinate is named w ; the renaming from the source's t is the identity $(x, y, t) \mapsto (x, y, w)$. The source expressions were read in full from the parent satellite and the independent fluid extension; their paths and SHA-256 hashes are in `source_hashes.json`. Every calculation used below is reproved here and replayed by the included exact checker.

6.1 The original map and its localized coordinate-ring isomorphism

Put $A = k[a, b, c]$, $B = k[x, y, w]$, and define $F^* : A \rightarrow B$ by

$$\begin{aligned} F^*(a) &= F_1 = (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), \\ F^*(b) &= F_2 = y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), \\ F^*(c) &= F_3 = 2x - 3x^2 y - x^3 w. \end{aligned}$$

The determinant retains the source orientation (x, y, w) and target orientation (a, b, c) and equals -2 . Here is a direct proof. On $x \neq 0$ set $v = 1 + xy$, $h = 2 - 3xy - x^2 w$, $A_0 = v^2 + v - v^3 h$ and $B_0 = 4v + 2 - 3v^2 h$. Their inverse coordinate formulas are $y = (v - 1)/x$ and $w = (5 - 3v - h)/x^2$, so these are actual coordinates on the stated open set. Substitution in the original expressions yields $F = (A_0/x^2, B_0/x, xh)$ and

$$\det \frac{\partial(x, v, h)}{\partial(x, y, w)} = -x^3, \quad \det \frac{\partial F}{\partial(x, v, h)} = x^{-3} \det \begin{pmatrix} -2A_0 & 2v + 1 - 3v^2 h & -v^3 \\ -B_0 & 4 - 6vh & -3v^2 \\ h & 0 & 1 \end{pmatrix} = 2x^{-3}.$$

Expansion of the last numerator gives

$$(4 - 6vh)(-2v^2 - 2v + 3v^3 h) + (2v + 1 - 3v^2 h)(4v + 2 - 6v^2 h) = 2.$$

Multiplication proves $\det DF = -2$ on $x \neq 0$. Both sides are polynomials, and their equality in the localization of the domain B proves equality in B as well.

Retain the exact polynomial and derivative

$$P = cr^3 - 2r^2 + br - 2a, \quad P' = 3cr^2 - 4r + b, \quad \alpha = P'/2.$$

Define the A -algebra

$$E = (A[r]/(P))[1/P'].$$

There is an isomorphism of A -algebras $\Phi : E \rightarrow B[1/x]$ given by

$$(a, b, c, r, 1/P') \mapsto (F_1, F_2, F_3, y + 1/x, x/2).$$

Its inverse $\Psi : B[1/x] \rightarrow E$ is specified on every generator by

$$x \mapsto 2/P', \quad y \mapsto r - P'/2, \quad w \mapsto 5(P'/2)^2 - 3r(P'/2) - c(P'/2)^3, \quad x^{-1} \mapsto P'/2.$$

To prove these assertions before quotienting, introduce independent coordinates (r, α, c) with $\alpha \neq 0$ and put

$$H(r, \alpha, c) = (\alpha^{-1}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3).$$

Solving $F_3 = c$ after setting $r = y + 1/x$, $\alpha = 1/x$ gives this formula and its inverse exactly. Direct substitution gives

$$F \circ H = (r^2 + r\alpha - cr^3, 4r + 2\alpha - 3cr^2, c).$$

The equation $F_2 = b$ says $2\alpha = P'$, and then $2(F_1 - a) = P$. Conversely those two relations give $F_1 = a, F_2 = b$. They show that the displayed maps respect $P = 0$, the inverse elements, and the A -algebra structure. The recovery equations $r = y + 1/x$, $\alpha = 1/x$ show both compositions are the identity on all their generators. This proves the ring isomorphism, including its excluded divisor $P' = 0$.

The morphism F^* is injective. Indeed $A[r]/(P)$ is isomorphic to $k[b, c, r]$ by eliminating $a = (cr^3 - 2r^2 + br)/2$; both substitution maps are inverse on the generators. A nonzero $h(a, b, c) = \sum_{j=0}^d h_j(b, c)a^j$, $h_d \neq 0$, has, after this substitution, highest r coefficient $h_d(b, c)c^d/2^d$ at degree $3d$, which is nonzero in $k[b, c]$. Thus A injects into the quotient and, since $P' \neq 0$ there, into E and then $B[1/x]$. Consequently it also injects into B . This proves dominance without treating it as an assumption.

6.2 Every specialized fibre and the finite free locus

For $(a_0, b_0, c_0) \in k^3$, set

$$B_{a_0, b_0, c_0} = k[x, y, w]/(F_1 - a_0, F_2 - b_0, F_3 - c_0), \quad E_{a_0, b_0, c_0} = k[r, 1/P']/(P),$$

where P and P' here have the specialized coefficients. If $c_0 \neq 0$, then $xh = c_0$ in the fibre ring, so x is a unit with inverse h/c_0 . The preceding isomorphism therefore proves $B_{a_0, b_0, c_0} \simeq E_{a_0, b_0, c_0}$ on the entire fibre. If $c_0 = 0$, the entire fibre ring is instead

$$B_{a_0, b_0, 0} \simeq k \times E_{a_0, b_0, 0}.$$

Here is a proof that includes its scheme structure. Work first in $R = k[x, y, w]/(F_1 - a_0, F_2 - b_0)$. The polynomials x, h generate the unit ideal since $1 = h/2 + x(3y + xw)/2$. Their ideals therefore have intersection (xh) : if $z = xu = hv$, multiplication by the displayed expression for 1 shows $z \in (xh)$. The map $R/(xh) \rightarrow R/(x) \times R/(h)$ is injective by that intersection; given classes modulo the two ideals, their lifts combine using the same two summands of 1 to prove surjectivity. This proves the asserted product decomposition. The first factor is k because $F(0, y, w) = (w + 4y^2, y, 0)$, with point $(0, b_0, a_0 - 4b_0^2)$. In the second factor, $x(3y + xw) = 2$ makes x a unit and the chart isomorphism identifies it with $E_{a_0, b_0, 0}$.

Over an algebraic closure, write $P = d \prod_j (r - \rho_j)^{m_j}$, with distinct ρ_j . The coprime factors yield a product of the rings $\bar{k}[r]/((r - \rho_j)^{m_j})$: the elementary two-ideal argument just given applies repeatedly. When $m_j = 1$, P' has nonzero value at ρ_j and is a unit in that factor. When $m_j > 1$, P' belongs to its nilpotent ideal $(r - \rho_j)$, so inverting P' turns that factor into the zero ring. Thus E retains exactly the simple roots, each with its reduced point. The complete finite inverse points are

$$\left(2/P'(\rho_j), \rho_j - P'(\rho_j)/2, 5(P'(\rho_j)/2)^2 - 3\rho_j P'(\rho_j)/2 - c_0(P'(\rho_j)/2)^3\right)$$

for $m_j = 1$, together with the boundary point when $c_0 = 0$. The original polynomial, its multiplicities, and its excluded roots remain recorded by (P, P') even when the localization removes a factor.

The coefficient discriminant is exactly

$$D = 4(b^2 - cb^3 + 18abc - 16a - 27a^2c^2).$$

For direct verification, expansion of the determinant

$$\det \begin{pmatrix} c & -2 & b & -2a & 0 \\ 0 & c & -2 & b & -2a \\ 3c & -4 & b & 0 & 0 \\ 0 & 3c & -4 & b & 0 \\ 0 & 0 & 3c & -4 & b \end{pmatrix} = -cD$$

gives the stated polynomial. This determinant is the resultant because its matrix represents $(u, v) \mapsto uP + vP'$ on the bases of polynomials of degrees less than 2 and 3, respectively, with the indicated descending coefficient order. Over a field it vanishes exactly when P, P' share a nonconstant factor: a nonzero kernel gives $uP = -vP'$ and the coprime case forces $P \mid v$, impossible at $\deg v < 3$; a common factor gives a kernel vector by dividing both polynomials by it. This also proves the simple-root interpretation.

Let $A_U = A[1/(cD)]$. Then $A_U[r]/(P)$ is free with basis $(1, r, r^2)$. Polynomial division by the unit leading coefficient c gives a spanning remainder of degree at most two, and a nonzero multiple of P has degree at least three, proving independence. No replacement of P is made here. The original relation gives the multiplication matrix

$$C = \begin{pmatrix} 0 & 0 & 2a/c \\ 1 & 0 & -b/c \\ 0 & 1 & 2/c \end{pmatrix}, \quad N = 3cC^2 - 4C + bI_3.$$

Its columns are exactly the coefficients of multiplying the ordered basis by r and P' , respectively. Direct determinant expansion gives $\det N = -D/c$, so $N^{-1} = -(c/D) \operatorname{adj}(N)$. This explicit inverse proves that P' is a unit in the quotient, and hence

$$B \otimes_A A_U \simeq A_U[r]/(P).$$

This is a finite free rank-three morphism. Its unramified behavior can also be proved without omitting the derivative: for a square-zero ideal I of an A_U -algebra T , a root r_0 in T/I has a lift $\tilde{r}$ in T , and the unique root lift is $\tilde{r} - P(\tilde{r})/P'(\tilde{r})$. The denominator is a unit because its reduction is; Taylor expansion has no terms of degree two in I . If two lifts differ by $e \in I$, subtraction gives $P'(\tilde{r})e = 0$, so $e = 0$. This is the unique infinitesimal lifting property, establishing the finite etale claim on precisely U .

At $c_0 = 0$, the polynomial still has degree two and its quadratic discriminant is $b_0^2 - 16a_0$, while $D = 4(b_0^2 - 16a_0)$. The boundary point remains necessary. In particular the fibre at $(-1/4, 0, 0)$ consists exactly of

$$(1, -3/2, 13/2), \quad (-1, 3/2, 13/2), \quad (0, 0, -1/4).$$

At $c_0 \neq 0$ the cubic is a triple-root polynomial exactly at $b_0 = 4/(3c_0)$, $a_0 = 4/(27c_0^2)$, with root $2/(3c_0)$. Indeed comparing $c_0(r - \rho)^3$ with all four coefficients gives these three equations and conversely verifies them. Its entire finite fibre is empty since its only root has $P' = 0$ and $c_0 \neq 0$ excludes $x = 0$.

6.3 Generic degree, the absence of a rational inverse, and nonfiniteness

In $k[a, b, c, r]$, (P) is a prime ideal since its quotient was explicitly identified with the domain $k[b, c, r]$. Thus P is irreducible there. It is primitive as a polynomial over $k[a, b, c]$ in r , since its coefficient -2 is a unit. The primitive-polynomial argument proves irreducibility over $K = k(a, b, c)$: clearing denominators in a proposed factorization and cancelling the common prime factors of the coefficients would produce a factorization in the unique-factorization ring $k[a, b, c][r]$. To justify that cancellation, the product of two primitive polynomials is primitive, since reduction modulo any prime leaves two nonzero polynomials over a domain and hence a nonzero product. This proves the required primitive-polynomial argument.

The chart identities now give

$$k(x, y, w) = k(F_1, F_2, F_3)(y + 1/x), \quad [k(x, y, w) : k(F_1, F_2, F_3)] = 3.$$

Both containments in the first equality follow from the displayed inverse formulas; irreducibility of the degree-three polynomial proves the degree. A rational inverse would make the two

fields equal, so none exists. There is also no rational section over a dense target open set: if its x coordinate were identically zero then its F_3 coordinate would be zero, contrary to the generic variable c ; otherwise its $r = y + 1/x$ would be an element of K satisfying the irreducible cubic. These proofs rule out those exact rational maps and retain the rank-three algebraic correspondence constructed above.

The morphism F is not finite. Define the following homomorphism to $k[t, t^{-1}]$ by its values on the original coordinates:

$$x = t^{-1}, \quad y = -3t/2, \quad w = 13t^2/2.$$

Direct substitution gives $(F_1, F_2, F_3) = (-t^2/4, 0, 0)$. If B were finite over A , then multiplication by x on finitely many A -module generators would give a monic relation for x : write $xm_i = \sum_j a_{ij}m_j$ and multiply the matrix $xI - (a_{ij})$ by its adjugate; its monic determinant annihilates every generator and therefore $1 \in B$. Under the displayed homomorphism such a relation would become

$$t^{-m} + \sum_{j=0}^{m-1} q_j(-t^2/4, 0, 0)t^{-j} = 0.$$

Multiplication by t^m gives a polynomial with constant term 1, while every summand in the sum is divisible by t . This is impossible. Thus nonfiniteness is proved directly from the original map and is compatible with its finite free restriction to U .

6.4 The projective binary cubic restores the entire boundary sheet

There is a stronger global presentation retaining the $x = 0$ sheet as a projective root. Define

$$Z = \{cR^3 - 2R^2T + bRT^2 - 2aT^3 = 0\} \subset \mathbb{P}_A^1, \quad [R : T] = [1 + xy : x].$$

The last expression defines a morphism on the entire source because $1 = (1 + xy) - yx$ proves its two homogeneous coordinates generate the unit ideal. Substitution of F_1, F_2, F_3 in the homogeneous equation gives zero: on $x \neq 0$ it is $x^3P(y + 1/x) = 0$, and equality in the localization of the source polynomial domain proves the polynomial identity everywhere.

Let Z^s be the open simple-root locus: on $T \neq 0$ its coordinates are $r = R/T$ and it is defined by $P = 0$, $P' \neq 0$; on $R \neq 0$ use the distinct root coordinate $h = T/R$ and put

$$Q(h) = c - 2h + bh^2 - 2ah^3, \quad d = Q'(h) = -2 + 2bh - 6ah^2.$$

Its simple-root condition is $d \neq 0$. On the overlap $h = 1/r$, $Q(h) = h^3P(1/h)$ and, on that zero scheme, $d = -hP'(1/h)$. Since h is a unit on the overlap, the two simple-root conditions agree exactly.

The morphism from the source to Z is an isomorphism onto Z^s . On the first chart this is the already proved inverse-root isomorphism. Here is the entire second chart, including $x = 0$:

$$x = -2h/d, \quad y = b - 3ah, \quad w = -ad^3/8 - y^2d^2/4 + 3y^2d/2.$$

To prove it, first keep a, b, h independent and define d, x, y, w by these displayed formulas with $d \neq 0$. Substitution and expansion give

$$1 + xy = -2/d, \quad F_1 = a, \quad F_2 = b, \quad F_3 = 2h - bh^2 + 2ah^3 = c - Q(h).$$

For example $1 + xy = (d - 2hb + 6ah^2)/d = -2/d$ directly. For the other formulas one may insert that identity into the original F_i : with $v = -2/d$, the first is $v^3w + y^2v(1 + 3v) = a$, since $4 + 3xy = 1 + 3v$; the second is $y + 3xv^2w + 3xy^2(1 + 3v) = y + 3ha = b$. The third follows by $F_3 = x(5 - 3v - x^2w)$, inserting the displayed w and $y = b - 3ah$, and expanding to

$2h - bh^2 + 2ah^3$. Thus the formulas define the inverse map on $Q = 0$, $d \neq 0$. Conversely on the source chart $v = 1 + xy \neq 0$, set $h = x/v$, $a = F_1$, $b = F_2$, $c = F_3$. The polynomial identity gives $Q = 0$; the two original equations for F_1, F_2 give $y = b - 3ah$ by subtracting $3hF_1$ from F_2 . Consequently $1 + xy = 1 + x(b - 3ah) = 1 + hv(b - 3ah)$ gives $v(1 - bh + 3ah^2) = 1$, equivalently $d = -2/v$. Then $x = hv = -2h/d$, and solving the original $F_1 = a$ for w gives exactly the stated w . These recover every generator, proving both compositions. The source charts $x \neq 0$ and $v \neq 0$ cover it; the two inverse maps agree on their overlap because both are inverse to the same forward coordinates. This proves the global isomorphism. At $h = 0$, $Q = 0$ requires $c = 0$, $d = -2$, and the formulas give $(x, y, w) = (0, b, a - 4b^2)$. Thus the boundary point is retained as $[R : T] = [1 : 0]$, with no discarded sheet.

The projective scheme Z is finite locally free of rank three over A . An elementary affine cover proves this. Evaluate its binary cubic at the three constant points $[j : 1]$, $j = 0, 1, -1$, obtaining

$$f_0 = -2a, \quad f_1 = c - 2 + b - 2a, \quad f_{-1} = -c - 2 - b - 2a.$$

They generate the unit ideal since $f_1 + f_{-1} - 2f_0 = -4$. On the base open set $f_j \neq 0$, the point $[j : 1]$ is absent from Z . Use the explicitly invertible homogeneous change

$$U = T, \quad V = R - jT, \quad R = V + jU, \quad T = U.$$

The removed point becomes $[U : V] = [1 : 0]$, so all of Z on this base open lies in the affine chart $V \neq 0$ with $s = U/V$. The affine coordinate formula is $R = 1 + js$, $T = s$; its polynomial is

$$c(1 + js)^3 - 2(1 + js)^2s + b(1 + js)s^2 - 2as^3,$$

whose leading coefficient is exactly f_j . It is a unit on this base open, so division with the original polynomial gives the free basis $1, s, s^2$ exactly as in the earlier cubic proof. The changes are invertible projective linear maps and retain all original coefficients. The three base opens cover, proving finite local freeness and its rank.

The branch locus is exactly $D = 0$. When $c \neq 0$ this is the resultant calculation already proved. When $c = 0$, $[1 : 0]$ is a simple root because $d(0) = -2$, and the remaining affine polynomial has degree two with repeated root exactly when $b^2 - 16a = 0$, equivalently $D = 0$. These arguments apply in the residue field at each base point and in its algebraic closure. Consequently over $D \neq 0$ every projective root is simple, so $Z^s = Z$ there and the original F is finite locally free of rank three on the larger open $\{D \neq 0\}$, including $c = 0$. The unique root-lifting argument on the two charts proves it is etale there. The algebra on $cD \neq 0$ used previously gives its concrete single-chart basis; that smaller open was sufficient and was not the entire finite etale locus.

6.5 A size-parameterized family with complete encoding

For each integer $n \geq 1$ retain independent variables (x_i, y_i, w_i) and targets (a_i, b_i, c_i) for $1 \leq i \leq n$. Define $F^{[n]}$ to apply the preceding three original polynomials to each block. This is a morphism $\mathbb{A}_k^{3n} \rightarrow \mathbb{A}_k^{3n}$ with coordinate homomorphism $a_i \mapsto F_1(x_i, y_i, w_i)$, $b_i \mapsto F_2(x_i, y_i, w_i)$, $c_i \mapsto F_3(x_i, y_i, w_i)$. Its Jacobian is block diagonal, so its determinant is exactly $(-2)^n$. Its complete fibre algebra is the tensor product over k of the single-block fibre algebras just proved. This assertion follows by presenting the tensor product as a polynomial ring on the disjoint sets of generators modulo the union of their defining relations. It includes every boundary factor at $c_i = 0$.

Over $\{\prod_i D_i \neq 0\}$ the block family is finite locally free of rank 3^n , by the preceding projective-cover proof and tensor products. Over its subopen $U_n = \{\prod_i c_i D_i \neq 0\}$, the algebra is finite free of rank 3^n with the explicit basis $\{\prod_i r_i^{e_i} : 0 \leq e_i \leq 2\}$. Spanning follows by repeated use of each original relation $c_i r_i^3 = 2r_i^2 - b_i r_i + 2a_i$. Independence follows by

taking tensor products of the proven single-block bases: applying the corresponding coordinate functionals to a putative relation extracts each coefficient. The generic algebra is a domain since eliminating the a_i identifies its polynomial presentation with $k[b_i, c_i, r_i : 1 \leq i \leq n]$ before localizing. A finite-dimensional domain over the target rational function field is a field, because multiplication by a nonzero element is an injective linear map and hence surjective. Thus the generic function-field degree is also exactly 3^n .

At the rational target with every block $(-1/4, 0, 0)$, the fibre contains exactly the 3^n ordered choices of the three displayed rational points. A format requiring one separate record for each point therefore has output length at least 3^n symbols. In contrast, the tuple of equations (P_i, P'_i) , their localizations, and the specified boundary rule has linear length in n with fixed-size coefficients at this target. The inequality follows from the mandated output format, and is not an inference about the difficulty of obtaining that compact representation.

Here is an exact arithmetic-circuit model. A gate is binary $+$, $-$, or $\times$, with fan-out allowed, and constants are 1, 2, 3, 4 for F . Input nodes, constant nodes, and output wires are not counted as gates. For one block the following list is a 26-gate circuit; each assignment is exactly one gate and no listed coefficient is suppressed:

$$\begin{array}{llll}
g_1 = xy, & g_2 = 1 + g_1, & g_3 = g_2g_2, & g_4 = g_3g_2, \\
g_5 = 3g_1, & g_6 = 4 + g_5, & g_7 = yy, & g_8 = g_7g_6, \\
g_9 = g_2g_8, & g_{10} = g_4w, & g_{11} = g_{10} + g_9, & g_{12} = xg_3, \\
g_{13} = g_{12}w, & g_{14} = 3g_{13}, & g_{15} = xg_8, & g_{16} = 3g_{15}, \\
g_{17} = y + g_{14}, & g_{18} = g_{17} + g_{16}, & g_{19} = xx, & g_{20} = g_{19}x, \\
g_{21} = g_{19}y, & g_{22} = 3g_{21}, & g_{23} = g_{20}w, & g_{24} = 2x, \\
g_{25} = g_{24} - g_{22}, & g_{26} = g_{25} - g_{23}.
\end{array}$$

The outputs are (g_{11}, g_{18}, g_{26}) ; substitution verifies all three original formulas. Thus $F^{[n]}$ uses $26n$ gates. Its component degrees remain 7, 6, 4 per block. The tuple is uniformly constructible: a loop on the binary index i writes the same 26 instructions with the input and gate indices of block i . Using binary indices, its bit description has length $O(n \log(n + 1))$, and the loop takes that many elementary output operations.

For bit evaluation, encode each signed integer by a sign and its binary magnitude, and assume magnitudes of the $3n$ inputs are less than 2^L , $L \geq 1$. Every displayed intermediate has magnitude less than 2^{7L+10} . For example $|g_1| < 2^{2L}$, $|g_2| < 2^{2L+1}$, $|g_6| < 2^{2L+2}$, $|g_{10}| < 2^{7L+3}$, and $|g_9| < 2^{6L+3}$; the remaining products in the list have smaller exponent bounds, while at most two sums require an extra bit. These inequalities also verify the loose common bound for all gates. Grade-school multiplication on $O(L)$ bits takes $O(L^2)$ bit operations by summing shifted partial products; addition and subtraction take $O(L)$ operations by one pass with carries. Evaluation therefore takes $O(n(L+1)^2)$ bit operations, plus the circuit description or output indexing cost. This is a uniform Boolean-time upper bound for this specified evaluation problem.

For target integers of magnitude less than 2^L , retain the coefficient tuples

$$(c_i, -2, b_i, -2a_i), \quad (3c_i, -4, b_i).$$

Their entries have at most $L + 3$ magnitude bits. Writing all polynomial and localization equations

$$P_i(r_i) = 0, \quad z_i P'_i(r_i) - 1 = 0$$

therefore costs $O(n(L+1))$ bits and bit operations up to the index encoding. The ring maps above identify z_i with $1/P'_i$ and hence $x_i = 2z_i$ on its stated chart. Testing $c_i = 0$ supplies the already proved boundary rule. No root selection is performed by writing these equations. When a rational root $r = p/q$, $q \neq 0$, is supplied as a certificate, with signed numerator and

denominator lengths at most M , integer cross multiplication verifies $P(r) = 0$ and $P'(r) \neq 0$ using a fixed number of products of $O(L + M + 1)$ -bit integers. The coordinate formulas then return exact rational numerators and denominators of $O(L + M + 1)$ bits in $O((L + M + 1)^2)$ operations. Reduction to coprime numerators and denominators is unnecessary: the output represents the same rational numbers and retains the actual expression. This proves the stated verification and lifting complexity while keeping the selection of an algebraic embedding distinct from its full algebra.

6.6 Exact determinant projections that retain every cubic coefficient

A single retained polynomial has the following affine-linear matrix:

$$M(a, b, c, r) = \begin{pmatrix} r & -1 & 0 & 0 \\ 0 & r & -1 & 0 \\ 0 & 0 & r & -1 \\ -2a & b & -2 & c \end{pmatrix}, \quad \det M = P.$$

Expansion along the last row gives the four signed contributions $-2a, br, -2r^2, cr^3$, respectively, proving the identity including signs. Introduce one new variable z and replace each constant entry by that constant times z . This gives the homogeneous linear matrix

$$M^h = \begin{pmatrix} r & -z & 0 & 0 \\ 0 & r & -z & 0 \\ 0 & 0 & r & -z \\ -2a & b & -2z & c \end{pmatrix}, \quad \det M^h = cr^3 - 2z^2r^2 + brz^2 - 2az^3 =: P^h.$$

Every monomial on the right has degree four; setting $z = 1$ gives the original polynomial exactly. On $z \neq 0$ its relation to the affine coordinates is the explicitly invertible substitution $(a, b, c, r, z) \mapsto (a/z, b/z, c/z, r/z, z)$ with inverse $(A, B, C, R, z) \mapsto (zA, zB, zC, zR, z)$, and $P^h = z^4 P(a/z, b/z, c/z, r/z)$. The divisor $z = 0$ is newly added: there $P^h = cr^3$. It is not identified with a finite original chart.

For n blocks, the block-diagonal determinant $\prod_i P_i$ has size $4n$. Its zero condition is a union of the individual hypersurfaces, since a product in a field is zero exactly when a factor is zero. It is not the simultaneous system $P_1 = \dots = P_n = 0$ when $n \geq 2$. For example set all $r_i = b_i = c_i = 0$, $a_1 = 0$, $a_2 = 1$, and for $i > 2$ set $a_i = 1$. The product vanishes but $P_2 = -2$.

The following determinant instead encodes every equation by a coefficient map. Introduce independent selectors $t_1, \dots, t_n$ and set

$$Q_n = \sum_{i=1}^n t_i P_i(r_i), \quad T_i = \begin{pmatrix} 1 & -r_i & 0 & 0 \\ 0 & 1 & -r_i & 0 \\ 0 & 0 & 1 & -r_i \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad v_i = \begin{pmatrix} -2a_i \\ b_i \\ -2 \\ c_i \end{pmatrix}.$$

Let $T = \text{diag}(T_1, \dots, T_n)$, $v = (v_1^\top, \dots, v_n^\top)^\top$, and $u = (t_1, 0, 0, 0, \dots, t_n, 0, 0, 0)$. Then

$$\mathcal{M}_n = \begin{pmatrix} T & v \\ -u & 0 \end{pmatrix}, \quad \det \mathcal{M}_n = Q_n, \quad \text{size}(\mathcal{M}_n) = 4n + 1.$$

For a complete proof, write $T_i = I - N_i$. Since $N_i^4 = 0$, $T_i^{-1} = I + N_i + N_i^2 + N_i^3$, whose first row is $(1, r_i, r_i^2, r_i^3)$. Thus $e_1^\top T_i^{-1} v_i = P_i$. Left multiplication of $\mathcal{M}_n$ by the determinant-one matrix $\begin{pmatrix} I & 0 \\ uT^{-1} & 1 \end{pmatrix}$ gives $\begin{pmatrix} T & v \\ 0 & uT^{-1}v \end{pmatrix}$. As $\det T = 1$, its determinant is $uT^{-1}v = Q_n$. Every entry of $\mathcal{M}_n$ is affine-linear in the original variables and selectors with coefficients in $\{0, 1, -1, -2\}$.

For an arbitrary coefficient ring R , the map $\iota : R^n \rightarrow R[t_1, \dots, t_n]_1$, $(f_1, \dots, f_n) \mapsto \sum_i t_i f_i$, is an R -module isomorphism onto the homogeneous selector-degree-one part, with inverse $Q \mapsto ([t_1]Q, \dots, [t_n]Q)$. Comparing the unique monomial coefficients proves both compositions are identities. In particular $(P_1, \dots, P_n) = ([t_1] \det \mathcal{M}_n, \dots, [t_n] \det \mathcal{M}_n)$ as ideals in $k[a_i, b_i, c_i, r_i]$. For a specialized point over any extension field, all fibre equations hold if and only if the determinant is identically zero as a polynomial in the selectors. Specializing the selectors first can lose information: when $t_1 = t_2 = 1$, the choices $P_1 = 2$, $P_2 = -2$ give zero sum without either equation holding. The coefficient map is an explicit reversible transport of the tuple, not an asserted isomorphism between the fibre scheme and one hypersurface in a larger space.

Replace each constant in $\mathcal{M}_n$ by that constant times z ; call the resulting homogeneous linear matrix $\mathcal{M}_n^h$. Its determinant is

$$\det \mathcal{M}_n^h = z^{4n-4} \sum_{i=1}^n t_i (c_i r_i^3 - 2z^2 r_i^2 + b_i r_i z^2 - 2a_i z^3).$$

To prove this without assigning an inverse at $z = 0$, first localize at z . The four diagonal entries of each T_i become z , and the first row of its inverse is $(z^{-1}, r_i z^{-2}, r_i^2 z^{-3}, r_i^3 z^{-4})$. The last vector becomes $(-2a_i, b_i, -2z, c_i)^T$, and $\det T = z^{4n}$. Applying the same determinant-one block multiplication gives the displayed formula in $k[\text{variables}, z, z^{-1}]$. Both sides belong to the polynomial subring, whose localization is injective, so the identity holds also at $z = 0$. The factor z^{4n-4} is explicit: for $n > 1$ the entire hyperplane $z = 0$ is an additional zero component of the padded determinant. At $z = 1$ all original fibre coefficients are recovered.

The determinantal projection is an exact coordinate-ring homomorphism. With $d = 4n + 1$ let $k[X_{pq} : 1 \leq p, q \leq d]$ be the coordinate ring of all $d \times d$ matrices. The homomorphism $\vartheta_n : k[X_{pq}] \rightarrow k[a_i, b_i, c_i, r_i, t_i, z]$ sends X_{pq} to the displayed (p, q) entry of $\mathcal{M}_n^h$. It sends the universal determinant to the stated padded polynomial. It is surjective: each a_i, b_i, c_i, r_i, t_i is a nonzero scalar multiple of a specified entry, and a diagonal entry gives z . Its kernel is generated by the linear relations among those entries. To prove the last claim constructively, for each target variable choose one entry mapping to it after its indicated scalar division. Subtract from every other X_{pq} the same linear combination of these chosen entries as its image. These linear differences lie in the kernel. Modulo them the source is the polynomial ring on the chosen entries; the induced map to the target polynomial ring has the inverse sending each target variable to its chosen entry. Therefore the kernel is exactly the ideal generated by those differences. This proves the map, its image, its kernel and the degree d of the determinant image.

Finally, Horner evaluation of the unaltered cubic uses seven gates: $c_i r_i$, then addition of -2 , multiplication by r_i , addition of b_i , multiplication by r_i , separately $(-2)a_i$, and addition of that last term. Computing all $t_i P_i$ and their sum uses $9n - 1$ gates. Thus Q_n has an explicitly uniform linear-size arithmetic circuit and a linear-size affine determinantal representation. The homogeneous padded form has a linear-size determinant representation as well. These are constructive upper bounds for a family derived from the original fibres. They retain their generic degree 3^n and their original exceptional loci. They prove no reduction from arbitrary Boolean satisfiability to these independent blocks, and they assert no Boolean complexity equality or separation. The concrete complexity transport established here is the coefficient-recoverable determinant projection and the fully specified uniform bit algorithms above.

7 Exact invariant-theory transports of the retained cubic and node

All schemes, coordinate rings and representations in this section are over $\mathbb{C}$. The computations are defined over $\mathbb{Q}$, except for the explicitly stated algebraic cube roots in the stabilizer. The

original inverse-fibre polynomial is retained as

$$P_{a,b,c}(r) = cr^3 - 2r^2 + br - 2a, \quad P_{a,b,c}^h(R, T) = cR^3 - 2R^2T + bRT^2 - 2aT^3.$$

In the source inverse chart, $r = y + 1/x$, $\alpha = P'(r)/2$, $x = \alpha^{-1}$, $y = r - \alpha$ and $w = 5\alpha^2 - 3r\alpha - c\alpha^3$; consequently the finite chart retains only simple roots. The constructions below keep the algebraic completion as a separate object. Their root at infinity when $c = 0$ is a homogeneous cubic root, not a licence to discard the source's additional point $(0, b, a - 4b^2)$.

The coordinate-ring restriction used in GCT is present explicitly in the primary treatment of Ikenmeyer and Panova [7, Sections 1(a) and 2(b)]. Here we derive the particular maps and representations directly. No determinant-permanent separation is inferred from the following fixed-dimensional calculations.

7.1 The binary representation, its coefficient slice and its discriminant

Let V be the four-dimensional space of binary cubics $f = AR^3 + BR^2T + CRT^2 + DT^3$. Write $G_{\text{bin}} = \text{GL}_2(\mathbb{C})$, $z = (R, T)^\top$, and define the left action

$$(g \cdot f)(z) = f(g^\top z).$$

Indeed $(g \cdot (h \cdot f))(z) = f(h^\top g^\top z) = ((gh) \cdot f)(z)$. For $g = \begin{pmatrix} p & q \\ s & t \end{pmatrix}$, $\delta = pt - qs \neq 0$, expansion gives

$$\begin{aligned} A_g &= Ap^3 + Bp^2q + Cpq^2 + Dq^3, \\ B_g &= 3Ap^2s + B(p^2t + 2psq) + C(2pqt + sq^2) + 3Dq^2t, \\ C_g &= 3Aps^2 + B(2pst + s^2q) + C(pt^2 + 2sqt) + 3Dqt^2, \\ D_g &= As^3 + Bs^2t + Cst^2 + Dt^3. \end{aligned}$$

The coefficient-slice embedding and its entire coordinate-ring map are

$$\begin{aligned} \iota : \mathbb{A}_{a,b,c}^3 &\longrightarrow V, & (a, b, c) &\longmapsto (c, -2, b, -2a), & \iota^* : \mathbb{C}[A, B, C, D] &\longrightarrow \mathbb{C}[a, b, c], \\ & & A &\longmapsto c, & B &\longmapsto -2, & C &\longmapsto b, & D &\longmapsto -2a. \end{aligned}$$

It is surjective: $a = -\iota^*D/2$, $b = \iota^*C$ and $c = \iota^*A$. Its kernel is $(B + 2)$, because evaluating a polynomial at $B = -2$ is division by the monic linear polynomial $B + 2$, and the remaining substitution $(A, C, D) = (c, b, -2a)$ is an invertible linear substitution. The original affine slice itself is not G_{bin} -stable.

Its saturation is the explicit equivariant morphism

$$\Phi : G_{\text{bin}} \times \mathbb{A}^3 \longrightarrow V, \quad (g, a, b, c) \longmapsto g \cdot \iota(a, b, c),$$

where h acts on the source by $(g, a, b, c) \mapsto (hg, a, b, c)$. Its pullback sends A, B, C, D to the four displayed expressions with $(A, B, C, D) = (c, -2, b, -2a)$, in $\mathbb{C}[p, q, s, t, \delta^{-1}, a, b, c]$. These formulas specify the map without eliminating any root or selecting a branch.

Proposition 7.1. *The image of Φ is $V \setminus \{0\}$, and $\ker \Phi^* = 0$.*

Proof. Every slice cubic has coefficient -2 and is nonzero; invertible substitution cannot send it to zero. Conversely, take a nonzero cubic f . There is a vector z_0 with $f(z_0) \neq 0$: a polynomial vanishing at every complex vector has all its coefficients zero, by successive one-variable interpolation. Choose w_0 independent of z_0 . Euler's homogeneous identity, obtained by differentiating $f(\lambda z_0) = \lambda^3 f(z_0)$ at $\lambda = 1$, gives $df_{z_0}(z_0) = 3f(z_0) \neq 0$. Consequently $df_{z_0}(w_0 + kz_0)$ is nonzero for all but one $k \in \mathbb{C}$. Choose one such k and replace w_0 by

$w = w_0 + kz_0$. The coefficient of R^2T in $f(Rz_0 + Tw)$ is exactly $df_{z_0}(w)$. Replace w by $-2w/df_{z_0}(w)$. The two chosen vectors remain independent and the resulting coefficient is -2 . Thus an explicitly specified invertible variable substitution sends f to the slice; its inverse shows $f \in \text{im } \Phi$. If a polynomial lies in $\ker \Phi^*$, it vanishes on every nonzero vector of V and hence also at zero, by restricting to each line. Successive interpolation again makes it the zero polynomial. $\square$

Define, with no division by the leading coefficient,

$$\Delta(A, B, C, D) = B^2C^2 - 4AC^3 - 4B^3D - 27A^2D^2 + 18ABCD.$$

The substitution in ι gives exactly

$$\iota^*\Delta = 4(b^2 - cb^3 + 18abc - 16a - 27a^2c^2). \quad (35)$$

For completeness, factor a binary cubic over $\mathbb{C}$ as $f(z) = \prod_{i=1}^3(\alpha_i R + \beta_i T)$, allowing one factor to carry any scalar. Direct expansion of the polynomial on the right gives

$$\Delta(f) = \prod_{1 \leq i < j \leq 3} (\alpha_i \beta_j - \alpha_j \beta_i)^2.$$

One may verify the identity first for $f = A \prod_i (R - r_i T)$ by expanding the squared Vandermonde product: it is $A^4 \prod_{i < j} (r_i - r_j)^2$ and expands to the displayed coefficient polynomial. Cubics with $A \neq 0$ are dense, so equality of these polynomials holds also at $A = 0$. This proves that $\Delta = 0$ is precisely a repeated projective root, including roots at infinity. Under $g \cdot f$, the coefficient column of each linear factor is multiplied by g . Each of the three two-by-two determinants is multiplied by $\det g$, proving

$$\Delta(g \cdot f) = (\det g)^6 \Delta(f). \quad (36)$$

The coordinate-ring representation is $(g \cdot H)(f) = H(g^{-1} \cdot f)$, so the one-dimensional span of Δ has character $(\det g)^{-6}$, not $(\det g)^6$.

7.2 The actual source orbit and all of its stabilizer

At the original target $(a, b, c) = (-1/4, 0, 0)$ put

$$q_0 = -2R^2T + \frac{1}{2}T^3, \quad \Delta(q_0) = 16.$$

Its three projective roots are $[1 : 0]$, $[1/2 : 1]$ and $[-1/2 : 1]$. We compute its stabilizer without identifying distinct scalar cubics. Introduce the auxiliary polynomial $h(U, V) = UV(U - V)$ and the exact invertible matrix

$$M = \begin{pmatrix} 1 & 1/2 \\ 0 & 1 \end{pmatrix}, \quad M^{-1} = \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix}, \quad q_0(z) = -2h(Mz).$$

This is an explicit comparison with an auxiliary polynomial; q_0 and its factor -2 remain fixed. Let

$$\begin{aligned} D_1 &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, & D_2 &= \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, & D_3 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \\ D_4 &= \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}, & D_5 &= \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, & D_6 &= \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}. \end{aligned}$$

Substitution in the three factors of h gives

$$h(D_j z) = \varepsilon_j h(z), \quad (\varepsilon_1, \dots, \varepsilon_6) = (1, 1, -1, 1, -1, -1).$$

Their fractional-linear maps are respectively t , $1 - t$, $1/t$, $t/(t - 1)$, $1/(1 - t)$ and $(t - 1)/t$. They give all permutations of $\{0, 1, \infty\}$: inspection gives six distinct permutations, and a projective transformation fixing all three must be the identity. To prove the latter assertion, fixing 0 and ∞ makes its matrix diagonal up to scalar; fixing 1 makes the two diagonal entries equal.

It follows that the full stabilizer H_0 is the following set of eighteen explicitly determined matrices:

$$H_0 = \left\{ \left(M^{-1} \kappa D_j M \right)^\top : 1 \leq j \leq 6, \quad \kappa^3 \varepsilon_j = 1 \right\}. \quad (37)$$

Indeed a stabilizer must permute the three roots. After conjugating its transpose by M , it is therefore κD_j for exactly one j and a nonzero scalar κ . Substitution into $-2h(Mz)$ makes the multiplier exactly $\kappa^3 \varepsilon_j$, which must be one. Conversely every listed matrix has that multiplier and stabilizes q_0 . Each of the six scalar equations has three different roots, and the six projective classes are different. The homomorphism to root permutations is explicitly $[z] \mapsto [g^{-\top} z]$: $(gh)^{-\top} = g^{-\top} h^{-\top}$ proves its composition law. The transposes used to compute stabilizer membership above give its inverse permutations. Consequently the root action gives the exact sequence

$$1 \longrightarrow \{\omega I : \omega^3 = 1\} \longrightarrow H_0 \longrightarrow \mathfrak{S}_3 \longrightarrow 1.$$

Every two unordered triples of distinct projective points are related by a projective transformation. Here is a constructive proof. Choose nonzero representatives v_1, v_2 for the first two points; they form a basis. Write a representative for the third as $av_1 + bv_2$; distinctness gives $a, b \neq 0$. The linear isomorphism sending av_1 and bv_2 to $(1, 0)^\top$ and $(0, 1)^\top$ sends the third point to $[1 : 1]$. Apply this construction twice and compose one map with the other's inverse. Consequently a squarefree binary cubic is an invertible substitution of q_0 times a nonzero scalar. Absorb that scalar by a scalar matrix with its specified cube root. We have proved

$$G_{\text{bin} q_0} = \{f : \Delta(f) \neq 0\}, \quad \overline{G_{\text{bin} q_0}} = V. \quad (38)$$

The closure equality also follows directly because the complement is the zero set of a nonzero polynomial. Its coordinate ring and the coordinate ring of the orbit are, respectively,

$$\mathbb{C}[\overline{G_{\text{bin} q_0}}] = \mathbb{C}[A, B, C, D], \quad \mathbb{C}[G_{\text{bin} q_0}] = \mathbb{C}[A, B, C, D, \Delta^{-1}].$$

The localization description is the defining coordinate ring of the principal affine open set: it is represented by the hypersurface $z\Delta - 1 = 0$, with projection inverse $f \mapsto (f, \Delta(f)^{-1})$.

We can compute the invariant functions as well. Scalar matrices act on V by $f \mapsto \lambda^3 f$. If a polynomial is G_{bin} -invariant, decompose it into its homogeneous terms. Comparing the distinct powers λ^{3d} in its value on $\lambda^3 f$ makes every positive-degree term zero. Hence $\mathbb{C}[V]^{G_{\text{bin}}} = \mathbb{C}$.

In contrast,

$$\mathbb{C}[V]^{\text{SL}_2} = \mathbb{C}[\Delta]. \quad (39)$$

To prove this equality explicitly, first note that $\det(H_0) = \{\zeta : \zeta^6 = 1\}$. Formula (36) gives containment. Scalar stabilizers give all cube roots of unity as determinants, and the matrix with D_2 and $\kappa = 1$ gives determinant -1 , so the reverse containment holds. If f_1 and f_2 are squarefree and have equal discriminant, choose g with $gf_1 = f_2$. Then $(\det g)^6 = 1$. The stabilizer of f_1 is conjugate to H_0 , so it has an element h with $\det h = (\det g)^{-1}$. Now

$gh \in \mathrm{SL}_2$ still sends f_1 to f_2 . For an SL_2 -invariant polynomial J , define the polynomial in one variable

$$j(t) = J\left(-2R^2T + \frac{t}{32}T^3\right).$$

The argument of J has discriminant t . Thus the just-proved orbit statement gives $J(f) = j(\Delta(f))$ on $\Delta \neq 0$, hence everywhere by polynomial density. Conversely (36) makes every polynomial in Δ an SL_2 -invariant.

For later representation comparisons one need not suppress any weights. For $g = \mathrm{diag}(\lambda, \mu)$ the full degree- d character of the coordinate ring of this orbit closure is

$$\sum_{i+j+k+\ell=d} \lambda^{-(3i+2j+k)} \mu^{-(j+2k+3\ell)}. \quad (40)$$

This follows from its monomial basis $A^i B^j C^k D^\ell$ and the four coefficient weights. In particular every degree, every weight and every multiplicity in this orbit closure agrees with the corresponding ambient coordinate-ring representation, because the two rings are equal by (38).

7.3 The retained node as a ring and a representation

On $b = c = 0$, the source root cover is $a = -r^2$. Its completed self-fibre product and its exact linear comparison are

$$\begin{aligned} B_{\mathrm{root}} &= \mathbb{C}[r_1, r_2]/(r_1^2 - r_2^2), & R_{\mathrm{node}} &= \mathbb{C}[u, v]/(uv), \\ u &= r_1 - r_2, & v &= r_1 + r_2, & r_1 &= (u + v)/2, & r_2 &= (v - u)/2. \end{aligned}$$

Substitution gives $uv = r_1^2 - r_2^2$, and the displayed inverse proves the ring isomorphism. In this ring,

$$a = -r_1^2 = -r_2^2 = -\frac{u^2 + v^2}{4}. \quad (41)$$

Removing $r_1 r_2 = 0$ removes the origin from both components, so the crossing and the modules supported there belong to the completed object.

Let $K \subset G_{\mathrm{node}} = \mathrm{GL}_2(\mathbb{C})$ be the monomial subgroup generated by the diagonal torus $T_{\mathrm{node}} = \{\mathrm{diag}(\lambda, \mu) : \lambda\mu \neq 0\}$ and $\sigma = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. It acts on node points by ordinary matrix multiplication, and on functions by $(k \cdot f)(z) = f(k^{-1}z)$. Thus u and v have torus characters λ^{-1} and μ^{-1} and σ exchanges them. The relations are preserved. This K does not fix the map to a : a diagonal point transformation sends the expression in (41) to $-(\lambda^2 u^2 + \mu^2 v^2)/4$. All uses of K below concern this specified node action, not an unstated automorphism over $\mathbb{A}_a^1$.

For clarity retain R_{node} and define its comparison ring

$$\tilde{R} = \mathbb{C}[u] \oplus \mathbb{C}[v], \quad j(f) = (f(u, 0), f(0, v)).$$

Every element of R_{node} has a unique expression $c + up(u) + vq(v)$: the relation kills all mixed monomials, and restriction to the two axes proves uniqueness. Thus j is injective, its image is exactly the pairs with equal constant terms, and the inverse on that image is $(p(u), q(v)) \mapsto p(u) + q(v) - p(0)$. Every element of $\tilde{R}$ is integral over the image, since $\tilde{R}$ is generated as an R_{node} -module by $(1, 1)$ and the idempotent $(1, 0)$, which satisfies $e^2 - e = 0$. The total ring of fractions is $\mathbb{C}(u) \oplus \mathbb{C}(v)$. To check this assertion, the non-zero-divisor $u + v$ allows the fractions $u/(u + v) = (1, 0)$ and $v/(u + v) = (0, 1)$; localizing within each component gives its rational-function field. Each polynomial ring in one variable is integrally closed: a rational function p/q with coprime p, q satisfying a monic equation makes q divide p^n , so q is

constant. Componentwise application proves $\tilde{R}$ integrally closed. It is therefore precisely the algebraic normalization comparison ring, with the injection retained.

Define $\epsilon : \tilde{R} \rightarrow \mathbb{C}$ by $\epsilon(p, q) = p(0) - q(0)$. Its codomain is the K -module $\mathbb{C}_{\text{sgn}}$ on which the torus acts trivially and σ acts by -1 . The preceding reconstruction proves the exact K -equivariant sequence

$$0 \longrightarrow R_{\text{node}} \xrightarrow{j} \tilde{R} \xrightarrow{\epsilon} \mathbb{C}_{\text{sgn}} \longrightarrow 0. \quad (42)$$

The conductor is precisely

$$\mathfrak{c} = (u, v), \quad j(\mathfrak{c}) = u\mathbb{C}[u] \oplus v\mathbb{C}[v].$$

Indeed a zero-constant pair remains an equal-constant pair after multiplication by any element of $\tilde{R}$. Conversely multiplying a conductor element by $(1, 0)$ forces its common constant to be zero. The quotient in (42) is annihilated by $\mathfrak{c}$, and conversely a nonzero constant cannot annihilate it.

Give u, v degree one. The degree-zero pieces of (42) are respectively the trivial representation, the sum of trivial and sign representations, and the sign representation. For every $n \geq 1$ the two rings have the same degree- n basis u^n, v^n , with torus character $\lambda^{-n} + \mu^{-n}$ and with σ swapping its basis. Therefore

$$R_{\text{node}}^{T_{\text{node}}} = \mathbb{C}, \quad \tilde{R}^{T_{\text{node}}} = \mathbb{C} \oplus \mathbb{C}, \quad R_{\text{node}}^K = \tilde{R}^K = \mathbb{C}.$$

These equalities follow by comparing the independent torus characters in the monomial bases and then taking the swap invariants. Thus the full K -invariant scalar rings coincide, while the exact sign cokernel in (42) remains nonzero.

The differential module and its comparison map are

$$\Omega_{R_{\text{node}}}^1 = (R_{\text{node}} du \oplus R_{\text{node}} dv) / R_{\text{node}}(v du + u dv), \quad \Omega_{\tilde{R}}^1 = \mathbb{C}[u]du \oplus \mathbb{C}[v]dv.$$

The presentation follows directly from the universal derivation of the polynomial ring and $d(uv) = vdu + u dv$. Its class $\theta = vdu = -udv$ is nonzero. Indeed if $(v, 0) = h(v, u)$ in the free module, then $hu = 0$, so the unique-form description gives $h \in (v)$. The equation $hv = v$ would then imply $h(0, v) = 1$, contradicting its zero constant. Multiplying the relation by u and v proves $u\theta = v\theta = 0$. The non-zero-divisor $u + v$ kills θ , so this is torsion in the literal module-theoretic sense.

If $Adu + Bdv$ maps to zero in $\Omega_{\tilde{R}}^1$, then $A(u, 0) = 0$ and $B(0, v) = 0$, giving $A = va(v)$ and $B = ub(u)$. The relations $v^2 du = 0$ and $u^2 dv = 0$ reduce its class exactly to $(a(0) - b(0))\theta$. Conversely θ maps to zero, and every element of $\Omega_{\tilde{R}}^1$ lifts by these two displayed summands. We have proved the exact sequence

$$0 \longrightarrow \mathbb{C}\theta \longrightarrow \Omega_{R_{\text{node}}}^1 \longrightarrow \Omega_{\tilde{R}}^1 \longrightarrow 0. \quad (43)$$

The original-coordinate identity is $\theta = r_2 dr_1 - r_1 dr_2$: substitute the two linear inverse formulas and use $r_1 dr_1 = r_2 dr_2$, the derivative of the original relation.

Differentiation commutes with the specified action on functions. It gives

$$\text{diag}(\lambda, \mu) \cdot \theta = \lambda^{-1} \mu^{-1} \theta, \quad \sigma \cdot \theta = u dv = -\theta.$$

Thus $\mathbb{C}\theta = \mathbb{C}_{\det^{-1}}$ as a K -module. With $\deg du = \deg dv = 1$, it lies in degree two, and

$$\text{Hilb}(\Omega_{R_{\text{node}}}^1, t) = \frac{2t}{1-t} + t^2.$$

More precisely, the degree- n character for $n \geq 1$ is $\lambda^{-n} + \mu^{-n}$ plus the additional character $\lambda^{-1} \mu^{-1}$ exactly at $n = 2$; the additional swap character is negative. No copy of $\det^{-1}$ occurs in R_{node} itself, since none of its torus weights is $\lambda^{-1} \mu^{-1}$. The exact module comparison (43), rather than replacement of the module by its coefficient ring, retains this difference.

7.4 An exact common torus and the discriminant pullback

Define the two homomorphisms from $\mathbb{G}_m$ by

$$\beta(\eta) = \text{diag}(\eta^{-1}, \eta^2) \in G_{\text{bin}}, \quad \nu(\eta) = \eta^3 I \in G_{\text{node}}.$$

The map from the completed node to binary cubics is

$$\psi(u, v) = -2R^2T + \frac{u^2 + v^2}{2}T^3 = P_{a,0,0}^h(R, T), \quad a = -\frac{u^2 + v^2}{4}. \quad (44)$$

Its coordinate-ring map is

$$\psi^* : A \mapsto 0, \quad B \mapsto -2, \quad C \mapsto 0, \quad D \mapsto (u^2 + v^2)/2.$$

Its kernel is exactly $(A, C, B + 2)$. To see the unasserted part of that equality, after division by those three generators any kernel element would be a polynomial $H(D)$ with $H(u^2/2) = 0$ on $v = 0$; its distinct powers of u force every coefficient to vanish. Its image is the polynomial subalgebra $\mathbb{C}[(u^2 + v^2)/2]$, not the entire node ring. Since $\beta(\eta)$ multiplies R^2T by $\eta^{-2}\eta^2 = 1$ and T^3 by η^6 , while $\nu(\eta)$ multiplies $u^2 + v^2$ by η^6 , this map is equivariant for the two stated homomorphisms. Its discriminant pullback is exactly

$$\psi^*\Delta = 16(u^2 + v^2) = -64a. \quad (45)$$

Both Δ restricted along β and θ restricted along ν have coordinate-module weight η^{-6} ; their domains and module structures remain different and have just been specified. The discriminant retains quadratic vanishing on each branch. Its zero scheme in the node has coordinate ring

$$R_{\text{node}}/(u^2 + v^2) = \mathbb{C}\{1, u, v, u^2\}, \quad v^2 = -u^2, \quad u^3 = v^3 = 0.$$

These relations span the quotient, and independence follows by homogeneous degree: degrees zero and one have no extra relation, while in degree two precisely the one relation $u^2 + v^2$ is imposed. It consequently has length four. The conductor quotient $R_{\text{node}}/\mathfrak{c} = \mathbb{C}$ has length one. Their support is the same origin because the radical of $(u^2 + v^2)$ in the node is (u, v) , as the displayed nilpotence proves. These are the exact scheme and module relationships, including the multiplicity retained by the discriminant.

7.5 Exact induction to the connected general linear group

We now turn the K -representations into explicitly defined rational G_{node} -representations. Put

$$L = \mathbb{C}[G_{\text{node}}] = \mathbb{C}[p, q, s, t, (pt - qs)^{-1}].$$

For any rational K -module M define

$$\mathcal{I}(M) = (L \otimes_{\mathbb{C}} M)^K,$$

where the diagonal action on a pure tensor is $k \cdot (f \otimes m) = (g \mapsto f(gk)) \otimes (k \cdot m)$. The left G_{node} -action is $h \cdot (f \otimes m) = (g \mapsto f(h^{-1}g)) \otimes m$. The two actions commute by direct substitution; hence the invariant subspace is a rational G_{node} -module. When M is a K -algebra, $\mathcal{I}(M)$ is an algebra with componentwise multiplication. These formulas define the invariant rings directly; no assertion that their spectra are geometric quotients is needed.

We prove exactness rather than assume it. A rational torus module is a sum of its integer weight spaces: expanding its coaction in the independent Laurent monomials $\lambda^i \mu^j$ and comparing the coaction identity proves that each coefficient is a vector of that weight and that

each vector has a finite such expansion. For a surjective rational T_{node} -module map and an invariant vector in its target, choose a lift and take its weight-zero component; the latter is an invariant lift. The normalizing swap preserves weight zero. If the target vector is also swap invariant, average that lift with its swap, dividing by two. This is a K -invariant lift. Injections and kernels commute with invariants simply by their definitions. Thus taking K -invariants is exact. Tensoring an exact sequence of complex vector spaces with L is exact, by choosing bases or a vector-space splitting. This proves exactness of $\mathcal{I}$ in full.

Apply it to the two proved sequences. With $\mathcal{A} = \mathcal{I}(R_{\text{node}})$ and $\tilde{\mathcal{A}} = \mathcal{I}(\tilde{R})$, we obtain

$$0 \longrightarrow \mathcal{A} \longrightarrow \tilde{\mathcal{A}} \xrightarrow{\mathcal{I}(\epsilon)} \mathcal{I}(\mathbb{C}_{\text{sgn}}) \longrightarrow 0, \quad (46)$$

$$0 \longrightarrow \mathcal{I}(\mathbb{C}_{\det^{-1}}) \longrightarrow \mathcal{I}(\Omega_{R_{\text{node}}}^1) \longrightarrow \mathcal{I}(\Omega_{\tilde{R}}^1) \longrightarrow 0. \quad (47)$$

The maps are the original inclusion, difference of constants and differential comparison applied coefficient by coefficient in L . The first sequence is a sequence of $\mathcal{A}$ -modules; its quotient carries the action induced from evaluation at the node origin. We have not identified the last two modules of (47) with absolute Kähler differentials of the invariant rings. Such an identification is unnecessary for the displayed exact transport.

The quotient and the kernel have explicit nonzero elements and finite-dimensional G_{node} -subrepresentations. Let $J = \text{diag}(1, -1)$ and define

$$Z(g) = gJg^{-1} = \frac{1}{pt - qs} \begin{pmatrix} pt + qs & -2pq \\ 2st & -(pt + qs) \end{pmatrix}. \quad (48)$$

For a diagonal k one has $Z(gk) = Z(g)$; for $k = \sigma$ one has $Z(g\sigma) = -Z(g)$, since $\sigma J \sigma = -J$. Consequently each entry of $Z(g)$ tensored with the fixed generator of $\mathbb{C}_{\text{sgn}}$ belongs to $\mathcal{I}(\mathbb{C}_{\text{sgn}})$. A lift through (46) is obtained by tensoring the same entry with $(1/2, -1/2) \in \tilde{R}$, whose difference of constants is one and whose K -character is sign. The diagonal entry is nonzero already at $g = I$.

The three coefficient functions $(pt + qs)/(pt - qs)$, $-2pq/(pt - qs)$ and $2st/(pt - qs)$ are linearly independent. If a linear combination were zero, multiplication by $pt - qs$ would give a polynomial linear combination of $pt + qs, pq, st$; comparison of their four distinct monomials forces its three coefficients zero. They span a G_{node} -stable three-dimensional module, because $Z(h^{-1}g) = h^{-1}Z(g)h$. More exactly, with $\mathfrak{sl}_2 = \{X : \text{tr } X = 0\}$, the map

$$\mathfrak{sl}_2 \longrightarrow \mathcal{I}(\mathbb{C}_{\text{sgn}}), \quad X \longmapsto (g \mapsto \text{tr}(XZ(g))) \otimes 1$$

is injective by the established coefficient independence and the nondegenerate trace pairing on trace-zero two-by-two matrices. It is equivariant for the adjoint action, since

$$\text{tr}(XZ(h^{-1}g)) = \text{tr}(hXh^{-1}Z(g)).$$

If desired its irreducibility can be checked without a classification theorem: under diagonal conjugation the matrices E_{12} , $\text{diag}(1, -1)$ and E_{21} have three distinct weights. A nonzero invariant subspace therefore contains one of these weight vectors, by coefficient projection. Differentiating conjugation by $I + tE_{12}$ and $I + tE_{21}$ gives brackets; the identities $[E_{12}, E_{21}] = J$, $[J, E_{12}] = 2E_{12}$ and $[J, E_{21}] = -2E_{21}$ then produce all three basis vectors.

For the torsion kernel use the regular function $\det g = pt - qs$. The tensor

$$e(g) = (\det g) \theta \in \mathcal{I}(\mathbb{C}_{\det^{-1}}) \quad (49)$$

is invariant because right translation multiplies its first factor by $\det k$ and its second factor by $(\det k)^{-1}$. It is nonzero, and the left action sends it to $(\det h)^{-1}e$. Thus the exact

connected-group transport retains a determinant-inverse line in degree two as well as the adjoint subrepresentation in the degree-zero conductor quotient. Degrees refer to the node variables; L has degree zero. The original annihilator information is retained by the module maps: an element of $\mathcal{A}$ whose coefficient functions vanish at the node origin annihilates both displayed defect modules, because each original defect is supported at that origin.

7.6 The exact coordinate-ring obstruction interface

For any two closed G -stable subsets X, Y of a finite-dimensional representation W , their polynomial ideals and coordinate rings are defined by $I(X) = \{f \in \mathbb{C}[W] : f|_X = 0\}$ and $\mathbb{C}[X] = \mathbb{C}[W]/I(X)$. Inclusion $X \subseteq Y$ gives $I(Y) \subseteq I(X)$, and the entire equivariant restriction map is

$$\mathbb{C}[Y] \twoheadrightarrow \mathbb{C}[X], \quad [f]_{I(Y)} \mapsto [f]_{I(X)}, \quad \ker = I(X)/I(Y).$$

Conversely, if a polynomial $f \in I(Y)$ has $f(x) \neq 0$ at some $x \in X$, that evaluation proves $x \notin Y$ and excludes the inclusion. A finite-dimensional G -submodule containing that polynomial may be used to record its representation content; it is finite-dimensional because substitution in a polynomial of bounded degree preserves that degree bound. This is the precise polynomial-map interface used by an orbit-closure obstruction.

For the squarefree source cubic, the full ideal is zero by (38), and its degree-wise characters are exactly (40). Thus this particular orbit closure supplies no nonzero polynomial vanishing on it. The saturation calculation proves exactly the same zero-kernel statement for the whole original coefficient slice. The conductor transport instead supplies the nonzero kernels and cokernels in (46)–(47), with explicit module elements and representations. Those sequences are actual correspondences with connected-group representation data; they do not assert that these modules are the coordinate ring of a permanent or determinant orbit closure. The proved common edge between the original node and the binary-cubic representation is (44), with its exact kernel, discriminant pullback, degrees, and torus homomorphisms. Every claim in this fixed-dimensional transport is thus attached to a specified map. No assertion about the complexity of an unbounded family or about a nonstandard quantum-group positivity conjecture has been substituted for one of these calculations.

8 Retaining specialization layers and arithmetic operators

8.1 The completed two-sheet algebra over its original base

Throughout this subsection k is a field of characteristic different from 2. Put $A = k[a]$ and retain $a = -r^2$ from the original $b = c = 0$ inverse-root cover. Its self-fibre product is

$$B = A[r_1, r_2]/(r_1^2 + a, r_2^2 + a) \simeq k[r_1, r_2]/(r_1^2 - r_2^2).$$

The isomorphism substitutes $a = -r_1^2$; its inverse sends the displayed r_i to their residue classes. In particular it keeps the base map. Successive division by the two monic quadratics gives the unique expression

$$f_0(a) + f_1(a)r_1 + f_2(a)r_2 + f_{12}(a)r_1r_2. \quad (50)$$

Uniqueness follows first in the free A -module $A[r_1]/(r_1^2 + a)$ and then in its monic extension in r_2 . Thus B is free of rank four over A .

The component ring is $N = k[t] \oplus k[t]$ with a acting as $(-t^2, -t^2)$. Define

$$\nu(r_1) = (t, t), \quad \nu(r_2) = (t, -t), \quad \delta(p, q) = \frac{p(0) - q(0)}{2}. \quad (51)$$

The relations are respected. Use the invertible linear substitution $u = r_1 - r_2$, $v = r_1 + r_2$, with inverse $r_1 = (u + v)/2$, $r_2 = (v - u)/2$ and relation $uv = 0$. Every class has a unique expression $c + up(u) + vq(v)$. Its restrictions to the two axes determine all three terms, so ν is injective. A pair of polynomials lies in the image precisely when its constants agree: the constant and two positive-degree axis parts reconstruct it. Hence

$$0 \longrightarrow B \xrightarrow{\nu} N \xrightarrow{\delta} k \longrightarrow 0 \quad (52)$$

is exact over A , with $ak = 0$. The conductor is $(r_1, r_2) = (u, v)$: pairs with zero constants remain in the image after multiplication by every element of N ; multiplication by $(1, 0)$ forces zero common constant in the converse.

In the ordered A -bases

$$\mathcal{B} = (1, r_1, r_2, r_1 r_2), \quad \mathcal{N} = ((1, 1), (t, t), (t, -t), (1, -1))$$

of B, N , respectively, the matrix is

$$[\nu]_{\mathcal{N} \leftarrow \mathcal{B}} = \text{diag}(1, 1, 1, -a). \quad (53)$$

The last column uses $(t^2, -t^2) = -a(1, -1)$. The four vectors of $\mathcal{N}$ are a basis by even-odd decomposition of each component and division by 2. Thus ν is an isomorphism over $a \neq 0$, with inverse last entry $-1/a$. No inverse is assigned at $a = 0$.

Proposition 8.1 (The exact specialization defect). *For $B_0 = B/aB$, $N_0 = N/aN$ there is an exact sequence*

$$0 \longrightarrow k \xrightarrow{\partial} B_0 \xrightarrow{\nu_0} N_0 \xrightarrow{\delta_0} k \longrightarrow 0, \quad \partial(1) = -r_1 r_2.$$

Here $B_0 = k[r_1, r_2]/(r_1^2, r_2^2)$, $N_0 = k[t]/(t^2) \oplus k[t]/(t^2)$, and the first k is $\text{Tor}_1^A(k, k)$ for the quotient in (52).

Proof. The free resolution $0 \rightarrow A \xrightarrow{a} A \rightarrow k \rightarrow 0$ becomes the zero map $k \rightarrow k$ after tensoring with k , so its first homology is k . The middle modules in (52) are free. Alternatively the diagonal matrix at $a = 0$ directly gives the exact kernel and cokernel. To compute the connecting sign, lift $1 \in k$ by $(1, -1) \in N$. Its product by a is $(-t^2, t^2) = \nu(-r_1 r_2)$. The class $r_1 r_2$ is nonzero because (50) stays a basis after tensoring. This proves every map and sign. $\square$

Let $\Gamma = \mu_2 \times \mu_2$ act by $(\epsilon_1, \epsilon_2)r_i = \epsilon_i r_i$ and fix a . Write $\chi_i(\epsilon_1, \epsilon_2) = \epsilon_i$. On N its first generator acts by $(p(t), q(t)) \mapsto (q(-t), p(-t))$, and the second by $(p(t), q(t)) \mapsto (q(t), p(t))$. Checking r_1, r_2 proves equivariance of ν . Both generators negate the constant difference, so the quotient character is $\chi_1 \chi_2$. The Tor term has the same character because a is fixed. With $\deg r_i = \deg t = 1$, $\deg a = 2$, its image has degree 2, whereas the cokernel has degree 0. The ungraded character of each special fibre is $1 + \chi_1 + \chi_2 + \chi_1 \chi_2$, but their graded characters are

$$\text{ch}_z B_0 = 1 + z(\chi_1 + \chi_2) + z^2 \chi_1 \chi_2, \quad \text{ch}_z N_0 = 1 + \chi_1 \chi_2 + z(\chi_1 + \chi_2).$$

The bases prove these statements completely. Moreover B_0 has unique maximal ideal $\mathfrak{m} = (r_1, r_2)$, $\mathfrak{m}^2 = (r_1 r_2)$, $\mathfrak{m}^3 = 0$. Every element with nonzero constant is invertible by the finite geometric series in its nilpotent part, which proves uniqueness. In contrast N_0 contains the nontrivial idempotent $(1, 0)$ and has two residue-field quotients. Thus identical ungraded multiplicities do not reconstruct the algebra or the specialization map.

8.2 An explicit finite-field zeta calculation

All preceding equations are defined over $\mathbb{Z}[1/2]$. Take $k = \mathbb{F}_q$ with q odd. Over $\mathbb{F}_{q^m}$ the two lines $r_2 = r_1$ and $r_2 = -r_1$ intersect in one point, so

$$|\mathrm{Spec} B(\mathbb{F}_{q^m})| = 2q^m - 1, \quad |\mathrm{Spec} N(\mathbb{F}_{q^m})| = 2q^m.$$

Define $Z_X(T) = \exp(\sum_{m \geq 1} |X(\mathbb{F}_{q^m})| T^m / m)$. The formal identity $-\log(1 - vT) = \sum_{m \geq 1} v^m T^m / m$ gives

$$Z_{\mathrm{Spec} B}(T) = \frac{1 - T}{(1 - qT)^2}, \quad Z_{\mathrm{Spec} N}(T) = \frac{1}{(1 - qT)^2}. \quad (54)$$

The conductor point contributes the exact factor $1 - T$. The reciprocal zero is 1 and reciprocal poles are q, q . This explicitly computed singular affine curve is not assigned the purity assertion for a smooth projective object. The original inverse chart requires $r_1 r_2 \neq 0$, equivalently $a \neq 0$. Its two lines each lose the origin and are disjoint, giving point count $2(q^m - 1)$ and zeta function $(1 - T)^2 / (1 - qT)^2$. This is the exact effect of restoring the chart exclusion.

8.3 The arithmetic operator and its finite matrix map

The arithmetic application uses the exact source quotient of Connes–Consani–Moscovici [4, Section 3.6, Proposition 3.6]:

$$Q_{\mathrm{ar}} = \mathcal{H}_{\leq 1} / \overline{\mathrm{span}\{\mathcal{F}_\mu \mathcal{E}(\psi_\ell^\pm)\}}, \quad (\mathcal{F}_\mu \mathcal{E}(\psi_\ell^\pm))(s) = R_\ell^\pm(s) \Xi(s).$$

Here $\mathcal{H}_{\leq 1}$ is their Hadamard topological ring of entire functions of order at most one. Multiplication by the original coordinate s induces S on this quotient. Their proposition identifies its spectrum with the s for which $1/2 + is$ is a nontrivial zeta zero. This arithmetic identification is a reported primary-source theorem. The following extension identities are proved here for this exact module and for every continuous module of the same type. The closed subspace is retained as stated; no replacement by an algebraic principal ideal or orthogonal complement is used.

Let Q be a complex topological vector space and S a continuous linear operator. Regard Q as a $\mathbb{C}[s]$ -module by $s \cdot v = Sv$. The base change $s \mapsto -r^2/2$ gives

$$\hat{Q} = \mathbb{C}[r] \otimes_{\mathbb{C}[s]} Q = Q \oplus rQ, \quad \mathcal{R} = \begin{pmatrix} 0 & -2S \\ I & 0 \end{pmatrix}, \quad -\mathcal{R}^2/2 = \mathrm{diag}(S, S).$$

The unique even–odd expression $p_0(-r^2/2) + rp_1(-r^2/2)$ proves freeness and all displayed maps. Use the finite direct-sum topology. The involution is $(v_0, v_1) \mapsto (v_0, -v_1)$; the trace is $2v_0$, with kernel rQ . This is $\mathbb{C}[s]$ -linear. Projection $\pi(v_0, v_1) = v_0$ has section $\iota(v) = (v, 0)$. Here spectrum means failure of a continuous two-sided inverse. If T inverts $S - \lambda I$, then $\mathrm{diag}(T, T)$ inverts $\mathrm{diag}(S, S) - \lambda I$. Conversely an inverse D of the latter gives the continuous inverse $\pi D \iota$ of $S - \lambda I$, by the two intertwining identities. Thus the spectrum in the original s coordinate is preserved exactly.

For each $d \geq 1$ define an actual polynomial map

$$\Phi_d : \mathrm{Mat}_d(\mathbb{C}) \longrightarrow \mathrm{Mat}_{2d}(\mathbb{C}), \quad A \longmapsto \begin{pmatrix} 0 & -2A \\ I_d & 0 \end{pmatrix}. \quad (55)$$

Its image is the affine linear slice with these three fixed blocks; its inverse there is $A = -M_{12}/2$. The map on coordinate rings is surjective with kernel generated by the entries of $M_{11}, M_{22}, M_{21} - I_d$. The remaining block maps freely to $-2A_{ij}$, proving the kernel. Conjugation by g is intertwined with conjugation by $\mathrm{diag}(g, g)$, by block multiplication.

Proposition 8.2 (All characteristic-coefficient invariants retained). *Write $\det(uI - A) = \sum_{j=0}^d (-1)^j e_j(A) u^{d-j}$, $e_0 = 1$. Then*

$$\det(tI - \Phi_d(A)) = \det(t^2 I + 2A) = \sum_{j=0}^d 2^j e_j(A) t^{2d-2j}.$$

Pullback on conjugation-invariant rings is the surjection

$$\mathbb{C}[e_1, \dots, e_{2d}] \longrightarrow \mathbb{C}[e_1, \dots, e_d], \quad e_{2j} \mapsto 2^j e_j, \quad e_{2j-1} \mapsto 0,$$

whose kernel is exactly the ideal of odd generators in the source.

Proof. For indeterminate $t \neq 0$, the Schur complement of tI_d in $\begin{pmatrix} tI & 2A \\ -I & tI \end{pmatrix}$ gives $t^d \det(tI + 2A/t) = \det(t^2 I + 2A)$. Polynomial identity extends to $t = 0$. Expansion gives every coefficient and sign.

We also prove the invariant-ring assertion. An invariant polynomial restricted to diagonal matrices is symmetric, by permutation matrices. Every symmetric polynomial is a polynomial in the e_j : in one homogeneous component its largest lexicographic monomial has exponents $b_1 \geq \dots \geq b_d$. Subtract its coefficient times $e_1^{b_1-b_2} \dots e_{d-1}^{b_{d-1}-b_d} e_d^{b_d}$. The leading monomial disappears without a larger one being introduced. The finite set of monomials of that degree makes this algorithm terminate. Matrices with distinct eigenvalues form the nonempty discriminant-open set and are diagonalizable. Independence of their eigenvectors follows by applying the product of the other eigenvalue factors to a dependence. The invariant and its polynomial in e_j agree there and hence everywhere: localization at a nonzero polynomial in a polynomial domain is injective. The e_j are independent since every monic complex polynomial factors and is the characteristic polynomial of a diagonal matrix. This proves the two invariant rings. Quotienting the source by its odd generators leaves a polynomial ring mapped isomorphically by the nonzero scalars 2^j , proving the exact kernel. $\square$

No unspecified finite truncation of the arithmetic quotient is used in (55). It is a natural finite-dimensional version of the same base change. The algebra B also embeds into $\text{End}_A(B)$ by left multiplication: an element acting as zero annihilates 1. In the basis $(1, r_1) \otimes (1, r_2)$ its two root operators are

$$R_1 = \begin{pmatrix} 0 & -a \\ 1 & 0 \end{pmatrix} \otimes I_2, \quad R_2 = I_2 \otimes \begin{pmatrix} 0 & -a \\ 1 & 0 \end{pmatrix}.$$

They commute, square to $-aI_4$ and retain their nonzero nilpotent specializations at $a = 0$. Setting $a = 2s$ reproduces the arithmetic factor $-2s$. These regular representations and invariant-ring maps connect the retained cover to concrete representation-theoretic objects.

9 Integral coefficient lattices and the ramified conductor

The following maps retain the arithmetic of the source-displayed coefficient (8). Put

$$O = \mathbb{Z}[1/2], \quad R = O[q, q^{-1}], \quad t = q - 1, \quad r = q - q^{-1}, \quad h = [7]_q [3]_q, \quad C = r^2 h.$$

The letters r and a in this section use the original $b = c = 0$ root coordinate and base map $a = -r^2$, respectively. The trace coordinate is still $p = q + q^{-1}$. None of $3, 7, h$ is inverted in R .

9.1 The exact lattice map over the integral base

The two-sheet calculation of Section 8 holds over $O[a]$: the displayed monic bases, the component restriction maps, and their inverses only use division by 2. Base change by $O[a] \rightarrow R$, $a \mapsto -r^2$, gives free modules B_R, N_R with their original ordered bases and map

$$V : B_R \longrightarrow N_R, \quad [V] = \text{diag}(1, 1, 1, r^2).$$

This assertion is also checked directly: the last component vector $r_1 r_2$ maps to $-a(1, -1) = r^2(1, -1)$. Thus the original sign is retained. Since R is a domain and $r \neq 0$, this map is injective, and projecting onto the last coordinate modulo r^2 proves

$$0 \longrightarrow B_R \xrightarrow{V} N_R \longrightarrow R/(r^2) \longrightarrow 0.$$

Let $E_C = R^4$ be a free module with ordered basis e_1, e_2, e_3, e_4 . Define all entries of the following maps:

$$H : E_C \longrightarrow B_R, \quad [H] = \text{diag}(1, 1, 1, h), \quad D_C : E_C \longrightarrow N_R, \quad [D_C] = \text{diag}(1, 1, 1, C).$$

Matrix multiplication proves $D_C = VH$; both maps are injective because h, C are nonzero. Define the two actual image lattices $L_C = D_C(E_C)$ and $L_r = V(B_R)$ inside N_R . Their first three coordinates equal R and their last coordinates are CR and $r^2 R$, respectively. Consequently there is an exact sequence

$$0 \longrightarrow R/(h) \xrightarrow{\iota} R/(C) \xrightarrow{\pi} R/(r^2) \longrightarrow 0, \quad \iota(\bar{f}) = \overline{r^2 f}, \quad \pi(\bar{f}) = \bar{f}. \quad (56)$$

Here $R/(h) = L_r/L_C$, $R/(C) = N_R/L_C$ and $R/(r^2) = N_R/L_r$ under the indicated last-coordinate maps. For well-definedness of ι , replacing f by $f + hg$ changes its image by Cg . If $r^2 f = Cg$, cancellation of the nonzero r^2 gives $f = hg$, proving injectivity. The kernel of π consists exactly of classes $r^2 f$, and π is surjective by lifting a representative. This proves every assertion.

The sign action $\Gamma = \mu_2 \times \mu_2$ on the four original basis vectors has characters $1, \chi_1, \chi_2, \chi_1 \chi_2$. Let it act trivially on R and by these same characters on E_C . All diagonal maps above commute with it, so every last-coordinate quotient in (56) has character $\chi_1 \chi_2$. The semilinear involution $q \mapsto q^{-1}$, fixing the four chosen basis vectors, also commutes with the maps: it sends r to $-r$ and fixes r^2, h, C . These are explicit symmetries of this lattice diagram. No unspecified Chevalley action on E_C is inferred.

9.2 The full specialization sequence and its factor 21

Specialization means tensoring over R with $O = R/(t)$. Write $u = (q+1)/q$, so $r = tu$ and $u(1) = 2$. We have $h(1) = 21$. Both V and D_C specialize to $\text{diag}(1, 1, 1, 0)$. Their kernels and cokernels are therefore free copies of O with basis given by the fourth coordinate. The induced kernel map from H is multiplication by 21; the induced cokernel map from the identity on N_R is the identity of O . In particular this diagram preserves more than the ranks of the maps.

Here is the complete derived specialization of (56), including the connecting map. For any R -module M , the two-term free resolution $0 \rightarrow R \xrightarrow{t} R \rightarrow O \rightarrow 0$ proves

$$\text{Tor}_1^R(M, O) = \{m \in M : tm = 0\}, \quad \text{Tor}_j^R(M, O) = 0 \quad (j \geq 2).$$

If $d = td_1 \neq 0$, cancellation in R shows

$$\text{Tor}_1^R(R/(d), O) = O \overline{d_1}.$$

Indeed $tf = dg = td_1g$ implies $f = d_1g$, and $d_1g \in (d)$ holds exactly when $g \in (t)$. The ideal (t) is prime because $R/(t) = O$ is a domain. It does not contain h , whose value is $21 \neq 0$ in O . Thus $tf = hg$ forces $g = tg_1$, then $f = hg_1$, and $\text{Tor}_1^R(R/(h), O) = 0$. Applying these facts to $d = C, r^2$ proves the exact specialized sequence

$$0 \longrightarrow O \xrightarrow{21} O \xrightarrow{\text{mod } 21} O/(21) \xrightarrow{0} O \xrightarrow{1} O \longrightarrow 0. \quad (57)$$

We verify the maps, so that no exactness assertion depends on a hidden choice of generators. The two Tor generators are $C/t = tu^2h$ and $r^2/t = tu^2$, respectively. Projection sends the first to h times the second, and hence gives 21 after reduction by t . To compute the connecting map, lift r^2/t from $R/(r^2)$ to $R/(C)$. Its product by t is $r^2 = \iota(1)$; the connecting class is therefore 1 in $R/(h, t) = O/(21)$, with positive sign. The next map is induced by multiplication by r^2 , which vanishes modulo t . The final map is the identity on constant classes. Multiplication by 21 is injective on O , with cokernel exactly $O/(21)$, verifying exactness of the displayed sequence directly as well.

Over $D = \mathbb{Q}[q, q^{-1}]_{(q-1)}$ the element h is a unit: its value 21 is nonzero in the residue field $\mathbb{Q}$. Thus H has the explicit inverse $\text{diag}(1, 1, 1, h^{-1})$ there, and it identifies the two injections over D . Over R the additional quotient $R/(h)$ and the residual $O/(21)$ remain. These are the additional comparison modules killed by passage to D . This is not an exhaustive inventory of localization effects: the same localization also removes the $q = -1$ branch of r^2 .

For completeness the entire t -saturation can be computed while retaining the integral constants. Work in $R_+ = O[q, q^{-1}, (q+1)^{-1}]$, where u is a unit and (t) is still prime. The ideal generated by C is (t^2h) and the ideal generated by r^2 is (t^2) . Their t -saturation are (h) and R_+ . For the first assertion, if $t^e f = t^2 h g$, then for $e \geq 2$ cancellation gives $t^{e-2} f = hg$. Primality of t and $t \nmid h$ imply successively that g is divisible by t^{e-2} , so $f \in (h)$. For $e < 2$ the same conclusion follows directly by cancellation. Conversely $t^2 h R_+$ lies in (C) , proving the asserted saturation; the second assertion follows from $t^2 R_+ = (r^2)$. Thus the saturated image lattices are $R_+^3 \oplus h R_+$ and R_+^4 , respectively.

9.3 The retained integral jets and the two exceptional primes

The exact equality $C = t^2 u^2 h$ gives

$$C \equiv 84t^2 \pmod{t^3 R_+}.$$

This follows by evaluating $u^2 h$ at $q = 1$: $2^2 \cdot 7 \cdot 3 = 84$. Over $\mathbb{Q}$ the coefficient has exact order two, as previously proved. The integral residue 84 also determines why that order changes in characteristics 3 and 7; these characteristics are allowed over O .

For an odd prime ℓ , in $\mathbb{F}_\ell[q, q^{-1}]$ one has the polynomial identity

$$[\ell]_q = (q - q^{-1})^{\ell-1}. \quad (58)$$

To prove it, multiply the finite sum defining $[\ell]_q$ by $q^{\ell-1}$. The result is $1 + q^2 + \dots + q^{2(\ell-1)}$. Its product with $q^2 - 1$ is $q^{2\ell} - 1 = (q^2 - 1)^\ell$ in characteristic ℓ . Cancellation in the polynomial domain gives $(q^2 - 1)^{\ell-1}$, and multiplication by $q^{-(\ell-1)}$ proves (58). In particular the original coefficient reduces exactly to

$$C = r^4 [7]_q \quad \text{over } \mathbb{F}_3, \quad C = r^8 [3]_q \quad \text{over } \mathbb{F}_7.$$

Because $r = tu$, $u(1) = 2$, $[7]_1 = 1$ in $\mathbb{F}_3$ and $[3]_1 = 3$ in $\mathbb{F}_7$, the exact orders of C at $q = 1$ are four and eight, respectively. Their first nonzero coefficients are $2^4 \cdot 1 = 1$ in $\mathbb{F}_3$ and $2^8 \cdot 3 = 5$ in $\mathbb{F}_7$. For every other odd characteristic ℓ , both 21 and 4 are nonzero, so the exact order is two and its first coefficient is $84 \bmod \ell$. The node map still has last entry r^2 , of exact order

two in every odd characteristic. The larger orders therefore belong to the retained residual factor of the GCT source-displayed polynomial, and the injection H records their relation to the node map.

These results construct an integral coefficient-lattice correspondence. They specify every map and every residual module in that correspondence. Section 3 supplies the actual Cartan-twisted divided-square endpoint from the original GCT-IV basis subquotient. Adding its three specified identity summands gives exactly D_C here, so $D_C = VH$ is also an explicit comparison of that generator endpoint with the original node. The trace obstruction proved there prevents extending the four-dimensional interval to the full raising action. Section 4 now constructs that full original source module, all its raising and lowering operators, and the exact integral quotient map to the retained interval. Its raw-word projection has a specified section and free kernel; the square comparing its full domain with the earlier restricted four-word lattice retains the latter's nonzero cokernel. Thus the larger source action and the conductor comparison here are related by proved integral maps.

10 The original Laurent coefficient and its geometric objects

Throughout this construction the quantum parameter is the indeterminate q , and Q denotes a prime power. They are different objects. The retained source-displayed Laurent polynomial, with its original four terms, is

$$C(q) = q^{10} - q^4 - q^{-4} + q^{-10}. \quad (59)$$

Define, as a Laurent polynomial over $\mathbb{Z}$,

$$[n]_q = q^{n-1} + q^{n-3} + \cdots + q^{-(n-1)}, \quad h(q) = [7]_q[3]_q,$$

and set

$$B(T) = (1 + T + \cdots + T^6)(1 + T + T^2).$$

We construct schemes over $\mathbb{Z}$:

$$U_n = \mathbb{A}_{\mathbb{Z}}^n \setminus V(x_0, \dots, x_{n-1}), \quad X = U_7 \times_{\mathbb{Z}} U_3, \quad Y = \mathbb{P}_{\mathbb{Z}}^6 \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^2.$$

Here $V(x_0, \dots, x_{n-1})$ is the entire zero section $\text{Spec } \mathbb{Z}$, not a chosen point over one field. Products and complements in what follows commute with base change to any field. These definitions build a geometry for the exact scalar (59); they do not specify a quantum-group representation or a canonical-basis convolution space. The source audit in the accompanying research record questions the assignment of this displayed polynomial to the printed target tableau. The construction here takes exactly the displayed polynomial as its input and asserts no resolution of that matrix-entry assignment.

Proposition 10.1. *The following identities hold in $\mathbb{Z}[q, q^{-1}]$:*

$$C(q) = q^{-10}(q^{14} - 1)(q^6 - 1), \quad (60)$$

$$h(q) = q^{-8}B(q^2), \quad (61)$$

$$C(q) = (q - q^{-1})^2 h(q). \quad (62)$$

Furthermore

$$B(T) = 1 + 2T + 3T^2 + 3T^3 + 3T^4 + 3T^5 + 3T^6 + 2T^7 + T^8. \quad (63)$$

Proof. The right side of (60) expands to $q^{-10}(q^{20} - q^{14} - q^6 + 1)$, preserving all four terms of (59). Directly from the definition, $[7]_q = q^{-6} \sum_{i=0}^6 q^{2i}$ and $[3]_q = q^{-2} \sum_{j=0}^2 q^{2j}$, which proves (61). The finite polynomial identities $(q^2 - 1) \sum_{i=0}^6 q^{2i} = q^{14} - 1$ and $(q^2 - 1) \sum_{j=0}^2 q^{2j} = q^6 - 1$, together with $(q - q^{-1})^2 = q^{-2}(q^2 - 1)^2$, prove (62) without inverting $q^2 - 1$. The coefficient of T^d in B is the number of ordered pairs (i, j) with $0 \leq i \leq 6$, $0 \leq j \leq 2$ and $i + j = d$. Listing these nine counts gives (63). $\square$

11 The quotient morphism, its inverses on charts, and its fibres

Proposition 11.1. *There is a morphism*

$$\pi_n : U_n \longrightarrow \mathbb{P}_{\mathbb{Z}}^{n-1}, \quad (x_0, \dots, x_{n-1}) \longmapsto [x_0 : \dots : x_{n-1}],$$

which is a Zariski locally trivial $\mathbb{G}_m$ -torsor for the action $\lambda \cdot (x_i)_i = (\lambda x_i)_i$. Its product

$$\pi = \pi_7 \times \pi_3 : X \longrightarrow Y$$

is a Zariski locally trivial $(\mathbb{G}_m)^2$ -torsor. For any field k and any point of $Y(k)$, its scheme fibre is isomorphic over k to $(\mathbb{G}_{m,k})^2$.

Proof. Let $D_+(z_i)$ be the standard chart on $\mathbb{P}^{n-1}$, with coordinates $u_j = z_j/z_i$ for $j \neq i$, and let $D(x_i) \subset U_n$. Define on these charts the ring homomorphism

$$\mathbb{Z}[u_j : j \neq i] \longrightarrow \mathbb{Z}[x_0, \dots, x_{n-1}, x_i^{-1}], \quad u_j \longmapsto x_j/x_i.$$

On $D(x_i) \cap D(x_k)$ the two projective coordinates obey $u_j^{(k)} = u_j^{(i)}/u_k^{(i)}$, with the i th coordinate equal to $1/u_k^{(i)}$. Thus the homomorphisms glue to π_n . An explicit isomorphism and inverse are

$$\begin{aligned} D(x_i) &\longrightarrow D_+(z_i) \times \mathbb{G}_m, & (x_j)_j &\longmapsto ((x_j/x_i)_{j \neq i}, x_i), \\ D_+(z_i) \times \mathbb{G}_m &\longrightarrow D(x_i), & ((u_j)_{j \neq i}, t) &\longmapsto x_i = t, \quad x_j = tu_j \ (j \neq i). \end{aligned}$$

Every expression is regular on the indicated chart. Substitution in each direction returns every original coordinate, and the action becomes $t \mapsto \lambda t$. The charts cover U_n and $\mathbb{P}^{n-1}$. On a double overlap the fibre coordinate changes by $t_k = t_i u_k^{(i)}$, with an invertible multiplier, so the local torsor identifications agree.

For completeness, the action criterion holds as an isomorphism of schemes:

$$\mathbb{G}_m \times U_n \longrightarrow U_n \times_{\mathbb{P}^{n-1}} U_n, \quad (\lambda, x) \longmapsto (\lambda x, x).$$

On the chart where x_i is invertible, its inverse sends (x', x) to $(x'_i/x_i, x)$. Equality of the projective coordinates gives $x'_j = (x'_i/x_i)x_j$ for every j , and x'_i/x_i is invertible. These ratios agree on overlaps, proving the global inverse. The local sections with $t_i = 1$ prove that π_n is locally surjective. Equally, an invariant morphism out of U_n factors on each chart through $(u, t) \mapsto u$, because its value at (u, t) equals its value at $(u, 1)$; these factorizations agree on overlaps. This proves the geometric quotient property for invariant morphisms to schemes.

Taking products gives all statements for π . A k -rational point has some nonzero projective coordinate in each factor, so it lies in one of these trivializing charts defined over k . Pulling that chart isomorphism back to the point gives the asserted scheme fibre and preserves the two separate scalar actions. $\square$

Remark 11.2. The chart proof is a scheme proof and covers arbitrary base rings. It does not identify arbitrary projective R -points with a single globally chosen coordinate vector. A point of $U_n(R)$ is represented by coordinates generating the unit ideal, and the chart trivializations hold after the corresponding localizations. For finite fields, ordinary nonzero vectors provide exactly the required representatives.

12 Classes, point counts and the quantum specialization map

Let k be a perfect field. The ring $K_0(\text{Var}_k)$ used here is generated by isomorphism classes $[V]$ of finite-type reduced separated k -schemes, with relations $[V] = [Z] + [V \setminus Z]$ for closed reduced Z , and product $[V][W] = [V \times_k W]$. All schemes used in the calculation and their products are reduced. Write $\mathbb{L} = [\mathbb{A}_k^1]$. We use the localization in which $\mathbb{L}$ is invertible only where explicitly displayed.

Proposition 12.1. *In this ring,*

$$[Y_k] = B(\mathbb{L}), \quad (64)$$

$$[X_k] = (\mathbb{L}^7 - 1)(\mathbb{L}^3 - 1) = (\mathbb{L} - 1)^2 B(\mathbb{L}). \quad (65)$$

The polynomial lift of these identities under the injective ring map

$$\iota : \mathbb{Z}[T, T^{-1}] \longrightarrow \mathbb{Z}[q, q^{-1}], \quad T \longmapsto q^2,$$

is precisely

$$\iota(T^{-5}(T^7 - 1)(T^3 - 1)) = C(q), \quad \iota(T^{-4}B(T)) = h(q).$$

Proof. In $\mathbb{P}^{n-1}$ stratify by the last nonzero coordinate. The stratum with last nonzero coordinate z_d has the explicit isomorphism

$$\mathbb{A}_k^d \longrightarrow \{z_{d+1} = \cdots = z_{n-1} = 0, z_d \neq 0\}, \quad (a_0, \dots, a_{d-1}) \longmapsto [a_0 : \cdots : a_{d-1} : 1 : 0 : \cdots : 0].$$

The inverse consists of the regular ratios z_i/z_d . The successive closed subspaces $\mathbb{P}^0 \subset \cdots \subset \mathbb{P}^{n-1}$ show by the defining relation that $[\mathbb{P}^{n-1}] = \sum_{d=0}^{n-1} \mathbb{L}^d$. Taking the two products proves (64). Removing the zero section over k gives $[U_n] = \mathbb{L}^n - 1$, proving the first equality for X . The torsor is trivial over each displayed stratum, so its inverse image has class $(\mathbb{L} - 1)\mathbb{L}^d$. Applying this separately in the two factors proves the second equality directly from the geometric map. The same equality also follows from finite polynomial multiplication. The map ι is injective because distinct powers T^d map to the distinct Laurent monomials q^{2d} , which are linearly independent over $\mathbb{Z}$. Substitution proves the two asserted lifts using Proposition 10.1. No injectivity of $\mathbb{Z}[T, T^{-1}] \rightarrow K_0(\text{Var}_k)[\mathbb{L}^{-1}]$ is asserted or needed. $\square$

Theorem 12.2. *For every prime power Q and every integer $m \geq 1$, set $N = Q^m$. Then*

$$\#Y(\mathbb{F}_N) = B(N), \quad (66)$$

$$\#X(\mathbb{F}_N) = (N^7 - 1)(N^3 - 1) = (N - 1)^2 B(N), \quad (67)$$

$$C(Q^{m/2}) = Q^{-5m} \#X(\mathbb{F}_{Q^m}), \quad (68)$$

$$h(Q^{m/2}) = Q^{-4m} \#Y(\mathbb{F}_{Q^m}). \quad (69)$$

Here $Q^{m/2}$ is the positive real square root of Q^m used to evaluate a Laurent polynomial. The values in the last two equations are rational and independent of the choice of sign of that square root.

Proof. There are exactly N^n ordered n -tuples over $\mathbb{F}_N$, one of which is zero; this proves the first expression for $\#X$. In the projective stratum with last nonzero coordinate indexed by d , the displayed isomorphism gives N^d rational points. Summing over d and multiplying the two factors proves (66). Every rational base point of Y has a fibre isomorphic to $(\mathbb{G}_m)^2$ over that same field by Proposition 11.1; the fibre has $(N - 1)^2$ rational points. The rational-point map is surjective, since each point lies in a trivializing chart with the section $(1, 1)$. Partitioning $X(\mathbb{F}_N)$ by those fibres proves the second expression in (67). Substituting $q^2 = N$ in (60) and

(61) gives the last two formulas with exactly the powers N^{-5} and N^{-4} . Every exponent of q in both expressions is even. Thus these substitutions require no square root in a finite field and their values do not change under $q \mapsto -q$.

The corresponding counting map is the ring homomorphism

$$\nu_m : K_0(\text{Var}_{\mathbb{F}_Q})[\mathbb{L}^{-1}] \longrightarrow \mathbb{Q}, \quad [V] \longmapsto \#V(\mathbb{F}_{Q^m}), \quad \mathbb{L}^{-1} \longmapsto Q^{-m}.$$

Indeed, a closed subvariety and its complement partition the finite set of rational points, and the rational points of a product are the Cartesian product of the two finite sets. These facts prove that the defining relations and multiplication are preserved. Since $\nu_m(\mathbb{L}) = Q^m$ is invertible in $\mathbb{Q}$, the displayed localization map exists. $\square$

13 Zeta functions with every sign retained

For a finite-type scheme $V/\mathbb{F}_Q$, its point-count zeta series is defined in $1 + t\mathbb{Q}[[t]]$ by

$$Z(V, t) = \exp \left(\sum_{m \geq 1} \#V(\mathbb{F}_{Q^m}) \frac{t^m}{m} \right).$$

All exponentials and logarithms in this section are formal power series; there is no analytic convergence assumption.

Theorem 13.1. *The schemes above have zeta functions*

$$Z(X, t) = \frac{(1 - Q^7 t)(1 - Q^3 t)}{(1 - Q^{10} t)(1 - t)}, \quad (70)$$

$$Z(Y, t) = \prod_{d=0}^8 (1 - Q^d t)^{-a_d}, \quad (71)$$

$$(a_0, \dots, a_8) = (1, 2, 3, 3, 3, 3, 3, 2, 1). \quad (72)$$

The series of the original Laurent coefficient is

$$\exp \left(\sum_{m \geq 1} C(Q^{m/2}) \frac{t^m}{m} \right) = Z(X, Q^{-5} t) = \frac{(1 - Q^2 t)(1 - Q^{-2} t)}{(1 - Q^5 t)(1 - Q^{-5} t)}. \quad (73)$$

Similarly the series of h equals $Z(Y, Q^{-4} t) = \prod_{d=0}^8 (1 - Q^{d-4} t)^{-a_d}$.

Proof. For any $a \in \mathbb{Q}$, termwise formal differentiation gives $\frac{d}{dt} \sum_{m \geq 1} a^m t^m / m = a / (1 - at)$. Both $\sum_{m \geq 1} a^m t^m / m$ and $-\log(1 - at)$ have that derivative and constant term zero; equality follows coefficient by coefficient in characteristic zero. The expression for $\#X$ is $Q^{10m} - Q^{7m} - Q^{3m} + 1$, so applying this identity to its four separate terms gives (70) with the two negative terms in the numerator. Formula (63) gives (71) by the same argument for all nine terms. Equation (68) replaces t^m by $(Q^{-5} t)^m$ in the defining series. Substitution in (70) gives every exponent in (73). Equation (69) proves the last statement identically. $\square$

14 Actual compact-support Frobenius groups

Fix $k = \mathbb{F}_Q$ of characteristic p , an algebraic closure $\bar{k}$, and a prime $\ell \neq p$. In this section $H_c^i(V)$ means the ℓ -adic group $H_c^i(V_{\bar{k}}, \mathbb{Q}_{\ell})$. We use geometric Frobenius Φ_Q , the inverse of the field automorphism $a \mapsto a^Q$, on the Galois representations. Thus Φ_Q acts on $\mathbb{Q}_{\ell}(-d)$ by Q^d . The four foundational cohomological results used here are affine-space acyclicity,

open–closed localization, Künneth, and Poincaré duality. Precise author-hosted references are Milne, *Lectures on Étale Cohomology*, v2.21, Example 16.3, Proposition 18.3 together with the open–closed sheaf sequence of Proposition 8.15, Theorem 22.4, and Theorem 24.1, <https://www.jmilne.org/math/CourseNotes/LEC.pdf>. These are established foundations of the cohomology theory, used with their stated hypotheses; none is a proposed GCT hypothesis.

We specify the coefficient passage in those references. Put $\Lambda_r = \mathbb{Z}/\ell^r\mathbb{Z}$ for $r \geq 1$. The affine acyclicity, localization, and duality statements first apply with Λ_r coefficients because $\bar{k}$ is algebraically closed and ℓ is invertible on every scheme here. Compact-support cohomology of constructible finite sheaves on these finite-type schemes is finite by Milne, Proposition 18.3(c). At each cohomological degree, the images in any fixed finite group along the inverse system form a descending sequence of subgroups and therefore stabilize. This is the Mittag–Leffler condition, so inverse limits preserve these long exact sequences. Taking the inverse limit over r and then tensoring over $\mathbb{Z}_\ell$ with its fraction field $\mathbb{Q}_\ell$ is precisely the cohomology convention of Milne, Section 19, pages 123–126; the latter tensor product is exact because localization is exact. The affine top groups obtained from the compatible duality trace maps are $\Lambda_r(-n)$, whose limit is $\mathbb{Z}_\ell(-n)$. Consequently the identifications used below hold as Galois representations with their full integral Tate indices before tensoring. The trace maps commute with Galois: Galois permutes the closed-point classes specifying the trace, and uniqueness in Theorem 24.1(a) preserves that trace. Finally the arithmetic automorphism acts on μ_{ℓ^r} by raising to the Q th power; its inverse acts on $\mathbb{Q}_\ell(1)$ by Q^{-1} , proving the stated action on $\mathbb{Q}_\ell(-d)$. This Frobenius convention agrees with Milne, Proposition 29.10 and Section 30.

Lemma 14.1. *For $n \in \{3, 7\}$ the only nonzero compact-support groups of U_n are*

$$H_c^1(U_n) = \mathbb{Q}_\ell, \quad H_c^{2n}(U_n) = \mathbb{Q}_\ell(-n).$$

The first identification is the localization connecting map from the origin; the second is the map induced by the open immersion into $\mathbb{A}^n$. Both are Galois-equivariant.

Proof. Affine-space acyclicity says that $H^0(\mathbb{A}^n) = \mathbb{Q}_\ell$ and that every positive-degree group vanishes. The scheme is smooth, geometrically connected, of dimension n . Its duality trace therefore gives $H_c^{2n}(\mathbb{A}^n) = \mathbb{Q}_\ell(-n)$ and $H_c^i(\mathbb{A}^n) = 0$ for $i \neq 2n$. The origin is a rational proper point; its only cohomology group is $H_c^0(\{0\}) = \mathbb{Q}_\ell$ with trivial Galois action. For the open immersion $j : U_n \rightarrow \mathbb{A}^n$ and closed immersion $i : \{0\} \rightarrow \mathbb{A}^n$, the sheaf sequence

$$0 \longrightarrow j_!\mathbb{Q}_\ell \longrightarrow \mathbb{Q}_\ell \longrightarrow i_*\mathbb{Q}_\ell \longrightarrow 0$$

is exact: a geometric stalk in U_n gives $0 \rightarrow \mathbb{Q}_\ell \xrightarrow{1} \mathbb{Q}_\ell \rightarrow 0$, and a stalk at the origin gives $0 \rightarrow 0 \rightarrow \mathbb{Q}_\ell \xrightarrow{1} \mathbb{Q}_\ell \rightarrow 0$. Its compact-support long exact sequence contains

$$0 = H_c^0(\mathbb{A}^n) \longrightarrow \mathbb{Q}_\ell \xrightarrow{\delta} H_c^1(U_n) \longrightarrow H_c^1(\mathbb{A}^n) = 0$$

and

$$0 = H_c^{2n-1}(\{0\}) \longrightarrow H_c^{2n}(U_n) \longrightarrow \mathbb{Q}_\ell(-n) \longrightarrow H_c^{2n}(\{0\}) = 0.$$

At every other index all adjacent groups are zero, forcing the remaining groups of U_n to vanish. All schemes and morphisms are defined over k , so the maps, including δ , commute with the Galois action. The assumptions $n = 3, 7$ in particular give $2n > 1$, keeping the two displayed degrees distinct. $\square$

Theorem 14.2. *The entire compact-support graded Galois representation of X is*

i	Künneth summand	$H_c^i(X)$	Φ_Q	$(-1)^i$
2	$H_c^1(U_7) \otimes H_c^1(U_3)$	$\mathbb{Q}_\ell$	1	+1
7	$H_c^1(U_7) \otimes H_c^6(U_3)$	$\mathbb{Q}_\ell(-3)$	Q^3	-1
15	$H_c^{14}(U_7) \otimes H_c^1(U_3)$	$\mathbb{Q}_\ell(-7)$	Q^7	-1
20	$H_c^{14}(U_7) \otimes H_c^6(U_3)$	$\mathbb{Q}_\ell(-10)$	Q^{10}	+1

Every group in an unlisted degree vanishes. After tensoring every group with $\mathbb{Q}_\ell(5)$, its alternating trace is exactly

$$\sum_i (-1)^i \operatorname{Tr} \left(\Phi_Q^m \mid H_c^i(X)(5) \right) = Q^{5m} - Q^{2m} - Q^{-2m} + Q^{-5m} = C(Q^{m/2}).$$

The alternating determinant of the same twisted representation is the right side of (73).

Proof. Over the field $\mathbb{Q}_\ell$, Künneth identifies $H_c^i(X)$ with the direct sum of the tensor products $H_c^a(U_7) \otimes H_c^b(U_3)$ for $a + b = i$; there is no torsion term. Lemma 14.1 leaves exactly the four pairs $(1, 1), (1, 6), (14, 1), (14, 6)$. Tensoring Tate representations adds their twist indices. This gives exactly the four degrees, representations and eigenvalues in the table. Künneth is natural over k , so these identifications preserve Frobenius, not only dimensions. The twist (5) multiplies every eigenvalue by Q^{-5} . Taking the indicated signed m th powers gives the original four terms in their original order. Finally each nonzero group is one dimensional, so $\det(1 - t\Phi_Q) = 1 - t\lambda$ for its listed eigenvalue λ . In $\prod_i \det(1 - t\Phi_Q \mid H_c^i(X)(5))^{(-1)^{i+1}}$, the even-degree factors occur in the denominator and the odd-degree factors in the numerator. This is (73) directly; no general trace-formula assertion is needed to match the independently proved point-count zeta function in this case. $\square$

Proposition 14.3. *The only nonzero groups of Y are*

$$H^{2d}(Y) = \mathbb{Q}_\ell(-d)^{\oplus a_d} \quad (0 \leq d \leq 8),$$

with a_d as in (63). Consequently $h(Q^{m/2})$ is the trace of Φ_Q^m on $\bigoplus_d H^{2d}(Y)(4)$.

Proof. We spell out the projective-space calculation from affine cohomology. The pair $\mathbb{P}^{n-1} \subset \mathbb{P}^n$, with open complement $\mathbb{A}^n$, gives the localization sequence. Since $\mathbb{P}^n$ and $\mathbb{P}^{n-1}$ are proper, compact-support cohomology is ordinary cohomology for them. The affine compact-support calculation in the proof of Lemma 14.1 applies to every $n \geq 1$. Hence restriction $H^i(\mathbb{P}^n) \rightarrow H^i(\mathbb{P}^{n-1})$ is an isomorphism for $i < 2n - 1$. At degree $2n - 1$, induction from $\mathbb{P}^0$ gives zero for the target and therefore for the source. The next part is

$$0 \longrightarrow \mathbb{Q}_\ell(-n) \longrightarrow H^{2n}(\mathbb{P}^n) \longrightarrow 0.$$

At degrees greater than $2n$, the same sequence forces vanishing. Induction gives $H^{2j}(\mathbb{P}^n) = \mathbb{Q}_\ell(-j)$ for $0 \leq j \leq n$, with odd groups zero; every map is defined over k . Künneth now gives one summand $\mathbb{Q}_\ell(-(i + j))$ for each pair $0 \leq i \leq 6, 0 \leq j \leq 2$. Counting the pairs gives a_d and all claimed Frobenius eigenvalues. The twist (4) changes Q^{dm} to $Q^{(d-4)m}$. Their sum is $Q^{-4m}B(Q^m)$, already identified with $h(Q^{m/2})$ in (69). $\square$

Remark 14.4. These explicit eigenvalues determine the weights without a general Riemann-hypothesis argument. On $H_c^i(X)$ the weights at $i = 2, 7, 15, 20$ are respectively $0, 6, 14, 20$, because an eigenvalue Q^d has complex absolute value $Q^{(2d)/2}$. On $H^{2d}(Y)$ the weight is $2d$. Thus the negative terms of C come from the two odd compact-support degrees of a nonproper variety. They do not describe a negative dimension or a failure of a Frobenius absolute-value law. The actual groups of X were computed by localization and Künneth on $U_7 \times U_3$; the torsor identity in the class ring does not assert that $H_c^*(X)$ is $H^*(Y) \otimes H_c^*((\mathbb{G}_m)^2)$.

15 Exact specialization order and the limits of the construction

Proposition 15.1. *The original Laurent coefficient has an exact order-two zero at $q = 1$:*

$$C(1) = 0, \quad C'(1) = 0, \quad C''(1) = 168, \quad \left. \frac{C(q)}{(q-1)^2} \right|_{q=1} = 84.$$

The residual factor has $h(1) = 21$. The full pre-specialization relation is

$$C(q) = (q - 1)^2 q^{-10} (q + 1)^2 B(q^2), \quad (74)$$

so these numbers arise from the same projective base and the two separately retained torus factors of π .

Proof. Equations (60)–(61) give $C(q) = q^{-10}(q^2 - 1)^2 B(q^2)$, which is exactly (74). The factor after $(q - 1)^2$ is a Laurent polynomial regular at $q = 1$ with value $1 \cdot 4 \cdot (7 \cdot 3) = 84$, which is nonzero in $\mathbb{Z}$. This proves the exact order, and differentiating the product twice gives $C'''(1) = 2 \cdot 84 = 168$. Directly differentiating the original four terms checks the preserved signs:

$$C'(1) = 10 - 4 + 4 - 10 = 0, \quad C''(1) = 90 - 12 - 20 + 110 = 168.$$

The formula for $h(1)$ follows from $[7]_1 = 7$ and $[3]_1 = 3$, or from the 21 affine strata of Y . The two factors $q^2 - 1$ are the polynomial images under $T \mapsto q^2$ of the two $\mathbb{G}_m$ classes $T - 1$ in Proposition 12.1. $\square$

The exact realization obtained here consists of the morphism $X \rightarrow Y$, its two scalar actions, the class identities, the counting homomorphisms, the even-Laurent substitution $T \mapsto q^2$, and the specified graded Frobenius representations and Tate twists. In particular, the original coefficient is realized as an alternating trace of an actual scheme's compact-support cohomology, while its residual $[7]_q[3]_q$ is the centered trace of the displayed projective product. All their maps and constants have been specified above. No half-integral Tate twist is used: the formal ring $\mathbb{Z}[q^2, q^{-2}]$ maps to the ring generated by integral Tate representations by $q^{2d} \mapsto [\mathbb{Q}_\ell(-d)]$. More precisely, the target is the Grothendieck ring of finite-dimensional continuous $\mathbb{Q}_\ell$ -representations of $\text{Gal}(\bar{k}/k)$, restricted to the subring generated by the Tate classes. Tensor products satisfy $\mathbb{Q}_\ell(-d) \otimes \mathbb{Q}_\ell(-e) = \mathbb{Q}_\ell(-(d + e))$, and the class of $\mathbb{Q}_\ell(d)$ is the multiplicative inverse of the class of $\mathbb{Q}_\ell(-d)$. These identities establish the stated ring homomorphism. On any invariant exact sequence, the matrix of Φ_Q^m is block triangular and its trace is additive; on a tensor product its trace is the product of the two traces. Thus taking this trace defines a ring homomorphism to $\mathbb{Q}_\ell$, and its composition on q^{2d} is exactly Q^{dm} . It agrees with the even-Laurent specialization already used in the point-count formulas.

The scheme and torsor realize this retained scalar with all stated cohomological degrees. Section 16 constructs actual additive generator functors, their integral decategorification map to the source interval, and a separately specified cross-path relation cone. Its explicit Frobenius-equivariant endpoint map uses precisely these cohomology groups after two-periodization; the integer grading and the chosen extra relation data are recorded there. Compatibility with canonical-basis convolution in the full original source module remains unproved. The source endpoint itself is resolved by the original action paths in Section 3, with the full source module in Section 4.

16 Graded generator functors and an explicit Frobenius relation cone

We retain the four actual source classes in their fixed order $e_0 = T, e_1 = T_1, e_2 = T_2, e_3 = Q$, with precisely the source rescalings and column words proved in the generator-interval calculation. Put

$$\begin{aligned} A &= \mathbb{Z}[q, q^{-1}], & \alpha &= [7] - [3], & \beta &= [8], & \gamma &= [6], & d &= [3], \\ \delta &= -d, & p &= [2], & C &= q^{10} - q^4 - q^{-4} + q^{-10}. \end{aligned}$$

The source quotient has lowering matrix

$$N = \begin{pmatrix} 0 & 0 & 0 & 0 \\ \alpha & 0 & 0 & 0 \\ \beta & 0 & 0 & 0 \\ 0 & \gamma & -d & 0 \end{pmatrix}, \quad D = CE_{30}, \quad N^2 = pD.$$

The construction below gives actual additive functors defined by tensor products, an integral map between their two source-labelled length-two branches, and a deformation retraction after two-periodization to the previously computed Frobenius object. Its relation map is specified explicitly; it is not inferred from the equality of Laurent polynomials.

16.1 The state category and its exact integral decategorification

Let $\mathcal{G}$ be the category of finite free internally $\mathbb{Z}$ -graded abelian groups, each furnished with a homogeneous basis. Write $\mathbb{Z}\langle w \rangle$ for a copy of $\mathbb{Z}$ whose basis vector has internal degree w . Morphisms preserve internal degree. Let $\mathcal{K} = K^b(\mathcal{G})$ be its bounded homotopy category, and let $\mathcal{S} = \mathcal{K}^4$. The tensor differential convention is

$$d(x \otimes y) = dx \otimes y + (-1)^i x \otimes dy \quad (x \text{ of cohomological degree } i).$$

Internal degrees add under tensor product. The internal shift $\langle w \rangle$ increases every internal degree by w . The cochain translation convention is $V[1]^i = V^{i+1}$, with translated differential $-d$; in particular a degree-zero group translated by $[1]$ lies in degree -1 .

Define the decategorification group using isomorphism classes in $\mathcal{K}$, direct-sum relations, and $[\text{Cone}(f)] = [Y] - [X]$ for every cochain map $f : X \rightarrow Y$. Its internal shift gives an A -module structure. There is an isomorphism

$$\chi_q : K_0(\mathcal{K}) \longrightarrow A, \quad [V] \longmapsto \sum_{i,w} (-1)^i \text{rank}_{\mathbb{Z}}(V_w^i) q^w. \quad (*)$$

Here and below sums are finite. To prove the assertion, a direct sum adds ranks, and $\text{Cone}(f)^i = Y^i \oplus X^{i+1}$ gives exactly the required subtraction. A contractible complex has Euler rank zero in each internal degree: its contracting homotopy splits its successive boundaries, so the adjacent boundary contributions cancel. Alternatively, the cone of a homotopy equivalence is contractible, and its finite alternating chain ranks vanish by the same splitting. Thus the formula is unchanged under isomorphism in the homotopy category. Filtering a bounded complex by its cohomological degrees and using the cone relation expresses its class as the alternating sum of the classes of its chain groups. Each such group is a direct sum of the specified $\mathbb{Z}\langle w \rangle$. This proves that the map $q^w \mapsto [\mathbb{Z}\langle w \rangle]$ is inverse to $(*)$. The tensor formula and the displayed tensor differential show that χ_q also preserves products. The same argument in the four factors gives a fixed isomorphism

$$\Psi : K_0(\mathcal{S}) \longrightarrow M_A = \bigoplus_{i=0}^3 Ae_i, \quad [(V_0, V_1, V_2, V_3)] \longmapsto \sum_{i=0}^3 \chi_q(V_i) e_i.$$

In particular the four labels have not been replaced or permuted.

16.2 Objects on the four original arrows

For each indicated finite set, retain the displayed label as part of the basis. Define degree-zero complexes

$$\begin{aligned} A &= \bigoplus_{a \in \{-6, -4, 4, 6\}} \mathbb{Z}a, & \deg_q(a) &= a, \\ B &= \bigoplus_{b \in \{-7, -5, -3, -1, 1, 3, 5, 7\}} \mathbb{Z}b, & \deg_q(b) &= b, \\ G &= \bigoplus_{g \in \{-5, -3, -1, 1, 3, 5\}} \mathbb{Z}g, & \deg_q(g) &= g, \\ J &= \bigoplus_{j \in \{-2, 0, 2\}} \mathbb{Z}j, & \deg_q(j) &= j. \end{aligned}$$

The equality $\chi_q(A) = [7] - [3]$ follows by retaining the seven terms of $[7]$ and subtracting its three terms with exponents $2, 0, -2$; the four surviving terms are exactly the four displayed summands. This equality does not rescale any source class. The other three positive characters are $[8], [6], [3]$, by their complete finite sums. The negative source sign is implemented by the cochain translation $J[1]$, which has character $-[3]$.

Define an exact additive functor $F : \mathcal{S} \rightarrow \mathcal{S}$ using tensor products by

$$F(V)_0 = 0, \quad F(V)_1 = A \otimes V_0, \quad F(V)_2 = B \otimes V_0, \quad F(V)_3 = (G \otimes V_1) \oplus (J[1] \otimes V_2).$$

Its action on a tuple of cochain maps is the corresponding identity tensor those maps. Tensoring by these finite free complexes preserves mapping cones, with the usual tensor-differential signs, so it induces an endomorphism of the stated decategorification group. Direct use of the preceding formulas proves the intertwining identity

$$\Psi[F] = N\Psi.$$

This is an integral identity over the original ring A .

Let K_V shift the four components internally by $(9, 7, 7, 5)$, respectively, and let K_W shift each component by 3. Their inverses are the opposite shifts. These are actual invertible functors. For every nonzero edge of F , the target V -shift is the source shift minus two. Identifying equal internal shifts on each tensor factor therefore gives a natural isomorphism

$$K_V F K_V^{-1} \simeq F\langle -2 \rangle.$$

The W -shifts are equal on both ends of each edge and commute with F . Set $F_W = 0$. Its commutation and Cartan relations hold as equal zero functors. The two Cartan functors commute componentwise. Each of the two original central total-weight generators is represented by the common internal shift $\langle 15 \rangle$ on all four components; these commute with every displayed functor. Finally $F^3 = 0$: the only length-two paths end in component three, from which there is no outgoing arrow. Thus the negative-Borel relations involving the first lowering generators and Cartan operators hold already as specified natural isomorphisms, not only as scalar equalities.

16.3 The retained geometric scalar object

Let C_X be the zero-differential integral bigraded complex with exactly four basis vectors

label	x_{00}	x_{03}	x_{70}	x_{73}
cohomological degree	2	7	15	20
internal degree	-10	-4	4	10

These labels retain, in order, the Künneth factors $(H_c^1(U_7), H_c^1(U_3))$, $(H_c^1(U_7), H_c^6(U_3))$, $(H_c^{14}(U_7), H_c^1(U_3))$, and $(H_c^{14}(U_7), H_c^6(U_3))$. The actual groups and their localization and trace identifications were computed for $X = (\mathbb{A}^7 \setminus 0) \times (\mathbb{A}^3 \setminus 0)$. Their Tate twists after the common twist (5) are respectively (5), (2), (-2), (-5). Thus the internal degrees retain twice the Frobenius exponents and $\chi_q(\mathbf{C}_X) = C$, with every cohomological degree and sign retained.

Define $\mathbf{D}(V)$ to have only component three nonzero, equal to $\mathbf{C}_X \otimes V_0$. Then $\Psi[\mathbf{D}] = D\Psi$. Its Cartan relation uses the original weight difference -4 . The support also gives $\mathbf{F}\mathbf{D} = \mathbf{D}\mathbf{F} = \mathbf{D}^2 = 0$. Put $\mathbf{H} = \mathbb{Z}\langle 1 \rangle \oplus \mathbb{Z}\langle -1 \rangle$ in cohomological degree zero, so that $\chi_q(\mathbf{H}) = p$.

The following obstruction is exact for these specified objects. On the state $(\mathbb{Z}, 0, 0, 0)$, the composite $\mathbf{F}^2$ has component-three complex

$$(\mathbf{G} \otimes \mathbf{A}) \oplus (\mathbf{J} \otimes \mathbf{B})[1],$$

whose zero differential leaves 24 even and 24 odd free generators. The complex $\mathbf{H} \otimes \mathbf{C}_X$ has four even and four odd generators. Tensoring with $\mathbb{Q}$, an isomorphism in either the bounded homotopy category or the two-periodic derived category would induce an isomorphism of its cohomology. The indicated dimensions are unequal. Therefore these specified functors have no natural isomorphism $\mathbf{F}^2 \simeq \mathbf{H} \otimes \mathbf{D}$. This obstruction proves the failure of that lift; the next map supplies an exact stronger correspondence between its two branches and the geometric endpoint.

16.4 A complete integral map between the two length-two branches

Write

$$\mathbf{P} = \mathbf{G} \otimes \mathbf{A}, \quad \mathbf{R} = \mathbf{J} \otimes \mathbf{B}.$$

For basis naming use pairs (a, g) in $\mathbf{P}$ and (b, j) in $\mathbf{R}$, preserving their source intermediate states T_1 and T_2 , respectively. At each internal weight w , list the pairs of either type in ascending lexicographic order and let their lengths be p_w, r_w . Define

$$\theta : \mathbf{R} \longrightarrow \mathbf{P}, \quad \theta(R_{w,k}) = \begin{cases} P_{w,k}, & 1 \leq k \leq \min(p_w, r_w), \\ 0, & k > p_w. \end{cases}$$

Here $R_{w,k}$ denotes the k -th source pair and $P_{w,k}$ the k -th target pair of the given weight. This defines an actual matrix over $\mathbb{Z}$ consisting of twenty distinct entries one and all remaining entries zero. It preserves internal and cohomological degree. Its exact ranks and unmatched vectors are

w	rank θ_w	unmatched (a, g)	unmatched (b, j)
11	0	(6, 5)	
9	1	(6, 3)	
7	2		
5	2		(7, -2)
3	2		(5, -2)
1	3		
-1	3		
-3	2		(-1, -2)
-5	2		(-3, -2)
-7	2		
-9	1	(-4, -5)	
-11	0	(-6, -5)	

For verification, the $\mathbf{P}$ -basis is the Cartesian product of the four explicitly given values of a and the six values of g ; the $\mathbf{R}$ -basis is the product of the eight values of b and the three values

of j . Adding the two entries gives exactly the twelve rows in the table. Listing and matching within one row cannot affect any other row. The rank column sums to twenty, and each side has 24 basis vectors, so each side has exactly the four listed residual generators. Because all matched entries are one on distinct rows and columns, the image is a direct summand and both the kernel and cokernel are free, with exactly those residual bases. There is no unrecorded integral torsion.

Let $\mathbf{L} = \text{Cone}(\theta)$. It has $\mathbf{R}$ in cohomological degree -1 , $\mathbf{P}$ in degree zero, and differential exactly θ . In this convention the sign of its character is

$$\chi_q(\mathbf{L}) = \gamma\alpha - d\beta = pC.$$

More strongly, the complex splits integrally as twenty two-term complexes $\mathbb{Z} \xrightarrow{1} \mathbb{Z}$ and eight isolated residual generators. The split is the displayed basis permutation, not a rational rank argument. Map each matched target basis vector back to its matched source basis vector and all other basis vectors to zero; call this cochain homotopy h . If ρ is projection to the eight residual vectors and ι is their inclusion, then on a matched pair $dh + hd$ is the identity on both vectors and on a residual vector it is zero. Consequently

$$1 - \iota\rho = dh + hd, \quad \rho\iota = 1.$$

This proves an explicit integral deformation retraction and all its signs.

Regard $\mathbf{P}, \mathbf{R}, \mathbf{L}$ also as endpoint functors on $\mathcal{S}$, tensoring their arguments from component zero into component three. The map $\theta \otimes 1$ is natural in every state and morphism. Its mapping-cone triangle is therefore a triangle of explicitly specified functors

$$\mathbf{R} \xrightarrow{\theta} \mathbf{P} \longrightarrow \mathbf{L} \longrightarrow \mathbf{R}[1].$$

The original composite $\mathbf{F}^2$ is the split sum $\mathbf{P} \oplus \mathbf{R}[1]$, not this cone with its nonzero cross-branch differential. Their equality in the decategorification group follows from this actual triangle. Their nonisomorphism follows from the already computed cohomology dimensions. In particular, the cross-branch relation has not silently been installed as the tensor differential of $\mathbf{F}^2$.

16.5 The endpoint map with all cohomological degrees recorded

Let Per send a bounded complex to its two-periodic complex by summing its even and odd cohomological degrees separately, retaining the differential, internal degree, and individual summand labels. This operation explicitly forgets the distinction between cohomological degrees of the same parity. The eight residual generators of $\mathbf{L}$ identify with the eight indicated tensor generators of $\mathbf{H} \otimes \mathbf{C}_X$ as follows. The symbol $+$ means the $\mathbf{H}$ -basis of internal degree one and $-$ means its basis of internal degree minus one:

residual path	$\mathbf{H} \otimes \mathbf{C}_X$	internal degree	original endpoint cohomological degree
(6, 5)	$+\otimes x_{73}$	11	20
(6, 3)	$-\otimes x_{73}$	9	20
(-4, -5)	$+\otimes x_{00}$	-9	2
(-6, -5)	$-\otimes x_{00}$	-11	2
(7, -2)	$+\otimes x_{70}$	5	15
(5, -2)	$-\otimes x_{70}$	3	15
(-1, -2)	$+\otimes x_{03}$	-3	7
(-3, -2)	$-\otimes x_{03}$	-5	7

The first four residual vectors are in cohomological degree zero in $\mathbf{L}$, while the last four are in degree -1 . Each row preserves parity and internal degree. Extend this table linearly to

an isomorphism of the residual two-periodic complex with $\text{Per}(\mathbf{H} \otimes \mathbf{C}_X)$. Composition with ρ , and in the other direction with ι , gives inverse homotopy equivalences

$$\text{Per}(\mathbf{L}) \simeq \text{Per}(\mathbf{H} \otimes \mathbf{C}_X),$$

with precisely the contraction h proved above. This supplies the actual morphism relating all surviving branch summands to the scalar geometry. The cohomological columns also prove that this map is not a degree-preserving map to the unperiodized scalar cohomology: for example its weight-eleven class has degree zero on the cone side and degree twenty on the geometric side. Moreover the differences for the weights eleven and minus eleven are twenty and two, respectively, so no single common cohomological shift repairs this failure.

16.6 Frobenius realization and the integral-Tate parity obstruction

Fix a prime power Q , a prime ℓ not dividing Q , and choose a root z of $z^2 = Q$ in $E = \mathbb{Q}_\ell(z)$. This is a specified extra coefficient choice for the odd internal degrees; it was unnecessary for the even-degree scalar object. The element z is a unit in the ring of integers of E . For each power of its maximal ideal, its reduction has finite multiplicative order, hence $n \mapsto z^n$ extends from $\mathbb{Z}$ to a continuous homomorphism from $\widehat{\mathbb{Z}}$ to that finite unit group. The extensions commute with reduction because they agree on the dense integer subgroup. Their inverse limit is a continuous character $\widehat{\mathbb{Z}} \rightarrow E^\times$. Identifying the Galois group of $\overline{\mathbb{F}}_Q/\mathbb{F}_Q$ with $\widehat{\mathbb{Z}}$, with geometric Frobenius sent to one, therefore defines an actual one-dimensional continuous representation $\mathcal{L}$ on which geometric Frobenius acts as z . Its square is identified, on their specified basis vectors, with $E(-1)$; both characters send geometric Frobenius to Q .

Define the tensor realization on every graded basis summand by

$$\mathbb{Z}\langle w \rangle \mapsto \mathcal{L}^{\otimes w},$$

using dual powers for negative w , extending scalars in the integer matrices to E , and retaining every cohomological degree. Maps preserving internal degree become Frobenius-equivariant maps, because every nonzero matrix entry joins the same character on source and target. In particular the twenty entries of θ , the projection, inclusion and contraction, and every entry of the residual table are Frobenius equivariant. The actual geometric object $H_c^*(X, E)(5)$ is the realization of $\mathbf{C}_X$ by its previously proved localization and Künneth identifications. Here H_c^* explicitly denotes the direct sum of the actual cohomology groups placed in their original cohomological degrees with zero differential. It does not denote a chosen cochain model for $R\Gamma_c(X, E)$. No canonical splitting of that derived object into its cohomology groups is asserted or used. We obtain a specified equivalence of two-periodic complexes of continuous Frobenius representations

$$\text{Per}(\mathbf{L}_E) \simeq \text{Per}((\mathcal{L} \oplus \mathcal{L}^{-1}) \otimes H_c^*(X, E)(5)).$$

Every entry of the map and its inverse has been given above. Its alternating trace of the m -th Frobenius power is

$$(z^m + z^{-m})(Q^{5m} - Q^{2m} - Q^{-2m} + Q^{-5m}).$$

If the chosen complex embedding of z is the positive square root, this is precisely $[2]_{Q^{m/2}} C(Q^{m/2})$. The two choices of z are retained choices of odd-degree realization; the four even-degree scalar summands are independent of that choice.

The need for this extension has a precise parity obstruction. Suppose one tries to realize all four arrow coefficients and the divided-square coefficient by only integral Tate degrees after replacing the four state generators by $q^{s_i} e_i$, for integers s_i . A coefficient from state i to state j becomes multiplied by $q^{s_i - s_j}$. Integral Tate characters have only even powers of q . The

nonzero exponent parities of $\alpha, \beta, \gamma, -d$ are respectively 0, 1, 1, 0. Thus all four arrows being even requires

$$s_0 - s_1 = 0, \quad 1 + s_0 - s_2 = 0, \quad 1 + s_1 - s_3 = 0, \quad s_2 - s_3 = 0 \pmod{2}.$$

The first and third imply $s_3 - s_0 = 1 \pmod{2}$. But every term of C has even exponent, so the divided-square arrow from zero to three being even would require $s_3 - s_0 = 0 \pmod{2}$, a contradiction. This proves that diagonal state regratings cannot provide that all-integral-Tate lift. The character $\mathcal{L}$ just constructed is an explicit alternative, not an unmentioned half-integral Tate twist in the original scalar calculation.

The construction therefore gives an integral four-state generator functor, all its first-generator negative-Borel relations, a precise decategorification morphism to the actual source subquotient, a complete rank-twenty relation map, and its two-periodic Frobenius endpoint equivalence. It also records the exact obstruction to the naive divided power functor isomorphism and to restoring the full cohomological grading by a common shift. It does not identify the added relation map with a canonical-basis convolution map in the full source module. Constructing that source geometric compatibility remains a further mathematical problem; no nonstandard Riemann-hypothesis or complexity conclusion is supplied by the finite interval alone.

17 The source GCT coefficient on the retained arithmetic cover

The GCT IV calculation quoted and checked in the preceding source section uses the Laurent parameter q and the original coefficient

$$C(q) = q^{10} - q^4 - q^{-4} + q^{-10} = (q - q^{-1})^2 [7]_q [3]_q, \quad [j]_q = \frac{q^j - q^{-j}}{q - q^{-1}}.$$

Here $[j]_q$ denotes its Laurent-polynomial expression at $q = \pm 1$ as well; division in the displayed expression is an identity away from these points, not deletion of those points from the polynomial. We now construct an exact map to the original retained cover.

Proposition 17.1 (A base change with all coordinates retained). *For a field k of characteristic different from 2, set*

$$D = k[a, r]/(r^2 + a), \quad E = k[p] \otimes_{k[a]} D, \quad a \mapsto 4 - p^2.$$

There is an isomorphism

$$E = k[p, r]/(p^2 - r^2 - 4) \simeq k[q, q^{-1}]$$

with maps

$$p \mapsto q + q^{-1}, \quad r \mapsto q - q^{-1}, \quad q \mapsto (p + r)/2, \quad q^{-1} \mapsto (p - r)/2.$$

Its original base coordinate is exactly $a = 4 - p^2 = -(q - q^{-1})^2$.

Proof. Tensoring the displayed presentation of D replaces a by $4 - p^2$, giving the stated relation. The Laurent expressions satisfy that relation. Conversely $(p + r)(p - r)/4 = 1$, so the proposed inverse images of q, q^{-1} have product one and define a Laurent ring map. The sums and differences of these expressions recover p, r . Starting with q , the same compositions recover q, q^{-1} . Thus the maps are inverse on generators, proving the isomorphism and every stated base-coordinate identity. $\square$

The variable p is new; it does not rename or replace the original third source coordinate w . On the original simple-root chart with $b = c = 0$, the retained formulas become

$$\alpha = -2r, \quad x = -\frac{1}{2r}, \quad y = 3r, \quad w = 26r^2, \quad a = -r^2, \quad r \neq 0.$$

Indeed substitution into $\alpha = P'/2$ and the full inverse formulas gives each equality with its original coefficient. Substitution into the original F then gives $(a, b, c) = (-r^2, 0, 0)$. After Proposition 17.1, these are rational functions of q defined precisely on $q \neq 0, \pm 1$: $q = 0$ was already absent from the Laurent parameter space, and $q = \pm 1$ is equivalent to $r = 0$. For each such q , the finite boundary point $(0, 0, a)$ in the original fibre is retained separately. It is not a point of this $x \neq 0$ chart.

The exact positive Laurent factor has the polynomial expression

$$[7]_q = p^6 - 5p^4 + 6p^2 - 1, \quad [3]_q = p^2 - 1.$$

For a direct verification, set $U_0 = 1, U_1 = p$ and $U_j = pU_{j-1} - U_{j-2}$. Substituting $p = q + q^{-1}$ makes $U_j = q^j + q^{j-2} + \dots + q^{-j}$, by cancellation of the interior terms in the recurrence. Applying this through $j = 6$ proves the two displayed polynomials. Consequently the original coefficient, without discarding its vanishing factor, is

$$\begin{aligned} C(q) &= -a((4-a)^3 - 5(4-a)^2 + 6(4-a) - 1)(3-a) \\ &= -a(7 - 14a + 7a^2 - a^3)(3-a). \end{aligned} \tag{75}$$

The second expression follows by expanding the first, which remains displayed to record the substitution. The coefficient $-a$ is part of the original object in both expressions. At $q = 1$, $a = -(q-1)^2(q+1)^2/q^2$. The other factors take values 7 and 3, so

$$\left. \frac{C(q)}{(q-1)^2} \right|_{q=1} = 4 \cdot 7 \cdot 3 = 84.$$

In characteristic zero this proves a zero of order exactly two. It proves a zero of order one in the original base coordinate a : the coefficient of a in (75) is -21 . The order change is precisely the displayed ramification, not a lost specialization layer.

17.1 The resulting rank-four arithmetic extension

Now take $k = \mathbb{C}$ and retain $a = 2s$. The same ring is finite free over $\mathbb{C}[s]$:

$$\mathbb{C}[q, q^{-1}] \simeq \mathbb{C}[s, q]/(q^4 + (2s-2)q^2 + 1). \tag{76}$$

To prove this, the relation gives $q^{-1} = -q^3 + (2-2s)q$, so q is already invertible in the quotient. Conversely the Laurent map $s = -(q - q^{-1})^2/2$ gives the quartic. The two substitutions recover s and q , proving inverse ring maps. Division by the monic quartic gives the free basis $(1, q, q^2, q^3)$. In the conic presentation the alternative basis $(1, r, p, rp)$ is free over $\mathbb{C}[s]$ by the two monic relations $r^2 = -2s, p^2 = 4 - 2s$. The conversion identities are

$$r = q^3 + (2s-1)q, \quad p = -q^3 + (3-2s)q, \quad rp = 2q^2 + 2s - 2, \quad q = (p+r)/2.$$

They follow from the expression for q^{-1} and prove the basis conversion in both directions (also $q^2 = (rp + 2 - 2s)/2$).

For the actual topological $\mathbb{C}[s]$ -module (Q, S) used in the retained arithmetic construction, form $Q^{(4)} = \mathbb{C}[q, q^{-1}] \otimes_{\mathbb{C}[s]} Q$. Give it the fourfold direct-sum topology in the monic basis.

Multiplication by q has the continuous block matrix

$$K = \begin{pmatrix} 0 & 0 & 0 & -I \\ I & 0 & 0 & 0 \\ 0 & I & 0 & 2I - 2S \\ 0 & 0 & I & 0 \end{pmatrix}, \quad \widehat{S} = \text{diag}(S, S, S, S). \quad (77)$$

The last column is exactly the quartic relation. Thus $K^4 + (2\widehat{S} - 2I)K^2 + I = 0$ and $K^{-1} = -K^3 + (2I - 2\widehat{S})K$. Both are continuous polynomial expressions in commuting block operators. The two original root and trace coordinates are

$$R = K - K^{-1}, \quad P = K + K^{-1}, \quad R^2 = -2\widehat{S}, \quad P^2 = 4I - 2\widehat{S}.$$

These identities follow either by the proved ring maps or by direct substitution of the inverse polynomial. In the alternative ordering $(1, r, p, rp)$, multiplication by r is the direct sum of two copies of $\begin{pmatrix} 0 & -2S \\ I & 0 \end{pmatrix}$. Thus this is the exact two-stage extension of the parent's two-sheet module, with its factors and topology retained. Projection to the constant coefficient and its section show, by the same inverse construction used earlier, that $\text{Spec } \widehat{S} = \text{Spec } S$.

Equation (75) also supplies a completely specified operator:

$$C(K) = -2\widehat{S}(7I - 28\widehat{S} + 28\widehat{S}^2 - 8\widehat{S}^3)(3I - 2\widehat{S}). \quad (78)$$

Laurent evaluation is legitimate because K has the explicit inverse. This proves a concrete transport of a source GCT coefficient into the retained arithmetic module. It does not assert that all nonstandard quantum-group generators act on $Q^{(4)}$: only this scalar Laurent parameter algebra and its proved action have been constructed.

17.2 The original coefficient heat flow on this cover

Use material time τ , which generates the forward coefficient heat flow of the retained polynomial, with $a' = 2$, $b' = 6c$, $c' = 0$. The arithmetic heat parameter t acts instead by $-\partial_s^2/4$; Section 18 proves its exact pullback and retains the original Euclidean diffusion coefficients. For the entire original cubic, $\partial_\tau(cr^3 - 2r^2 + br - 2a) = 6cr - 4 = \partial_r^2 P$. On $b = c = 0$ this is $P = -2(r^2 + a)$ with $a' = 2$. Implicit differentiation at a simple root gives $r' = -1/r$. On the conic the second relation $p^2 = 4 - a$ gives $p' = -1/p$ when $p \neq 0$. Therefore the exact lifted coefficient derivation is

$$D_\tau = -\frac{q}{rp} \frac{d}{dq}, \quad D_\tau a = 2, \quad D_\tau s = 1, \quad D_\tau r = -\frac{1}{r}, \quad D_\tau p = -\frac{1}{p}.$$

Indeed $dr/dq = 1 + q^{-2} = p/q$ and $dp/dq = 1 - q^{-2} = r/q$; multiplication by the displayed coefficient proves all four identities. Its domain is the Laurent curve with $rp \neq 0$, equivalently $q \notin \{0, 1, -1, i, -i\}$ over $\mathbb{C}$. The points $q = \pm 1$ have the original excluded root $r = 0$. At $q = \pm i$ one has $p = 0$ and $r = \pm 2i$; these are the new ramification points of the additional p -cover over $s = 2$, while the original r -chart remains finite. Thus both exceptional loci and their different coordinate causes are recorded by explicit equations.

For the original source coordinates this heat derivative is

$$D_\tau x = -\frac{1}{2r^3}, \quad D_\tau y = -\frac{3}{r}, \quad D_\tau w = -52.$$

Differentiating the proved identity gives

$$F(-1/(2r), 3r, 26r^2) = (-r^2, 0, 0), \quad DF(D_\tau x, D_\tau y, D_\tau w)^\top = (2, 0, 0)^\top.$$

Since $\det DF = -2$, this is exactly the original inverse-Jacobian heat lift on the stated slice. The boundary point has derivative $(0, 0, 2)$ separately. For the GCT source-displayed coefficient polynomial the chain rule, with the full polynomial $C_a(a)$ in (75), gives $D_\tau C(q) = 2C'_a(a)$. No analytic unit or vanishing factor is deleted in this derivation. This is an action on the localized coefficient ring. A differentiation action on the topological arithmetic quotient is not inferred from its multiplication action; the latter is exactly the operator action already proved in (77)–(78).

The source identifies the classical spectral coordinate through $\Xi(s) = \xi(1/2 + is) = 8H_0(2s)$. Nothing in the finite-free extension changes that original coordinate. A certified off-real spectral point of S would remain off-real in each direct summand; a nonreal eigenvalue of K alone has no such implication. The exact identity $\hat{S} = -(K - K^{-1})^2/2$ gives $S = \pi[-(K - K^{-1})^2/2]\iota$, with constant-coefficient projection $\pi : Q^{(4)} \rightarrow Q$ and section $\iota : Q \rightarrow Q^{(4)}$. This is the required test for the original coordinate. None of these identities supplies an off-real s , a positivity proof for all GCT coefficients, or a complexity lower bound. They supply the exact common algebra and the specific transported coefficient on which further calculations can act.

17.3 The exact positivity test under two real structures

On $\mathbb{C}[q, q^{-1}]$ define two conjugate-linear involutions by $q^{*\text{real}} = q$ and $q^{*\text{circle}} = q^{-1}$, conjugating complex coefficients in each case. Multiplicativity on Laurent monomials and the fact that applying either map twice fixes q prove both involution identities on the entire commutative algebra. The conic isomorphism gives

$$p^{*\text{real}} = p, \quad r^{*\text{real}} = r, \quad p^{*\text{circle}} = p, \quad r^{*\text{circle}} = -r.$$

Both fix $a = -r^2$ and $s = a/2$. Thus they extend the same real structure on the arithmetic base with different root-coordinate adjoints.

Evaluation at $q_0 \neq 0$ preserves $*_{\text{real}}$ exactly when $\overline{q_0} = q_0$, and preserves $*_{\text{circle}}$ exactly when $\overline{q_0} = q_0^{-1}$. These say, respectively, that q_0 is nonzero real and that $|q_0| = 1$. For positive real q_0 , all monomials of $[7]_q[3]_q$ are positive and $(q - q^{-1})^2 \geq 0$, so $C(q_0) \geq 0$. For a precise circle character instead take

$$q_0 = \frac{\sqrt{14} + i\sqrt{2}}{4}, \quad q_0^{-1} = \frac{\sqrt{14} - i\sqrt{2}}{4}, \quad p_0 = \frac{\sqrt{14}}{2}, \quad r_0 = \frac{i}{\sqrt{2}}.$$

Products and sums verify $|q_0| = 1$, $a_0 = 1/2$, $s_0 = 1/4$. The full coefficient in (75) then gives

$$C(q_0) = -\frac{1}{2} \left(7 - 7 + \frac{7}{4} - \frac{1}{8} \right) \frac{5}{2} = -\frac{65}{32}.$$

This exactly evaluates the source's mixed-sign coefficient, retaining its positive residual Laurent factor and signed vanishing factor. It rules out the particular claim of nonnegativity at every circle character. It proves neither a failure of the source's stated canonical-basis positivity nor a negative value on an established zeta spectral point: $s_0 = 1/4$ has not been asserted to belong to that spectrum. The maps and involutions above specify the entire comparison.

18 The full material generator and the extended coefficient connection

The following calculation incorporates the parent's complete material generator at its original coefficients. Its source path, reading scope and hash are recorded in `sources.json`. We rederive every algebraic and differential identity used here from the displayed F . The material coordinate is denoted $\mathbf{z} = (x, y, w)$ in this section; the letter q continues to denote the GCT Laurent parameter.

18.1 The drift, its exact pullback, and the Euclidean operator

Put $s(\mathbf{z}) = F_1(\mathbf{z})/2$, $b(\mathbf{z}) = F_2(\mathbf{z})$, $c(\mathbf{z}) = F_3(\mathbf{z})$. Since $\det DF = -2$, the vector fields $W_j = (DF)^{-1}e_j$ are defined everywhere and satisfy $W_j F_k = \delta_{jk}$. Set $U = 2W_1 + 6F_3W_2$. For a C^2 function ϕ on an open real target set, or a holomorphic function on an open complex target set, the ordinary chain rule gives the complete intertwining identity

$$U\phi(s, b, c) = (\partial_s\phi + 6c\partial_b\phi)(s, b, c), \quad Us = 1, \quad Ub = 6c, \quad Uc = 0. \quad (79)$$

Every coefficient is retained. In particular $U(b - 6cs) = 6c - 6c = 0$. Applying the same product rule twice gives

$$U^2\phi(s, b, c) = (\partial_s^2\phi + 12c\partial_s\partial_b\phi + 36c^2\partial_b^2\phi)(s, b, c).$$

There are no missing derivatives of the coefficients: the derivative of c along U is zero by (79).

The algebra map $\mathcal{I}f = f \circ s$ on entire functions is injective: the polynomial section $\sigma(s) = (0, 0, 2s)$ satisfies $s \circ \sigma = s$, so $\sigma^*\mathcal{I} = I$. Both maps are continuous for the topology of uniform convergence on compact subsets, because the continuous image of a compact set is compact. The preceding chain rule proves inductively

$$U^k\mathcal{I}f = \mathcal{I}f^{(k)}, \quad -\frac{1}{4}U^2\mathcal{I}f = -\frac{1}{4}\mathcal{I}f''.$$

Thus the original arithmetic differential operator has this exact material pullback on its specified image. This proves an operator identity, without asserting that differentiation descends to a particular quotient by a closed subspace. The material time translates s at unit speed; the arithmetic heat operator is its negative second derivative divided by four.

The full inverse-root polynomial in these same coordinates is $P = cr^3 - 2r^2 + br - 4s$, with $P_r = 3cr^2 - 4r + b$. On the localization at P_r , the derivation

$$\mathcal{D} = \partial_s + 6c\partial_b + \frac{4 - 6cr}{P_r}\partial_r, \quad \mathcal{D}c = 0$$

annihilates P : its three contributions are -4 , $6cr$ and $4 - 6cr$. It therefore defines a derivation of the localized quotient ring. It is the unique lift with the stated derivatives of s, b, c , because differentiating $P = 0$ determines $P_r\mathcal{D}r = 4 - 6cr$. Through the previously proved inverse-chart isomorphism it equals U , since both have the same derivatives of all target coordinates and DF is invertible. The $b = c = 0$ derivation used earlier is exactly its restriction, for which $P_r = -4r$ and $\mathcal{D}r = -1/r$.

For the original real Euclidean metric put $v = 1 + xy$. The original polynomial is $s = (v^3w + y^2(v + 3v^2))/2$. Differentiation gives

$$\begin{aligned} s_x &= \frac{1}{2}(3yv^2w + y^3(1 + 6v)), \\ s_y &= \frac{1}{2}(3xv^2w + 2y(v + 3v^2) + xy^2(1 + 6v)), \\ s_w &= v^3/2, \\ s_{xx} &= 3y^2vw + 3y^4, \\ s_{yy} &= 3x^2vw + v + 3v^2 + 2xy(1 + 6v) + 3x^2y^2, \quad s_{ww} = 0. \end{aligned}$$

These identities follow by the product rule with $v_x = y$, $v_y = x$; in particular the two displayed second derivatives sum to

$$B(\mathbf{z}) := \Delta_E s = 3wv(x^2 + y^2) + 18x^2y^2 + 21xy + 3y^4 + 4.$$

Define $A(\mathbf{z}) = s_x^2 + s_y^2 + s_w^2$ using the three full expressions above. Applying the scalar chain rule in each of the three source directions and summing proves

$$\Delta_E \mathcal{I}f = A\mathcal{I}f'' + B\mathcal{I}f'. \quad (80)$$

For real coordinates A is the squared Euclidean norm of ds . Its holomorphic extension is the indicated sum of squares, with no complex conjugation inserted.

The original common-fibre points $(1, -3/2, 13/2)$ and $(0, 0, -1/4)$ both map under F to $(-1/4, 0, 0)$, as already proved. Substitution in the last formulas gives, respectively,

$$(s_x, s_y, s_w) = (-9/32, -3/16, -1/16), \quad (A, B) = (121/1024, -7/2),$$

and

$$(s_x, s_y, s_w) = (0, 0, 1/2), \quad (A, B) = (1/4, 4).$$

Consequently no operator on functions of s can satisfy $\Delta_E \mathcal{I} = \mathcal{I}L$ on all twice differentiable functions: applying such an identity to $f(s) = s$ would require $-7/2 = 4$. The full relation that continues the transport is (80), retaining its two source-dependent coefficients. It is compatible with the separate exact identity for $-U^2/4$ above and does not change that identity.

18.2 The conic extension with the original parameter b retained

All the constructions in this section are over $\mathbb{C}$. The polynomial variables b, s remain independent until a scalar value of b is explicitly fixed for the topological module construction. The original coefficient is $a = 2s$, and the original cubic on $c = 0$ is $-2r^2 + br - 4s$. Set

$$A = \mathbb{C}[b, s], \quad B = A[r]/(r^2 - br/2 + 2s), \quad \beta = b/4, \quad \eta = r - \beta,$$

and retain the inverse substitution $r = \eta + \beta$. Thus

$$s = br/4 - r^2/2 = b^2/32 - \eta^2/2, \quad \delta = b^2 - 32s = 16\eta^2, \quad L = 4 + b^2/16 - 2s = 4 + \eta^2. \quad (81)$$

Here η, β, δ, L record explicit expressions in the original objects; none changes the meanings of b, r, s or a .

18.2.1 The finite cover and its inverse maps

Form the rank-two extension of B

$$E = B[p]/(p^2 - L).$$

Successive division by the two monic quadratic polynomials gives the free A -basis $(1, r, p, rp)$: division first gives coefficients of $1, p$ in B , and division in B gives coefficients of $1, r$; uniqueness at each step proves linear independence. There are inverse $\mathbb{C}[b]$ -algebra maps $E \longleftrightarrow \mathbb{C}[b, q, q^{-1}]$, given on generators by

$$\begin{aligned} r &\mapsto \beta + q - q^{-1}, & p &\mapsto q + q^{-1}, \\ s &\mapsto b^2/32 - (q - q^{-1})^2/2, & b &\mapsto b, \\ q &\mapsto (p + r - \beta)/2, & q^{-1} &\mapsto (p - r + \beta)/2. \end{aligned} \quad (82)$$

To prove that these are well-defined, the two Laurent expressions have $p^2 - \eta^2 = 4$, and the proposed images of q, q^{-1} have product $(p^2 - \eta^2)/4 = 1$. The image of s then satisfies (81), hence both defining relations of E . Conversely the sums and differences of the images of q, q^{-1} recover p, η , and hence r and s ; the reverse compositions recover q, q^{-1} and b . This proves every map and its inverse.

For a temporary displayed coefficient put $h = 2s - b^2/16$. Then there are inverse A -algebra maps

$$E \simeq A[q]/(q^4 + (2s - b^2/16 - 2)q^2 + 1), \quad q^{-1} = -q^3 + (2 - 2s + b^2/16)q. \quad (83)$$

Indeed multiplying the proposed inverse of q by q gives one by the quartic relation. In this quotient $(q - q^{-1})^2 = -h$, which follows by multiplying the quartic by q^{-2} . Substituting in (82) therefore gives the inverse map from E , and both compositions fix their generators. In particular the quartic quotient is already a Laurent algebra. Monic division gives its free basis $(1, q, q^2, q^3)$.

The complete conversion between these two ordered bases is

$$\begin{aligned} r &= \beta + q^3 + (h-1)q, & p &= -q^3 + (3-h)q, \\ rp &= h-2 + \beta(3-h)q + 2q^2 - \beta q^3, & q &= (p+r-\beta)/2, \\ q^2 &= (rp - \beta p + 2 - h)/2, & q^3 &= r - \beta - (h-1)(p+r-\beta)/2. \end{aligned} \quad (84)$$

The first two identities follow from the inverse of q in (83). The identity $\eta p = q^2 - q^{-2} = 2q^2 + h - 2$ proves the third and fifth. The first identity proves the sixth after the fourth is inserted. Thus no inverse basis map is omitted. The matrix whose columns are $(1, r, p, rp)$ in the quartic basis is

$$T_b = \begin{pmatrix} 1 & \beta & 0 & h-2 \\ 0 & h-1 & 3-h & \beta(3-h) \\ 0 & 0 & 0 & 2 \\ 0 & 1 & -1 & -\beta \end{pmatrix}, \quad \det T_b = 4.$$

Expanding along its first column and then the row containing 2 gives $-2(-(h-1)-(3-h)) = 4$.

18.2.2 Every original source coordinate on the cover

Retain the original map, with its third source coordinate called w :

$$\begin{aligned} F_1 &= (1+xy)^3w + y^2(1+xy)(4+3xy), \\ F_2 &= y + 3x(1+xy)^2w + 3xy^2(4+3xy), \\ F_3 &= 2x - 3x^2y - x^3w. \end{aligned}$$

The simple-root inverse on $c = 0$ is

$$\alpha = (b-4r)/2 = -2\eta, \quad H_b(\eta) = \left(-\frac{1}{2\eta}, \frac{b}{4} + 3\eta, 26\eta^2 + \frac{3b\eta}{2}\right), \quad \eta \neq 0. \quad (85)$$

These are the original formulas $x = 1/\alpha$, $y = r - \alpha$, $w = 5\alpha^2 - 3r\alpha$ after the displayed invertible substitution. Here p is an additional coordinate, and never replaces w . For a direct check of the full map set $v = 1 + xy = -(\eta + \beta)/(2\eta)$ and note $4 + 3xy = -(\eta + 3\beta)/(2\eta)$. The polynomial identity

$$(\beta + 3\eta)^2(\eta + 3\beta) - (\eta + \beta)^2(13\eta + 3\beta) = 4\eta^2(\beta - \eta)$$

follows by expanding both degree-three polynomials. Since $w = 2\eta(13\eta + 3\beta)$, substitution in F_1 gives $(\eta + \beta)(\beta - \eta) = \beta^2 - \eta^2 = 2s$. Substitution in F_2 gives $\beta + 3\eta + 3(\beta - \eta) = 4\beta = b$. Finally

$$F_3 = -\frac{1}{\eta} - \frac{3(\beta + 3\eta)}{4\eta^2} + \frac{26\eta^2 + 6\beta\eta}{8\eta^3} = 0.$$

This proves $F(H_b(\eta)) = (2s, b, 0)$ with all coefficients retained. Conversely the original inverse-chart coordinate is $r = y + 1/x$, so $\eta = y + 1/x - b/4$; adjoining p then recovers $q = (p + y + 1/x - b/4)/2$. The finite point

$$(x, y, w) = (0, b, 2s - 4b^2)$$

lies on the same original fibre, because direct substitution gives $F = (w + 4y^2, y, 0) = (2s, b, 0)$. It remains separate from the $x \neq 0$ chart, including at $s = b^2/32$.

The original inverse chart requires $\eta \neq 0$, equivalently $\delta \neq 0$, or $q \neq \pm 1$. The Laurent coordinate already requires $q \neq 0$. The extra quadratic extension has its additional branch at $p = 0$, equivalently $L = 0$, or $q = \pm i$. For completeness the relative differential module is exactly

$$\Omega_{E/A} = (E/(\eta)) d\eta \oplus (E/(p)) dp.$$

Indeed $E = A[\eta, p]/(\eta^2 - b^2/16 + 2s, p^2 - L)$; the presentation of relative differentials gives precisely the relations $2\eta d\eta = 0$ and $2p dp = 0$, and 2 is invertible. Thus these are all the ramification loci, not just exhibited ones. Their base values are

$$s = b^2/32 \quad \text{and} \quad s = b^2/32 + 2,$$

respectively. They are disjoint since their difference is 2. At the latter points $\eta = \pm 2i$, so the original inverse chart is still defined. At the former points $p = \pm 2$; it is exactly the original chart which excludes the point. These identities also specify the domains of the derivative calculation below.

18.2.3 Multiplication on the same arithmetic module

Fix any scalar $b \in \mathbb{C}$, retaining its chosen value. Let Q be the original topological $\mathbb{C}[s]$ -module with continuous multiplication S by its original coordinate s . Give Q the A -module structure $b \mapsto bI$, $s \mapsto S$ and form $Q_b^{(4)} = E \otimes_A Q$. Since E is free in the quartic basis, the coefficient map gives an algebraic isomorphism $Q_b^{(4)} \simeq Q^4$; we equip it with the fourfold direct-sum topology through this map. With $\hat{S} = \text{diag}(S, S, S, S)$, multiplication by q is

$$K_b = \begin{pmatrix} 0 & 0 & 0 & -I \\ I & 0 & 0 & 0 \\ 0 & I & 0 & (2 + b^2/16)I - 2S \\ 0 & 0 & I & 0 \end{pmatrix}. \quad (86)$$

The last column is the original quartic relation; the other three columns shift the monic basis. Therefore

$$K_b^4 + (2\hat{S} - (b^2/16 + 2)I)K_b^2 + I = 0, \quad K_b^{-1} = -K_b^3 + ((2 + b^2/16)I - 2\hat{S})K_b.$$

Multiplication by this proposed inverse on either side gives I by the quartic, because all its coefficients commute with K_b . Both operators are continuous, being finite matrices of polynomials in S . Define the original-root and added-coordinate operators by

$$\mathcal{R}_b = \frac{b}{4}I + K_b - K_b^{-1}, \quad \mathcal{P}_b = K_b + K_b^{-1}.$$

The proved ring maps give

$$\begin{aligned} \mathcal{R}_b^2 - \frac{b}{2}\mathcal{R}_b + 2\hat{S} &= 0, & \mathcal{P}_b^2 &= (4 + b^2/16)I - 2\hat{S}, \\ \hat{S} &= \frac{b}{4}\mathcal{R}_b - \frac{1}{2}\mathcal{R}_b^2 = \frac{b^2}{32}I - \frac{1}{2}(K_b - K_b^{-1})^2. \end{aligned} \quad (87)$$

In the basis $(1, r, p, rp)$ multiplication by r is

$$R_b \oplus R_b, \quad R_b = \begin{pmatrix} 0 & -2S \\ I & bI/2 \end{pmatrix}.$$

Indeed $r \cdot 1 = r$ and $r \cdot r = -2s + (b/2)r$; multiplying both equations by p gives the second block. The operator obtained by substituting S in T_b is a continuous isomorphism: the inverse formulas (84) are finite polynomial expressions in S with scalar denominators 2. Thus this is an exact extension of the original two-sheet arithmetic module.

The continuous-invertibility spectrum of $\hat{S}$ is precisely that of S . If $S - \lambda I$ has a continuous two-sided inverse, four copies give an inverse for $\hat{S} - \lambda I$. Conversely let V be a continuous two-sided inverse of the latter. For the first-coordinate inclusion $\iota : Q \rightarrow Q^4$ and projection $\pi : Q^4 \rightarrow Q$, the identities $\hat{S}\iota = \iota S$ and $\pi\hat{S} = S\pi$ show that $\pi V \iota$ is a continuous left and right inverse of $S - \lambda I$. Also (87) gives the typed identity

$$S = \pi \left[\frac{b^2}{32} I - \frac{1}{2} (K_b - K_b^{-1})^2 \right] \iota.$$

Thus the recovered spectral coordinate is the original s . An eigenvalue or spectral value of K_b alone does not assert that its recovered s is off the original real axis.

Retain the GCT source-displayed Laurent coefficient polynomial

$$C(q) = q^{10} - q^4 - q^{-4} + q^{-10} = (q - q^{-1})^2 [7]_q [3]_q.$$

The recurrence $U_0 = 1, U_1 = p, U_j = pU_{j-1} - U_{j-2}$, after $p = q + q^{-1}$, gives $U_j = q^j + q^{j-2} + \dots + q^{-j}$ by cancellation of the interior terms. Hence $[7]_q = p^6 - 5p^4 + 6p^2 - 1$ and $[3]_q = p^2 - 1$. The factorization follows directly by multiplying $(q - q^{-1})[7]_q = q^7 - q^{-7}$ and $(q - q^{-1})[3]_q = q^3 - q^{-3}$; their product is exactly the four-term Laurent expression above. Every factor of the coefficient therefore descends by the explicit formula

$$\begin{aligned} C(q) &= (b^2/16 - 2s)(L^3 - 5L^2 + 6L - 1)(L - 1) \\ &= -(2s - b^2/16)(3 - 2s + b^2/16) \\ &\quad \cdot (7 - 14(2s - b^2/16) + 7(2s - b^2/16)^2 - (2s - b^2/16)^3). \end{aligned} \quad (88)$$

The second equality substitutes $L = 4 - (2s - b^2/16)$ and expands only that polynomial; the first formula preserves its direct conic substitution. In particular the coefficient has not become the previous $b = 0$ polynomial at the same s unless $b = 0$ is imposed. For $\hat{L} = (4 + b^2/16)I - 2\hat{S}$, the actual transported operator is exactly

$$C(K_b) = (b^2 I / 16 - 2\hat{S})(\hat{L}^3 - 5\hat{L}^2 + 6\hat{L} - I)(\hat{L} - I).$$

Evaluation is legitimate by the explicit continuous inverse of K_b ; the identity follows from the proved Laurent algebra map. No factor vanishing at a branch point has been divided out.

18.2.4 The complete localized coefficient connection and its square

Localize the base at both branch factors: $A^\circ = A[1/(\delta L)]$ and $E^\circ = E \otimes_A A^\circ$. The original material derivation on the requested $c = 0$ slice has $Db = 0$, $Ds = 1$, and consequently $Da = 2$. There is exactly one derivation of E° extending these values, and it satisfies

$$D\eta = -1/\eta, \quad Dr = \frac{4}{b - 4r} = \frac{4b - 16r}{\delta}, \quad Dp = -1/p, \quad Dq = -q/(\eta p). \quad (89)$$

In fact differentiating the two monic quadratic relations forces the first and third values because η and p are invertible on this localization. Defining them this way annihilates both relations, proving existence as well as uniqueness. Since $Dr = D\eta$ and $b - 4r = -4\eta$, the second formula follows, including its expression in the free basis. In Laurent coordinates $d\eta/dq = p/q$ and $dp/dq = \eta/q$; thus

$$D = -\frac{q}{\eta p} \frac{\partial}{\partial q} \quad (b \text{ fixed})$$

has precisely these values and also gives $Ds = 1$ in (81). Its Laurent domain is $q \notin \{0, 1, -1, i, -i\}$. This records separately the absence of $q = 0$, the original inverse-chart branch, and the added p branch.

For a column $f = (f_0, f_1, f_2, f_3)^\top$ of coefficients in A° , interpret it as $f_0 + rf_1 + pf_2 + rpf_3$. The product rule and (89) give exactly

$$Df = (\partial_s I_4 + \Gamma_b)f, \quad \Gamma_b = \begin{pmatrix} A_b & 0 \\ 0 & A_b - I_2/L \end{pmatrix}, \quad A_b = \begin{pmatrix} 0 & 4b/\delta \\ 0 & -16/\delta \end{pmatrix}. \quad (90)$$

Here the left side denotes the coefficient column of the derived element, rather than component-wise differentiation alone. The first two columns follow from $D1 = 0$ and $Dr = (4b - 16r)/\delta$. For the lower columns, $Dp = -p/L$ because $p^2 = L$, and $D(rp) = p(4b - 16r)/\delta - rp/L$. These verify all four columns. Thus the p contribution $-I_2/L$ is present in the full connection.

All zeroth-order terms of its square are

$$\begin{aligned} D^2 &= \partial_s^2 I_4 + 2\Gamma_b \partial_s + Z_b, \\ Z_b &= \begin{pmatrix} (16/\delta)A_b & 0 \\ 0 & (16/\delta - 2/L)A_b - I_2/L^2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 64b/\delta^2 & 0 & 0 \\ 0 & -256/\delta^2 & 0 & 0 \\ 0 & 0 & -1/L^2 & 64b/\delta^2 - 8b/(\delta L) \\ 0 & 0 & 0 & -256/\delta^2 + 32/(\delta L) - 1/L^2 \end{pmatrix}. \end{aligned} \quad (91)$$

For a full proof, $\partial_s \delta = -32$ and $\partial_s L = -2$ give $A'_b = (32/\delta)A_b$ and $(-I_2/L)' = -2I_2/L^2$. Direct matrix multiplication gives $A_b^2 = -(16/\delta)A_b$. It follows that the upper block of $\Gamma'_b + \Gamma_b^2$ is $(16/\delta)A_b$. The lower block is

$$(32/\delta)A_b - 2I_2/L^2 - (16/\delta)A_b - 2A_b/L + I_2/L^2 = (16/\delta - 2/L)A_b - I_2/L^2.$$

Applying the product rule twice gives the displayed differential operator and multiplying the scalars into A_b gives every entry of Z_b . In particular neither $-2A_b/L$ nor $-I_2/L^2$ disappears.

As a further compatibility check, the original root multiplication obeys $R_b^2 - (b/2)R_b + 2sI_2 = 0$ and

$$[\partial_s I_2 + A_b, R_b] = R'_b + A_b R_b - R_b A_b = 4(bI_2 - 4R_b)/\delta.$$

The last equality follows by substituting its four matrix entries: the result is $\delta^{-1} \begin{pmatrix} 4b & 32s \\ -16 & -4b \end{pmatrix}$. It is exactly multiplication by Dr , establishing compatibility with the original quadratic relation rather than only with its shift.

Restriction to $b = 0$ is a quotient of this entire coefficient calculation. Then $\eta = r$, $\delta = -32s$, $L = 4 - 2s$, and the connection becomes

$$\Gamma_0 = \text{diag} \left(0, \frac{1}{2s}, -\frac{1}{4-2s}, \frac{1}{2s} - \frac{1}{4-2s} \right).$$

Its full zeroth-order square term becomes

$$Z_0 = \text{diag} \left(0, -\frac{1}{4s^2}, -\frac{1}{(4-2s)^2}, -\frac{1}{4s^2} - \frac{1}{s(4-2s)} - \frac{1}{(4-2s)^2} \right).$$

These follow by direct substitution in (91). The restricted domain is $s \neq 0, 2$, and the quartic, inverse source map, original-root operator and coefficient in the preceding formulas all specialize by setting $b = 0$. Thus the earlier two-sheet connection is exactly the first block of this four-sheet connection.

Finally the original coordinates in (85) have the retained derivatives

$$Dx = -\frac{1}{2\eta^3}, \quad Dy = -\frac{3}{\eta}, \quad Dw = -52 - \frac{3b}{2\eta}.$$

For example $Dw = (52\eta + 3b/2)(-1/\eta)$; differentiation of the other two displayed source functions proves their values. Differentiating the already proved complete identity $F(H_b(\eta)) = (2s, b, 0)$ now gives

$$DF(H_b(\eta)) (Dx, Dy, Dw)^\top = (2, 0, 0)^\top.$$

The separate finite point has derivative $(0, 0, 2)$. For the original GCT coefficient, (88) gives the exact identity $DC(q) = \partial_s C_b(s)$ with b fixed, including all its factors and the b -dependence. This derivation is on E° . The topological construction above proves multiplication on $Q_b^{(4)}$ and its spectral-coordinate recovery; it does not by itself define ∂_s on Q , prove the invertibility there of $\delta(S)$ or $L(S)$, or descend this localized connection to an arithmetic quotient. No such derivative action is used in any operator identity proved here.

19 An explicit Boolean-to-algebraic edge with its encoding

The inclusion $P \subseteq NP$ follows directly: a deterministic machine is a nondeterministic machine with one available move at each step and the same running time. The reverse inclusion is the equality question. We retain a concrete edge by arithmetizing a three-literal conjunctive formula's verifier.

For $n, m \geq 1$, use $6mn$ commuting variables $E_{j,k,i,\sigma}$, where $1 \leq j \leq m$, $1 \leq k \leq 3$, $1 \leq i \leq n$, $\sigma \in \{+, -\}$, and auxiliary variables $b_1, \dots, b_n$. Over $\mathbb{Z}$ define

$$\begin{aligned} L_{j,k}(E, b) &= \sum_{i=1}^n (E_{j,k,i,+} b_i + E_{j,k,i,-} (1 - b_i)), \\ V_{n,m}(E, b) &= \prod_{j=1}^m \left(1 - \prod_{k=1}^3 (1 - L_{j,k}(E, b)) \right), \\ Q_{n,m}(E) &= \sum_{b \in \{0,1\}^n} V_{n,m}(E, b). \end{aligned}$$

A formula φ with n explicitly named variables and m ordered three-literal clauses gives the zero-one array $E(\varphi)$ with one 1 in each slot (j, k) , at its literal's index and sign. Repeated literals are allowed. Input consists of explicitly named variables and ordered signed binary indices with delimiters. Writing $E(\varphi)$ uses $O(mn)$ bits and operations. This convention does not allow exponentially many unused variables to be specified solely by a binary upper bound: alternatively relabel the r occurring variables and record the unused count separately. The projection from all n Boolean variables to those r coordinates has exactly 2^{n-r} elements in each fibre and preserves the verifier. Thus the original count is 2^{n-r} times the relabelled count. A binary specification of an enormous unused count would require this factored output format; writing all bits of the expanded count need not be polynomial in that shorter input encoding.

Proposition 19.1 (Exact counting reduction). *The integer $Q_{n,m}(E(\varphi))$ is exactly the number of satisfying assignments of φ . Deciding satisfiability of these formulas therefore reduces by this polynomial array map and a nonzero test to evaluation of this family on the specified arrays.*

Proof. For Boolean b , each selected $L_{j,k}$ is exactly the literal's truth value, b_i or $1 - b_i$. The product of their three complements is 1 exactly when the clause is false. Its complement

is its truth value. The outer product is 1 exactly when all clauses hold, and 0 otherwise. Summation proves the integer equality and the bound $0 \leq Q_{n,m}(E(\varphi)) \leq 2^n$. The stated array construction and zero test give the reduction. $\square$

The displayed construction gives $V_{n,m}$ a uniform division-free arithmetic circuit of $O(mn)$ gates, constants 0, 1, total degree at most $6m$ in (E, b) and degree at most $3m$ in E . Each literal has degree at most two, each clause at most six, and the products give these bounds. The loops over the displayed indices construct the circuit in polynomial time. To obtain the usual one-index family, assign $t = (n + m - 2)(n + m - 1)/2 + n$. The consecutive diagonals of pairs (n, m) show this is a bijection from positive pairs to positive integers. Trying consecutive diagonals through $n + m \leq t + 1$ decodes it in polynomial time, and $n, m \leq t$, so all variable, degree and gate bounds remain polynomial in t . If a definition requires exactly t witness bits, multiply the verifier by $\prod_{i=n+1}^t (1 - b_i)$ before summing; only the all-zero added bits contribute, leaving $Q_{n,m}$ unchanged. By the definition of VNP as Boolean exponential summation of a polynomial-size, polynomially bounded-degree arithmetic circuit family, this proves $Q_{n,m} \in \text{VNP}_{\mathbb{Q}}$ by construction.

There is also an explicit bit interface for a supplied, topologically ordered, fan-in-two division-free circuit C over $+, -, \times$, with signed binary integer constants. On input $E(\varphi)$, evaluate its gates modulo $M = 2^{n+1}$. For L gates and total constant-bit length B_C , this takes $O(B_C + L(n + 1)^2)$ arithmetic bit operations, plus graph and input reading and the chosen machine's memory-access overhead, by binary addition and school multiplication. Reduction to low $n + 1$ bits handles products, long constants and signs. Induction over gates proves exact agreement with the integer value modulo M . If the supplied circuit computes $Q_{n,m}$, its residue is its exact value on this input since $0 \leq Q \leq 2^n < M$. This gives the precise evaluation reduction without a bound on intermediate integer magnitudes.

No polynomial-size circuits for $Q_{n,m}$ have been constructed here. Polynomial-time generation of their full binary descriptions would make the displayed evaluation uniform and bound both gate count and constant-bit length. Descriptions supplied as size-dependent advice give a polynomial-advice interface only when their total binary description length is polynomial. Arbitrary complex constants do not admit this binary modular operation without additional input data. The retained inverse-fibre selector polynomial already has the explicit small determinant proved earlier; the present counting polynomial has the different, fully specified joint verifier product and Boolean summation. The next reduction must transport those joint constraints with their parameter growth. None is erased by calling both objects polynomial families.

20 Shared Boolean constraints on the full original retained fibre

This section constructs the joint constraints and their counting map for arbitrarily many witnesses. The base field k has characteristic zero. The integral and bit statements use $\mathbb{Z}$ and $\mathbb{Q}$ with the encodings specified below. A witness variable is denoted B_i ; it is different from the original target coordinate b_i .

20.1 The full three-point source fibre and its Boolean corner

Retain the original source map, including all coefficients,

$$\begin{aligned} F_1 &= (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 w, \end{aligned}$$

with source orientation (x, y, w) and target (a, b, c) . To retain the Jacobian claim with a direct proof, on $x \neq 0$ put $v = 1 + xy$, $h = 2 - 3xy - x^2 w$, $A_0 = v^2 + v - v^3 h$, $B_0 = 4v + 2 - 3v^2 h$. The inverse coordinates are $y = (v - 1)/x$, $w = (5 - 3v - h)/x^2$, and substitution gives $F = (A_0/x^2, B_0/x, xh)$. The coordinate-change determinant is $-x^3$, while the determinant of the latter map is x^{-3} times

$$(4 - 6vh)(-2v^2 - 2v + 3v^3 h) + (2v + 1 - 3v^2 h)(4v + 2 - 6v^2 h) = 2.$$

Their product is -2 . The identity holds everywhere because the polynomial coordinate ring embeds in its localization at x . Thus $\det DF = -2$ in the original orientations. Its retained polynomial is $P = cr^3 - 2r^2 + br - 2a$ and $P' = 3cr^2 - 4r + b$. We use the explicitly specified target $(-1/4, 0, 0)$ and set

$$A_1 = k[x, y, w]/(F_1 + 1/4, F_2, F_3), \quad T_1 = k[X]/(X^3 - X).$$

Proposition 20.1. *There is an isomorphism $A_1 \longrightarrow T_1$ given by*

$$x \longmapsto X, \quad y \longmapsto -3X/2, \quad w \longmapsto (27X^2 - 1)/4.$$

Evaluation at $X = -1, 0, 1$ identifies T_1 with k^3 in that order. The three idempotents, in the same order, are

$$e_- = (X^2 - X)/2, \quad e_0 = 1 - X^2, \quad e_+ = (X^2 + X)/2.$$

The factor e_0 is the original projective boundary point $(x, y, w) = (0, 0, -1/4)$.

Proof. Put $h = 2 - 3xy - x^2 w$ and $R = k[x, y, w]/(F_1 + 1/4, F_2)$. The identity $1 = h/2 + x(3y + xw)/2$ proves $(x) + (h) = R$. Consequently $(x) \cap (h) = (xh)$: for $z = xu = hv$ multiply the displayed expression for 1 by z . As $F_3 = xh$, the map $R/(F_3) \rightarrow R/(x) \times R/(h)$ is an isomorphism. Its injectivity is the intersection equality; multiplying lifts of the two components by the two displayed summands of 1 proves surjectivity. The first factor is k , at $(0, 0, -1/4)$.

In the second factor $x(3y + xw) = 2$ makes x invertible. Put $r = y + 1/x$ and $\alpha = 1/x$. Solving $F_3 = 0$ gives $x = \alpha^{-1}$, $y = r - \alpha$, $w = 5\alpha^2 - 3r\alpha$. Direct insertion into the original map gives

$$(F_1, F_2, F_3) = (r^2 + r\alpha, 4r + 2\alpha, 0).$$

Thus $F_2 = 0$ gives $\alpha = -2r$, and $F_1 = -1/4$ gives $r^2 = 1/4$. Conversely these equations, with $\alpha = -2r$, give the original three target equations and recover $r = y + 1/x$, $\alpha = 1/x$. The factor is therefore exactly $k[r]/(r^2 - 1/4)$. In it $x = -2r$, $y = 3r = -3x/2$, $x^2 = 1$, and $w = 13/2 = (27x^2 - 1)/4$. The two factors together prove the stated map, since $k[X]/(X^3 - X) \simeq k[X]/(X) \times k[X]/(X^2 - 1)$ by the same coprime-ideal argument. The displayed idempotents have values $(1, 0, 0), (0, 1, 0), (0, 0, 1)$; their values show that they sum to one, are orthogonal, and recover every triple. This also proves evaluation is an isomorphism, without discarding the boundary factor. $\square$

For $n \geq 1$ let $A_n = A_1^{\otimes n}$ and write X_i for its corresponding source coordinate. This is the full original fibre of $F^{[n]}$ and has the basis $\{X_1^{d_1} \cdots X_n^{d_n} : 0 \leq d_i \leq 2\}$. Repeated division by $X_i^3 - X_i$ proves spanning; evaluations at $\{-1, 0, 1\}^n$ and products of the three idempotents prove independence. In particular its dimension is exactly 3^n . Put

$$e_i = X_i^2, \quad \beta_i = (X_i^2 + X_i)/2, \quad e = \prod_{i=1}^n e_i.$$

The elements e_i, β_i, e are idempotents. The element β_i has values $0, 0, 1$ at $X_i = -1, 0, 1$, respectively, while e_i has values $1, 0, 1$. The quotient and corner maps are

$$A_n/(1-e) \xrightarrow{\sim} eA_n, \quad [f] \mapsto ef, \quad eA_n \xrightarrow{\sim} k[B_1, \dots, B_n]/(B_i^2 - B_i), \quad eX_i \mapsto 2B_i - 1.$$

The corner has identity e . For the first map $ef = 0$ implies $f = (1-e)f$, proving exactly its kernel and injectivity; its image is its definition. For the second map the inverse sends B_i to $e\beta_i = e(1 + X_i)/2$. The relations $eX_i^2 = e$ prove both compositions on the generators, including the identities. Also $(1-e) = (1-e_1, \dots, 1-e_n)$ as ideals: $1-e = \sum_i (1-e_i) \prod_{h < i} e_h$, while $1-e_i = (1-e_i)(1-e)$. Thus this corner keeps exactly the 2^n choices with $X_i \neq 0$; the complementary boundary factors remain recorded by $1-e$ in the full algebra.

There is a useful exact integral distinction here. The evaluation map

$$\mathbb{Z}[X]/(X^3 - X) \longrightarrow \mathbb{Z}^3$$

has image $\{(u, v, w) : u \equiv w \pmod{2}\}$. Indeed $A + BX + CX^2$ has values $(A - B + C, A, A + B + C)$; conversely use $A = v$, $B = (w - u)/2$, $C = (u + w - 2v)/2$. The inclusion has index two, and its conductor in $\mathbb{Z}^3$ is $2\mathbb{Z} \times \mathbb{Z} \times 2\mathbb{Z}$: multiplication by $(1, 0, 0)$ and $(0, 0, 1)$ proves necessity of the two evenness conditions, and the image criterion proves sufficiency. The split lattice $\mathbb{Z}e_- \oplus \mathbb{Z}e_0 \oplus \mathbb{Z}e_+$ is consequently an explicit integral lattice inside A_1 ; tensor products give a lattice $\mathbb{Z}^{3^n}$ in A_n . This retains the factor two and the original rational source coordinates. It does not reduce their denominators modulo an even modulus.

20.2 A trace that carries actual shared verifier constraints

Let $n, m \geq 1$ and retain the $6mn$ independent variables $E_{j,k,i,\sigma}$, where $1 \leq j \leq m$, $1 \leq k \leq 3$, $1 \leq i \leq n$, and $\sigma \in \{+, -\}$. In $\mathbb{Z}[E, B]$ put

$$\begin{aligned} L_{j,k}(E, B) &= \sum_{i=1}^n (E_{j,k,i,+} B_i + E_{j,k,i,-} (1 - B_i)), \\ H_j(E, B) &= \prod_{k=1}^3 (1 - L_{j,k}(E, B)), \\ V_{n,m}(E, B) &= \prod_{j=1}^m (1 - H_j(E, B)), \\ Q_{n,m}(E) &= \sum_{B \in \{0,1\}^n} V_{n,m}(E, B). \end{aligned}$$

Every occurrence of a witness index uses the same B_i , across all clauses and positions. Over $k[E]$, define

$$u(E) = e V_{n,m}(E, \beta) \in A_n \otimes_k k[E], \quad m_u : f \mapsto uf.$$

Theorem 20.2 (Full-fibre counting trace). *The following identity is an identity of polynomials in all the independent E coordinates:*

$$\mathrm{Tr}_{A_n \otimes k[E]/k[E]}(m_u) = Q_{n,m}(E).$$

For a three-literal conjunctive formula φ , let $E(\varphi)$ have a single 1 in each (j, k) slot at its selected signed literal, and zero in every other position. Then $u(E(\varphi))$ is an idempotent, its multiplication map is a projection of rank exactly $\#\text{SAT}(\varphi)$, and its trace is that same integer.

Proof. The product idempotents from Proposition 20.1 are a basis after extending scalars to $k[E]$. Multiplication by u is diagonal in this basis, with entry

$$\left(\prod_i X_i^2\right) V_{n,m} \left(E, \left((X_i^2 + X_i)/2\right)_i\right) \quad \text{at } (X_1, \dots, X_n) \in \{-1, 0, 1\}^n.$$

The entry is zero if any $X_i = 0$. On the remaining points $B_i = (X_i + 1)/2$ runs bijectively over $\{0, 1\}^n$. Adding the diagonal entries proves the polynomial identity. On the specified array each $L_{j,k}$ has its literal's truth value. Thus $H_j = 1$ exactly when all three literals are false, and $\prod_j (1 - H_j)$ has value one exactly on the satisfying assignments. All eigenvalues of m_u are therefore zero or one, and its idempotence, rank and trace assertions follow in the displayed basis. The equality with the integer count uses the canonical embedding $\mathbb{Z} \rightarrow k$, which is injective in characteristic zero. $\square$

This also provides the integral matrix needed for reduction modulo an integer. In the split lattice's product-idempotent basis, m_u is diagonal with entries in $\mathbb{Z}[E]$, by the preceding display: each β_i has the integer value zero or one. Its trace can thus be reduced modulo 2^{n+1} entry by entry without interpreting $1/2$ in that quotient ring. The lattice still has rank 3^n , and its Boolean corner has rank 2^n .

The construction respects explicit witness symmetries. Permuting the X_i and simultaneously permuting the index i of E permutes the product-idempotent basis and leaves the trace unchanged. On the corner, $X_i \mapsto -X_i$ sends B_i to $1 - B_i$; exchanging $E_{j,k,i,+}$ and $E_{j,k,i,-}$ then leaves every literal sum invariant. These maps commute for distinct i , square to the identity, and permutations conjugate the i th map to the permuted one. They give the exact action of $(\mathbb{Z}/2\mathbb{Z})^n \rtimes \mathfrak{S}_n$ on the corner and the parameter ring, under which the trace polynomial is invariant. On the full algebra the bit flip sends β_i to $e_i - \beta_i$. The difference from $1 - \beta_i$ is $e_i - 1$, annihilated by e . Thus the same trace invariance holds on the full algebra with its boundary factors retained. This proves the action and the map it preserves, rather than identifying two presentations by notation.

20.3 Universal incidence rings retaining the original cubics

Use the polynomial ring

$$R_{n,m} = k[E, a_i, b_i, c_i, r_i, B_i : 1 \leq i \leq n]$$

and, before imposing any target equations, retain

$$P_i = c_i r_i^3 - 2r_i^2 + b_i r_i - 2a_i, \quad P'_i = 3c_i r_i^2 - 4r_i + b_i.$$

Let $I_{n,m}$ be the ideal generated, for each i , by the six specified expressions

$$P_i, \quad a_i + 1/4, \quad b_i, \quad c_i, \quad B_i^2 - B_i, \quad 2r_i + 2B_i - 1,$$

and additionally by all $H_j(E, B)$, $1 \leq j \leq m$. All target specializations are therefore equations in the original coordinate ring; no original coefficient is silently replaced.

Proposition 20.3. *There are explicit isomorphisms of $k[E]$ -algebras*

$$\begin{aligned} R_{n,m}/I_{n,m} &\simeq k[E, B]/(B_i^2 - B_i, H_j(E, B)) \\ &\simeq (A_n \otimes k[E])/(1 - e, H_j(E, \beta)). \end{aligned}$$

The inverse-fibre localization by $\prod_i P'_i$ changes neither ring. Specializing $E = E(\varphi)$ gives a reduced algebra with one copy of k for each satisfying assignment, and no other factors.

Proof. The first map sends

$$a_i \mapsto -1/4, \quad b_i, c_i \mapsto 0, \quad r_i \mapsto (1 - 2B_i)/2,$$

and fixes E, B . Under it the original polynomial becomes $-2(B_i^2 - B_i)$, so every defining relation maps to zero. The inverse fixes E, B and is inverse on a, b, c, r by their listed relations, proving the first isomorphism and its entire kernel. The second is the previously proved Boolean-corner map, with $B_i \leftrightarrow e\beta_i$; it sends the clause relations to their displayed counterparts. In the first quotient $P'_i = 4B_i - 2$ and

$$(4B_i - 2)(B_i - 1/2) = 1 + 4(B_i^2 - B_i) = 1.$$

Thus its inverse is exactly $B_i - 1/2$, proving the localization claim. The original source recovery formulas give

$$\begin{aligned} x_i &= 2/P'_i = 2B_i - 1, & y_i &= r_i - P'_i/2 = 3/2 - 3B_i, \\ w_i &= 5(P'_i/2)^2 - 3r_i P'_i/2 - c_i(P'_i/2)^3 = 13/2. \end{aligned}$$

They are the selected source sheets of the original fibre. Their inverse on this quotient is $B_i = (x_i + 1)/2$, as required.

For completeness the Boolean algebra over $k[E]$ splits by the orthogonal idempotents $\prod_i B_i^{\epsilon_i} (1 - B_i)^{1 - \epsilon_i}$ into $\prod_{\epsilon \in \{0,1\}^n} k[E]$. The clause ideal is sent componentwise to $(H_j(E, \epsilon) : 1 \leq j \leq m)$. After specializing the one-hot array, each generator has value zero or one. A component survives as k exactly when all of these values are zero; otherwise it becomes the zero ring. These are exactly the satisfying assignments. This proves reducedness and the full factor description, including inconsistent and repeated clauses. $\square$

20.4 A single determinant with reversible coefficient recovery

Introduce independent selectors $t_i, \lambda_i, \mu_i, \nu_i, \rho_i, \tau_i$ for the six expressions in their displayed order, and s_j for the clauses. Put

$$\begin{aligned} S_{n,m} &= \sum_{i=1}^n (t_i P_i + \lambda_i (a_i + 1/4) + \mu_i b_i + \nu_i c_i + \rho_i (B_i^2 - B_i) + \tau_i (2r_i + 2B_i - 1)) \\ &\quad + \sum_{j=1}^m s_j H_j(E, B). \end{aligned}$$

The coefficient map from the homogeneous selector-degree-one part of $R_{n,m}[t, \lambda, \mu, \nu, \rho, \tau, s]$ to $R_{n,m}^{6n+m}$ is an isomorphism of $R_{n,m}$ -modules: its inverse forms the sum of each tuple entry times its own selector. The uniqueness of polynomial coefficients proves both compositions. In particular the coefficient ideal of $S_{n,m}$ is exactly $I_{n,m}$. At any point over an extension field, the incidence relations hold if and only if $S_{n,m}$ is identically zero in all selectors. This retains the shared variables, not merely a product of independent hypersurface equations.

We give a complete determinant construction with a polynomial size bound. Write the displayed expression as a binary arithmetic formula with operations addition and multiplication and signed rational constants; retain every occurrence. For the cubic use $c_i r_i r_i r_i + (-2)r_i r_i + b_i r_i + (-2)a_i$. Write $B_i^2 - B_i = B_i B_i + (-1)B_i$, and in each literal write $1 - B_i = 1 + (-1)B_i$. Each clause complement is $1 + (-1)L_{j,k}$. Parenthesize sums and products in their displayed index order. This formula has exactly

$$\ell = 30n + (18n + 7)m$$

leaves. For each i the six selected terms have respectively 12, 3, 2, 2, 5, 6 leaves. Each $L_{j,k}$ has $6n$ leaves, its complement has $6n + 2$, and a selected three-factor product has $18n + 7$. This proves the count without hiding repeated witnesses. Let M be this formula's number of multiplication nodes. The six selected terms for each i contribute respectively 8, 1, 1, 1, 3, 3 multiplications, while each literal sum contributes $3n$, each complement contributes one more, and combining the selector and its three complements contributes three. Thus

$$M = 17n + (9n + 6)m \leq \ell - 1.$$

Construct a directed graph with a source v_s and sink v_t as follows. To implement a sum between given endpoints, implement both summands with the same endpoints and fresh internal vertices. To implement a product, create one fresh middle vertex and implement its first factor from the source to that vertex and its second factor from there to the sink. To implement a leaf with label z_0 , create a fresh private vertex, with edges labelled z_0 from source to private vertex and 1 from private vertex to sink. Here z_0 is one original indeterminate or one signed rational constant, not the homogenizing variable introduced below. Induction proves the graph is acyclic and that the sum of products of labels over all source-to-sink paths is exactly the formula: sums partition paths between the two subgraphs, and products uniquely concatenate paths at their middle vertex. Its number of vertices is $v = 2 + \ell + M$ and its number of edges is 2ℓ .

Order vertices topologically, let N be the strictly upper triangular matrix of edge labels, and let e_s, e_t be the corresponding coordinate vectors. No two leaf-labelled edges are merged: each has its own private target vertex. With

$$d = v + 1 = 3 + \ell + M = 3 + 47n + (27n + 13)m \leq 2\ell + 2, \quad \mathcal{D}_{n,m} = \begin{pmatrix} I_v - N & e_t \\ -e_s^\top & 0 \end{pmatrix},$$

every entry is affine-linear over $\mathbb{Q}$ and

$$\det \mathcal{D}_{n,m} = S_{n,m}.$$

Indeed $N^v = 0$, so $(I_v - N)^{-1} = \sum_{j=0}^{v-1} N^j$, whose (s, t) entry is the path sum. Multiplication on the left by $\begin{pmatrix} I & 0 \\ e_s^\top (I - N)^{-1} & 1 \end{pmatrix}$ has determinant one and produces the upper block triangular matrix with last diagonal entry $e_s^\top (I - N)^{-1} e_t$. Its other diagonal block has determinant one. This proves the exact determinant identity and sign.

Let Z be a new homogenizing variable and replace each constant matrix entry by that constant times Z , leaving all variable entries linear. Call the resulting matrix $\mathcal{D}_{n,m}^h$. There are $q = 11n + 6mn + m$ affine variables in $S_{n,m}$, including all selectors. Let w denote their ordered tuple, with order fixed by the above index loops, and put

$$H_{n,m}(w, Z) = \det \mathcal{D}_{n,m}^h(w, Z) = Z^d S_{n,m}(w/Z).$$

The last identity follows on $Z \neq 0$ by factoring Z from each matrix row. Both sides are polynomials, since $\deg S_{n,m} \leq 7 < d$, so injectivity of localization proves equality at $Z = 0$ also. In fact $\deg S_{n,m} = 7$: for $n, m \geq 1$ the monomial using $s_1 E_{1,1,1,+} E_{1,2,1,+} E_{1,3,1,+} B_1^3$ has nonzero coefficient. Consequently the explicit padding factor is Z^{d-7} times the degree-seven homogenization of $S_{n,m}$. The hyperplane $Z = 0$ is an added zero component; at $Z = 1$ every original incidence coefficient is recovered.

The coordinate-ring map is fully specified:

$$\Theta : \mathbb{Q}[Y_{ab} : 1 \leq a, b \leq d] \longrightarrow \mathbb{Q}[w, Z], \quad Y_{ab} \longmapsto (\mathcal{D}_{n,m}^h)_{ab}.$$

Every indeterminate in w occurs as a leaf, hence as an individual matrix entry with coefficient -1 ; a diagonal entry is Z . Thus Θ is surjective. Choose one such entry for each variable, divide

by its nonzero scalar, and subtract from every remaining Y_{ab} the exact linear combination of chosen entries given by its image. The differences generate the kernel: modulo them the source is the polynomial ring on the chosen entries, and the map has the inverse sending each target variable to its chosen entry. This proves the image, kernel, polynomial degree and all variable identifications.

20.5 The orbit and representation consequence of these joint constraints

Work now over $\mathbb{C}$. Set $V_d = \text{Mat}_{d \times d}(\mathbb{C})$, $W_d = \text{Sym}^d(V_d^*)$, and $G_d = \text{GL}(V_d)$, with $(g \cdot f)(Y) = f(g^{-1}Y)$. Let $L : \mathbb{C}^{q+1} \rightarrow V_d$ be the linear map with value $\mathcal{D}_{n,m}^h$. The coordinate calculation proves L is injective. Also $q+1 \leq \ell+1 < d^2$, since each different variable occurs among the ℓ leaves and $d \geq \ell+3$. Let $p : V_d \rightarrow \mathbb{C}^{q+1}$ select the first $q+1$ entries in row order, let i fill those entries and set the rest to zero, and put $A = Lp$. Then $pi = 1$,

$$\tilde{H} = H_{n,m} \circ p = \det_d \circ A, \quad i^* \tilde{H} = H_{n,m}.$$

The map p^* is injective and $i^*p^* = 1$, while $L^* = \Theta$.

Write $\Omega_d = \overline{G_d \cdot \det_d}$. Then $\tilde{H} \in \Omega_d$. Indeed $A + \epsilon I$ is invertible except at the finitely many zeros of its monic determinant polynomial, and $\det_d \circ (A + \epsilon I)$ is an orbit point there. Every polynomial equation vanishing on the orbit evaluates to a polynomial in ϵ vanishing away from that finite set, hence vanishes identically and at zero. This is exactly closure membership. It is a strict boundary point. The universal determinant has zero directional-annihilator subspace, because its cofactors are linearly independent: each cofactor has a distinct pair of row and column multidegrees. The chain rule preserves the dimension of this subspace under G_d , whereas $\ker p$ is a nonzero subspace of the annihilator of $\tilde{H}$. The polynomial is nonzero because the recovered selector coefficient of λ_1 is $a_1 + 1/4$.

Let $Z_{n,m} = \overline{G_d \cdot \tilde{H}}$. Both this and Ω_d are cones: scalar coordinate substitutions realize every nonzero scalar multiple of a degree- d form. The containment $Z_{n,m} \subseteq \Omega_d$ gives, in every coordinate-ring degree $h \geq 0$, a surjective G_d -equivariant map

$$\mathbb{C}[\Omega_d]_h \longrightarrow \mathbb{C}[Z_{n,m}]_h.$$

Here the map is restriction, and its kernel is exactly $I(Z_{n,m})_h / I(\Omega_d)_h$. Homogeneity follows by evaluating each equation on scalar multiples and comparing coefficients in the scalar. This proves the exact representation quotient. In particular, for every finite-dimensional irreducible representation U , precomposing with this surjection is an injection

$$\text{Hom}_{G_d}(\mathbb{C}[Z_{n,m}]_h, U) \hookrightarrow \text{Hom}_{G_d}(\mathbb{C}[\Omega_d]_h, U),$$

because a linear map vanishing on the surjection's image vanishes everywhere. This inequality uses the exact modules without importing an unproved multiplicity calculation or a semisimplicity assertion.

20.6 Unbounded family, bit evaluation, and the size of the counting map

Index all positive pairs by $t = (n+m-2)(n+m-1)/2 + n$. Consecutive diagonals $n+m = 2, 3, \dots$ give consecutive disjoint intervals of integers of lengths $1, 2, \dots$, and n labels positions within each interval. This proves a bijection with the positive integers. Sequentially testing the diagonals up to $t+1$ decodes it in polynomial time, and $n, m \leq t$. The construction of the formula, graph, topological order, matrix, variable list and selector coefficient map is uniform: the displayed loops write $O(mn+n)$ formula symbols and vertices, and their binary labels have $O(\log(mn+n+1))$ bits. Constants belong to $\{0, 1, -1, 2, -2, 1/4, -1/4\}$; the sign in $I_v - N$

accounts for the entry $-1/4$. The sparse matrix description has $O((mn+n)\log(mn+n+1))$ bits; a dense description has $O((mn+n)^2\log(mn+n+1))$ bits. Both are polynomial in t .

The verifier $V_{n,m}$ has a uniformly generated division-free arithmetic circuit with $O(mn)$ gates, degree at most $6m$ in (E, B) and degree at most $3m$ in E . Each literal sum has total degree at most two, a clause at most six, and the product has the stated bounds. Therefore the one-index family $Q_t = Q_{n,m}$ is in $\text{VNP}_{\mathbb{Q}}$ by the definition as a Boolean sum of such a polynomial-size, bounded-degree circuit family. If exactly t witness bits are required, multiply the verifier by $\prod_{i=n+1}^t (1 - B_i)$; only the all-zero additional bits survive, so its sum stays exactly Q_t and its degree and size stay polynomial.

Represent a three-literal formula by its explicitly listed variables and ordered triples of signed binary variable indices, with delimiters. Its one-hot array can be written in $O(mn)$ bit positions plus index processing; this is polynomial in that representation. The equality $Q_{n,m}(E(\varphi)) = \#\text{SAT}(\varphi)$ and the test for a nonzero integer give the exact reduction to evaluating this one-index family. This convention does not encode exponentially many unused variables by a binary upper bound. If only r of the explicitly listed n variables occur, restricting an assignment has exactly 2^{n-r} lifts, so the original count is precisely 2^{n-r} times the relabelled count, with the factor retained.

For a supplied topologically ordered division-free circuit with fan-in at most two and integer constants that computes $Q_{n,m}$, let its gate count be G and total constant-bit length be C . Evaluate its gates modulo $M_0 = 2^{n+1}$. Induction over $+$, $-$, $\times$ proves the result is the integer value modulo M_0 . Since the value on a formula array lies in $[0, 2^n]$, its least nonnegative residue is its exact value. Binary addition and school multiplication give $O(C + G(n+1)^2)$ bit operations, plus reading the circuit graph and the chosen machine's indexing costs. Integer constants can be reduced by reading their low $n+1$ bits and their sign. This bound does not assume bounds on unreduced intermediate integers. A rational or complex circuit requires its actual constant representation; arbitrary complex constants cannot be interpreted by this binary operation.

The determinant $S_{n,m}$ and its padding have the proved polynomial size, and coefficient extraction from them recovers the actual coupled incidence ideal. To obtain the count, the exact operation proved here forms $\prod_j (1 - H_j)$, maps it to the Boolean corner of the original fibre, and takes the multiplication trace. That full algebra has dimension 3^n and its corner has dimension 2^n . A direct diagonal computation therefore visits exponentially many basis entries; the small determinant for the constraints does not supply a smaller matrix for this trace. The construction proves an explicit bridge from the original retained fibres to the unbounded shared-verifier VNP family and its incidence orbit modules. It proves neither polynomial circuits for Q_t nor a Boolean or algebraic complexity separation. The next size question is the exact complexity of this now specified trace family, not the loss of the joint constraints.

21 A direct permanent projection of the shared-verifier trace

The polynomial $Q_{N,M}(E)$ and its full-fibre trace now have a concrete relation to the permanent, with all witness extensions and sizes specified. For an integer $n \geq 1$ introduce the n^2 independent variables Z_{ij} , $1 \leq i, j \leq n$, and define

$$\text{per}_n(Z) = \sum_{\sigma \in \mathfrak{S}_n} \prod_{i=1}^n Z_{i, \sigma(i)} \quad \text{in } \mathbb{Z}[Z_{ij}].$$

We will use exactly

$$N = 3n^2, \quad M_0 = n^3 + 5n^2, \quad M = n^3 + 6n^2 \quad (92)$$

witness bits and clauses. The original source map in Proposition 20.1 and its target $(-1/4, 0, 0)$ remain unchanged in every one of these N factors.

21.1 The uniquely extended Boolean witness

Order the N bits by the following three groups, each in lexicographic index order:

$$B_{ij} \ (1 \leq i, j \leq n), \quad R_{ik} \ (1 \leq i, k \leq n), \quad C_{jk} \ (1 \leq j, k \leq n).$$

The bit B_{ij} specifies whether the permutation uses entry (i, j) . For each row i , impose the equivalences

$$R_{i1} \leftrightarrow B_{i1}, \quad R_{ik} \leftrightarrow (R_{i,k-1} \vee B_{ik}) \ (2 \leq k \leq n), \quad R_{in} = 1.$$

Also impose $\neg B_{ij} \vee \neg B_{il}$ for every $j < l$. For each column j , impose the equivalences

$$C_{j1} \leftrightarrow B_{1j}, \quad C_{jk} \leftrightarrow (C_{j,k-1} \vee B_{kj}) \ (2 \leq k \leq n), \quad C_{jn} = 1,$$

and $\neg B_{ij} \vee \neg B_{lj}$ for every $i < l$.

We specify every clause in these equivalences. Encode $w \leftrightarrow v$ as $(\neg w \vee v)$ and $(w \vee \neg v)$. Encode $w \leftrightarrow (u \vee v)$ as

$$(\neg u \vee w), \quad (\neg v \vee w), \quad (u \vee v \vee \neg w).$$

The first two clauses say that either true input forces $w = 1$; the third says that $w = 1$ forces at least one true input. These two implications prove the equivalence for every Boolean triple. The two clauses for equality prove both implications in the same way. Encode the final truth requirement by the unary clause w . Pad a binary clause $(a \vee b)$ to $(a \vee b \vee b)$ and a unary clause a to $(a \vee a \vee a)$. Since $0, 1$ satisfy $v^2 = v$, this padding preserves each clause's truth value under its polynomial arithmetization $1 - \prod_{k=1}^3 (1 - L_k)$.

Order clauses first by rows, then by columns, and within each row or column by the equivalences in increasing prefix index, the final truth requirement, and the exclusion pairs in lexicographic order. This fixes an ordered list of three-literal clauses without an unspecified choice. Each row or column contributes

$$2 + 3(n-1) + 1 + \binom{n}{2} = 3n + \binom{n}{2}$$

clauses, including when $n = 1$. There are $2n$ such lists, yielding $M_0 = 2n(3n + \binom{n}{2}) = n^3 + 5n^2$.

For each assignment to the B_{ij} , the equivalences force the unique extension

$$R_{ik} = \bigvee_{j=1}^k B_{ij}, \quad C_{jk} = \bigvee_{i=1}^k B_{ij}.$$

The formulas follow by induction on k starting with the equality clause; they also satisfy every equivalence, proving existence and uniqueness. The final truth requirements and pair exclusions then say that each row and each column has exactly one true entry. Reading the true entry of row i as $\sigma(i)$ gives a map to $\mathfrak{S}_n$: no two rows choose the same column, and every column is chosen. Conversely every permutation yields exactly those entries and exactly the displayed prefix flags. These maps are inverse. In particular auxiliary witnesses do not multiply the number of contributions.

21.2 The exact parameter-ring map

Use the $6MN$ independent parameters $E_{a,k,h,\pm}$ of Section 20, with clause index $1 \leq a \leq M$, slot $1 \leq k \leq 3$, and witness index $1 \leq h \leq N$ in the order above. Define a ring homomorphism

$$\Psi_n : \mathbb{Z}[E_{a,k,h,\pm}] \longrightarrow \mathbb{Z}[Z_{ij}] \quad (93)$$

by giving the image of every parameter. In the first M_0 clauses, put 1 at the unique parameter corresponding to the signed literal in that slot and 0 at all other parameters of that slot. For each (i, j) , append one clause, in lexicographic order. In its first slot, put Z_{ij} at the positive parameter for the witness B_{ij} , put 1 at the negative parameter for that same witness, and put 0 at all its other parameters. Put 0 at every parameter in its second and third slots. These n^2 appended clauses give exactly the value of M in (92). The map uses only variables and the integer constants 0, 1.

This map is surjective: Z_{ij} is the image of its chosen appended positive parameter, denoted E_{ij}^* . Its kernel is explicit. For each of the other parameters whose image is 0 or 1, take the relation $E - \Psi_n(E)$. These relations generate the kernel: the quotient is the polynomial ring on the E_{ij}^* , and its inverse to the induced map sends Z_{ij} to $[E_{ij}^*]$. Thus (93) is the coordinate map of a closed affine embedding, with the indicated linear ideal.

Write $\mathcal{A}(B, R, C)$ for the product of the first M_0 arithmetized clauses. The appended clause has first literal $Z_{ij}B_{ij} + (1 - B_{ij})$ and other two literals zero. Therefore its arithmetized value is exactly

$$1 - (1 - Z_{ij}B_{ij} - (1 - B_{ij}))(1 - 0)(1 - 0) = 1 - B_{ij} + Z_{ij}B_{ij}.$$

This calculation is a polynomial identity before Boolean substitution. It proves

$$\Psi_n(V_{N,M}) = \mathcal{A}(B, R, C) \prod_{i,j} (1 - B_{ij} + Z_{ij}B_{ij}).$$

When evaluated at Boolean witnesses, $\mathcal{A}$ is one precisely at the uniquely extended permutation assignments, and zero at all others. At the permutation σ the remaining product is $\prod_i Z_{i,\sigma(i)}$: the selected entries contribute Z_{ij} and every unselected entry contributes 1. Hence

$$\Psi_n(Q_{N,M}(E)) = \text{per}_n(Z). \quad (94)$$

All these equalities hold over $\mathbb{Z}$. Indeed both sides are the same finite sum of the explicitly computed integer-coefficient polynomials, term by term. Base change gives the identity over any commutative ring; the interpretation of an integer count as a rank uses the characteristic-zero fibre in the earlier trace theorem.

21.3 Transport through the original full fibre and its trace

Work over a characteristic-zero field k and retain $A_N = A_1^{\otimes N}$, its original source coordinates, its product idempotent e , and the Boolean idempotents β_h . Apply Ψ_n to the coefficients of $u(E) = eV_{N,M}(E, \beta)$, and call its image $u_n(Z)$. There is a commutative square of actual linear maps

$$\begin{array}{ccc} A_N \otimes_k k[E] & \xrightarrow{m_{u(E)}} & A_N \otimes_k k[E] \\ \downarrow 1 \otimes \Psi_n & & \downarrow 1 \otimes \Psi_n \\ A_N \otimes_k k[Z] & \xrightarrow{m_{u_n(Z)}} & A_N \otimes_k k[Z]. \end{array}$$

For a pure tensor the equality says $\Psi_n(u(E)f) = \Psi_n(u(E))\Psi_n(f)$, which is the multiplicativity of the coefficient map. Linearity proves it on all inputs. In the retained product-idempotent

basis the multiplication matrices are diagonal, and coefficient substitution acts on each diagonal entry. Thus taking the finite sum of those entries commutes with Ψ_n , giving the fully specified equality

$$\mathrm{Tr}_{A_N \otimes k[Z]/k[Z]}(m_{u_n}(Z)) = \mathrm{per}_n(Z). \quad (95)$$

The full algebra has dimension 3^{3n^2} ; its Boolean corner has dimension 2^{3n^2} . The $x = 0$ factors of the original source fibre are still present and receive weight zero through e . For each nonzero contribution, the B entries encode a permutation and the R, C entries are the forced flags. This describes every contributing original fibre point, not merely the final trace value.

The same formulas give a complete finite-support generating series. In the split idempotent basis, m_{u_n} has one diagonal entry $\prod_i Z_{i,\sigma(i)}$ for each $\sigma \in \mathfrak{S}_n$ and zero at all other basis positions. Consequently

$$\det(I - Tm_{u_n}) = \prod_{\sigma \in \mathfrak{S}_n} \left(1 - T \prod_i Z_{i,\sigma(i)} \right),$$

and, for each integer $d \geq 1$,

$$\mathrm{Tr}(m_{u_n}^d) = \mathrm{per}_n((Z_{ij}^d)_{ij}).$$

These follow by multiplying or summing diagonal entries, retaining their multiplicities even when two monomials acquire the same value after a later specialization of Z . The first coefficient of the displayed characteristic determinant is $-\mathrm{per}_n(Z)$, with that sign fixed by choosing the single $-T$ factor in the product.

21.4 The corresponding class-variety morphism

The affine parameter projection also gives a precise homogeneous orbit-closure map. Retain N, M from (92), let $a = 6MN$, and set $D = 3M$. The total degree of $Q_{N,M}$ in its a parameters is exactly D over a characteristic-zero field. The upper bound follows because each of the M clauses has parameter degree at most three. For the lower bound, take the monomial

$$\mathfrak{m} = \prod_{b=1}^M \prod_{k=1}^3 E_{b,k,1,-}.$$

At a fixed Boolean witness its coefficient in $V_{N,M}$ is $(1 - B_1)^{3M}$: the degree-three part of each clause is $L_{b,1}L_{b,2}L_{b,3}$, with positive sign. Summing gives coefficient 2^{N-1} in $Q_{N,M}$, since exactly that many witnesses have $B_1 = 0$. This coefficient is nonzero in characteristic zero, proving the exact degree without any possible cancellation being suppressed.

Introduce a new coordinate Z_0 and define

$$f(E, Z_0) = Z_0^D Q_{N,M}(E/Z_0).$$

Every term has nonnegative Z_0 exponent because the degree is at most D , so this defines a degree- D polynomial on $V = \mathbb{C}^{a+1}$. Let $W = \mathbb{C}^{n^2+1}$ have coordinates (Z_{ij}, Z_0) . Homogenize every substitution in (93): replace its image 0 by 0, 1 by Z_0 , and Z_{ij} by itself. Together with the identity on Z_0 , these formulas define a linear map $L : W \rightarrow V$. The projection $p : V \rightarrow W$ reads the chosen coordinates (E_{ij}^*, Z_0) , so $pL = 1_W$. In particular L is injective and $A = Lp$ is an explicitly specified idempotent endomorphism of V .

On $Z_0 \neq 0$ the proven permanent projection yields

$$f(L(Z, Z_0)) = Z_0^D \mathrm{per}_n(Z/Z_0) = Z_0^{D-n} \mathrm{per}_n(Z).$$

Since $D \geq n$, both sides are polynomials; their difference vanishes after localization at the nonzero polynomial Z_0 in a polynomial domain, and hence is zero everywhere. Put

$$g = f \circ A = p^*(Z_0^{D-n} \text{per}_n(Z)) \in \text{Sym}^D(V^*).$$

The section L^* gives $L^*p^* = 1$, so no coefficients of the padded permanent are lost when it is written on this larger space.

Let $G = \text{GL}(V)$ act by $(h \cdot u)(v) = u(h^{-1}v)$. Then

$$\overline{G \cdot g} \subseteq \overline{G \cdot f}.$$

To prove the inclusion, write $v = Lp(v) + (v - Lp(v))$. The two summands belong to $\text{im } L$ and $\ker p$; their intersection is zero because $pL = 1$. Thus $V = \text{im } L \oplus \ker p$, and A acts by 1 and 0 on those summands, respectively. In particular

$$\det(A + \epsilon I_V) = (1 + \epsilon)^{n^2+1} \epsilon^{a-n^2}.$$

This proves invertibility for $\epsilon \neq 0, -1$, retaining both exceptional values and their full multiplicities. At every other scalar value, $f \circ (A + \epsilon I_V)$ belongs to $G \cdot f$, by taking $h = (A + \epsilon I_V)^{-1}$. A polynomial equation of that orbit evaluated on this polynomial family vanishes at infinitely many ϵ , so vanishes identically and at $\epsilon = 0$. This proves $g \in \overline{G \cdot f}$. The latter is closed and G -stable: multiplication by a fixed element of G is an invertible linear map of the ambient space and sends the orbit to itself, hence its closure to itself. It therefore contains the orbit closure of g , proving the full inclusion.

Both orbit closures are affine cones. Nonzero scalar multiples of a degree- D form are obtained from scalar coordinate substitutions, because every nonzero complex number has a D th root. The zero form is the limit of the polynomial family $\epsilon^D u$, so also belongs to each closure. Comparing powers of a scalar in an equation on either cone shows that each homogeneous component of that equation vanishes there. Thus restriction gives, in every coordinate-ring degree $j \geq 0$, the exact G -equivariant sequence

$$0 \longrightarrow \frac{I(\overline{G \cdot g})_j}{I(\overline{G \cdot f})_j} \longrightarrow \mathbb{C}[\overline{G \cdot f}]_j \longrightarrow \mathbb{C}[\overline{G \cdot g}]_j \longrightarrow 0.$$

The maps are induced by the identity on the ambient polynomial coordinate ring; inclusion of the two ideals proves their kernels and surjectivity. Equivariance follows because both ideals are G -stable. For every finite-dimensional G -module U , precomposition with the last surjection is an injection

$$\text{Hom}_G(\mathbb{C}[\overline{G \cdot g}]_j, U) \hookrightarrow \text{Hom}_G(\mathbb{C}[\overline{G \cdot f}]_j, U).$$

A map in the kernel vanishes on the entire image of a surjection and hence on its whole domain, proving injectivity directly. This is an exact representation consequence for the stated padded permanent and verifier class varieties. It does not assert an inclusion of either class variety in a determinant class variety.

21.5 Uniform sizes and the exact circuit implication

The one-index parameter for the preceding family is exactly

$$t_n = (N + M - 2)(N + M - 1)/2 + N, \quad N + M = n^3 + 9n^2.$$

In particular $t_n = O(n^6)$. The clause construction writes $M = O(n^3)$ clauses with $O(\log(n + 1))$ -bit witness indices; the full dense parameter map lists $6MN = O(n^5)$ entries, each either 0,

1, or a named Z_{ij} . All lists are generated by the explicit bounded loops above in polynomial time. A sparse list uses only the three selected literal positions per ordinary clause and two nonzero positions per appended clause. This proves uniformity as well as the numerical bounds.

Substitution of (93) in any division-free arithmetic circuit computing $Q_{N,M}$ replaces input leaves by $0, 1, Z_{ij}$ and leaves every gate intact. Induction on the gate order proves the resulting circuit computes the permanent, with no increase in the number of arithmetic-operation gates. Existing constants remain unchanged; at most two shared constant vertices for $0, 1$ add $O(1)$ constant storage. In a description that repeats constants at every input leaf, the additional total constant-bit length is at most a constant times the number of replaced leaves. In either convention the maximum bit length of a constant becomes at most the maximum of the previous bound and 1. If the supplied circuit family is polynomial in its one-index parameter t , its restriction at $t = t_n$ has polynomial size in n because the displayed t_n is polynomial. An algorithm producing those descriptions remains polynomial after this explicit substitution. These are exact maps of supplied circuit descriptions; no small circuit for Q or for the permanent has been assumed or constructed by them.

This permanent projection makes the trace complexity question concrete. For a binary input matrix Z , every surviving diagonal entry is 0 or 1, so the integer permanent lies in $[0, n!]$. Also $n! \leq n^n \leq 2^{n^2} < 2^{N+1}$: the inequality $n \leq 2^n$ follows by induction from $1 \leq 2$, since $n+1 \leq 2n$ for $n \geq 1$. Consequently evaluating a supplied integer-constant circuit for the projected trace modulo 2^{N+1} recovers the exact count as its least nonnegative residue. The preceding gate-substitution map and the earlier school-multiplication bound apply with this unchanged modulus, including its exact $N+1 = 3n^2 + 1$ bits. The polynomial-size incidence determinant from the preceding section still encodes the shared equations. The trace and characteristic determinant above retain their explicitly exponential ambient sizes. Obtaining smaller representations or proving the required lower bounds is additional mathematics, and no Boolean or algebraic complexity equality or separation follows solely from these exact morphisms.

22 Exact obstructions for two specified trace representations

Retain the parameter projection, source fibre, and multiplication operator of the preceding section. Write $\mathcal{A}_N$ for its full split source algebra and u_Z for the projected multiplier, where $N = 3n^2$, $M = n^3 + 6n^2$. In its original point-idempotent basis the diagonal entries of multiplication by u_Z are zero except at the uniquely flagged permutation witnesses. Their entries are $\prod_i Z_{i, \sigma(i)}$. Thus the already constructed polynomial is

$$P_n(Z) = \text{Tr}_{\mathcal{A}_N}(m_{u_Z}) = \sum_{\sigma \in S_n} \prod_{i=1}^n Z_{i, \sigma(i)}.$$

The following calculations concern this same trace. No change of source coordinates or removal of its zero eigenvalues is needed. The original rational-coordinate fibre retains its characteristic-zero base here. The integral split idempotent lattice constructed above carries the same diagonal multiplication matrix with entries in $\mathbb{Z}[Z]$; its trace polynomial therefore has integer coefficients. It is that integral matrix and polynomial, and hence their coefficient matrices, which can be base changed to any field, including characteristic two. This does not reduce the source coordinates containing $1/2$ modulo two.

22.1 Coefficient contraction and its exact rank

Fix a field k , $n \geq 1$, and $0 \leq k_0 \leq n$. To avoid using the field symbol as an integer index, denote the cut by k_0 throughout. Let H_{k_0} be the vector space with basis

$$h_j = \prod_{i=1}^{k_0} Z_{i,j_i}, \quad j \in \{1, \dots, n\}^{k_0},$$

and let T_{k_0} have basis $t_l = \prod_{i=k_0+1}^n Z_{i,l_i}$ indexed by the remaining $n - k_0$ choices. Empty products give the one-dimensional spaces at the two endpoints. Since the variable sets are disjoint, multiplication identifies $H_{k_0} \otimes_k T_{k_0}$ with the space of polynomials homogeneous of degree one in each row of Z . Indeed, multiplication maps the displayed tensor basis bijectively onto its distinct monomial basis. Define the coefficient contraction

$$\mathcal{F}_{k_0} : H_{k_0}^* \longrightarrow T_{k_0}, \quad \mathcal{F}_{k_0}(h_j^*) = \sum_l [h_j t_l] P_n t_l.$$

Here $[m]$ denotes the coefficient of the exact monomial m .

Theorem 22.1. *For every field and every displayed cut, $\text{rank } \mathcal{F}_{k_0} = \binom{n}{k_0}$. There is an integral factorization through the free module on the k_0 -element subsets of $\{1, \dots, n\}$, and an integral identity minor of that size.*

Proof. Let U_{k_0} have basis e_S for those subsets. Define $a : H_{k_0}^* \rightarrow U_{k_0}$ by $a(h_j^*) = e_{\{j_1, \dots, j_{k_0}\}}$ if the indices j_i are distinct, and by zero otherwise. Define

$$b(e_S) = \sum_{\substack{\{l_{k_0+1}, \dots, l_n\} = S^c \\ \text{all indices distinct}}} t_l.$$

A monomial in P_n uses each column exactly once. Conversely every choice using each row and column exactly once is the monomial of a unique permutation and has coefficient one. These two facts give $\mathcal{F}_{k_0} = ba$ on every displayed basis vector, including those with repeated columns. The factorization proves the upper bound.

For each S , take $j(S)$ to be the elements of S in increasing order assigned to the first k_0 rows, and take $l(S)$ to be the elements of S^c in increasing order assigned to the remaining rows. The coefficient of $h_{j(S)} t_{l(S')}$ is one exactly when $S = S'$: disjointness and cardinalities force $S = (S'^c)^c$. It is zero otherwise. Thus these rows and columns give the identity matrix over $\mathbb{Z}$, proving the lower bound over every field. No division by a factorial or by a characteristic-dependent integer has occurred. $\square$

This is also an exact map from the original trace construction. For a matrix $B(Z)$ over a polynomial ring, finite coefficient extraction commutes with its trace: $[m] \text{Tr } B = \sum_v [m] B_{vv} = \text{Tr}([m]B)$. This follows directly from the definition of the diagonal sum, for every coefficient and over every base ring. Apply it to the full $3^N \times 3^N$ matrix m_{u_Z} . The projection to its uniquely flagged permutation diagonal entries therefore gives precisely the maps a, b above, while every retained zero diagonal entry contributes zero. This proves the coefficient contraction correspondence; the matrix size 3^N is unchanged.

For every integer $d \geq 1$ the same argument applied to the previously proved power trace $\text{Tr}(m_{u_Z}^d) = \text{per}_n((Z_{ij}^d))$ gives the same rank. Use the head and tail bases with each exponent 1 replaced by d ; these are still distinct monomials in every characteristic. Their coefficient matrix is the same integral zero-one matrix.

22.2 Optimal widths of row-ordered branching programs

A row-ordered algebraic branching program here means a finite directed layered graph with layers $0, 1, \dots, n$, one source at layer zero, one sink at layer n , and edges only from layer $i - 1$ to layer i . Every such edge is labelled by a homogeneous linear form in $Z_{i1}, \dots, Z_{in}$ over k . The output is the sum over all source-to-sink paths of the product of their labels. The width w_i is the number of vertices at layer i ; distinct parallel edges may equivalently be combined by adding their labels.

Theorem 22.2. *Every such program for P_n has $w_i \geq \binom{n}{i}$ at every layer. These bounds hold simultaneously with equality. The minimum total number of vertices, including both endpoints, is 2^n . An attaining program has exactly $n2^{n-1}$ edges.*

Proof. For a vertex v in layer i , let p_v be the sum of path products from the source to v , and q_v the corresponding sum from v to the sink. Each full path has exactly one such v and decomposes uniquely into its prefix and suffix. Distributivity therefore gives $P_n = \sum_{v \text{ in layer } i} p_v q_v$, with $p_v \in H_i$ and $q_v \in T_i$. Its coefficient contraction factors as

$$H_i^* \longrightarrow k^{w_i} \longrightarrow T_i, \quad \lambda \longmapsto (\lambda(p_v))_v, \quad (c_v)_v \longmapsto \sum_v c_v q_v.$$

Its rank is at most w_i . The preceding theorem proves the lower bound.

For attainment, put at layer i all subsets $S \subseteq \{1, \dots, n\}$ of size i . For every $j \notin S$ put an edge from S to $S \cup \{j\}$ labelled $Z_{i+1,j}$. A full path chooses the columns $j_1, \dots, j_n$ without repetition, so determines one permutation. Every permutation determines exactly this path. Multiplying its labels and then summing gives P_n . The number of vertices is $\sum_i \binom{n}{i} = 2^n$, because each subset has one cardinality and there are two membership choices for each of the n elements. The number of edges is $\sum_{S \subseteq [n]} (n - |S|) = n2^{n-1}$: count pairs (S, j) with $j \notin S$, choosing j in n ways and an arbitrary subset of its $n - 1$ other elements. Summing the width lower bounds proves minimality of the stated total vertex number. $\square$

The row condition is part of this precise representation class. The factorization just proved does not apply to a branching program that may reuse a row at several layers or intermix row variables inside a layer. Thus the result gives no lower bound for that larger class from an unproved identification of its states with H_i . The proved exact map is the prefix–suffix contraction above.

22.3 An exponential obstruction for nonnegative arithmetic

Now work over $\mathbb{R}$. A monotone arithmetic circuit means a finite acyclic circuit with leaves that are variables Z_{ij} or nonnegative real constants, and binary addition and multiplication gates. Its size s is the number of arithmetic-operation gates. Fan-out is allowed. No subtraction or division is allowed. A parse tree at a gate is defined recursively: at an addition choose one of its two children; at a multiplication take both; copy occurrences when a gate is used more than once. A nonzero parse tree is one whose constant-leaf product is positive. It contributes that positive coefficient times the product of its variable leaves. Induction in a topological ordering proves that a gate polynomial is the sum of its parse-tree contributions. There are only finitely many parse trees, since the circuit is finite and acyclic.

Lemma 22.3. *Suppose such a circuit computes P_n . A gate occurrence is called live if it occurs in a nonzero output parse tree, and a gate is live if it has such an occurrence. For every live gate g there are uniquely determined subsets $I_g, J_g \subseteq [n]$ of the same size such that every monomial at g with positive coefficient is a matching from I_g onto J_g . In particular its degree is the fixed integer $d_g = |I_g| = |J_g|$.*

Proof. Fix one nonzero output parse tree and one occurrence of g in it. Remove the subtree rooted at that occurrence and retain the rest, including every other occurrence of g if there is one. Let cm be the product of the retained constant and variable leaves, so $c > 0$. Every monomial t with positive coefficient at g has a nonzero parse tree there, by the parse expansion. Substituting that tree in the retained occurrence produces a nonzero output parse tree contributing a positive multiple of mt . Nonnegative coefficients cannot cancel, so mt occurs in P_n .

Every monomial of P_n has total exponent one in each row and column, and exponent at most one on each variable. Consequently m is a partial matching; all such t use exactly the rows and columns missing from m , once each. These two complementary sets are I_g and J_g . They have the same size because any monomial t uses one variable per used row and column. At least one t exists, since the fixed occurrence was nonzero. That monomial already determines its sets of rows and columns, showing uniqueness and independence of the chosen live occurrence. $\square$

At a live addition, every child chosen in a nonzero parse tree has the same profile and degree as the parent: each of its monomials also occurs at the parent. At a live multiplication with nonzero children, both children are live by extending the fixed context, their row sets and their column sets are disjoint, and their unions are those of the parent. This follows by multiplying any positive monomial of the first child by one of the second and using the lemma. In particular their degrees add. These conclusions establish the needed homogeneity and disjointness from positivity; they have not been added as circuit hypotheses.

Theorem 22.4. *For every $n \geq 4$, every monotone arithmetic circuit computing the original projected trace P_n satisfies*

$$s \geq \binom{n}{\lceil n/3 \rceil} \geq 2^{\lceil n/3 \rceil}.$$

Proof. Each of the $n!$ permutation monomials has positive coefficient, so choose one of its nonzero output parse trees, using any fixed ordering of the finite tree set to make the choice definite. Start at its output occurrence, of degree n . At an addition follow its chosen child. At a multiplication follow a child of larger degree, with a fixed rule for ties. Continue until reaching the first occurrence of degree at most $2n/3$.

This procedure terminates: it follows a descending path in a finite tree, and leaf degrees are zero or one. An addition cannot be the first degree decrease across the threshold, because its chosen child has the same degree. Hence that first crossing follows a multiplication of degree greater than $2n/3$. Its selected child has at least half the parent degree. The selected gate g thus has

$$n/3 < d_g \leq 2n/3, \quad \text{and in particular} \quad \lceil n/3 \rceil \leq d_g \leq \lfloor 2n/3 \rfloor.$$

For $n \geq 4$ this degree is at least two, so g is an arithmetic-operation gate, not an input leaf.

Assign the original permutation to this gate. Its chosen parse tree contains a monomial from g , which matches precisely I_g to J_g . Therefore every permutation assigned to g sends I_g onto J_g . For $d_g = d$ the number of all permutations with this property is exactly $d!(n-d)!$: choose independently a bijection $I_g \rightarrow J_g$ and one $I_g^c \rightarrow J_g^c$. Each has $d!$ and $(n-d)!$ choices, respectively, by successive choices of images. The assignments partition the $n!$ permutations among at most s gates. Thus

$$n! \leq s \max_{\lceil n/3 \rceil \leq d \leq \lfloor 2n/3 \rfloor} d!(n-d)!.$$

The ratio $\binom{n}{d+1}/\binom{n}{d} = (n-d)/(d+1)$ proves that the binomial coefficients increase up to their middle and then decrease symmetrically, since $\binom{n}{d} = \binom{n}{n-d}$. The two endpoints of the

displayed range are complementary integers, so its minimum binomial coefficient is $\binom{n}{\lceil n/3 \rceil}$. This proves the first bound. Finally set $r_0 = \lceil n/3 \rceil$. For $n \geq 4$ we have $r_0 \leq n/2$: write $n = 3a + b$ with $b = 0, 1, 2$ and check the three inequalities directly, with $a \geq 1$ and the case $b = 0$ having $a \geq 2$ if necessary. For $0 \leq i < r_0$, $(n - i)/(r_0 - i) \geq n/r_0 \geq 2$, since $r_0(n - i) - n(r_0 - i) = i(n - r_0) \geq 0$. Multiplication gives

$$\binom{n}{r_0} = \prod_{i=0}^{r_0-1} \frac{n-i}{r_0-i} \geq 2^{r_0},$$

as claimed. $\square$

This theorem constrains positive arithmetic representations of the specific original-fibre trace after the proved permanent projection. The universal verifier polynomial before that projection contains signed coefficients, and the support lemma was proved only for the non-negative circuit just defined. The projection is nevertheless the exact gate-preserving input-substitution map of the preceding section. If a nonnegative circuit description survives that map as a circuit of this class, the displayed lower bound applies to its arithmetic gate count. For signed circuits the cancellation step in the proof is unavailable; the present theorem makes no assertion that it holds. The row-ordered and nonnegative bounds provide two explicit asymptotic obstructions with their maps and domains. The broader unrestricted obstruction in the research programme remains to be constructed.

23 A smaller sign-algebra trace and its exact source correspondence

The permanent trace of Section 21 admits a second, smaller finite-algebra representation. We retain the same n^2 coordinates Z_{ij} and the same permanent polynomial; we construct the correspondence between the two representations explicitly. The sign sum used below is the Ryser–Glynn expression recorded in [8, Equation (8)]; the original Glynn article is [6]. All identities and maps needed here are proved below, independently of those references.

23.1 The sign algebra, its integral basis, and its regular trace

Fix an integer $n \geq 1$. Put $O = \mathbb{Z}[1/2]$, $R = O[Z_{ij}]$, $r_n = 2^{n-1}$, and

$$B_n = O[d_2, \dots, d_n] / (d_j^2 - 1 : 2 \leq j \leq n), \quad d_1 = 1.$$

For $n = 1$ this means $B_1 = O$. For each subset $S \subseteq \{2, \dots, n\}$ write $d_S = \prod_{j \in S} d_j$, including $d_\emptyset = 1$. The elements d_S form an O -basis. Indeed, successive division by the monic polynomials $d_j^2 - 1$ gives a unique remainder with exponents zero or one: viewing the ring as a polynomial ring in the last variable over the preceding quotient, monic division gives a free module with basis $1, d_j$. Induction proves the assertion over O and equally over $\mathbb{Z}$. Multiplication is

$$d_S d_T = d_{S \triangle T},$$

where $S \triangle T = (S \setminus T) \cup (T \setminus S)$. There are exactly r_n basis elements, one for each independent membership choice in a set of size $n - 1$.

Write $\tau_B(b) = \text{Tr}_{B_n/O}(m_b)$ for the regular trace, where $m_b(x) = bx$. Multiplication by d_S permutes the basis by $T \mapsto S \triangle T$. A basis vector is fixed if and only if $S = \emptyset$, since cancelling T under symmetric difference gives that equivalence. Thus

$$\tau_B\left(\sum_S b_S d_S\right) = r_n b_\emptyset. \tag{96}$$

This identity holds first over $\mathbb{Z}$ and then over every coefficient ring by taking the same finite diagonal sum.

There is also an explicit split basis. Let $\Delta_n = \{\delta \in \{-1, 1\}^n : \delta_1 = 1\}$ and set

$$\epsilon_\delta = \prod_{j=2}^n \frac{1 + \delta_j d_j}{2}.$$

Each factor is idempotent, since $(1 + \delta_j d_j)^2/4 = (1 + \delta_j d_j)/2$. Opposite factors multiply to $(1 - d_j^2)/4 = 0$, and summing the two factors for one coordinate gives one. Multiplying these identities shows that the ϵ_δ are orthogonal idempotents summing to one. Evaluation at $d_j = \eta_j$ gives $\epsilon_\delta(\eta) = 1$ when $\eta = \delta$ and zero otherwise. The identity $d_S = \sum_\delta (\prod_{j \in S} \delta_j) \epsilon_\delta$ follows by multiplying $d_j \epsilon_\delta = \delta_j \epsilon_\delta$ and summing the idempotents. It proves the mutually inverse algebra maps

$$B_n \longrightarrow O^{\Delta_n}, \quad b \longmapsto (b(\delta))_\delta, \quad (c_\delta)_\delta \longmapsto \sum_\delta c_\delta \epsilon_\delta.$$

In particular $\tau_B(b) = \sum_\delta b(\delta)$.

For completeness the factor inverted here has exact integral content. Before localizing, the evaluation matrix from the d_S basis to the integral point basis is $H_{\delta, S} = \prod_{j \in S} \delta_j$. For two subsets S, T , the sum $\sum_\delta H_{\delta, S} H_{\delta, T}$ is r_n if $S = T$ and zero otherwise: in the latter case choose a coordinate in $S \triangle T$ and pair each δ with the sign vector obtained by flipping that coordinate. Therefore

$$H^\top H = r_n I, \quad H^{-1} = r_n^{-1} H^\top \quad \text{over } \mathbb{Q}.$$

The image of $\mathbb{Z}[d_2, \dots, d_n]/(d_j^2 - 1)$ in $\mathbb{Z}^{\Delta_n}$ consists exactly of integer vectors v such that every component of $H^\top v$ is divisible by r_n : necessity follows from the first identity, and sufficiency follows by applying the displayed inverse. Thus the integral evaluation map is injective and becomes an isomorphism after inverting two. For $n \geq 2$ it is not surjective: the coordinate vector supported at a single δ has $H^\top v$ with every component 1 or -1 , which is not divisible by $r_n \geq 2$. No integral identification of these two lattices has been made.

23.2 The exact parity computation, with both orientations

Put $\chi_n = \prod_{j=1}^n d_j = d_{\{2, \dots, n\}}$ and define row-linear elements and their multiplier by

$$L_i(Z) = \sum_{j=1}^n d_j Z_{ij}, \quad v_n(Z) = r_n^{-1} \chi_n \prod_{i=1}^n L_i(Z) \quad \text{in } B_n \otimes_O R. \quad (97)$$

Theorem 23.1. *The regular multiplication trace on the rank- r_n algebra satisfies*

$$\text{Tr}_{B_n \otimes R/R}(m_{v_n(Z)}) = \sum_{\sigma \in \mathfrak{S}_n} \prod_{i=1}^n Z_{i, \sigma(i)} = \text{per}_n(Z).$$

The identity holds over O and after any homomorphism from O to a commutative ring, in particular over $\mathbb{Q}$ and every field of odd characteristic.

Proof. Expand the product in (97) without changing its row order:

$$v_n(Z) = r_n^{-1} \sum_{(j_1, \dots, j_n) \in [n]^n} \chi_n d_{j_1} \cdots d_{j_n} \prod_{i=1}^n Z_{ij_i}.$$

For a tuple let m_j be the number of its entries equal to j . The group-basis monomial $\chi_n \prod_i d_{j_i}$ is one exactly when $m_j + 1$ is even for each $j \geq 2$, or equivalently when each of $m_2, \dots, m_n$ is odd. In that case write $m_j = 1 + 2a_j$ for $j \geq 2$, with $a_j \geq 0$. The equality $\sum_j m_j = n$ becomes

$$m_1 + 2 \sum_{j=2}^n a_j = 1.$$

It forces $m_1 = 1$ and every $a_j = 0$. Conversely, if every column is chosen once, all the exponents are even as required. Thus exactly the permutation tuples contribute to the coefficient of $d_\emptyset$, each with coefficient r_n^{-1} times its original row monomial. Equation (96) multiplies that coefficient by the exact factor r_n , proving the formula. For $n = 1$ the same equation says $m_1 = 1$; directly $r_1 = \chi_1 = d_1 = 1$ and $v_1(Z) = Z_{11}$ in the rank-one algebra. Finally all the calculations are identities in the displayed free algebra over O , so applying a coefficient homomorphism preserves them, including their traces. $\square$

The split-basis form of this equality is the fully signed expression

$$\text{per}_n(Z) = 2^{1-n} \sum_{\delta \in \Delta_n} \left(\prod_{j=1}^n \delta_j \right) \prod_{i=1}^n \left(\sum_{j=1}^n \delta_j Z_{ij} \right).$$

The prefactor and product of signs are retained. To give the other orientation explicitly, use a second copy of the sign algebra with generators $e_2, \dots, e_n$, $e_1 = 1$, and put

$$\hat{v}_n(Z) = 2^{1-n} \left(\prod_{i=1}^n e_i \right) \prod_{j=1}^n \left(\sum_{i=1}^n e_i Z_{ij} \right).$$

Let the coefficient-ring involution send Z_{ij} to Z_{ji} , and send d_j in the first sign algebra to e_j in the second. This is an isomorphism of the underlying O -algebras with polynomial coefficients; its square, with the reverse generator renaming, is the identity. It sends $v_n(Z)$ to $\hat{v}_n(Z)$ by direct substitution. The trace of the latter is $\text{per}_n(Z^\top) = \text{per}_n(Z)$: the term $\prod_i Z_{\sigma(i),i}$ becomes $\prod_j Z_{j,\sigma^{-1}(j)}$, and inversion permutes $\mathfrak{S}_n$ bijectively. Thus transposition is a specified coordinate map, not an implicit interchange of the original rows and columns in (97).

Over $\mathbb{Z}$ the unscaled numerator $w_n(Z) = \chi_n \prod_i L_i(Z)$ is an element of the integral group algebra and satisfies $\text{Tr}(m_{w_n}) = r_n \text{per}_n$. For $n \geq 2$ the scaled element v_n is not in that integral group algebra with polynomial coefficients. The monomial $Z_{11} \cdots Z_{n1}$ occurs exactly once in the expansion and has coefficient χ_n/r_n , which is not integral in the d_S basis. The resulting trace polynomial nevertheless belongs to $\mathbb{Z}[Z]$ by the proved identity. It can be reduced to characteristic two through that integer polynomial or the original split integral trace matrix. There is no homomorphism $O \rightarrow k$ for a field k of characteristic two, because the unit two would map to zero. The scaled sign-algebra operator is therefore not specialized by such a nonexistent coefficient map. Its integer numerator has trace zero there for $n \geq 2$, which cannot by itself recover the permanent; at $Z = I_n$ the permanent is one.

23.3 The original source fibre over the same coefficient ring

Retain the original map with its source orientation (x, y, w) and target orientation (a, b, c) :

$$\begin{aligned} F_1 &= (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 w. \end{aligned}$$

It still has $\det DF = -2$, and its retained polynomial and derivative are $P = cr^3 - 2r^2 + br - 2a$ and $P' = 3cr^2 - 4r + b$. Neither expression is altered by the sign algebra. The original fibre at $(-1/4, 0, 0)$ is defined over O , and the isomorphism

$$O[x, y, w]/(F_1 + 1/4, F_2, F_3) \longrightarrow O[X]/(X^3 - X), \quad (x, y, w) \longmapsto (X, -3X/2, (27X^2 - 1)/4)$$

continues to hold there. To check the base ring explicitly, put $h = 2 - 3xy - x^2w$. The identity $1 = h/2 + x(3y + xw)/2$ makes (x) and (h) comaximal after imposing $F_1 + 1/4, F_2$. As $F_3 = xh$, the quotient splits into its $x = 0$ and $h = 0$ factors by the same two-ideal intersection argument. The first is O , with $(x, y, w) = (0, 0, -1/4)$. In the second, $x(3y + xw) = 2$ makes x a unit; putting $r = y + 1/x$, $\alpha = 1/x$ gives $F_1 = r^2 + r\alpha$, $F_2 = 4r + 2\alpha$, hence $\alpha = -2r$, $r^2 = 1/4$. Conversely these equations make r a unit and recover the original source coordinates and target equations. This factor is $O[r]/(r^2 - 1/4)$, with $x = -2r$, $y = -3x/2$, $w = 13/2$. All inverses used are powers of two or the stated source unit. The two factors give exactly $O[X]/(X^3 - X)$, with its original $x = 0$ point retained.

The idempotents $e_- = (X^2 - X)/2$, $e_0 = 1 - X^2$, $e_+ = (X^2 + X)/2$ give evaluation at $X = -1, 0, 1$ over O as in Proposition 20.1. Take $N = 3n^2$ tensor factors and write $\mathcal{A}_N$ for the resulting full source algebra over O . Its point set is $\Omega_N = \{-1, 0, 1\}^N$; write E_ω for its product idempotents. It has rank 3^N , and $\Lambda_N = \bigoplus_{\omega \in \Omega_N} \mathbb{Z}E_\omega$ is the original split integral lattice, with its coordinatewise multiplication. In particular $\mathcal{A}_N = \Lambda_N \otimes O$. This lattice differs from the unsplit integral presentation with source-coordinate denominators, as already proved for each factor.

For a permutation σ , let $\omega(\sigma)$ be the original source point with $B_{ij} = 1$ exactly when $j = \sigma(i)$ and with the unique row and column prefix flags of the preceding construction. In every witness coordinate h this point has

$$X_h = 2B_h - 1, \quad y_h = -3X_h/2, \quad w_h = (27X_h^2 - 1)/4.$$

Here B_h denotes the appropriate original witness bit, including the two groups of flags. Distinct permutations have distinct B -arrays and therefore distinct points. Put

$$f_n = \sum_{\sigma \in \mathfrak{S}_n} E_{\omega(\sigma)}, \quad A_{\text{perm}} = f_n \mathcal{A}_N.$$

Orthogonality proves $f_n^2 = f_n$ and gives the algebra decomposition $\mathcal{A}_N = A_{\text{perm}} \oplus (1 - f_n)\mathcal{A}_N$. The first ideal has identity f_n and rank $n!$. The map $a \mapsto f_n a$ is a surjective algebra map to this ideal with kernel $(1 - f_n)\mathcal{A}_N$. It retains the complementary ideal as its explicitly specified kernel, including every boundary point with some $X_h = 0$.

The projected original multiplier is exactly

$$u_n(Z) = \sum_{\sigma \in \mathfrak{S}_n} E_{\omega(\sigma)} \prod_{i=1}^n Z_{i, \sigma(i)}. \quad (98)$$

Indeed the unweighted constraint-clause product is one precisely on those uniquely flagged assignments, each appended weighted clause supplies the specified entry, and the original product idempotent kills every boundary factor. These are precisely the evaluations of the projected multiplier established above; the split evaluation isomorphism proves equality as an element, not only as a trace. Since $u_n = f_n u_n$, its multiplication is zero on the full complementary ideal. Consequently

$$\text{Tr}_{\mathcal{A}_N}(m_{u_n}) = \text{Tr}_{A_{\text{perm}}}(m_{u_n}) = \text{per}_n(Z).$$

The full original rank 3^{3n^2} and all its zero eigenvalues remain present in the first trace.

23.4 A common tensor module and an exact sequence of traces

Let $T_n = (O^n)^{\otimes n}$, with factor i assigned to row i . For $\mathbf{j} = (j_1, \dots, j_n) \in [n]^n$ denote its tensor basis vector by $t_{\mathbf{j}}$. Define three O -linear maps on every basis vector by

$$\begin{aligned} S_n : T_n &\longrightarrow A_{\text{perm}}, & S_n(t_{\mathbf{j}}) &= \begin{cases} E_{\omega(\sigma)}, & j_i = \sigma(i) \text{ for all } i, \\ 0, & \text{otherwise,} \end{cases} \\ K_n : T_n &\longrightarrow B_n, & K_n(t_{\mathbf{j}}) &= r_n^{-1} \chi_n \prod_i d_{j_i}, \\ \varepsilon_n : T_n &\longrightarrow O, & \varepsilon_n(t_{\mathbf{j}}) &= \begin{cases} 1, & \mathbf{j} \text{ is a permutation tuple,} \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Writing τ_A for regular trace on A_{perm} , we have the equalities of actual linear maps

$$\tau_A S_n = \varepsilon_n = \tau_B K_n. \quad (99)$$

The first follows because a coordinate idempotent has trace one; the second is exactly the parity computation for one tuple. The universal row-choice tensor

$$t(Z) = \bigotimes_{i=1}^n \left(\sum_{j=1}^n Z_{ij} t_j \right) = \sum_{\mathbf{j}} \left(\prod_i Z_{ij_i} \right) t_{\mathbf{j}} \quad \text{in } T_n \otimes_O R$$

is sent by S_n to (98), by K_n to (97), and by ε_n to the permanent. Thus (99) relates every coefficient before taking the trace of the polynomial.

The kernels and surjections in this correspondence are explicit. The map S_n is surjective, and its kernel has as basis the nonpermutation tuples, because the remaining images are the distinct basis idempotents. For a tuple put

$$p(\mathbf{j}) = \{j \in \{2, \dots, n\} : j \text{ occurs an odd number of times in } \mathbf{j}\}.$$

For each subset $U \subseteq \{2, \dots, n\}$ choose $\mathbf{a}(U)$ to consist first of the elements of U in increasing order and then of $n - |U|$ entries equal to one. Its parity set is U . Thus $K_n(t_{\mathbf{a}(U)}) = r_n^{-1} \chi_n d_U$, and these elements form a basis of B_n because multiplication by the unit χ_n/r_n permutes and rescales its basis. This proves surjectivity with a specified section. The differences

$$t_{\mathbf{j}} - t_{\mathbf{a}(p(\mathbf{j}))} \quad (\mathbf{j} \neq \mathbf{a}(p(\mathbf{j})))$$

form a basis of $\ker K_n$: subtracting these differences reduces an arbitrary vector to the selected representatives, whose images are independent; each difference has coefficient one on its own unselected basis vector and zero on the other unselected vectors, which proves independence.

The parity proof also shows that $p(\mathbf{j}) = \{2, \dots, n\}$ holds exactly for permutation tuples. Every other selected representative $\mathbf{a}(U)$ is therefore nonpermutational. The regular trace kernel in B_n is the free span of d_V with $V \neq \emptyset$, by (96) and the invertibility of r_n . The preceding representatives consequently prove

$$K_n(\ker S_n) = \ker \tau_B. \quad (100)$$

Proposition 23.2. *There is an exact sequence of O -modules*

$$T_n \xrightarrow{(S_n, -K_n)} A_{\text{perm}} \oplus B_n \xrightarrow{(a, b) \mapsto \tau_A(a) + \tau_B(b)} O \longrightarrow 0.$$

The first image has rank $n! + r_n - 1$, and its kernel $\ker S_n \cap \ker K_n$ is free of rank $n^n - n! - r_n + 1$.

Proof. Equation (99) says the composite is zero. The second map is surjective because it sends $(E_{\omega(\text{id})}, 0)$ to one. Suppose $\tau_A(a) + \tau_B(b) = 0$. Choose $t_0 \in T_n$ with $S_n(t_0) = a$. Then $b + K_n(t_0)$ has trace $\tau_B(b) + \varepsilon_n(t_0) = \tau_B(b) + \tau_A(a) = 0$. Equation (100) supplies $t_1 \in \ker S_n$ such that $K_n(t_1) = -b - K_n(t_0)$. It follows that $(S_n, -K_n)(t_0 + t_1) = (a, b)$, proving exactness.

There is an explicit basis for the kernel of the second arrow: take $(E_{\omega(\sigma)} - E_{\omega(\text{id})}, 0)$ for $\sigma \neq \text{id}$, together with $(0, d_V)$ for $V \neq \emptyset$ and the single vector $(-r_n E_{\omega(\text{id})}, 1)$. Their traces vanish. Subtracting the nonidentity A_{perm} coefficients and the nonempty B_n coefficients from any trace-zero pair reduces it to $(a_0 E_{\omega(\text{id})}, b_0)$ with $a_0 + r_n b_0 = 0$, which is the last vector multiplied by b_0 . The same coordinates prove independence. There are $n! + r_n - 1$ such vectors. Choose lifts of these basis vectors under the first surjection just proved; their linear extension gives a section. Then T_n is the direct sum of their free span and the first kernel. For freeness of that kernel without an unstated theorem, decompose T_n into permutation and nonpermutation basis vectors. An element of $\ker S_n$ uses only the latter. Group those vectors by their parity sets $U \neq \{2, \dots, n\}$. On each group K_n sends every vector to the same nonzero distinct basis multiple. Its kernel therefore has basis the differences against the selected representative in that group, by the argument above. The total number is $(n^n - n!) - (r_n - 1)$, as asserted. $\square$

For $n \geq 2$ this common-tensor construction cannot be replaced by a linear map making either of its two surjections factor through the other. The tuple $(1, \dots, 1)$ is nonpermutational, so its S_n image is zero whereas its K_n image is the nonzero unit χ_n/r_n . This disproves a factorization $K_n = LS_n$. For two distinct permutations, the difference of their tensor vectors has K_n image zero, since each image is $1/r_n$, whereas its S_n image is a nonzero difference of basis idempotents. This disproves a factorization $S_n = MK_n$. The exact sequence and coefficient trace maps above supply the proved relation despite these specific obstructions. For $n = 1$ both coefficient maps are isomorphisms between rank-one modules, and the original full source still has rank $3^3 = 27$ with its stated complement. More explicitly, at $n = 1$ the unique flagged point is $(X_{B_{11}}, X_{R_{11}}, X_{C_{11}}) = (1, 1, 1)$. Evaluation there is a surjective O -algebra map $\mathcal{A}_3 \rightarrow O = B_1$; its kernel is the span of the other 26 point idempotents, and it sends $u_1(Z) = Z_{11}E_{(1,1,1)}$ to $v_1(Z) = Z_{11}$.

23.5 A multiplication obstruction and the unequal power traces

Equality of the first traces does not give equality of their spectra. Specialize the original coordinates to the identity matrix I_n . In (98) only $\sigma = \text{id}$ survives, so $u_n(I_n) = E_{\omega(\text{id})}$ is an idempotent in the full original algebra and its permutation ideal. On the sign side $L_i(I_n) = d_i$, and hence

$$v_n(I_n) = r_n^{-1} \chi_n \prod_i d_i = r_n^{-1} \chi_n^2 = r_n^{-1} 1.$$

For $n \geq 2$ over $\mathbb{Q}$ this element is not idempotent, since $r_n^{-2} - r_n^{-1} = (1 - r_n)/r_n^2 \neq 0$. Every multiplicative map, whether unital or not, sends an idempotent to an idempotent. Thus no $\mathbb{Q}[Z]$ -linear multiplicative map from the original full source algebra or its permutation ideal to $B_n \otimes \mathbb{Q}[Z]$ sends $u_n(Z)$ to $v_n(Z)$: such a map would specialize at $Z = I_n$ to the impossible map just described. This is an obstruction for the stated compatible base map. No assertion about arbitrary changes of the Z coordinates is needed. In odd characteristic the specific test may disappear when $r_n = 1$ in the field; the rational assertion retains its domain.

There is a stronger generic obstruction that covers every field k of characteristic different from two, even after passing to the rational function field $K = k(Z_{ij})$. Both source algebras and the sign algebra split as products of copies of K , by their stated point idempotents. Compose a K -linear multiplicative map from either source algebra to the sign algebra with any sign-coordinate evaluation. Each primitive source idempotent must map to an idempotent of the field K , hence to zero or one: the equation $x(x - 1) = 0$ in a field forces these two

possibilities. Orthogonality allows at most one of them to map to one. Therefore the image of u_n in this coordinate is either zero or one of its permutation monomials. At a fixed sign vector δ , however, the coefficient of the nonpermutation monomial $\prod_i Z_{i1}$ in $v_n(\delta)$ is $r_n^{-1} \prod_j \delta_j \neq 0$, and the coefficient of every permutation monomial is $r_n^{-1} \neq 0$. These statements follow by selecting the indicated unique term in each row sum; all row monomials are distinct. For $n \geq 2$ this polynomial is consequently neither zero nor a single permutation monomial. The injection $k[Z] \hookrightarrow K$ preserves these inequalities. This proves that no such K -linear multiplicative map sends u_n to v_n , including nonunital maps. It also proves the compatible polynomial base-map obstruction over every such field, since a map over $k[Z]$ would extend to K by scalar extension.

For every integer $d \geq 1$ the original power trace and the new one are, respectively,

$$\mathrm{Tr}(m_{u_n}^d) = \sum_{\sigma} \prod_i Z_{i,\sigma(i)}^d, \quad \mathrm{Tr}(m_{v_n}^d) = r_n^{-d} \sum_{\delta \in \Delta_n} \left(\prod_j \delta_j \right)^d \prod_i \left(\sum_j \delta_j Z_{ij} \right)^d.$$

These follow by taking powers of the diagonal entries in their specified point bases. At $Z = I_n$ they equal 1 and r_n^{1-d} . In particular $d = 2$ distinguishes them over $\mathbb{Q}$ for $n \geq 2$. Their characteristic determinants at this same specialization are $1 - T$ and $(1 - T/r_n)^{r_n}$, respectively: the full original operator has one eigenvalue one and all other eigenvalues zero, whereas the sign operator has r_n copies of $1/r_n$. The common first trace, the exact coefficient maps, and these different higher traces are all retained simultaneously.

23.6 The exact dimension obstruction for a fixed algebra

Let k now be any field and D a finite-dimensional associative unital k -algebra of dimension q_D . Fix $c \in D$ and elements $a_{ij} \in D$, all independent of the polynomial coordinates Z . No commutativity of D is assumed. Define

$$A_i(Z) = \sum_{j=1}^n a_{ij} Z_{ij}, \quad Q_D(Z) = \mathrm{Tr}_{D \otimes k[Z]/k[Z]}(m_{cA_1(Z) \cdots A_n(Z)}),$$

where the product has the displayed order, the Z coordinates are central, and $m_b(x) = bx$ is left multiplication. Associativity implies $m_b m_{b'} = m_{bb'}$ directly on every $x \in D$.

Theorem 23.3. *For every cut $0 \leq k_0 \leq n$, the row coefficient contraction of Q_D has rank at most q_D . Consequently, if $Q_D(Z) = \mathrm{per}_n(Z)$ as polynomials, then*

$$q_D \geq \binom{n}{k_0} \quad (0 \leq k_0 \leq n), \quad q_D \geq \binom{n}{\lfloor n/2 \rfloor}.$$

Proof. Use the head and tail monomial spaces H_{k_0}, T_{k_0} of Theorem 22.1. Define a linear map on the dual head basis by

$$a : H_{k_0}^* \longrightarrow D, \quad h_j^* \longmapsto a_{1j_1} \cdots a_{k_0 j_{k_0}},$$

with the empty product equal to 1_D . Define

$$b : D \longrightarrow T_{k_0}, \quad z \longmapsto \sum_l \mathrm{Tr}_D(m_{c z a_{k_0+1, l_{k_0+1}} \cdots a_{n, l_n}}) t_l.$$

Both are k -linear, since multiplication by a fixed factor is linear and the matrix trace is linear. Expanding the ordered product in Q_D shows that its coefficient at $h_j t_l$ is exactly the displayed trace with $z = a_{1j_1} \cdots a_{k_0 j_{k_0}}$. Its coefficient contraction is therefore ba and has rank at most $\dim_k D = q_D$. If it is the permanent contraction, the integral identity minor of Theorem 22.1 gives rank $\binom{n}{k_0}$ over this same field. The middle choice gives the last inequality. $\square$

For any field of odd characteristic, or of characteristic zero, (97) is a representation of this precise form with $D = B_n \otimes k$, $c = \chi_n/r_n$, $a_{ij} = d_j$, and $q_D = 2^{n-1}$. Thus the construction gives an upper bound 2^{n-1} for this algebra dimension, while the theorem gives its universal lower bound $\binom{n}{\lfloor n/2 \rfloor}$. This lower bound is exponential: the binomial coefficients count all subsets by their cardinalities and sum to 2^n ; the ratio $\binom{n}{k+1}/\binom{n}{k} = (n-k)/(k+1)$ shows the middle coefficient is maximal, so it is at least $2^n/(n+1)$. The factorization assumes a fixed algebra with coordinate-independent structure and coefficients. It places no restriction on representing an arbitrary polynomial as a single unconstrained element of a rank-one algebra; in that larger description the element itself could already contain the permanent.

The sign algebra is smaller than the original source algebra for every $n \geq 1$, since $2^{n-1} \leq 3^{n-1} < 3^{3n^2}$, and smaller than its Boolean corner 2^{3n^2} . Its stated dimension remains exponential. Enumerating its 2^{n-1} signs and directly evaluating the displayed n row sums and product uses $O(n^2 2^{n-1})$ arithmetic operations: each sign requires n^2 signed entries, $n(n-1)$ additions for the row sums, $n-1$ multiplications for their product, and a bounded number of sign and scale operations; the resulting values are summed over that many signs. These loops give a uniform explicit algorithm. For integer $L \geq 0$ and entries with $|Z_{ij}| < 2^L$, each row sum has absolute value at most $n2^L$, and the integer signed numerator is bounded in absolute value by $2^{n-1} n^n 2^{nL}$. Its exact division by 2^{n-1} yields the integer permanent by the proof above; a sufficient signed bit bound for that numerator is $n-1 + n\lceil \log_2 n \rceil + nL + 2$. No polynomial-size representation or unrestricted circuit separation is inferred from this compressed trace or its fixed-algebra bound.

24 The determinant orbit interface, with its exact obstruction scope

All varieties and representations in this section are over $\mathbb{C}$. Put $V_D = \text{Mat}_{D \times D}(\mathbb{C})$, $W_D = \text{Sym}^D(V_D^*)$, and $G_D = \text{GL}(V_D)$. The action, including its inverse, is

$$(g \cdot f)(X) = f(g^{-1}X), \quad \Omega_D = \overline{G_D \cdot \det_D} \subset W_D.$$

Closures below are Zariski closures. The affine coordinate ring of W_D is $B_D = \text{Sym}(W_D^*)$, with degree- e part $\text{Sym}^e(\text{Sym}^D V_D)$. Here D is the matrix size; e is a coordinate-ring degree and is not the polynomial degree D .

24.1 Original padding and stabilizer conventions

GCT I, Section 4, Proposition 4.1, constructs the determinant orbit-closure map from a formula through a possibly singular linear substitution; GCT II, Section 3.1, uses an independent padding variable [13, 12]. Explicitly, for $m < D$, let X be the bottom-right $m \times m$ submatrix of $Y \in V_D$ and $y = Y_{11}$, which is outside that submatrix. The linear map on forms is

$$j_{D,m} : \text{Sym}^m(V_m^*) \longrightarrow W_D, \quad h \longmapsto y^{D-m} h(X).$$

It is injective: the ring substitution retaining X and putting $y = 1$ and all other entries of Y equal to zero is a left inverse on its image. There is also a homogeneous coefficient left inverse, obtained by taking the coefficient of y^{D-m} and then setting the other outside variables to zero. Thus this map has no kernel and explicitly retains every coefficient of h . In the notation of GCT II, its n is the present m , and its m is the present D .

Those papers use projective $\text{SL}(V_D)$ orbit closures. The precise comparison with the affine convention above is as follows. For $g \in G_D$, choose $\mu \in \mathbb{C}^*$ with $\mu^{D^2} = \det(g)$ and set $s = \mu^{-1}g$. Then $s \in \text{SL}(V_D)$ and $g \cdot f = \mu^{-D}(s \cdot f)$. Hence the projective orbits agree.

Moreover every nonzero scalar multiple of f is in $G_D \cdot f$, because λI acts by λ^{-D} . The affine closure is therefore a cone, and its homogeneous vanishing ideal is precisely the homogeneous ideal of the projective orbit closure: a homogeneous equation vanishes on the projective orbit exactly when it vanishes on every nonzero vector above it. This proves the comparison of coordinate rings, with their degrees retained.

The original stabilizer discussion distinguishes a form from its line. For example the linear substitution $T_{A,B}(Y) = AYB^{-1}$, with $A, B \in \mathrm{GL}_D$, satisfies

$$\det(T_{A,B}Y) = \frac{\det A}{\det B} \det Y, \quad T_{A,B} \cdot \det = \frac{\det B}{\det A} \det.$$

The first identity follows by multiplication of determinants and the second by inserting $T_{A,B}^{-1}$. Thus every such substitution fixes the determinant line, and it fixes the form exactly when $\det A = \det B$. Transposition fixes the form as well. To intersect these substitutions with the original ambient $\mathrm{SL}(V_D)$, retain the additional equation $(\det A / \det B)^D = 1$; with transposition the left side is multiplied by $(-1)^{D(D-1)/2}$. Indeed left multiplication acts independently on D columns, right multiplication on D rows, and transposition exchanges the $D(D-1)/2$ off-diagonal coordinate pairs. These equations prevent identifying the line stabilizer with the form stabilizer. For the permanent, row and column permutations and transposition preserve the form, and row scalars α_i and column scalars β_j multiply it by $\prod_i \alpha_i \prod_j \beta_j$: each defining monomial uses each row and column once. These verified subgroups supply the stabilizer context used here; no classification of a new fibre stabilizer is inferred from them.

24.2 An explicit endomorphism and a strict boundary point

For the retained-fibre family take $n \geq 1$, $D = 4n+1$, and $E_n = \mathbb{C}^{5n+1}$ with ordered coordinates

$$(a_1, b_1, c_1, r_1, t_1, \dots, a_n, b_n, c_n, r_n, t_n, z).$$

Write $L_n : E_n \rightarrow V_D$ for the linear map whose matrix value is the previously constructed $\mathcal{M}_n^h$, retaining all entries:

$$T_i^h = \begin{pmatrix} z & -r_i & 0 & 0 \\ 0 & z & -r_i & 0 \\ 0 & 0 & z & -r_i \\ 0 & 0 & 0 & z \end{pmatrix}, \quad v_i^h = \begin{pmatrix} -2a_i \\ b_i \\ -2z \\ c_i \end{pmatrix}, \quad \mathcal{M}_n^h = \begin{pmatrix} \mathrm{diag}(T_i^h) & v^h \\ -u & 0 \end{pmatrix},$$

where $u = (t_1, 0, 0, 0, \dots, t_n, 0, 0, 0)$ and v^h stacks the v_i^h . Its determinant, already proved by block multiplication, is

$$H_n = z^{4n-4} \sum_{i=1}^n t_i (c_i r_i^3 - 2z^2 r_i^2 + b_i r_i z^2 - 2a_i z^3).$$

It has degree D . Every coordinate of E_n is a nonzero scalar multiple of an entry of L_n , so L_n is injective.

Order the D^2 entries of $Y \in V_D$ row by row. Let $p_n : V_D \rightarrow E_n$ take the first $5n+1$ entries in exactly the displayed order, and let $i_n : E_n \rightarrow V_D$ fill those entries and put zero in the remaining positions. The inequality $5n+1 < D^2$ holds for $n \geq 1$, and $p_n i_n = I$. These are specified linear maps, rather than an identification of two variable spaces. Set

$$A_n = L_n p_n \in \mathrm{End}(V_D), \quad \tilde{H}_n = H_n \circ p_n = \det \circ A_n \in W_D.$$

On coordinate rings, p_n^* is injective, $i_n^* p_n^* = I$, and $L_n^*(X_{pq}) = (\mathcal{M}_n^h)_{pq}$. The latter is the previously proved surjection ϑ_n , with its stated ideal of linear relations as kernel. Thus $A_n^* = p_n^* L_n^*$ retains that kernel, and $i_n^*(\tilde{H}_n) = H_n$ recovers the full determinant polynomial.

Proposition 24.1. *For every $n \geq 1$, $\tilde{H}_n \in \Omega_{4n+1}$, and $\tilde{H}_n \notin G_{4n+1} \cdot \det_{4n+1}$.*

Proof. Let $A_n(\epsilon) = A_n + \epsilon I_{V_D}$. Its determinant as a $D^2 \times D^2$ endomorphism is a monic polynomial of degree D^2 in ϵ . Away from its finite root set it is invertible, and

$$f_\epsilon(Y) = \det_D(A_n(\epsilon)Y) = A_n(\epsilon)^{-1} \cdot \det_D(Y)$$

belongs to the determinant orbit. The coefficients of f_ϵ are polynomials in ϵ . Every equation vanishing on that orbit, when evaluated on f_ϵ , is consequently a polynomial vanishing for all but finitely many complex values of ϵ . It is zero identically and hence at $\epsilon = 0$. This proves the claimed closure membership, without assigning an inverse to $A_n(0)$.

For strictness, define the linear subspace $\text{Ann}(f) = \{v : \partial_v f = 0\}$, using formal directional derivatives. The chain rule gives $\text{Ann}(g \cdot f) = g \text{Ann}(f)$. For the universal determinant, $\partial_v \det_D = \sum_{i,j} v_{ij} \text{cof}_{ij}(Y)$. The cofactors are linearly independent: each is nonzero and has row multidegree one except in row i , and column multidegree one except in column j . Distinct pairs (i, j) have distinct combined multidegrees, so their monomials cannot cancel. Therefore $\text{Ann}(\det_D) = 0$. In contrast, $\ker p_n \subseteq \text{Ann}(\tilde{H}_n)$ and $\dim \ker p_n = D^2 - (5n + 1) > 0$. The orbit membership is impossible. Finally H_n is nonzero: set $z = t_1 = c_1 = r_1 = 1$, $a_1 = b_1 = 0$ and $t_i = 0$ for $i > 1$, obtaining $H_n = -1$. $\square$

24.3 The coordinate quotient and the obstruction direction

Put $Z_n = \overline{G_D \cdot \tilde{H}_n}$. The proposition and G_D -invariance of Ω_D give $Z_n \subseteq \Omega_D$. Both are cones by the scalar calculation above. For any such closed cone Z , its ideal $I(Z) \subset B_D$ is homogeneous: if $F = \sum_e F_e$ vanishes on Z , then for $z \in Z$ the polynomial $\sum_e t^e F_e(z)$ vanishes for every $t \in \mathbb{C}$, so every $F_e(z) = 0$. Consequently restriction is the explicit graded map

$$\rho_n : \mathbb{C}[\Omega_D] = B_D/I(\Omega_D) \longrightarrow B_D/I(Z_n) = \mathbb{C}[Z_n], \quad F + I(\Omega_D) \longmapsto F + I(Z_n).$$

It is well defined since $I(\Omega_D) \subseteq I(Z_n)$, is surjective in each degree by taking a polynomial representative, and has kernel $I(Z_n)/I(\Omega_D)$. For the action on functions $(g \cdot F)(f) = F(g^{-1} \cdot f)$, stability of both sets proves $\rho_n(g \cdot F) = g \cdot \rho_n(F)$ by evaluation at every point.

In each degree the quotient also gives the exact multiplicity inequality

$$\dim \text{Hom}_{G_D}(S, \mathbb{C}[Z_n]_e) \leq \dim \text{Hom}_{G_D}(S, \mathbb{C}[\Omega_D]_e)$$

for every irreducible finite-dimensional rational G_D -module S . Here is the splitting argument. Choose a linear section of $(\rho_n)_e$ and average $u^{-1}su$ over the compact unitary group $U(D^2)$ with its invariant probability measure. The averaged map is a section since ρ_n is equivariant, and is unitary-equivariant by change of integration variable. It is G_D -equivariant: differentiating its intertwining identity on skew-Hermitian matrices and taking complex linear spans gives the identity on all matrices, and exponentiating gives it on $\text{GL}_{D^2}(\mathbb{C})$. Every invertible complex matrix has a logarithm, as follows block by block from its Jordan form and the finite logarithm series on the nilpotent part. Composing maps $S \rightarrow \mathbb{C}[Z_n]_e$ with this section is an injection into the other Hom space, proving the inequality. This proof also shows exactly why a multiplicity in the smaller orbit closure exceeding the corresponding determinant multiplicity would obstruct inclusion in a different instance.

The stabilizer restriction has its own exact map. For $f \in W_D$ let $H_f = \{h \in G_D : h \cdot f = f\}$. The orbit map $\alpha_f(g) = g \cdot f$ induces

$$\alpha_f^* : \mathbb{C}[\overline{G_D \cdot f}] \hookrightarrow \mathbb{C}[G_D]^{H_f, \text{right}}, \quad F \longmapsto (g \mapsto F(g \cdot f)).$$

Regularity follows from the entries of g^{-1} being regular on G_D ; right invariance follows from $gh \cdot f = g \cdot f$; injectivity follows because a function vanishing on the orbit vanishes on its

closure. For e -homogeneous F and a line stabilizer with $h \cdot f = \chi_f(h)f$, the precise identity instead is $\alpha_f^* F(gh) = \chi_f(h)^e \alpha_f^* F(g)$. Thus form stabilizers give invariants and line stabilizers give this specified character. No surjectivity onto all right-invariant functions is asserted; that would require extension across the orbit boundary.

24.4 The later no-occurrence result at its stated scope

For a separate pair of matrix sizes $N > m$, Bürgisser–Ikenmeyer–Panova use the top-left projection $\pi : V_N \rightarrow V_m$ and the form

$$p_{N,m}(Y) = Y_{11}^{N-m} \text{per}_m(\pi Y), \quad Z_{N,m}^{\text{BIP}} = \overline{\text{GL}(V_N) \cdot p_{N,m}}.$$

Here Y_{11} is one of the permanent variables. Their source theorem, [3, Theorem 1.4], states that for positive integers N, e, m with $N \geq m^{25}$ and $\lambda \vdash Ne$, occurrence of λ in $\mathbb{C}[Z_{N,m}^{\text{BIP}}]$ implies occurrence in $\mathbb{C}[\Omega_N]$. This is their proved theorem, not a result derived from the present fibres. We retain their same-variable padding and exponent 25; the independent padding above is not silently substituted into that statement. The central scalar tI acts on $\mathbb{C}[W_N]_e$ by t^{Ne} , explaining the partition size Ne . The theorem compares occurrence, and does not assert the multiplicity inequality needed to force a quotient.

A later concrete multiplicity example has a different, specified ambient space. In $\mathbb{C}[x_1, x_2, x_3]_6$ put $\text{Ch}_3^6 = \{\ell_1 \cdots \ell_6\}$ and $\text{Pow}_{3,k}^6 = \{\ell_1^6 + \cdots + \ell_k^6\}$, where all ℓ_i are homogeneous linear forms. Dörfler–Ikenmeyer–Panova prove, in their partition labeling, at coordinate degree 7,

$$\text{mult}_{(34,6,2)} \mathbb{C}[\text{Ch}_3^6]_7 = 7 < 8 = \text{mult}_{(34,6,2)} \mathbb{C}[\text{Pow}_{3,4}^6]_7.$$

Their Main Theorem (2a) and Section 3 additionally prove that every partition occurring in the ambient coordinate ring occurs on Ch_3^6 , in every coordinate degree. Thus no occurrence obstruction separates $\text{Pow}_{3,k}^6$ from this Chow variety, for any k , although the displayed multiplicities separate them for $k = 4$ [5]. For the quantifiers we use the proof’s equation labeled `eq:posplethimpliesposchow`, which retains the same partition and degree on both sides; the theorem’s final sentence has inconsistent indices. This is a source theorem at fixed $(m, n) = (3, 6)$, not a statement in all dimensions or about permanent orbit closures. Our finite plethysm checker does not certify the paper’s Chow ranks or its all-degree no-occurrence proof.

Our concrete family has the stronger quotient ρ_n already proved, so neither an occurrence obstruction nor a multiplicity obstruction in the direction $Z_n \not\subseteq \Omega_{4n+1}$ can arise. This conclusion concerns precisely the coefficient-recoverable family $\tilde{H}_n$: setting $z = 1$ and taking selector coefficients still recovers every original cubic, while forgetting the selectors or replacing the tuple by one product has the information losses proved earlier. The displayed maps provide an actual endpoint in GCT orbit geometry for the retained-fibre construction. They do not identify this family with padded permanents or supply a Boolean satisfiability reduction.

25 Exact tensor-square transport on the full quantum source

Throughout this section $A = \mathbb{Z}[q, q^{-1}]$, $K = \mathbb{Q}(q)$, $D = \mathbb{Q}[q, q^{-1}]_{(q-1)}$, and X_A is the original 1260-dimensional honest lattice of Section 4. Put $X_D = D \otimes_A X_A$. The rational specialization is the already proved map

$$\sigma : X_D / (q-1)X_D \xrightarrow{\sim} S_{(7,5,3)}(V_{\mathbb{Q}} \otimes W_{\mathbb{Q}}).$$

All 15 boxes and both degree-fifteen central characters remain. The passage to D here is explicit: integral defects at the primes 2, 3, 7 are retained in the earlier integral chapters and are not asserted to survive this localization.

The coproduct used in the original GCT-IV source is

$$\Delta(K) = K \otimes K, \quad \Delta(E) = E \otimes K^{-1} + 1 \otimes E, \quad \Delta(F) = F \otimes 1 + K \otimes F. \quad (101)$$

These are the source's equation `e U_q coproduct` in [1], TeX lines 1766–1770 in the archived April 2013 edition. They apply separately to the two commuting colours V, W . The formulas also directly show coassociativity: for example both iterated coproducts of F equal $F \otimes 1 \otimes 1 + K \otimes F \otimes 1 + K \otimes K \otimes F$. Every tensor action below uses these formulas, with this orientation.

25.1 A failure on actual source vectors

Let $\tau(x \otimes y) = y \otimes x$ be the ordinary flip. Use the original vector T of Section 3, put $z = T$, and set $v = F_V T$. The full-source calculation proves $E_V z = 0$, $K_V z = q^9 z$, and $E_V v = [9]z$. Thus $v \neq 0$. The vectors z, v have different weights and are independent over K , and at $q = 1$ they remain independent over $\mathbb{Q}$ since $E_V v = 9z$ there. The coproduct gives exactly

$$\Delta(F_V)(z \otimes z) = v \otimes z + q^9 z \otimes v, \quad (102)$$

$$\begin{aligned} (\tau \Delta(F_V) - \Delta(F_V) \tau)(z \otimes z) &= (q^9 - 1)(v \otimes z - z \otimes v), \\ \Delta(E_V)(z \otimes v - v \otimes z) &= [9](1 - q^{-9})z \otimes z. \end{aligned} \quad (103)$$

All displayed nonconstant coefficients are nonzero in K . Consequently the usual symmetric subspace of $X_K^{\otimes 2}$ is not a submodule: its vector $z \otimes z$ has the nonsymmetric image in (102). The usual antisymmetric subspace is not a submodule either, by (103). Over characteristic zero the symmetric-square quotient kills the antisymmetric subspace, so that quotient does not inherit the original tensor action. Likewise the exterior-square quotient does not inherit it. These are statements about the specified tensor action, not about the possible existence of other quantum symmetric or exterior powers.

At $q = 1$ the two coproducts become $E \otimes 1 + 1 \otimes E$ and $F \otimes 1 + 1 \otimes F$. They commute with every ordinary permutation of tensor factors; the diagonal Cartan actions do also. For the classical polynomial group action the same assertion follows directly from $g^{\otimes d} \tau_i = \tau_i g^{\otimes d}$. Thus for every partition λ of d , applying its row symmetrizations and column antisymmetrizations to $\sigma^{\otimes d}$ gives an actual classical Schur-functor isomorphism. The inverse is induced by $(\sigma^{-1})^{\otimes d}$: both tensor maps commute with the same permutation operators, so their two compositions on the corresponding images are the identity. This specifies the classical morphism and why the same construction does not directly supply its generic quantum version.

25.2 Explicit Clebsch–Gordan maps, with their specialization

Write L_m for the ratio-weight module with divided-power basis $e_0, \dots, e_m$, actions

$$K e_i = q^{m-2i} e_i, \quad F e_i = [i+1] e_{i+1}, \quad E e_i = [m-i+1] e_{i-1}.$$

Terms outside the basis range are zero. Its central degree is supplied separately by its original $\mathfrak{gl}_2$ label; no central character is changed by this ratio-weight notation. For L_n use $f_0, \dots, f_n$. For $0 \leq r \leq \min(m, n)$ define

$$a_i(m, n, r) = (-1)^i q^{i(n-2r+i+1)} \prod_{j=0}^{i-1} \frac{[n-r+j+1]}{[m-j]}, \quad (104)$$

$$\begin{aligned} z_{mnr} &= \sum_{i=0}^r a_i(m, n, r) e_i \otimes f_{r-i}, & h_r &= m + n - 2r, \\ w_{mnr,s} &= \Delta(F)^{(s)} z_{mnr}, & 0 \leq s &\leq h_r. \end{aligned} \quad (105)$$

An empty product equals one. All denominators here are units of D , since $[j](1) = j \neq 0$ for $j > 0$.

The coefficient of $e_i \otimes f_{r-1-i}$ in $\Delta(E)z_{mnr}$ is

$$a_{i+1}[m-i]q^{-n+2(r-1-i)} + a_i[n-r+i+1] = 0.$$

The equality follows by dividing the explicit consecutive coefficients in (104). Thus z_{mnr} is a highest vector of weight h_r ; it is nonzero since $a_0 = 1$. The string proof of Lemma 5.1 now gives exactly

$$\Delta(F)w_{mnr,s} = [s+1]w_{mnr,s+1}, \quad \Delta(E)w_{mnr,s} = [h_r - s + 1]w_{mnr,s-1}.$$

The boundary statements include $\Delta(F)w_{mnr,h_r} = 0$. The resulting strings are independent: the central Casimir $(q - q^{-1})^2 FE + qK + q^{-1}K^{-1}$ has distinct eigenvalues $q^{h_r+1} + q^{-h_r-1}$ for the distinct nonnegative integers h_r , and each individual string has distinct weights. Finally

$$\sum_{r=0}^{\min(m,n)} (m+n-2r+1) = (m+1)(n+1).$$

For example, when $m \leq n$ the sum is $(m+1)(m+n+1) - m(m+1) = (m+1)(n+1)$; the other case interchanges m, n . These vectors therefore give a basis over K .

They also give a basis over D , not only over its fraction field. Specializing the coefficient recurrence at $q = 1$ gives the same highest-vector proof for the ordinary $\mathfrak{sl}_2$ action. The ordinary Casimir $FE + H(H+2)/4$ is central (substitution of $[H, E] = 2E$, $[H, F] = -2F$, $[E, F] = H$ verifies its commutators) and has eigenvalue $h(h+2)/4$ on a highest vector of weight h . These numbers are distinct for $h \geq 0$, so the specialized strings are independent by the same argument. The defining commutator gives $EF^s z = s(h-s+1)F^{s-1}z$, hence their lengths and nonzero coefficients are exactly as stated. Their dimension sum is unchanged. The square matrix of the $w_{mnr,s}$ thus has determinant with nonzero value at $q = 1$. Its determinant is a unit of the local ring D , and its inverse is explicitly its adjugate divided by that determinant. This proves both inverse compositions over D .

25.3 An equivariant involution on all 1260^2 tensor coordinates

Set the nonzero rational constant

$$\gamma_{mnr} = (-1)^r \frac{n!(m-r)!}{(n-r)!m!}.$$

Define the D -linear map on the preceding basis by

$$J_{m,n} : L_m \otimes L_n \longrightarrow L_n \otimes L_m, \quad w_{mnr,s} \longmapsto \gamma_{mnr} w_{nmr,s}. \quad (106)$$

This is a completely specified finite matrix: take the matrix of the target strings, multiply each r block by γ_{mnr} , and multiply by the adjugate inverse of the source string matrix. The string action formulas prove that it intertwines E, F, K ; the retained total central degrees agree on source and target. Moreover $\gamma_{mnr}\gamma_{nmr} = 1$, which proves $J_{n,m}J_{m,n} = 1$ on every basis vector.

At $q = 1$ the ordinary flip sends z_{mnr} to $\gamma_{mnr}z_{nmr}$. Indeed it is a highest vector of the same weight; that highest space is one-dimensional in this tensor product, as follows from the string basis and the distinct h_r . Its coefficient of $f_0 \otimes e_r$ is $a_r(m, n, r)|_{q=1}$, which is exactly γ_{mnr} . Commutation with the classical lowering operator proves the same assertion for every descendant. Hence $J_{m,n}|_{q=1}$ is precisely the flip.

Now use the actual source isomorphism Φ of (30). Its explicit projectors and inverse have denominators nonzero at $q = 1$, so they extend over D . Writing $H_{ab,D} = X_D \cap \ker E_V \cap \ker E_W$ in weight $(a, 15 - a; b, 15 - b)$ gives

$$X_D = \bigoplus_{a,b=8}^{12} L_{2a-15} \otimes L_{2b-15} \otimes H_{ab,D}$$

via this same Φ_D ; the original determinant twists on each L factor remain those in its full highest-weight label. To justify base change of the highest spaces, the displayed projector and raising inverse extract their components over D . The two inverse identities proved on strings thus hold over D . In particular the $H_{ab,D}$ are direct summands of finite free D -modules; a submodule of a finite free module over this DVR is free, so the same multiplicities g_{ab} apply.

For a pair of summands $(a, b), (c, d)$ in $X_D \otimes X_D$, first regroup their V, W , and multiplicity factors in their indicated order. Apply

$$J_{2a-15,2c-15} \otimes J_{2b-15,2d-15} \otimes (z \otimes z' \mapsto z' \otimes z),$$

then undo the regrouping into the swapped pair of summands. The regrouping commutes with the action because the two colours act only on their own factors and the multiplicity factors have trivial action. Conjugate the resulting direct-sum map by $\Phi_D \otimes \Phi_D$. Denote the resulting actual full-source operator by J_X .

All its entries are fixed by the formulas above and the original generator/projector matrices. It preserves the full tensor action, including both degree-thirty central characters, and

$$J_X^2 = 1, \quad J_X|_{q=1} = \tau_{X_1, X_1}.$$

The inverse identities for Φ_D and $J_{m,n}$ prove the first statement on every direct summand; their specialization statements prove the second. Thus

$$P_+ = (1 + J_X)/2, \quad P_- = (1 - J_X)/2 \tag{107}$$

are orthogonal equivariant idempotents with sum one. Their images are finite free D -modules, and specialization commutes with the direct-sum decomposition. Counting the symmetric basis $b_i \otimes b_i, b_i \otimes b_j + b_j \otimes b_i$ for $i < j$, and the antisymmetric basis for $i < j$, proves

$$\text{rank}_D \text{im } P_+ = 794430, \quad \text{rank}_D \text{im } P_- = 793170.$$

Under $\sigma^{\otimes 2}$ their specializations are exactly the symmetric and exterior squares of $S_{(7,5,3)}(V_{\mathbb{Q}} \otimes W_{\mathbb{Q}})$. This constructs actual quantum-equivariant square modules on the full source, not just modules with the same character.

25.4 The degree-three coherence calculation and a braided repair

The involution just constructed must not be assigned ordinary Young symmetrizers in higher degree without checking the permutation relations. Here is an exact failure inside the actual source. The entry $g_{8,8} = 1$ of (26) supplies a summand with V and W ratio weights both one. Choose a D -basis of its rank-one multiplicity space. Fix the W highest vector in each of three tensor positions. On this subspace the two W commutators act as identity, so the following V calculation embeds in $X_D^{\otimes 3}$ by the injective map $\Phi_D^{\otimes 3}$.

In the ordered $L_1 \otimes L_1$ basis $(00, 01, 10, 11)$, the construction gives

$$J = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & (q^2 - 1)/(q^2 + 1) & 2q/(q^2 + 1) & 0 \\ 0 & 2q/(q^2 + 1) & (1 - q^2)/(q^2 + 1) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{108}$$

Indeed the weight-zero highest vector of the trivial string is $01 - q10$, with eigenvalue -1 , and the weight-zero vector of the length-two string is $q01 + 10$, with eigenvalue $+1$. In the order $r = 0, r = 1$, these columns have determinant $-(1 + q^2)$ and yield exactly the displayed middle block. On the triple tensor product put $J_{12} = J \otimes 1$, $J_{23} = 1 \otimes J$. Direct multiplication gives

$$(J_{12}J_{23}J_{12} - J_{23}J_{12}J_{23})(001) = \frac{2(q^2 - 1)^2}{(q^2 + 1)^3}(-001 + q010) \neq 0. \quad (109)$$

The two output basis tensors are independent and the rational coefficient is nonzero. Therefore this specific full-source involution does not give a symmetric-group action in degree three. Its exact defect has order two at $q = 1$, with leading term $(q - 1)^2(-001 + 010)$. This follows from $(q^2 - 1)^2 = (q - 1)^2(q + 1)^2$ and evaluation of the residual factors.

There is also an explicit braided repair of this test sector. Put

$$B = \frac{q - q^{-1}}{2}I + \frac{q + q^{-1}}{2}J = \begin{pmatrix} q & 0 & 0 & 0 \\ 0 & q - q^{-1} & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & q \end{pmatrix}.$$

It is an intertwiner for the V -coloured $L_1 \otimes L_1$ action because J and I are. The fixed W highest slice used to embed the test is not asserted to be stable under W lowering. Its two eigenvalues are q and $-q^{-1}$, so $(B - qI)(B + q^{-1}I) = 0$, and $B^{-1} = B - (q - q^{-1})I$. To verify the braid relation without an unproved matrix assertion, put $a = q - q^{-1}$. On $(001, 010, 100)$ the common value of $B_{12}B_{23}B_{12}$ and $B_{23}B_{12}B_{23}$ is respectively

$$q^2a \, 001 + qa \, 010 + q \, 100, \quad qa \, 001 + q \, 010, \quad q \, 001.$$

The equality in the first column uses $a^2q + a = q^2a$, obtained by substituting $a = q - q^{-1}$. On $(011, 101, 110)$ the same three formulas hold in that ordered basis. On 000 and 111 both products act by q^3 . This proves equality on all eight basis vectors. Also $B|_{q=1} = \tau$.

Thus the actual source admits the stated equivariant square decomposition, its proposed ordinary permutation coherence fails by (109), and a Hecke braid is constructed on the precise sector detecting that failure. No extension of this last 4-by-4 braid to the full nonstandard Schur/plethysm construction has been assumed. These exact maps specify the next representation-level input needed to transport the conductor filtration to higher quantum plethysm. No GCT obstruction or Boolean complexity separation follows merely from the nonzero coherence defect.

26 A diagonal-gauge obstruction and the full signed source cover

This section concerns the original honest source basis of $\check{X}_{(7,5,3)}$, not an unspecified prospective canonical basis. It gives a pointwise obstruction for every real $q > 0$, including $q = 1$, and an integral positive cover with an exact quotient to all four source generators. The cover's nonzero quantum-relation defect is also computed. The ordinary sign-readjustment obstruction is already stated in GCT IV [1], immediately before its Example **ex not positive**. The additional statements here are the arbitrary diagonal-gauge invariant, its precise effectiveness consequence, and the full source cover and defect maps.

26.1 Original coefficients and a pointwise invariant

Retain $A = \mathbb{Z}[q, q^{-1}]$, $K = \mathbb{Q}(q)$ and $[m] = \sum_{j=0}^{m-1} q^{m-1-2j}$ for positive integers m . Put $[0] = 0$ and $[-m] = -[m]$. The fixed basis of the honest lattice X_A is $b_0, \dots, b_{1259}$. In the original

source indexing,

$$T = b_{126}, \quad T_1 = b_{146}, \quad T_2 = b_8, \quad Q = b_{144}.$$

Their actual column expressions, with the original scaling, are

$$\begin{aligned} T &= (123)(123)(124)(12)(12)(1)(1), \\ T_1 &= (123)(123)(124)(23)(12)(1)(1), \\ T_2 &= -[2]^{-1}(123)(123)(123)(12)(12)(4)(1), \\ Q &= -[2]^{-1}(123)(123)(124)(13)(12)(4)(1). \end{aligned}$$

The quotient calculation of the original words supplies these identities inside X_A ; the displayed $[2]^{-1}$ is part of the representative, not an inversion newly made in A . The W -weight of each vector is $(9, 6)$. Their V -weights are, in order, $(12, 3), (11, 4), (11, 4), (10, 5)$. The printed endpoint with a (23) in place of (13) has W -weight $(8, 7)$ and is a different vector; it is not used as Q .

The following are entries in the full 1260 by 1260 matrix F_V , before any four-state quotient:

source	target	coefficient
126	146	$\alpha = [7] - [3] = q^6 + q^4 + q^{-4} + q^{-6}$
126	8	$\beta = [8]$
146	144	$\gamma = [6]$
8	144	$-d = -[3]$

Other full-matrix entries are retained. In particular the argument below does not require these four vectors to form a full-group submodule. Define

$$p = [2], \quad C = q^{10} - q^4 - q^{-4} + q^{-10}.$$

Expansion of the finite Laurent sums gives

$$\alpha\gamma - \beta d = pC, \quad C = (q - q^{-1})^2 [7][3]. \quad (110)$$

For example $[6] = [3](q^3 + q^{-3})$ and direct multiplication gives $(q^3 + q^{-3})([7] - [3]) - [8] = [2](q - q^{-1})^2 [7]$; these two identities prove the first formula with no specialization. The second formula expands to exactly the four terms defining C .

Fix a real $t > 0$ and evaluate $q = t$. Every term of $[m](t)$ is positive for $m > 0$, and the four displayed terms of $\alpha(t)$ are positive. Hence the loop invariant of the two original length-two paths is

$$\mathcal{H}(t) = \frac{\alpha(t)\gamma(t)}{-\beta(t)d(t)} = -1 - \frac{p(t)C(t)}{\beta(t)d(t)} \leq -1. \quad (111)$$

Equality holds exactly at $t = 1$, since $[7](t)[3](t) > 0$ and $(t - t^{-1})^2 = 0$ exactly there. Thus the invariant is strictly negative on the entire positive real axis. At $t = 1$ the four coefficients are 4, 8, 6, -3 , respectively, and the two products are 24, -24 . The vanishing of $C(1)$ therefore retains the opposite signs of these two nonzero paths.

Theorem 26.1. *For every $t > 0$ there is no diagonal matrix $D = \text{diag}(d_0, \dots, d_{1259})$ with $d_i \in \mathbb{C}^\times$ such that all entries of $D^{-1}F_V(t)D$ are nonnegative real numbers. In particular there is no such simultaneous diagonal gauge for the four original Chevalley generators. The assertion still holds after a permutation of the source basis. No diagonal gauge over $\mathbb{C}(q)$ can make all these entries nonnegative Laurent polynomials, or rational functions that are nonnegative wherever defined on a nonempty positive real interval.*

Proof. The entry from i to j is multiplied by d_i/d_j . Each of the two path products from 126 to 144 is consequently multiplied by the same nonzero factor d_{126}/d_{144} . Their quotient remains $\mathcal{H}(t) < 0$. If all four transformed entries were nonnegative real numbers, they would be strictly positive because none vanishes, so the quotient would be positive. This is a contradiction even when the d_i are complex. A permutation merely relocates the same four entries. For a rational diagonal gauge, choose t in the proposed interval away from the finite set of poles and zeros of its diagonal entries. The preceding pointwise contradiction then applies. A nonzero Laurent polynomial with nonnegative coefficients is strictly positive at every positive real t , which proves the polynomial assertion as a special case. $\square$

26.2 The exact sign class, and categorical effectiveness

For a family of real matrices with fixed nonzero support, form an undirected multigraph: each nonzero matrix entry gives a labelled edge between its source and target, retaining its matrix colour and sign. Loops and parallel edges are retained. Let the edge sign be $\sigma_e \in \{1, -1\}$. A traversal in either direction uses the same sign. A closed walk has sign equal to the product of these edge signs. In additive notation $\sigma_e = (-1)^{\epsilon_e}$, the assignment ϵ is a $\mathbb{Z}/2$ -valued graph one-cochain. Since a graph has no two-cells, its class modulo vertex coboundaries is an element of $H^1(\Gamma; \mathbb{Z}/2)$, whose definition here is precisely that quotient of edge assignments.

Lemma 26.2. *All entries of a family can be made nonnegative by a common diagonal complex gauge if and only if every closed walk of its sign graph has positive sign. When this holds, a diagonal gauge with entries in $\{1, -1\}$ already suffices.*

Proof. Write a putative complex diagonal entry as $d_i = r_i z_i$, where $r_i > 0$ and $|z_i| = 1$. A real nonzero coefficient of sign σ_e from i to j becomes positive only if $\sigma_e z_i/z_j = 1$, or $z_j = \sigma_e z_i$. Multiplying these identities around a closed walk gives product sign one. Conversely, in each connected component choose a root and set its sign to one. Define the sign of a vertex to be the product of edge signs along any root-to-vertex walk. Two choices have quotient the sign of a closed walk, so the value is independent of the walk. The endpoint signs then satisfy $s_j = \sigma_e s_i$ on every edge. The diagonal entries s_i make each coefficient positive. This also proves that the obstruction is exactly $[\epsilon]$, not merely a necessary condition extracted from one failed gauge. $\square$

All 8162 nonzero entries of the four original matrices are coefficientwise sign coherent, in agreement with GCT IV's theorem **t positivity**. Thus their signs do not vary with $t > 0$. The exact census in the original basis is

	positive entries	negative entries	total
F_V	1648	350	1998
E_V	1837	246	2083
F_W	1848	150	1998
E_W	1963	120	2083

The graph is connected. Its 8162 labelled edges give cycle rank $8162 - 1260 + 1 = 6903$; it has 6146 unordered vertex pairs after parallel edges are identified. These finite statements are certified by a spanning traversal of every original matrix entry, retained in the accompanying independent graph certificate.

The negative cycle displayed in (111) has minimum possible length. Indeed every edge changes exactly one of the first weight coordinates by one. It therefore reverses the parity of their sum, so there are no loops or odd cycles. Direct comparison of all parallel entries gives the same sign for every edge on a fixed unordered pair, so a two-edge closed walk has positive

sign. Our four-edge walk is negative, proving minimality. An additional two-colour witness uses the actual entries

$$(F_V)_{20,0} = [7], \quad (F_W)_{20,10} = [5], \quad (F_W)_{136,10} = [3], \quad (F_V)_{136,0} = -[3].$$

The oriented loop $0 \rightarrow 20 \leftarrow 10 \rightarrow 136 \leftarrow 0$ has coefficient quotient $-[7]/[5] < 0$ for $t > 0$, and value $-7/5$ at $t = 1$. Cancellation in a divided-power endpoint is not needed for this second witness.

For an exact formulation of effectiveness, call a labelled graded category effective when its object classes are finite sums $\sum_{i,w} n_{i,w} q^w [B_i]$ with $n_{i,w} \in \mathbb{Z}_{\geq 0}$, and these labels and shifts freely generate its split Grothendieck group. This includes finite direct sums of labelled graded free groups, and a graded semisimple category with the stated simples. An additive endofunctor compatible with the internal grading shifts and taking objects to objects then has an A -linear matrix over $\mathbb{Z}_{\geq 0}[q, q^{-1}]$: its i -th column is the decomposition of the image of B_i .

Corollary 26.3. *There is no such effective category and grading-shift-compatible additive source-generator functor whose induced F_V is identified with the original F_V by $[B_i] \mapsto d_i(q)b_i$ for invertible diagonal rational factors. This remains impossible after arbitrary internal shifts, parity changes of individual source labels, or permutations of those labels, provided the images of objects are required to remain effective in the resulting labels. At a fixed $t > 0$ the coordinate simplicial cone generated by any individually rescaled source basis is not preserved by all the generator images.*

Proof. An effective matrix evaluates to a nonnegative matrix at each positive real t . Apply Theorem 26.1 at a nonsingular value. Internal shift and individual parity change give diagonal factors q^{w_i} and $(-1)^{a_i}$, covered by the same theorem. The cone assertion is the real matrix statement: a linear map preserves the coordinate nonnegative cone exactly when every one of its columns lies in that cone. $\square$

This is an effectiveness obstruction with an explicitly stated basis constraint. It is not a statement that a Frobenius trace must be positive: an alternating trace retains its cohomological signs. Nor does it contradict the nonstandard positivity formulations that explicitly retain signs and powers of $q - q^{-1}$.

26.3 An integral positive cover of every original generator

For each of the four full source matrices M , split its entries coefficientwise as $M = M^+ - M^-$ with $M^+, M^- \in \text{Mat}_{1260}(\mathbb{Z}_{\geq 0}[q, q^{-1}])$ and disjoint coefficient supports. Define

$$\tilde{X}_A = X_A^+ \oplus X_A^-, \quad \tilde{M} = \begin{pmatrix} M^+ & M^- \\ M^- & M^+ \end{pmatrix}, \quad |M| = M^+ + M^-. \quad (112)$$

Here b_i^+, b_i^- are new labelled copies of every original vector; the signs in their names specify the projection below. No change is made to the underlying original b_i . The formula uses all the original entries and all their exponents. Since they are sign coherent, an original edge of positive sign stays on its sheet and an edge of negative sign switches sheets, always with its original absolute Laurent coefficient.

Define A -linear maps

$$\iota : X_A \longrightarrow \tilde{X}_A, \quad x \longmapsto (x, x), \quad \pi : \tilde{X}_A \longrightarrow X_A, \quad (x, y) \longmapsto x - y.$$

The following exact diagram holds for each of the four generators:

$$0 \longrightarrow (A^{1260}, |M|) \xrightarrow{\iota} (A^{2520}, \tilde{M}) \xrightarrow{\pi} (X_A, M) \longrightarrow 0. \quad (113)$$

To prove this, ι is injective, $\pi(x, 0) = x$, and $\pi(x, y) = 0$ exactly when $x = y$. Direct block multiplication gives

$$\pi\widetilde{M} = M\pi, \quad \widetilde{M}\iota = \iota|M|.$$

The underlying sequence of free A -modules is split by $s(x) = (x, 0)$, with no division by two; the section is not asserted to commute with the generators. Thus there is neither an unmentioned 2-inversion nor an integral torsion kernel.

Give both copies of b_i its original four weight coordinates. Every original Cartan diagonal matrix has Laurent monomial entries, so it lifts to the same diagonal matrix on both sheets. Its inverse lifts in the same way. The two original total degree-fifteen central characters remain q^{15} . The displayed inclusion and projection commute with these diagonal operators. The Cartan conjugation rules hold on the cover because each nonzero lifted edge has exactly the original source and target weights. In particular, the sign cover does not replace a raising edge by a lowering edge.

To describe its exact algebraic scope, let $\mathcal{T}$ be the free associative A -algebra on the four Chevalley symbols, the original Cartan symbols and their named inverses, and the integral Cartan operators H_V, H_W defined below. On the two sheets and on the kernel these last operators act by the same original diagonal matrices. Acting by the specified matrices makes all three terms of (113) actual $\mathcal{T}$ -modules, and the displayed maps are $\mathcal{T}$ -linear. For any polynomial relation $r \in \mathcal{T}$ whose original source matrix is zero,

$$\pi r(\widetilde{M}) = 0.$$

Consequently there is a unique, explicitly defined A -linear defect map $\Delta_r : \widetilde{X}_A \rightarrow A^{1260}$ with

$$r(\widetilde{M}) = \iota\Delta_r, \quad \Delta_r = \text{pr}_+ r(\widetilde{M}). \quad (114)$$

Existence follows from the equality of the plus and minus output coordinates, and uniqueness from injectivity of ι . If $r(\widetilde{M})\iota$ is evaluated instead, the same identities give $\Delta_r\iota = r(M_{\text{ker}})$, where M_{ker} denotes the four absolute Chevalley matrices and the original Cartan diagonals. Thus the complete relation loss is an actual map into the displayed kernel. The quotient by that kernel recovers all original source relations.

This doubled graph is the covering determined by the sign class. It is connected: paths from a root lift to one sheet at any target; a negative closed walk switches the root sheet, and then the same paths reach the other sheet at every target. A vertex sign on the cover, positive on the plus sheet and negative on the minus sheet, trivializes the pullback of the sign class. Conversely a one-sheet cover cannot trivialize its nonzero class. A finite graph covering of a connected graph has a constant number of vertices in each fibre: lift any path and its inverse to obtain bijections of fibres. Therefore this degree-two, 2520-vertex cover has the least possible degree among nonempty graph covers that trivialize this sign class. This minimality statement is about such graph covers, not arbitrary changes of representation basis or unrestricted categorical models.

26.4 A nonzero full-source quantum-relation defect

Let H_V be the original diagonal operator with entry $[w_{V,1}(b_i) - w_{V,2}(b_i)]$ on b_i ; equivalently over K it is $(K_V - K_V^{-1})/(q - q^{-1})$. Its integral diagonal formula, with its signs, defines the operator already over A . Let $\widetilde{H}_V = \text{diag}(H_V, H_V)$, the actual same Cartan expression on both sheets.

The exact source matrices give

$$(\widetilde{E}_V \widetilde{F}_V - \widetilde{F}_V \widetilde{E}_V - \widetilde{H}_V) b_{132}^+ = [2][3](b_{148}^+ + b_{148}^-). \quad (115)$$

Here the original source labels are

$$b_{132} = (123)(123)(124)(12)(12)(3)(3), \quad b_{148} = (123)(123)(124)(12)(23)(1)(3).$$

Both have V -weight $(10, 5)$ and W -weight $(9, 6)$. For completeness, all paths from 132 to 148 contributing to the two compositions are the following; coefficients are recorded in traversal order:

composition	intermediate index	first coefficient	second coefficient
$E_V F_V$	152	$-[3]$	$[2]$
$E_V F_V$	158	$[7]$	$[2]$
$F_V E_V$	128	$[2]$	$[7] - [3]$

On the original source their sum is $-[3][2] + [7][2] - [2]([7] - [3]) = 0$. On the cover the first path ends on the minus sheet and the second on the plus sheet. The subtracted third path ends on the plus sheet. Each output sheet therefore has coefficient $[2][3]$. The full column calculation verifies that all other target entries in the commutator defect vanish, proving (115). The diagonal term cannot affect target $148 \neq 132$. The original full matrices, including the same diagonal, satisfy the zero commutator relation in every column; thus this is precisely a failure upstairs, with zero image under π . Its coefficient expands to

$$[2][3] = q^3 + 2q + 2q^{-1} + q^{-3},$$

which is nonzero over A and strictly positive at every $t > 0$. The W -coloured calculation gives

$$(\tilde{E}_W \tilde{F}_W - \tilde{F}_W \tilde{E}_W - \tilde{H}_W) b_{256}^+ = [2][3](b_{263}^+ + b_{263}^-).$$

Each colour has exactly 52 even-sheet input columns with nonzero commutator defect. These claims use all target coordinates, not a principal submatrix. The independent Laurent checker verifies them column by column. It follows that the cover in (112) is not a full quantum-group representation, although its exact quotient is the original one.

The failure already includes the commuting lowering generators:

$$[\tilde{F}_V, \tilde{F}_W] b_0^+ = -[3]^2(b_{146}^+ + b_{146}^-), \quad b_0 = (123)(123)(123)(12)(12)(1)(1). \quad (116)$$

All contributing original paths from 0 to 146 are

composition	intermediate index	first coefficient	second coefficient
$F_V F_W$	126	$[3]$	$[7] - [3]$
$F_W F_V$	20	$[7]$	$[3]$
$F_W F_V$	136	$-[3]$	$[3]$

The original difference is $[3]([7] - [3]) - [7][3] + [3]^2 = 0$. On the cover the third path ends on the minus sheet and is subtracted with its positive absolute coefficient. The plus-sheet difference is also $-[3]^2$. Every other target entry cancels, as the full original-path enumeration verifies. In total the numbers of nonzero even-sheet input columns for the six commutator defects are

commutator	$[E_V, F_V]$	$[E_W, F_W]$	$[E_V, E_W]$	$[E_V, F_W]$	$[F_V, E_W]$	$[F_V, F_W]$
columns	52	52	68	52	52	50

For the first two columns the stated Cartan diagonal is subtracted; for the four mixed columns the original commutator is zero. The independent checker verifies all six original identities and all their exact diagonal-kernel defects over A .

26.5 An actual categorical map and the retained cone

Let $\mathcal{G}$ be the category of finite free internally graded abelian groups with finite support. A Laurent polynomial $a(q) = \sum_w a_w q^w$ with nonnegative integer coefficients defines the actual group $L_a = \bigoplus_w \mathbb{Z}^{a_w} \langle w \rangle$. Use the ordered basis labels $(w, 1), \dots, (w, a_w)$ and set $L_0 = 0$. Define an additive functor on $\mathcal{G}^{2520}$ by

$$(\tilde{M}V)_i = \bigoplus_j L_{\tilde{M}_{ij}} \otimes_{\mathbb{Z}} V_j.$$

On morphisms use the identity on each first tensor factor. Its Grothendieck matrix is exactly $\tilde{M}$, since tensor products add degrees and multiply ranks. This formula realizes every one of the positive-cover coefficients as actual objects, over $\mathbb{Z}$.

Let $\mathcal{S}$ be internally graded finite free super abelian groups, with even morphisms, and let Π exchange parity. It satisfies $\Pi^2 = 1$ on objects and maps. Its super Grothendieck group is defined by direct-sum relations and $[\Pi V] = -[V]$; the signed rank character identifies that group with A . The parity-summing functor

$$Q : \mathcal{G}^{2520} \rightarrow \mathcal{S}^{1260}, \quad (V_i^+, V_i^-)_i \mapsto (V_i^+ \oplus \Pi V_i^-)_i$$

has induced map π , with every plus component placed in even parity before taking the direct sum. Define

$$(MW)_i = \bigoplus_j \left(L_{M_{ij}^+} \otimes W_j \right) \oplus \left(L_{M_{ij}^-} \otimes \Pi W_j \right).$$

There is a natural parity-preserving isomorphism

$$Q\tilde{M} \simeq MQ. \tag{117}$$

To verify it, expand the two sides. The even output summands on each side are $L_{M_{ij}^+} \otimes V_j^+$ and $L_{M_{ij}^-} \otimes V_j^-$; the odd output summands are $L_{M_{ij}^+} \otimes V_j^-$ and $L_{M_{ij}^-} \otimes V_j^+$. Match identical labelled tensor factors by the identity. This specifies every component, its inverse, and naturality. Thus the signed source action has an actual integral super-object correspondence with the full effective cover, not only an equality of matrix dimensions.

The quantum relations have not been imposed as functor isomorphisms by this definition. The nonzero defect just computed precludes that interpretation for the positive cover. The full generators on super Grothendieck groups do satisfy the source relations, through (117) and the proven matrix action. No assertion of a canonical-basis convolution model follows merely from the existence of these additive functors.

There is also a precise relation to the previously constructed four-state Frobenius cone. Let A, B, G, J be the graded groups with respective exponent sets

$$\begin{aligned} \{-6, -4, 4, 6\}, & \quad \{-7, -5, -3, -1, 1, 3, 5, 7\}, \\ \{-5, -3, -1, 1, 3, 5\}, & \quad \{-2, 0, 2\}. \end{aligned}$$

The two lifted paths from T^+ end in Q^+ with $P = G \otimes A$ and in Q^- with $R = J \otimes B$. Applying Q gives $P \oplus \Pi R$, with character $\alpha\gamma - d\beta = pC$ and with the intermediate labels T_1, T_2 still attached. The actual integral map $\theta : R \rightarrow P$ used in the Frobenius cone matches, in each weight, equal-position basis pairs in lexicographic order. Its twenty unit entries cancel twenty matched pairs; the cone has four residual even and four residual odd basis vectors. This cross-path differential is additional map data, not the differential of the split path object. The cover and the parity map therefore recover exactly its two branch objects before that differential is installed. More explicitly, place an arbitrary input V at T^+ and project the square functor

to the two labelled output sheets at Q . The extracted endpoint functors are $V \mapsto \mathbf{P} \otimes V$ and $V \mapsto \mathbf{R} \otimes V$. The map $\theta \otimes 1_V : \mathbf{R} \otimes V \rightarrow \mathbf{P} \otimes V$ is natural, and its cone is exactly the specified endpoint cone functor. Its underlying graded parity object is the image under $\mathbf{Q}$ of the two extracted paths. The identity of those graded objects is not claimed to be a chain map from the split zero-differential path complex to the cone.

26.6 What changes when the basis is allowed to mix

For each fixed $t > 0$ an effective cone for the four full generators does exist after an explicit type of nondiagonal source change. Here the exact source commutator is read as the integral formula $[E, F]|_{X_h} = [h](t)$, including its value at $t = 1$. This assertion can be proved without assuming a positive basis in the original labels.

Consider one colour on any of the present finite-dimensional specializations. If z is highest of weight n , the commutator gives by induction

$$EF^j z = [j](t)[n - j + 1](t)F^{j-1}z.$$

The induction uses the finite-sum identity $[j + 1][n - j] = [j][n - j + 1] + [n - 2j]$, verified by multiplying the Laurent sums, so it also holds at $t = 1$. Let k be the largest integer with $F^k z \neq 0$. Applying the formula to $F^{k+1}z = 0$ gives $[k + 1](t)[n - k](t)F^k z = 0$. For integral h , $[h](t) = 0$ exactly when $h = 0$. Thus $k = n \geq 0$. It follows that the divided string $z_i = F^i z / [i]!(t)$, $0 \leq i \leq n$, has all its stated vectors nonzero and satisfies

$$Fz_i = [i + 1](t)z_{i+1}, \quad Ez_i = [n - i + 1](t)z_{i-1},$$

where the out-of-range terms vanish. All nonzero coefficients are strictly positive.

To prove these strings span the entire representation, take its largest occurring weight n and a basis of its highest space. Their strings are independent: different levels have different weights, and applying the corresponding power of E to a relation at one level reduces it to a relation in that highest basis with a nonzero common coefficient. Their sum N is a submodule. Induct on dimension in the quotient, whose largest weight is smaller. For a quotient highest vector of weight m , lift it to z of weight m . Then $Ez \in N_{m+2}$. On every top string, $E : N_m \rightarrow N_{m+2}$ is surjective when the target occurs, because the coefficient just displayed is nonzero; the quotient string has $m \geq 0$, so there is no endpoint exception. Subtract a preimage to make the lift highest. Its length- m string maps isomorphically to the selected quotient string. These lifted strings are independent modulo N and split off the quotient. This completes the induction.

Apply the construction first to the V generators. The commuting W generators preserve each V highest subspace, where the same argument applies. If $H_{n,m}(t)$ is the simultaneous highest subspace of weight ratios (n, m) and a real basis of it is chosen, define $L_n(t)$ to have the divided-string basis e_i , $0 \leq i \leq n$, the displayed E, F action, and original $\mathfrak{gl}_2$ weights

$$\left(\frac{15 + n}{2} - i, \frac{15 - n}{2} + i \right).$$

The occurring n are odd, since all original total weights are fifteen, so these coordinates are integers. Define $L_m(t)$ in the same way for W . Then the map

$$\Phi_t : \bigoplus_{n,m} L_n(t) \otimes L_m(t) \otimes H_{n,m}(t) \longrightarrow X_A \otimes_{q \rightarrow t} \mathbb{R}, \quad e_i \otimes f_j \otimes z \longmapsto F_V^{(i)} F_W^{(j)} z$$

is therefore an isomorphism. The multiplicity factor has trivial quantum-group action in the domain. The formulas above prove that the map intertwines all raising and lowering generators; the original weight coordinates prove the Cartan and degree-fifteen central characters

as well. This is the specialization of the actual string construction, now justified at every positive t .

The nonnegative span of these string vectors is a simplicial cone preserved by all four generators and their Cartan diagonals: their matrices have nonnegative entries in these strings, and each Cartan eigenvalue is a positive power of t . Its matrix Φ_t is obtained from actual simultaneous highest kernels and the original divided powers, so it is an exact correspondence with the source. Theorem 26.1 proves that Φ_t cannot be a diagonal matrix times a permutation. Thus the fixed-source obstruction and this effective mixed-basis cone are related by an explicit proved intertwiner. The construction depends on choices in the highest multiplicity spaces and is not asserted to be the source-compatible canonical geometric realization sought in the nonstandard Riemann-hypothesis programme.

Reproduction and scope. The complete original matrices and basis, with their original coefficient and word data, are the frozen inputs in `agents/full_source_generators`. The accompanying `check_positive_cover.py` verifies their hashes, every edge sign, all four retained coefficients, both Laurent identities, every column of the inclusion and projection intertwiners, the two full original commutator identities, and every cover commutator defect. The independent sign audit checks graph connectivity, the cycle class, the shortest-cycle assertion and the same retained entries. The primary-source loci are GCT IV TeX lines 7240–7323 for sign coherence, 7324–7326 for the already stated sign-readjustment obstruction, and 7328–7350 for the printed example and its two paths; its frozen source hash is recorded in the machine receipt. The new obstruction is to a stated effective realization preserving the original basis up to diagonal gauge. The exact cover and mixed-string maps do not prove a nonstandard Riemann hypothesis, an asymptotic class-variety obstruction, or a Boolean or algebraic complexity-class separation.

27 An integral tableau basis for the exact plethystic maps

All constructions in this section are over an arbitrary commutative ring R with identity. In particular, neither an integer nor a determinant is inverted. A partition is a finite sequence $\lambda = (\lambda_1 \geq \dots \geq \lambda_\ell > 0)$. Write $d = |\lambda|$, $t = \lambda_1$, and $h_j = \#\{r : \lambda_r \geq j\}$ for its column lengths. The empty partition is allowed, with $d = t = \ell = 0$.

27.1 The intrinsic image and its polynomial realization

For a finite free R -module E define

$$C_\lambda(E) = \bigotimes_{j=1}^t \bigwedge^{h_j} E, \quad P_\lambda(E) = \bigotimes_{r=1}^\ell \text{Sym}^{\lambda_r} E. \quad (118)$$

An empty tensor product in this section is R . For each column, send an exterior product to the alternating tensor

$$v_1 \wedge \dots \wedge v_h \longmapsto \sum_{\sigma \in \mathfrak{S}_h} \text{sgn}(\sigma) v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(h)}.$$

This formula is multilinear and alternating over R : when two inputs are equal, transposing their labels pairs equal tensors with opposite coefficients. It therefore defines a map out of the exterior power, also in characteristic two. Tensor these maps in the column order $1, \dots, t$, relabel their tensor positions by the boxes (r, j) , reorder the positions into row order, and take the symmetric quotient in each row. Denote the resulting map by

$$\delta_{\lambda, E} : C_\lambda(E) \longrightarrow P_\lambda(E), \quad S_\lambda(E) = \text{im } \delta_{\lambda, E}. \quad (119)$$

Equation (119) is the Schur module convention used throughout this section. There is no additional row symmetrizer, factorial, or Young-idempotent normalization.

Let $e_1, \dots, e_n$ be an ordered basis of E . The module $\text{Sym}^a(E)$ has the basis $e_1^{\alpha_1} \dots e_n^{\alpha_n}$ with $\sum_i \alpha_i = a$. Indeed, the tensor-word basis is partitioned into permutation orbits, and the defining symmetry relations identify the words in each orbit; the free module on these orbits gives an inverse to this quotient description. The module $\bigwedge^a E$ has the basis $e_{i_1} \wedge \dots \wedge e_{i_a}$ for $i_1 < \dots < i_a$. To see this without an assumption on R , sort each word with its permutation sign and send a word with repeated indices to zero. This alternating multilinear rule gives an inverse to the map from the free module on the displayed increasing words.

Introduce independent variables $X_{r,i}$ for $1 \leq r \leq \ell$ and $1 \leq i \leq n$. The row monomial bases identify $P_\lambda(E)$ with the component of $R[X_{r,i}]$ having degree λ_r in the variables with first index r . For an ordered list $I = (i_1, \dots, i_a)$, where $a \leq \ell$, put

$$[I]_a = \det(X_{r,i_s})_{1 \leq r, s \leq a}, \quad [\emptyset]_0 = 1.$$

Repeated entries give zero, and permuting the list multiplies this polynomial by the permutation sign. For a filling T of the diagram of λ , write $I_j(T)$ for its j th column, read from top to bottom, and set

$$b_T = \prod_{j=1}^t [I_j(T)]_{h_j}. \quad (120)$$

Expanding the alternating tensors in the definition of δ shows term by term that b_T is the image of the column exterior tensor indexed by T . Consequently the b_T with strictly increasing columns generate $S_\lambda(E)$. A filling is called *semistandard* when its columns strictly increase and its rows weakly increase, using the alphabet $1, \dots, n$.

27.2 An alternating identity and an integral straightening proof

The following identity is proved in the polynomial ring, so it remains valid after every specialization of its coefficients.

Lemma 27.1 (The tail-head shuffle identity). *Let $\ell \geq a \geq b \geq k \geq 1$ and let $m = a - k + 1$. Let A be a list of column indices of length $k - 1$, let $U = (u_1, \dots, u_{a+1})$, and let B be a list of length $b - k$. For a subset $S = \{s_1 < \dots < s_m\} \subseteq \{1, \dots, a + 1\}$, let U_S be the sublist in the original position order; the complementary sublist has the same convention. Set*

$$\epsilon(S) = (-1)^{\sum_{v=1}^m (s_v - v)}.$$

Then

$$\sum_{\substack{S \subseteq \{1, \dots, a+1\} \\ |S|=m}} \epsilon(S) [A, U_S]_a [U_{S^c}, B]_b = 0. \quad (121)$$

Here the minors use initial row segments of the same matrix.

Proof. First let every entry of the lists denote a vector of R^a , and let the second determinant use its projection onto the first b coordinates. Keep A and B fixed. The left side is a multilinear function of the $a + 1$ vectors of U . It is alternating, as follows. Suppose two of these vectors have equal values. A term putting both in the first determinant, or both in the second determinant, is zero. Pair each remaining subset with the subset obtained by exchanging those two positions between the subset and its complement. The two determinant products in such a pair agree up to the signs needed to restore the position orders inside the two sublists. The shuffle sign is the sign of the permutation that lists S first and S^c second. Including the two internal reordering signs, the exchange is a single transposition of the full

list. Thus the two paired terms have opposite coefficients and cancel. This proves vanishing on equal inputs by an explicit pairing, without deducing it from skew symmetry or dividing by two.

Expand each of the $a + 1$ vectors in the a standard basis vectors of R^a . Multilinearity expresses the value as a sum of values on standard basis vectors. Each of these tuples repeats a vector, by the pigeonhole principle, so each value is zero by the preceding argument. The multilinear function therefore vanishes identically. Apply this result over the polynomial ring to the column vectors $(X_{1,i}, \dots, X_{a,i})^t$. Its projection onto the first b coordinates is precisely the column occurring in the second flag minor. This gives (121) with its stated signs and row sets. $\square$

Theorem 27.2 (Integral semistandard basis). *For every finite free R -module E with its specified ordered basis, the elements b_T indexed by semistandard tableaux of shape λ form an R -basis of $S_\lambda(E)$. The image in (119) is a direct summand as an R -module of $P_\lambda(E)$; the splitting need not be natural. For every ring homomorphism $R \rightarrow R'$, the natural base-change map*

$$R' \otimes_R S_\lambda(E) \longrightarrow S_\lambda(R' \otimes_R E) \quad (122)$$

is an isomorphism. No flatness hypothesis is needed.

Proof. If λ is empty, all three claims concern the identity map on R and follow directly. If $h_1 > n$, the first exterior factor is zero, and there are no strictly increasing first columns. Both the image and the claimed basis are empty, giving the zero module. We may therefore suppose that λ is nonempty and $h_1 \leq n$.

First prove spanning. Order all strictly column-increasing fillings by lexicographic order of the word obtained by reading their columns from left to right, and each column from top to bottom. This is a finite totally ordered set. A filling T that is not semistandard has adjacent columns

$$I = (i_1 < \dots < i_a), \quad J = (j_1 < \dots < j_b), \quad a \geq b,$$

and a row $k \leq b$ at which $i_k > j_k$. Apply Lemma 27.1 with

$$A = (i_1, \dots, i_{k-1}), \quad U = (i_k, \dots, i_a, j_1, \dots, j_k), \quad B = (j_{k+1}, \dots, j_b).$$

The subset $S_0 = \{1, \dots, a - k + 1\}$ has sign $+1$ and gives exactly $[I]_a[J]_b$. Move all the other summands to the other side of (121), and multiply by the unchanged minors in the remaining columns of T .

Every member of $(i_k, \dots, i_a)$ is larger than every member of $(j_1, \dots, j_k)$. Thus the entries of U are distinct. A subset $S \neq S_0$ replaces at least one entry of the left-column tail by a strictly smaller entry from the right-column head. If a resulting column has a repeated entry, its determinant is zero and the term is discarded. In any other term, sort the two columns numerically and multiply by the two sorting signs. The fixed left-column prefix is unchanged as a multiset; all inserted values are smaller than all removed values. At the first place where the sorted old and new left columns differ, the new column therefore has the smaller entry. All preceding columns of the entire filling are unchanged. The resulting filling is strictly smaller than T in the specified total order. The coefficient of the original term was exactly one, so this replacement uses only integer addition and subtraction. Induction on the finite total order proves that every generator b_T is an integral linear combination of the semistandard b_T . This proof works after passage to R as well.

For independence, order the variables by

$$X_{1,1} > X_{1,2} > \dots > X_{1,n} > X_{2,1} > \dots > X_{2,n} > \dots > X_{\ell,n},$$

and use lexicographic monomial order. If $i_1 < \cdots < i_a$, the leading monomial of $[I]_a$ is $X_{1,i_1} \cdots X_{a,i_a}$, with coefficient $+1$. Indeed, among determinant permutations the largest monomial must choose the smallest available column index for its first row, then the smallest remaining index for its second row, and so on. Monomial order is preserved by multiplication, so

$$\text{in}(b_T) = \prod_{(r,j) \in \lambda} X_{r,T(r,j)}, \quad \text{with coefficient } +1. \quad (123)$$

For a semistandard filling, this monomial determines each row's multiset of entries. A weakly increasing row is uniquely recovered from that multiset. Hence distinct semistandard tableaux have distinct leading monomials. In a nonzero linear combination over R , choose the largest leading monomial among the tableaux whose coefficients are nonzero. No other summand has a monomial that large. Its coefficient in the sum is the chosen nonzero coefficient times 1, and cannot vanish. This proves independence even when R has zero divisors.

For completeness, this monicity also produces a splitting. Work first over $\mathbb{Z}$ and list the semistandard polynomials in decreasing order of their distinct leading monomials. Given any polynomial in the finite row-degree component $P_\lambda(\mathbb{Z}^n)$, process this list in order. At each stage, extract the coefficient of the current leading monomial, record it, and subtract that coefficient times the corresponding b_T . Later subtractions cannot change a leading monomial already processed, since all their monomials are smaller. Return the sum of the recorded multiples of b_T . This is a $\mathbb{Z}$ -linear map into their span, and it is the identity on each b_T . It is therefore a retraction of the inclusion of that span. All its matrix entries are integers, so base change gives the stated retraction over every R .

Finally the map (122) sends $1 \otimes b_T$ to the polynomial b_T with coefficients in R' . The basis just proved on both sides shows that it is an isomorphism. The splitting explains in particular why taking this image commutes with arbitrary, possibly nonflat, base change. $\square$

27.3 Functoriality, all matrix entries, and the exact diagonal

Proposition 27.3 (Functorial maps without denominators). *Every R -linear map $f : E \rightarrow F$ between finite free modules induces a map $S_\lambda(f) : S_\lambda(E) \rightarrow S_\lambda(F)$. These maps preserve identities and composition and commute with arbitrary base change. In the tableau bases, every matrix entry is an integer polynomial, homogeneous of degree $|\lambda|$ in the entries of f .*

Proof. The alternating tensor map commutes with $f^{\otimes h}$ term by term, tensor reordering commutes with applying f in every slot, and symmetric quotient maps are natural. Thus the square

$$\begin{array}{ccc} C_\lambda(E) & \xrightarrow{\delta_{\lambda,E}} & P_\lambda(E) \\ \downarrow C_\lambda(f) & & \downarrow P_\lambda(f) \\ C_\lambda(F) & \xrightarrow{\delta_{\lambda,F}} & P_\lambda(F) \end{array}$$

commutes. Restricting its right arrow to the images defines the claimed map. The identities and composition laws hold on the ambient row-symmetric modules and hence on their submodules. All constructions use tensor, exterior, and symmetric operations given by the displayed integral formulas. Together with Theorem 27.2, this proves the base-change statement.

Here is a formula that also proves the claim about every matrix entry. Choose bases e_i and f_j , and write a matrix M for the map, so that $f(e_i) = \sum_j M_{j,i} f_j$. For a strictly increasing column I of length a , direct multilinear expansion and signed sorting in the exterior power give

$$\left(\bigwedge^a f \right) (e_{i_1} \wedge \cdots \wedge e_{i_a}) = \sum_{\substack{J \text{ increasing} \\ |J|=a}} \det(M_{J,I}) f_{j_1} \wedge \cdots \wedge f_{j_a}. \quad (124)$$

Terms with repeated row indices vanish. For each distinct row set the terms with the possible row orders sum, with their sorting signs, to the determinant in (124). For the columns $I_c(T)$ of a tableau T , take the product of these expansions and then apply $\delta_{\lambda,F}$. This gives

$$S_\lambda(f)b_T = \sum_{J_1, \dots, J_t} \left(\prod_{c=1}^t \det(M_{J_c, I_c(T)}) \right) b_{(J_1, \dots, J_t)}. \quad (125)$$

Apply the terminating integer straightening procedure already proved to each filling on the right. Every product of minors in its coefficient has total degree $\sum_c h_c = |\lambda|$. Straightening multiplies by integers and adds these products, so each final tableau coefficient is an integer homogeneous polynomial of that degree. This is a finite formula: there are $\prod_c \binom{\text{rank } F}{h_c}$ choices in the sum, and straightening strictly decreases an order on the finite set of column-increasing fillings. No complexity bound beyond this finite procedure is asserted here. $\square$

Theorem 27.4 (The exact signed diagonal map). *Let E and F have ordered bases of the same cardinality n , and let $D : E \rightarrow F$ have the specified matrix $\text{diag}(d_1, \dots, d_n)$, with arbitrary $d_i \in R$. For a semistandard tableau T , let $c_i(T)$ count its entries equal to i , and put*

$$w_T = \prod_{i=1}^n d_i^{c_i(T)}.$$

With the same tableau order on the two bases, the exact matrix of $S_\lambda(D)$ is $\text{diag}(w_T)_T$. In particular, there are the following specified maps and modules:

$$\ker S_\lambda(D) = \bigoplus_T \text{Ann}_R(w_T) b_T(E), \quad (126)$$

$$\text{im } S_\lambda(D) = \bigoplus_T (w_T) b_T(F), \quad (127)$$

$$\text{coker } S_\lambda(D) \cong \bigoplus_T R/(w_T). \quad (128)$$

The last isomorphism sends the class of $\sum_T a_T b_T(F)$ to $(a_T \bmod (w_T))_T$.

Proof. Apply D in the column exterior tensor defining b_T . Each occurrence of e_i contributes exactly the scalar d_i , and the product of these scalars is w_T . The commutative square in Proposition 27.3 therefore gives $S_\lambda(D)b_T(E) = w_T b_T(F)$ before any straightening. The tableau basis theorem proves the asserted matrix identity. A vector is in its kernel precisely when every coordinate a_T satisfies $w_T a_T = 0$, which gives the first formula. Each target coordinate of an image is independently an element of (w_T) , giving the second. The displayed residue map is surjective and has exactly this image as its kernel, proving the third. This also treats $d_i = 0$ and nonzero zero divisors. Signed elements d_i have not been replaced by associates or absolute values. $\square$

27.4 Iteration and the plethystic exponent matrix

Let $\mathcal{T}_\lambda(n)$ be the tableau set in Theorem 27.2, ordered by its column-reading words. Put $N = |\mathcal{T}_\lambda(n)|$. Regard this ordered set as the alphabet for semistandard tableaux U of a second partition μ . Write $c_T(U)$ for the number of entries of U equal to the alphabet letter T , and define

$$\Gamma_i(U) = \sum_{T \in \mathcal{T}_\lambda(n)} c_T(U) c_i(T). \quad (129)$$

Here the plethysm is the composed functor $S_\mu \circ S_\lambda$, with this integral Schur-image convention at both stages.

Corollary 27.5 (Nested Schur maps over the original ring). *The outer tableau elements b_U form an R -basis of $S_\mu(S_\lambda(E))$. For the specified diagonal map D of Theorem 27.4, the exact nested map is*

$$S_\mu(S_\lambda(D))b_U(E) = \left(\prod_{i=1}^n d_i^{\Gamma_i(U)} \right) b_U(F). \quad (130)$$

Every exponent is a nonnegative integer, and $\sum_i \Gamma_i(U) = |\mu||\lambda|$. The kernel, image, and cokernel are given by (128) with the index U and scalar $\prod_i d_i^{\Gamma_i(U)}$. For a general matrix M , every matrix entry of the composed functor is an integer polynomial homogeneous of degree $|\mu||\lambda|$ in M .

Proof. The inner module is finite free with the explicitly ordered basis $(b_T)_{T \in \mathcal{T}_\lambda(n)}$ by the basis theorem. Apply the same basis theorem to that free module and μ . The inner map is diagonal, with entries $w_T = \prod_i d_i^{c_i(T)}$. Apply the diagonal theorem to this map and the outer shape μ . Its scalar on b_U is exactly

$$\prod_T w_T^{c_T(U)} = \prod_T \prod_i d_i^{c_T(U)c_i(T)} = \prod_i d_i^{\Gamma_i(U)}.$$

The number of letters of an inner tableau is $|\lambda|$, so $\sum_i c_i(T) = |\lambda|$; the number of letters of an outer tableau is $|\mu|$, so $\sum_T c_T(U) = |\mu|$. Summing (129) proves the degree identity. The coordinate kernel and quotient formulas follow from the diagonal theorem, with no additional ring hypotheses. For a general M , apply the integer homogeneous matrix formula first to λ , and then substitute those degree- $|\lambda|$ entries into the degree- $|\mu|$ formula for the outer functor. This proves the last assertion. Empty shapes and zero inner rank follow directly from the conventions for empty tensor products and the absence of nonempty tableaux on an empty alphabet. $\square$

The construction above retains the entire composed functor and its tableau matrix. It does not require a decomposition into irreducible Schur summands. The proof gives ordinary integral Schur operations on the specified linear maps; identifying them with a particular nonstandard quantum-group construction requires the corresponding specified representation map. For a primary comparison with the flag-minor standard monomial model, see D. C. Lax, *Accessible Proof of Standard Monomial Basis for Coordinatization of Schubert Sets of Flags*, [arXiv:1508.02744](https://arxiv.org/abs/1508.02744), Sections 3–4. All basis, straightening, functoriality, and diagonal assertions used here were proved above over R .

28 Exact Schur transport of the integral generator endpoint

Retain the coefficient ring, original coefficient and original endpoint factorization of Section 9:

$$\begin{aligned} O &= \mathbb{Z}[1/2], & R &= O[q, q^{-1}], & t &= q - 1, & r &= q - q^{-1}, & h &= [7]_q[3]_q, \\ C &= q^{10} - q^4 - q^{-4} + q^{-10} = r^2 h. \end{aligned}$$

The ordered rank-four modules and maps are

$$E_C \xrightarrow{H} B_R \xrightarrow{V} N_R, \quad H = \text{diag}(1, 1, 1, h), \quad V = \text{diag}(1, 1, 1, r^2), \quad D_C = VH.$$

The fourth coordinate of E_C is the twisted source top line, and that of N_R is the bottom line, with the three specified identity summands retained. Thus D_C is the stabilized actual divided-square endpoint, not a new full quantum-group representation. The displayed bases identify each of the three modules with R^4 as an R -module; we keep their names to distinguish domains and codomains.

All Schur functors in this section are the integral, column-exterior to row-symmetric Schur functors described in Theorem 27.2. No factorial or hook length is inverted. We apply these functors to ordinary module maps. They retain the stated μ_2^2 characters and the semilinear bar involution, by functoriality. They also retain the explicitly diagonal Cartan action and zero lowering actions on the stabilized endpoint modules. This assertion does not apply an ordinary symmetric-group action to a generic full quantum tensor power.

28.1 The exact diagonal map for every partition

For a partition λ , let $\text{Tab}_d(\lambda)$ be its tableaux with entries in $\{1, \dots, d\}$ weakly increasing along rows and strictly increasing down columns. The empty shape has one empty tableau. A shape with more than d rows has no such tableaux and its Schur module is zero. For $T \in \text{Tab}_4(\lambda)$ write $w(T)$ for the number of entries equal to 4. Put

$$d_\lambda = |\text{Tab}_4(\lambda)|, \quad m_\lambda(k) = |\{T : w(T) = k\}|, \quad P_\lambda(z) = \sum_{k \geq 0} m_\lambda(k) z^k.$$

These definitions are finite and effective; they do not presume a plethysm decomposition or a positivity conjecture.

Theorem 28.1. *In the corresponding tableau bases of the three modules, the maps $S_\lambda(H)$, $S_\lambda(V)$ and $S_\lambda(D_C)$ have diagonal entries $h^{w(T)}$, $r^{2w(T)}$ and $C^{w(T)}$, respectively. Their composition is exactly $S_\lambda(D_C) = S_\lambda(V)S_\lambda(H)$. Each map is injective over R , and*

$$\text{coker } S_\lambda(D_C) = \bigoplus_{k \geq 1} (R/(C^k))^{m_\lambda(k)}.$$

Writing $\mathcal{L}_C, \mathcal{L}_r$ for their actual image lattices inside $S_\lambda(N_R)$ gives the exact sequence

$$0 \longrightarrow \bigoplus_{k \geq 1} (R/(h^k))^{m_\lambda(k)} \xrightarrow{i} \bigoplus_{k \geq 1} (R/(C^k))^{m_\lambda(k)} \xrightarrow{\pi} \bigoplus_{k \geq 1} (R/(r^{2k}))^{m_\lambda(k)} \longrightarrow 0, \quad (131)$$

where $i(\bar{f}_T) = \overline{r^{2w(T)} f_T}$ and $\pi(\bar{f}_T) = \bar{f}_T$. The three terms are, in that order, $\mathcal{L}_r/\mathcal{L}_C$, $S_\lambda(N_R)/\mathcal{L}_C$, and $S_\lambda(N_R)/\mathcal{L}_r$.

Proof. Each term in the column alternation followed by row multiplication has exactly the content of T . A diagonal map therefore multiplies the tableau vector by the product of its diagonal entries, giving the three stated scalars. The integral tableau basis ensures these are matrices over R itself. Naturality, or multiplication of their individual diagonal entries, gives the composition. Nonzero diagonal entries are non-zero-divisors in the domain R , proving injectivity. The cokernel of multiplication by a scalar a on R is $R/(a)$; the coordinates with $w = 0$ are identity maps and contribute zero. For each $k > 0$, multiplication by r^{2k} sends $R/(h^k)$ into $R/(C^k)$ since $r^{2k}h^k = C^k$. If its image is zero, the equality $r^{2k}f = C^k g$ cancels to $f = h^k g$. Projection onto $R/(r^{2k})$ is onto and its kernel consists exactly of these multiples of r^{2k} . Taking the finite direct sum proves every map and exactness assertion, including the image-lattice interpretations. $\square$

Pad λ with zero rows to length four. The exact smallest and largest exponents, for a nonzero Schur module, are

$$\min_T w(T) = \lambda_4, \quad \max_T w(T) = \lambda_1. \quad (132)$$

Every column of height four must contain 4, so the lower bound is λ_4 . Fill row i constantly with i to attain it. At most one 4 occurs per column, giving the upper bound λ_1 . In a column

of height a , place $5 - a, \dots, 4$ in order. As column heights weakly decrease from left to right, these entries weakly increase along each row. This is a tableau attaining the upper bound.

The complete multiplicities also have an explicit smaller formula. Deleting the entries 4 leaves a shape μ with at most three rows and

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \lambda_3 \geq \mu_3 \geq \lambda_4.$$

Conversely each tableau of that shape on $1, 2, 3$ extends uniquely by filling λ/μ with 4. Indeed the displayed inequalities say precisely that the added boxes lie at row ends and no two share a column; the row and column conditions then follow. Thus

$$m_\lambda(k) = \sum_{\substack{\lambda_i \geq \mu_i \geq \lambda_{i+1} \ (1 \leq i \leq 3) \\ |\lambda| - |\mu| = k}} \frac{(\mu_1 - \mu_2 + 1)(\mu_2 - \mu_3 + 1)(\mu_1 - \mu_3 + 2)}{2}. \quad (133)$$

The displayed summand is an integer count, not a division in R . To prove it directly, delete the 3 entries next. The remaining two-row shape (a, b) satisfies $\mu_2 \leq a \leq \mu_1$ and $\mu_3 \leq b \leq \mu_2$. Deleting its entries 2 leaves a one-row shape of any length from b to a , giving $a - b + 1$ possibilities. Hence its total is

$$\sum_{a=\mu_2}^{\mu_1} \sum_{b=\mu_3}^{\mu_2} (a - b + 1) = \frac{(\mu_1 - \mu_2 + 1)(\mu_2 - \mu_3 + 1)(\mu_1 - \mu_3 + 2)}{2},$$

by the finite arithmetic-progression sums. This proves (133) without a dimension formula assumption.

28.2 Nested Schur maps and the precise plethysm character

Fix another partition μ with $d_\mu > 0$, enumerate its integral tableau basis $T_1, \dots, T_{d_\mu}$ so that the weights $w_i = w(T_i)$ weakly increase, and choose any fixed order within a tie. A tableau $U \in \text{Tab}_{d_\mu}(\lambda)$ indexes an integral basis vector of $S_\lambda(S_\mu(E_C))$. Its exponent is

$$W(U) = \sum_{\text{boxes } b \text{ of } \lambda} w_{U(b)}, \quad M_{\lambda, \mu}(k) = |\{U : W(U) = k\}|.$$

Apply Theorem 28.1 to a diagonal map whose entries now are $C^{w_1}, \dots, C^{w_{d_\mu}}$. It proves, over the unchanged ring R , the exact matrices

$$S_\lambda(S_\mu(D_C)) = \text{diag}_U(C^{W(U)}), \quad S_\lambda(S_\mu(H)) = \text{diag}_U(h^{W(U)}), \quad S_\lambda(S_\mu(V)) = \text{diag}_U(r^{2W(U)}). \quad (134)$$

All injectivity, cokernel and conductor formulas above hold with $m_\lambda(k)$ replaced by $M_{\lambda, \mu}(k)$. This is a direct application of the proved basis and scalar calculation; it assumes neither exactness of Schur functors on arbitrary short exact sequences nor an integral direct-sum decomposition of a plethysm.

For completeness define the symmetric polynomial $s_\mu(x_1, \dots, x_4) = \sum_T \prod_b x_{T(b)}$ by its tableau character. The multivariable diagonal action proved above shows that the character of the nested module is obtained by evaluating s_λ on the list of monomials $\prod_b x_{T_i(b)}$, with each tableau contributing its own entry, including repeated monomials. This substitution is precisely $s_\lambda[s_\mu]$ in four variables. Consequently the exact one-variable identity is

$$\sum_k M_{\lambda, \mu}(k) z^k = s_\lambda(z^{w_1}, \dots, z^{w_{d_\mu}}) = s_\lambda[s_\mu](1, 1, 1, z). \quad (135)$$

Each coefficient on this restriction is a proved nonnegative integer count. This equation does not determine the decomposition into irreducible GL_4 characters: it is the restriction

of a character along one specified one-parameter subgroup. In particular this computation supplies a transport map and its invariant factors, not a plethysm-multiplicity obstruction for a class variety.

If λ has at most d_μ rows, the exact extreme exponents are

$$a_{\lambda,\mu} = \sum_i \lambda_i w_i, \quad b_{\lambda,\mu} = \sum_i \lambda_i w_{d_\mu+1-i}. \quad (136)$$

In a column of height a , the entry in its i th box is at least i and at most $d_\mu - a + i$. Summing the corresponding ordered weights proves the lower and upper bounds. Filling row i by i attains the lower bound. Filling each height- a column with $d_\mu - a + 1, \dots, d_\mu$ attains the upper bound and is row weakly increasing, for the same reason as in (132). Grouping the lower and upper column sums by the row lengths gives exactly (136).

Let $d_0 = m_\mu(0) = |\text{Tab}_3(\mu)|$. Weight zero in the nested module occurs precisely when every entry labels an inner tableau of weight zero. Therefore

$$M_{\lambda,\mu}(0) = |\text{Tab}_{d_0}(\lambda)|. \quad (137)$$

This also handles $d_0 = 0$: only the empty outer shape has a nonzero rank. At $C = 0$ the matrices are diagonal with an identity block of this size and zero elsewhere, giving their exact rank, kernel and cokernel over every base-change ring, not only over fields.

For example $S_{(2)}(R^4)$ has inner weights $0^6, 1^3, 2^1$. An outer two-box row chooses an unordered pair with repetitions. The five weight counts for $S_{(2)}(S_{(2)}(D_C))$ are

$$(21, 18, 12, 3, 1) : \quad \binom{7}{2}, \quad 6 \cdot 3, \quad 6 \cdot 1 + \binom{4}{2}, \quad 3 \cdot 1, \quad 1.$$

Its diagonal therefore has 21 identity entries, 18 entries C , 12 entries C^2 , 3 entries C^3 , and one entry C^4 . No irreducible decomposition was needed for this exact matrix.

There is also a uniform determinant formula. Let F be either S_λ or $S_\lambda \circ S_\mu$, let n be its polynomial degree and d its rank on R^4 . Then

$$\det F(D_C) = C^{nd/4}, \quad \sum_k k \text{ rank } F(R^4)_k = nd/4 \in \mathbb{Z}. \quad (138)$$

To prove the divisibility and exponent, grade the integral tableau basis by the four original contents. Every vector has total content n . A permutation matrix exchanging any two original basis vectors induces, by functoriality, an integral isomorphism exchanging the corresponding weight spaces. The sums of the four content coordinates over all basis vectors are thus equal integers, whose total is nd . The fourth sum is the sum of the exponents in the diagonal matrix. Taking its determinant proves the formula, including its positive sign in the matched ordered bases.

28.3 Integral specialization, residual primes and exact orders

For a weight $k > 0$, put $Q_{h,k} = R/(h^k)$, $Q_{C,k} = R/(C^k)$ and $Q_{r,k} = R/(r^{2k})$. The resolution $0 \rightarrow R \xrightarrow{t} R \rightarrow O \rightarrow 0$ computes every derived specialization used here. The ideal (t) is prime, and $h(1)^k = 21^k \neq 0$ in the domain O . Hence $\text{Tor}_1^R(Q_{h,k}, O) = 0$: if $tf = h^k g$, reduction modulo t forces $g(1) = 0$, and cancellation gives $f \in (h^k)$. The two other first Tor groups are free of rank one over O , with their exact generators C^k/t and r^{2k}/t . Indeed $tf = C^k g$ forces $f = (C^k/t)g$, and this representative vanishes modulo C^k exactly when $g \in (t)$; the same proof applies to r^{2k} . All higher Tor groups vanish by the length-one resolution. The specialized sequence for this weight is therefore

$$0 \longrightarrow O \xrightarrow{21^k} O \xrightarrow{\text{mod } 21^k} O/(21^k) \xrightarrow{0} O \xrightarrow{1} O \longrightarrow 0. \quad (139)$$

Projection sends C^k/t to h^k times r^{2k}/t , giving the first displayed scalar. Lift r^{2k}/t into $Q_{C,k}$ and multiply by t : the result r^{2k} is the image of $1 \in Q_{h,k}$. Thus the connecting map has the stated positive sign and sends the chosen generator to 1. The next map multiplies by r^{2k} and vanishes after specialization; the last is the identity. The entire Schur or nested Schur sequence is the direct sum of these sequences with the exact weight multiplicities already proved.

The specialized map $F(D_C)_0$ has kernel and cokernel $O^{d-m(0)}$ in the positive-weight coordinates. The induced map between its kernel and that of $F(V)_0$ is $\text{diag}_{k>0}(21^k)$ with repetitions $m(k)$. It is injective and its cokernel is

$$\bigoplus_{k>0} (O/(21^k))^{m(k)} \simeq \bigoplus_{k>0} ((O/(3^k)) \oplus (O/(7^k)))^{m(k)}.$$

For the final isomorphism use the map taking a class to its two residues. Since 3^k and 7^k are coprime, choose integers a, b with $a3^k + b7^k = 1$; an inverse sends (x, y) to $b7^k x + a3^k y$ modulo 21^k . Its two compositions are the identity. This proves the two residual prime modules without discarding the original factor 21^k .

It is essential to specify which valuation is being used. At the height-one vertical prime (ℓ) of R , for $\ell = 3$ or 7 , the polynomial C is a unit of $R_{(\ell)}$, since its Laurent coefficients include 1. Every transported map is consequently an isomorphism there; its vertical valuations are zero. The ideal (ℓ, t) gives a two-dimensional local ring and does not itself specify a discrete valuation. The exceptional orders below instead refer to the discrete valuation rings

$$D_0 = \mathbb{Q}[q, q^{-1}]_{(t)}, \quad D_\ell = \mathbb{F}_\ell[q, q^{-1}]_{(t)} \quad (\ell \text{ odd}).$$

Write $u = (q+1)/q$, so $r = tu$ and $u(1) = 2$. The exact orders and leading coefficients of the original C are

coefficient field	$\text{ord}_t C$	$t^{-\text{ord}_t C} C \bmod t$
$\mathbb{Q}$	2	84
$\mathbb{F}_3$	4	1
$\mathbb{F}_7$	8	5
$\mathbb{F}_\ell, \ell \neq 2, 3, 7$	2	$84 \bmod \ell$.

Here $C = t^2 u^2 h$ gives the characteristic-zero and ordinary-prime rows. For an odd prime ℓ , multiply the finite sum for $[\ell]_q$ by $q^{\ell-1}$ and then by $q^2 - 1$. In characteristic ℓ the result is $q^{2\ell} - 1 = (q^2 - 1)^\ell$. Cancellation in the polynomial domain gives $[\ell]_q = (q - q^{-1})^{\ell-1}$. Hence $C = r^4 [7]_q$ in characteristic 3 and $C = r^8 [3]_q$ in characteristic 7. Their leading residues are $2^4 \cdot 7 = 1$ in $\mathbb{F}_3$ and $2^8 \cdot 3 = 5$ in $\mathbb{F}_7$, proving the table exactly.

Let v be the relevant middle-column integer 2, 4, or 8, and c the last-column residue. Raising the retained equality to the k th power proves

$$C^k = t^{vk} \varepsilon_k, \quad \varepsilon_k \in D_\ell^\times, \quad \varepsilon_k \bmod t = c^k, \quad \text{coker } F(D_C) \simeq \bigoplus_{k>0} (D_\ell/(t^{vk}))^{m(k)}. \quad (140)$$

This is an explicit Smith form: multiply each matrix row by the inverse unit ε_k^{-1} , equivalently use target basis vector ε_k times the original vector, retaining that unit and its specified leading residue as part of the original map. Ordered by k , consecutive powers of t divide one another. Its length is $v \sum_k km(k)$, and the exact least and largest matrix-entry orders are v times the exponents in (132) or (136). More precisely, the j th filtration quotient of the cokernel, for $j \geq 0$, has residue-field dimension

$$\dim(t^j \text{coker} / t^{j+1} \text{coker}) = \sum_{k: vk > j} m(k).$$

For a single cyclic summand the classes $1, t, \dots, t^{vk-1}$ are a basis, proving both length and filtration formulas by direct sum.

The mixed-prime arithmetic can also be retained before reduction. In $\mathbb{Z}_\ell[[t]]$, with $q = 1+t$ and $\ell = 3$ or 7 , define

$$E_\ell(t) = \Phi_\ell(1+t) = \frac{(1+t)^\ell - 1}{t} = \sum_{j=0}^{\ell-1} \binom{\ell}{j+1} t^j.$$

Here the quotient is the displayed polynomial identity. Its constant term is ℓ , its leading term is $t^{\ell-1}$, and each other coefficient is divisible by ℓ ; the constant term is not divisible by ℓ^2 . The exact quantum factor identity

$$[\ell]_q = q^{1-\ell} \Phi_\ell(q) \Phi_{2\ell}(q), \quad \Phi_{2\ell}(q) = \sum_{j=0}^{\ell-1} (-q)^j,$$

follows by multiplying the two geometric sums; $\Phi_{2\ell}(1) = 1$. The other quantum factor in h has value 7 when $\ell = 3$ and 3 when $\ell = 7$. Thus, writing that factor as $h_{\neq \ell}$, the full equality is

$$C^k = t^{2k} E_\ell(t)^k U_\ell(t)^k, \quad U_\ell = u^2 q^{1-\ell} \Phi_{2\ell}(q) h_{\neq \ell} \in \mathbb{Z}_\ell[[t]]^\times. \quad (141)$$

The constant terms of U_3, U_7 are respectively 28 and 12 , which are units in the stated coefficient rings. This retains the horizontal factor t^{2k} and the separate arithmetic factor E_ℓ^k ; it does not replace a two-dimensional local ring by an unspecified valuation. Modulo ℓ , $E_\ell = t^{\ell-1}$, recovering exactly the orders $4k$ and $8k$.

There is an additional first Tor term if the conductor sequence is first reduced to characteristic 3 or 7 and then specialized at $t = 0$. Over the corresponding Laurent polynomial ring put $\bar{h} = h \bmod \ell$; it is nonzero and divisible by t . The three first Tor generators are now $\bar{h}^k/t$, $\bar{C}^k/t$, and $\bar{r}^{2k}/t$. Multiplication by $\bar{r}^{2k}$ sends the first exactly to the second; projection sends the second to $\bar{h}^k$ times the third, which is zero on the residue field. The connecting map again sends the third generator to 1 . Therefore, for every positive weight, the full sequence over $\mathbb{F}_\ell$ is

$$0 \longrightarrow \mathbb{F}_\ell \xrightarrow{1} \mathbb{F}_\ell \xrightarrow{0} \mathbb{F}_\ell \xrightarrow{1} \mathbb{F}_\ell \xrightarrow{0} \mathbb{F}_\ell \xrightarrow{1} \mathbb{F}_\ell \longrightarrow 0.$$

These descriptions follow from the same length-one resolution; in particular the extra Tor term has been calculated, not omitted.

Finally, all exponents in the original integral maps are nonnegative. A negative valuation does occur for the actual inverse after the specified localization: $F(D_C)^{-1}$ has entries C^{-k} over $R[C^{-1}]$ and orders $-vk$ over the fraction field of D_ℓ . If a positive-weight coordinate occurs, this inverse cannot be an R -linear inverse on the original lattices: an inverse image of its unit target vector would require $C^{-k} \in R$, which is impossible since C vanishes at $q = 1$. Its precise failure to be integral is the cokernel already computed in (140). This supplies an exact pole/defect correspondence without identifying a negative Laurent coefficient with a negative plethysm multiplicity.

28.4 Exact negative inertia at the retained circle character

Retain the circle real structure $q^* = q^{-1}$, with conjugation on coefficients, and the original exact character from the conic calculation:

$$q_0 = \frac{\sqrt{14} + i\sqrt{2}}{4}, \quad q_0^{-1} = \frac{\sqrt{14} - i\sqrt{2}}{4}, \quad p_0 = \frac{\sqrt{14}}{2}, \quad r_0 = \frac{i}{\sqrt{2}}, \quad a_0 = \frac{1}{2}, \quad s_0 = \frac{1}{4}.$$

The coefficient field is $K_0 = \mathbb{Q}(\sqrt{14}, i\sqrt{2}) \subset \mathbb{C}$ with its displayed embedding and usual conjugation. The product of the two displayed values for q_0 is $(14 + 2)/16 = 1$, so evaluation is a homomorphism respecting the circle involution. To check every factor, put $x_0 = q_0^2 + q_0^{-2} = p_0^2 - 2 = 3/2$. Then

$$[3]_{q_0} = x_0 + 1 = \frac{5}{2}, \quad [7]_{q_0} = x_0^3 + x_0^2 - 2x_0 - 1 = \frac{13}{8}.$$

The second identity expands $[7]_q$ as $(q^6 + q^{-6}) + (q^4 + q^{-4}) + (q^2 + q^{-2}) + 1$ and uses $q^6 + q^{-6} = x^3 - 3x$, $q^4 + q^{-4} = x^2 - 2$. Consequently the full factorization retains the exact signs

$$r_0^2 = -\frac{1}{2}, \quad h(q_0) = \frac{65}{16} > 0, \quad C(q_0) = -\frac{65}{32}. \quad (142)$$

Let F be either Schur functor already constructed, and identify its source and target *as based vector spaces* by the map taking each source tableau basis vector to the equally labelled target vector. This specifies the endomorphism A_F representing $F(D_C)(q_0)$ on the source; it does not infer an equality of the original source and target modules. Give this source the positive Hermitian form with its specified tableau basis orthonormal. Because every diagonal entry of A_F is the real number $(-65/32)^k$, the endomorphism is selfadjoint for this form. The Hermitian form that it defines has the complete formula

$$B_F(u, v) = \sum_T (-65/32)^{w(T)} \overline{u_T} v_T$$

for a single Schur functor, and the same formula with $U, W(U)$ for a nested one. In particular its positive, negative and zero indices are exactly

$$n_+ = \sum_{k \text{ even}} m(k), \quad n_- = \sum_{k \text{ odd}} m(k), \quad n_0 = 0. \quad (143)$$

Indeed the even and odd coordinate spans give a direct orthogonal sum of positive and negative definite spaces. A subspace of dimension larger than the even-coordinate count intersects the odd-coordinate space nontrivially by the dimension formula, so cannot be positive definite. Interchanging even and odd proves maximality for the negative index. Every diagonal coefficient is nonzero, so the radical is zero. This proves the inertia statement over the specified embedded field and after extension to $\mathbb{C}$.

The signature is thus the exact character evaluation

$$n_+ - n_- = P_\lambda(-1) = s_\lambda(1, 1, 1, -1)$$

for a single Schur functor, and

$$n_+ - n_- = s_\lambda[s_\mu](1, 1, 1, -1) = s_\lambda(\underbrace{1, \dots, 1}_{m_\mu(\text{even})}, \underbrace{-1, \dots, -1}_{m_\mu(\text{odd})})$$

for a nested one. The last equality follows by evaluating each inner-tableau monomial at $(1, 1, 1, -1)$; it is $+1$ or -1 according to its weight parity. The scalar $h(q_0)^k$ is positive for every coordinate. Therefore $F(V)(q_0)$, whose corresponding coefficients are $(-1/2)^k$, has exactly the same inertia. The actual conductor factor $F(H)(q_0)$ is positive definite in these matched bases and their matrices still multiply to $F(D_C)(q_0)$. This proves the propagation of the negative directions through the actual endpoint factorization, without discarding the positive residual factor.

For $S_{(2)} \circ S_{(2)}$ the previously computed profile gives $n_+ = 21 + 12 + 1 = 34$, $n_- = 18 + 3 = 21$, and signature 13. This also shows that the signature need not stay negative under iteration even when some negative directions survive: the exact parity-weight character, not an unsigned dimension, determines it. No entry of $m(k)$ has become negative. These Hermitian forms are specified forms on the endpoint Schur modules; no positivity form of the nonstandard RH programme or spectral membership of s_0 has been identified with them.

28.5 The original shape and an actual specialized source projector

For the original shape $\nu = (7, 5, 3)$, formula (133) is the following explicit finite sum:

$$\begin{aligned} & \sum_{a=5}^7 \sum_{b=3}^5 \sum_{c=0}^3 \frac{(a-b+1)(b-c+1)(a-c+2)}{2} z^{15-a-b-c} \\ &= 27 + 81z + 162z^2 + 270z^3 + 300z^4 + 252z^5 + 126z^6 + 42z^7. \end{aligned}$$

The exponents of $S_\nu(D_C)$, with multiplicities, are therefore exactly the eight displayed coefficients. Their sum is 1260 and their weighted sum is 4725. In characteristic zero at $q = 1$, the map has rank 27, kernel and cokernel dimension 1233, and cokernel length 9450 over D_0 . The corresponding lengths over D_3, D_7 are 18900 and 37800. These follow by multiplying 4725 by the proved orders 2, 4, 8. They describe this particular integral Schur transport and its stated base changes. At the distinct circle character q_0 of (142), its inertia is

$$\begin{aligned} (n_+, n_-, n_0) &= (27 + 162 + 300 + 126, 81 + 270 + 252 + 42, 0) = (615, 645, 0), \\ \text{signature} &= -30. \end{aligned}$$

This is a negative signature with 645 individually specified negative tableau directions, and is an immediate exact instance of (143).

There is also a proved map to the full original source, over its already justified rational specialization. Put $X = V \otimes W$ with the unchanged alphabet

$$x_1 = v_1 \otimes w_1, \quad x_2 = v_1 \otimes w_2, \quad x_3 = v_2 \otimes w_1, \quad x_4 = v_2 \otimes w_2.$$

Section 4 constructs the isomorphism $s_\nu : L_{\mathbb{Q}} \rightarrow Q_\nu(X)$ from the rational specialized full source lattice to the exterior-column classical relation quotient, and proves its inverse j_ν . Here $Q_\nu(X)$ denotes that precise quotient presentation, retaining its original column order. To compare it with the integral Schur-image convention in this section, define

$$\eta_\nu : Q_\nu(X) \longrightarrow S_\nu(X), \quad [\text{column exterior tensor of } T] \longmapsto b_T.$$

Every classical adjacent column relation is the alternating shuffle relation: the free letters form an alternating list of length $a + 1$ across two columns with left height a , with the other letters fixed. Its row-polynomial image vanishes by Lemma 27.1; the unchanged other columns multiply this zero polynomial. Thus η_ν is well defined. It is onto because column tensors generate the Schur image. Its source has dimension 1260 by the proved source isomorphism, and its target has dimension 1260 by the explicit tableau count above. Its kernel is therefore zero. Its inverse has a basis formula: send b_T for each semistandard T to its column-tensor class. This right inverse is also a left inverse by injectivity, proving both compositions. Alternation, row multiplication and quotient maps are equivariant, so both maps are equivariant as well. Put

$$\sigma_\nu = \eta_\nu s_\nu : L_{\mathbb{Q}} \longrightarrow S_\nu(X), \quad \sigma_\nu^{-1} = j_\nu \eta_\nu^{-1}.$$

Use the explicit coordinate isomorphisms $\theta_E : E_C \otimes_R \mathbb{Q} \rightarrow X$ and $\theta_N : N_R \otimes_R \mathbb{Q} \rightarrow X$ taking their i th displayed basis vector to x_i . Since $C(1) = 0$, their square is

$$\theta_N(D_C)_0 = P_X \theta_E, \quad P_X(x_i) = x_i \ (1 \leq i \leq 3), \quad P_X(x_4) = 0.$$

Applying S_ν and composing with the proved source maps gives the actual endomorphism

$$\Pi_\nu = \sigma_\nu^{-1} S_\nu(P_X) \sigma_\nu : L_{\mathbb{Q}} \longrightarrow L_{\mathbb{Q}}. \quad (144)$$

It is idempotent: $\sigma_\nu \sigma_\nu^{-1} = 1$, functoriality, and $P_X^2 = P_X$ give $\Pi_\nu^2 = \Pi_\nu$. The Schur tableau formula shows its image is $\sigma_\nu^{-1} S_\nu(\langle x_1, x_2, x_3 \rangle)$ with its 27-element tableau basis, and its kernel

is the span of the 1233 tableaux containing x_4 , pulled back by σ_ν^{-1} . These are explicit inverse descriptions of both subspaces, proving the ranks independently of an arbitrary choice of 1260 vectors in the source.

The original diagonal $GL(V) \times GL(W)$ torus preserves each x_i , so it commutes with P_X . The proved equivariance of $\sigma_\nu, \sigma_\nu^{-1}$ and Schur functoriality prove that Π_ν commutes with this torus. No full raising/lowering equivariance is claimed: already $F_V x_2 = x_4$, so $P_X F_V x_2 = 0$ whereas $F_V P_X x_2 = x_4$. The construction provides the exact coordinate and source maps under which the endpoint specialization acts. The rational source isomorphism is used only over $\mathbb{Q}$; the integral torsion and the characteristic-3, 7 calculations above have not been transferred through an unproved integral or modular source isomorphism.

The complete transport thus reaches ordinary Schur and nested Schur modules, their exact conductor quotients, their residual prime modules and their specified source specialization. It provides no class-variety coordinate-ring obstruction, canonical-basis Frobenius model, or Boolean complexity reduction by itself. Those maps must be supplied by the continuing GCT construction; no unproved complexity conclusion is a hypothesis in any result here.

29 The published concentration coordinates and an exact conic heat transport

This section uses the coordinate formulas of the public manuscript *Finite Time Blowup for Navier–Stokes*, whose author line is OpenAI. The source reading receipt records the preserved revisions and the audited source pin. The manuscript’s Theorem 1.1 claims finite-time unbounded velocity for every viscosity $\nu > 0$, zero initial velocity, a smooth compactly supported force, fixed compact spatial support before time one, and uniformly bounded kinetic energy. That global claim is not independently verified in this section. All statements proved below are explicit coordinate, differential-algebra, and scalar-germ identities. In particular, the leading field appearing in the source is retained as a leading field, rather than substituted for its completed construction. The current 166-page PDF has SHA-256

0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f.

The earlier 165-page PDF has SHA-256

8c8a94ad9ac824c8b605b9827cadf7beaca48bd10b380de3cfc872a2c37afa81.

The source audit compares the cited passages in both versions: the coordinates are (3.2), current p. 7 and (4.1), current p. 24; the differential identities are Lemma 4.1 and (4.2), current p. 25; the profiles and balances are (4.3)–(4.7), current pp. 25–26; the growth and viscosity formulas are (10.21)–(10.23), current p. 124. The corresponding earlier passages are on pp. 7, 24–25, and 124. The selected equations agree; complete revision equivalence is not asserted. The preserved source commit is

8937a8f4cbc7abaab5e9e97d1cc7f5d2319d9538.

The public manuscript supplies these source objects [14]. Its revised historical account credits Diego Córdoba, Luis Martínez-Zoroa, and the indicated joint work with Fan Zheng for preceding forced-flow constructions. Tristan Buckmaster’s primary statement separately identifies his joint work with Levent Alpöge on announced smooth-forcing incompressible-porous-media, Boussinesq, and three-dimensional Euler blowup, and describes an unfinished verification of a hypodissipative Navier–Stokes claim [2]. These are the stated attributions and mathematical scopes. Neither source observation establishes priority or a technical dependency of the positive-viscosity manuscript on that joint work. The new conic calculations below are proved here from the displayed objects.

29.1 The exact source coordinates and differential operators

To prevent collisions of source notation, write ρ for the source's cylindrical radius r , Q for its concentration variable q , ξ for its similarity variable η , and $\sigma = \rho^2/2$ for its squared-radius variable s . The arithmetic coordinates r, s, b, q retain their previous meanings. These are names for the original variables, not changes of their values. Retain

$$\begin{aligned} A &= \frac{1}{2} + h, & D &= \frac{1}{2} - h, & \tau &= 1 - t, \\ z &= Q^D \xi, & \tau &= Q(1 - \xi^2), & X &= \frac{\rho^2}{2Q}, \\ d &= 1 - \xi^2, & \ell &= 1 - 2h\xi^2. \end{aligned}$$

The coordinate identities hold for $0 < h < 1/2$, $\tau > 0$, $Q > 0$, $-1 < \xi < 1$; the source's profile construction uses $0 < h < 1/100$. Powers of Q denote their positive real values. For each $(z, \tau) \in \mathbb{R} \times (0, \infty)$ there is exactly one $Q > |z|^{1/D}$ with

$$Q - z^2 Q^{2h} = \tau.$$

Indeed the left side is zero at the stated endpoint, tends to infinity, and has derivative $1 - 2hz^2 Q^{2h-1} \geq 1 - 2h > 0$ on this interval. The implicit-function theorem and $\ell > 0$ prove smoothness. Inverting gives $\xi = zQ^{-D}$ and $\rho = \sqrt{2QX}$, the latter for $X > 0$. The time-axial Jacobian is

$$\det \frac{\partial(\tau, z)}{\partial(Q, \xi)} = Q^D (d + 2D\xi^2) = Q^D \ell > 0.$$

Differentiating these exact relations at fixed physical coordinates gives

$$\begin{aligned} Q_t &= -\ell^{-1}, & \xi_t &= D\xi/(Q\ell), & X_t &= X/(Q\ell), \\ Q_z &= 2\xi Q^{1-D}/\ell, & \xi_z &= d/(Q^D \ell), & X_z &= -2\xi X/(Q^D \ell), \\ Q_\rho &= 0, & \xi_\rho &= 0, & X_\rho &= \rho/Q. \end{aligned}$$

For example $Q_\tau = 1/\ell$, $\xi_\tau = -D\xi/(Q\ell)$, and $X_\tau = -X/(Q\ell)$, so the first line retains the time reversal. The ξ_z calculation uses $\ell - 2D\xi^2 = d$.

For any real w and smooth profile $f(X, \xi)$ define $\mathcal{P}_w f = Q^w f(X, \xi)$ and

$$\begin{aligned} T_w f &= \ell^{-1}(-wf + D\xi f_\xi + X f_X), \\ Z_w f &= \ell^{-1}(2w\xi f + df_\xi - 2\xi X f_X), & Rf &= 2X f_{XX} + 2f_X. \end{aligned}$$

The chain and product rules, with the preceding six derivatives, give

$$\partial_t \mathcal{P}_w = \mathcal{P}_{w-1} T_w, \quad \partial_z \mathcal{P}_w = \mathcal{P}_{w-D} Z_w, \quad (\partial_\rho^2 + \rho^{-1} \partial_\rho) \mathcal{P}_w = \mathcal{P}_{w-1} R. \quad (145)$$

For the last identity, the two terms from differentiating ρ/Q and the cylindrical term each contribute $Q^{w-1} f_X$, and $\rho^2 Q^{w-2} f_{XX} = 2X Q^{w-1} f_{XX}$. Thus all radial constants are retained. Applying the axial identity twice gives

$$(\partial_t - \nu \Delta) \mathcal{P}_w f = \mathcal{P}_{w-1} (T_w - \nu R) f - \nu \mathcal{P}_{w-1+2h} Z_{w-D} Z_w f, \quad (146)$$

$$-\frac{1}{4} \partial_t^2 \mathcal{P}_w f = -\frac{1}{4} \mathcal{P}_{w-2} T_{w-1} T_w f. \quad (147)$$

Here Δ is the three-dimensional scalar Laplacian on axisymmetric functions. These equations keep both distinct similarity weights $w - 1$ and $w - 1 + 2h$, as well as the arithmetic factor $-1/4$.

The actual source leading profiles satisfy

$$u_\theta^{(0)} = Q^{-A} E, \quad u_z^{(0)} = Q^{-A} U, \quad \rho u_\rho^{(0)} = V_0, \quad p^{(0)} = Q^{-2A} \Pi, \quad E = C_0^{-1} \sqrt{2X} \phi, \quad C_0 > 1,$$

where C_0 denotes the source's amplitude constant, not the GCT coefficient. Its incompressibility and pressure identities are

$$V_0 = \frac{X}{\ell} (2\xi U - 2D\xi \mathcal{A}_X U - d\partial_\xi \mathcal{A}_X U), \quad \mathcal{A}_X U = X^{-1} \int_0^X U(y, \xi) dy, \quad \Pi_X = E^2/(2X).$$

The first follows by integrating $(V_0)_X = -Z_{-A}U$ from zero, using $A + D = 1$ and integrating XU_X exactly. The second is the source radial-pressure balance $\partial_\rho p^{(0)} = (u_\theta^{(0)})^2/\rho$. The regular axis interpretation is the one stated by the source: $E/\sqrt{2X}$, U , and V_0/X are smooth at $X = 0$.

For this particular leading field its physical material derivative on axisymmetric scalars has the exact representation

$$(\partial_t + u_\rho^{(0)}\partial_\rho + u_z^{(0)}\partial_z)\mathcal{P}_w f = \mathcal{P}_{w-1}M_w f, \quad M_w = T_w + V_0\partial_X + UZ_w. \quad (148)$$

Indeed $u_\rho^{(0)}\partial_\rho \mathcal{P}_w f = Q^{w-1}V_0 f_X$, while the axial term has weight $w - A - D = w - 1$. This proves the correspondence for the actual source ansatz, including its nonlinear dependence on the chosen profiles.

For completeness this also determines its full momentum residual, including the cylindrical frame terms. Put $f_\rho = V_0/\sqrt{2X}$ and $R_1 = R - (2X)^{-1}$. On $X > 0$ the three components of $(\partial_t + u^{(0)} \cdot \nabla)u^{(0)} - \nu\Delta u^{(0)} + \nabla p^{(0)}$ are

$$\begin{aligned} \mathbf{r}_\rho &= \mathcal{P}_{-3/2}(M_{-1/2}f_\rho - \nu R_1 f_\rho) - \nu \mathcal{P}_{-3/2+2h}Z_{-1/2-D}Z_{-1/2}f_\rho, \\ \mathbf{r}_\theta &= \mathcal{P}_{-A-1}(M_{-A}E + V_0E/(2X) - \nu R_1 E) - \nu \mathcal{P}_{-A-1+2h}Z_{-A-D}Z_{-A}E, \\ \mathbf{r}_z &= \mathcal{P}_{-A-1}(M_{-A}U - \nu RU + Z_{-2A}\Pi) - \nu \mathcal{P}_{-A-1+2h}Z_{-A-D}Z_{-A}U. \end{aligned}$$

To verify these formulas, differentiation of the cylindrical unit vectors gives the radial convective term $-(u_\theta^{(0)})^2/\rho$, the azimuthal convective term $u_\rho^{(0)}u_\theta^{(0)}/\rho$, and the radial and azimuthal vector-Laplacian terms $-u_\rho^{(0)}/\rho^2$ and $-u_\theta^{(0)}/\rho^2$. The radial convective term cancels $\partial_\rho p^{(0)}$ by the exact displayed pressure balance. The azimuthal additional term is $\mathcal{P}_{-A-1}(V_0E/(2X))$. Finally $-2A - D = -A - 1$ gives the axial pressure weight. Substituting (145) proves every remaining term. No assertion that this residual vanishes is needed or made.

The source constructs its leading profiles at viscosity one. Its actual passage to arbitrary $\nu > 0$ is the spatial map

$$y = x/\sqrt{\nu}, \quad u_\nu(x, t) = \sqrt{\nu} u(y, t), \quad p_\nu(x, t) = \nu p(y, t), \quad f_\nu(x, t) = \sqrt{\nu} f(y, t).$$

Every momentum term acquires the factor $\sqrt{\nu}$: time differentiation preserves the velocity prefactor; the two velocity factors in advection and the spatial derivative give $\nu\nu^{-1/2}$; viscosity and two spatial derivatives give $\nu\sqrt{\nu}\nu^{-1}$; and the pressure gradient gives $\nu\nu^{-1/2}$. The divergence is unchanged. The Jacobian $dx = \nu^{3/2}dy$ gives $\|u_\nu\|_2^2 = \nu^{5/2}\|u\|_2^2$ and $\nu \int \|\nabla u_\nu\|_2^2 dt = \nu^{5/2} \int \|\nabla u\|_2^2 dt$ on the same time interval. In these rescaled fields the physical coordinates satisfy $z = \sqrt{\nu}Q^D\xi$, $\rho^2 = 2\nu QX$ and $\tau = Q(1 - \xi^2)$. Thus ∂_z gains $\nu^{-1/2}$ and the radial scalar Laplacian gains ν^{-1} in (145). This is the source's specified viscosity transport, with its time coordinate unchanged.

29.2 The material base and the complete two-variable conic connection

Work first over $\mathbb{C}$. Retain the original coordinates s, b, c, r and $P = cr^3 - 2r^2 + br - 4s$. The original material base derivation and its simple-root lift are

$$U_B = \partial_s + 6c\partial_b, \quad \mathcal{D} = \partial_s + 6c\partial_b + \frac{4 - 6cr}{3cr^2 - 4r + b}\partial_r.$$

The three contributions to $\mathcal{D}P$ are -4 , $6cr$, and $4-6cr$, so this is a derivation of the localization at P_r . Under the real base identification

$$s = \tau = 1 - t, \quad b = z, \quad c = c, \quad (149)$$

U_B becomes $-\partial_t + 6c\partial_z$. This identity concerns the stated material base; the leading physical material derivative has instead the explicitly proved profile formula (148). On $c = 0$ the original arithmetic operator is exactly $-\partial_s^2/4 = -\partial_t^2/4$. The previously established pullback by the original polynomial F composes with (149); its two original Euclidean diffusion coefficients are unchanged.

On $c = 0$, retain the complete cover

$$\begin{aligned} \beta &= b/4, & \eta &= r - \beta, & m &= b^2/16 - 2s, & n &= m + 4, \\ \mathcal{B} &= \mathbb{C}[s, b, (mn)^{-1}], & \mathcal{E} &= \mathcal{B}[\eta, p]/(\eta^2 - m, p^2 - n). \end{aligned}$$

The original branch factors are $\delta = 16m = b^2 - 32s$ and $L = n = 4 + b^2/16 - 2s$. The localization removes both and only these two factors. The ring is free of rank four, with bases $(1, \eta, p, \eta p)$ and $(1, r, p, rp)$; monic division proves existence and uniqueness of the four coefficients. There are inverse maps to the localized original Laurent presentation, given by

$$q = (p + \eta)/2, \quad q^{-1} = (p - \eta)/2, \quad \eta = q - q^{-1}, \quad p = q + q^{-1}, \quad s = b^2/32 - \frac{1}{2}(q - q^{-1})^2. \quad (150)$$

The two displayed images of q, q^{-1} have product $(n - m)/4 = 1$; sums and differences prove both inverse compositions. Thus the arithmetic q in this formula is never set equal to Q .

There are unique commuting derivations ∇_s, ∇_b extending ∂_s, ∂_b to $\mathcal{E}$. On generators they are

$$\begin{aligned} \nabla_s \eta &= -1/\eta, & \nabla_s p &= -1/p, & \nabla_s q &= -q/(\eta p), \\ \nabla_b \eta &= b/(16\eta), & \nabla_b p &= b/(16p), & \nabla_b q &= bq/(16\eta p). \end{aligned}$$

Differentiating the two defining quadratic relations proves these formulas and their uniqueness, since 2η and $2p$ are units. Their commutator is a derivation killing s, b ; differentiation of the same quadratic relations then forces it to kill η, p . This proves flatness without assuming that arbitrary differentiation descends to an unlocalized quotient.

Here are the full matrices in the retained basis $(1, r, p, rp)$. Column vectors are coefficient vectors, and $\nabla_i = \partial_i I_4 + \Gamma_i$:

$$\begin{aligned} \Gamma_s &= \begin{pmatrix} 0 & \beta/m & 0 & 0 \\ 0 & -1/m & 0 & 0 \\ 0 & 0 & -1/n & \beta/m \\ 0 & 0 & 0 & -1/m - 1/n \end{pmatrix}, \\ \Gamma_b &= \begin{pmatrix} 0 & 1/4 - b\beta/(16m) & 0 & 0 \\ 0 & b/(16m) & 0 & 0 \\ 0 & 0 & b/(16n) & 1/4 - b\beta/(16m) \\ 0 & 0 & 0 & b/(16m) + b/(16n) \end{pmatrix}. \end{aligned}$$

For example $\nabla_b r = 1/4 + b\eta/(16m)$ and $\nabla_b(rp) = p/4 + (b/(16m) + b/(16n))\eta p + b\beta p/(16n)$, which give the second and fourth columns. The other columns follow directly from the generator formulas. The inverse change of coefficient bases is explicit:

$$T = \begin{pmatrix} 1 & \beta & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \beta \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T^{-1} = \begin{pmatrix} 1 & -\beta & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -\beta \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

In the $(1, \eta, p, \eta p)$ basis write

$$H_s = \text{diag}(0, -m^{-1}, -n^{-1}, -m^{-1} - n^{-1}), \quad H_b = -\frac{b}{16} H_s.$$

Then $\Gamma_i = T^{-1} H_i T + T^{-1} (\partial_i T)$, which also verifies that the b -dependent basis derivative has been retained. For clarity the entire second axial derivative in this basis is

$$\nabla_b^2 = \partial_b^2 I_4 + 2H_b \partial_b + J_b,$$

where J_b is diagonal with entries

$$\begin{aligned} (J_b)_{00} &= 0, \\ (J_b)_{11} &= \frac{1}{16m} - \frac{b^2}{256m^2}, \quad (J_b)_{22} = \frac{1}{16n} - \frac{b^2}{256n^2}, \\ (J_b)_{33} &= \frac{1}{16} \left(\frac{1}{m} + \frac{1}{n} \right) - \frac{b^2}{256} \left(\frac{1}{m^2} + \frac{1}{n^2} \right) + \frac{b^2}{128mn}. \end{aligned}$$

These entries are $\partial_b H_b + H_b^2$, using $\partial_b m = \partial_b n = b/8$. Similarly

$$\nabla_s^2 = \partial_s^2 I_4 + 2H_s \partial_s + J_s, \quad J_s = \text{diag} \left(0, -m^{-2}, -n^{-2}, -m^{-2} - n^{-2} + 2/(mn) \right).$$

The formulas follow from $\partial_s m = \partial_s n = -2$. Returning to the original basis replaces each second-order constant matrix by $\partial_i \Gamma_i + \Gamma_i^2$ and each first-order matrix by $2\Gamma_i$, so no basis-dependent second derivative is lost.

Let the physical real base be

$$\Omega = \{(\rho, z, t) : \rho > 0, t < 1, m = z^2/16 - 2(1-t), mn \neq 0\}.$$

Tensoring $\mathcal{E}$ over its indicated base map with complex smooth functions on Ω gives a rank-four complex algebra bundle. On this bundle physical $\partial_t = -\nabla_s$, $\partial_z = \nabla_b$, and ∂_ρ differentiates the smooth coefficients and kills η, p . Consequently its complete scalar heat and arithmetic operators are

$$\mathcal{N}_\nu = -\nabla_s - \nu(\partial_\rho^2 + \rho^{-1} \partial_\rho + \nabla_b^2), \quad \mathcal{H} = -\frac{1}{4} \nabla_s^2, \quad [\mathcal{N}_\nu, \mathcal{H}] = 0. \quad (151)$$

The commutator vanishes because the two lifted derivations commute, all transverse derivatives commute with them, and ν is constant. The coefficient inclusion $f \mapsto f \cdot 1$ is an algebra map and intertwines every displayed physical derivative. Inserting the actual profiles into it therefore carries (146), (147), and the three exact residual identities to this bundle. This is the explicit differential correspondence; it does not identify the four cover coordinates with four independently constructed physical velocities.

29.3 The physical real structure and a finite negative coefficient

On the open physical region $-4 < m < 0$, put $a_* = \sqrt{-m}$ and $p_* = \sqrt{m+4}$ using the positive real square roots. The four sections of the complex cover are exactly

$$\eta = \epsilon i a_*, \quad p = \epsilon' p_*, \quad r = b/4 + \epsilon i a_*, \quad q = \frac{1}{2}(\epsilon' p_* + \epsilon i a_*), \quad \epsilon, \epsilon' \in \{1, -1\}.$$

Their squared moduli are $(m+4-m)/4 = 1$, and the maps (150) verify every equation. The four sections are distinct since $a_*, p_* > 0$. Ordinary complex conjugation fixes s, b , fixes p , changes η to $-\eta$, and sends q to q^{-1} . This is the real structure over the physical real base. There is no real value of q on this region: a real nonzero q would give $m = (q - q^{-1})^2 \geq 0$. The

branch boundaries $m = 0$ and $m = -4$ remain excluded from the connection, even though limits of the scalar polynomials below exist there.

On the source's actual growth path $z = 0$ and $\rho = \sqrt{2X_{\text{in}}\tau}$, retain $Q = \tau$, $\xi = 0$, $X = X_{\text{in}} > 0$. The base map gives $b = 0, s = \tau$, so $m = -2\tau$. For $0 < \tau < 2$ choose the sheet

$$\theta = \arcsin \sqrt{\tau/2}, \quad q = e^{i\theta}, \quad \eta = i\sqrt{2\tau}, \quad p = \sqrt{4-2\tau}, \quad r = i\sqrt{2\tau}. \quad (152)$$

Here $\theta \in (0, \pi/2)$ and the formulas agree with the four-sheet description. In particular the source concentration $Q \downarrow 0$ maps to the finite arithmetic point $q \rightarrow 1$, with the original branch point retained as a limit.

For the original GCT coefficient, every factor is retained:

$$\begin{aligned} C(q) &= q^{10} - q^4 - q^{-4} + q^{-10} = (q - q^{-1})^2 [7]_q [3]_q, \\ C(e^{i\theta}) &= -4 \sin(7\theta) \sin(3\theta), \\ C|_{b=0, s=\tau} &= -2\tau(3-2\tau)(7-28\tau+28\tau^2-8\tau^3) = \tau B(\tau), \\ B(\tau) &= -42 + 196\tau - 280\tau^2 + 160\tau^3 - 32\tau^4. \end{aligned} \quad (153)$$

The trigonometric equality follows by substituting $q = e^{i\theta}$ and applying $2 \cos(10\theta) - 2 \cos(4\theta) = -4 \sin(7\theta) \sin(3\theta)$. For the polynomial equality, $[3]_q = p^2 - 1$ and $[7]_q = p^6 - 5p^4 + 6p^2 - 1$ follow by multiplying the corresponding finite Laurent sums by $q - q^{-1}$. Substituting $p^2 = 4 - 2\tau$ and $(q - q^{-1})^2 = -2\tau$ gives the displayed factored polynomial and its full expansion. Therefore

$$C(q) < 0 \quad \text{for} \quad 0 < \tau < 2 \sin^2(\pi/7), \quad \lim_{\tau \downarrow 0} \frac{C(q)}{\tau} = -42.$$

Both sines are strictly positive on the stated interval because $0 < \theta < \pi/7$ and $0 < 3\theta < 3\pi/7 < \pi$. This proves a finite negative value and its exact vanishing order.

The precise scalar-germ transport of this coefficient is also explicit. Fix $0 < h < 1/100$, $e_0 > 0$, and any bounded real function ε on $(0, \tau_0)$. Define the actual scalar function

$$g_\varepsilon(\tau) = \tau^{-A}(e_0 + \tau^{2h}\varepsilon(\tau)).$$

Multiplication by the computed GCT coefficient gives the exact identity

$$C(q)g_\varepsilon(\tau) = \tau^D B(\tau)(e_0 + \tau^{2h}\varepsilon(\tau)), \quad \lim_{\tau \downarrow 0} \tau^{-D} C(q)g_\varepsilon(\tau) = -42e_0. \quad (154)$$

This follows from $1 - A = D$ and (153). Since $B(\tau) + 42$ is divisible by τ , and $0 < 2h < 1$, the right side equals $-42e_0\tau^D + O(\tau^{D+2h})$. It is negative for sufficiently small τ and tends to zero, since $D > 0$. Multiplication raises the displayed leading exponent by exactly one. In the formal power-series ring $\mathbb{R}[[\tau]]$, the same exact polynomial has order one, because $B(0) = -42 \neq 0$. No identification of τ with a prime number or of this order with a prime-adic valuation is made.

The source's equation (10.21) reports its completed velocity component as $\tau^{-A}(e_0 + O(\tau^{2h}))$ on this path. The exact map (154) acts on the full stated class of such scalar germs; identifying a member with that completed solution retains the independent-verification status of the source theorem. The negative coefficient result above uses only the proved conic map and polynomial identities. It gives no asymptotic circuit-size bound, plethysm multiplicity obstruction, or nonstandard-RH eigenvalue claim.

Finally, the source pressure normalization supplies a second exact sign statement with a different specified observable: $\Pi(X, \xi) = -\int_X^\infty E(y, \xi)^2/(2y) dy \leq 0$ whenever that integral is finite. It is strictly negative if the nonnegative integrand is positive on a set of positive

measure. Replacing p by $p + k(t)$ leaves its spatial gradient and the momentum equation unchanged, but need not preserve a chosen compact-support or infinity normalization. Thus the displayed pressure sign belongs to that normalization. The coefficient sign (153) has instead been proved for the fixed original Laurent polynomial on the stated unit-circle sheet.

30 The concentration path through the actual Schur conductor map

This section composes the previously proved conic path with the integral Schur endpoint, retaining the original source and target bases. Put

$$I = (0, 2 \sin^2(\pi/7)), \quad B(\tau) = -42 + 196\tau - 280\tau^2 + 160\tau^3 - 32\tau^4,$$

and use exactly the sheet in (152). The concentration parameter is $Q = \tau$, whereas the GCT parameter is $q = e^{i \arcsin \sqrt{\tau/2}}$. These parameters remain distinct. The endpoint factorization of Section 28 pulls back to the three based maps

$$\begin{aligned} D_\tau &= \text{diag}(1, 1, 1, \tau B(\tau)), \\ H_\tau &= \text{diag}(1, 1, 1, h_\tau), \quad V_\tau = \text{diag}(1, 1, 1, -2\tau), \\ h_\tau &= -B(\tau)/2 = 21 - 98\tau + 140\tau^2 - 80\tau^3 + 16\tau^4, \quad D_\tau = V_\tau H_\tau. \end{aligned} \quad (155)$$

Here h_τ denotes the retained conductor factor $[7]_q[3]_q$, and is not the source profile parameter h . Indeed $r^2 = -2\tau$ and $C = r^2[7]_q[3]_q = \tau B(\tau)$ prove every entry. On I both C and r^2 are negative, so $h_\tau > 0$. The pointwise maps have their earlier endpoint, intermediate, and node modules as source and target after this base change; the displayed matrices do not identify those three original modules.

For the formal series statement below, the exact ring map must also be specified. Set $R_m = \mathbb{Z}[m]$ and

$$h_m = (m + 3)((m + 4)^3 - 5(m + 4)^2 + 6(m + 4) - 1), \quad C_m = m h_m.$$

Take three free R_m -modules with the respective original endpoint, intermediate, and node basis labels. Define their maps to have matrices $\text{diag}(1, 1, 1, h_m)$ and $\text{diag}(1, 1, 1, m)$. Their composition has fourth entry C_m . The homomorphism $R_m \rightarrow R = \mathbb{Z}[1/2][q, q^{-1}]$ sending m to $(q - q^{-1})^2$ takes $m + 4$ to $(q + q^{-1})^2$ and hence h_m to $[7]_q[3]_q$. Mapping each free basis symbol to its stated original basis vector is an isomorphism after this base change: the inverse sends the original coordinate tuple back to the same symbols. The three matrices show that these isomorphisms intertwine every map. The same argument holds after the localizations used earlier. This proves descent of this based endpoint diagram to R_m . Its separate base change $R_m \rightarrow \mathbb{R}[[\tau]]$, $m \mapsto -2\tau$, is the formal diagram used here. In particular, no homomorphism sending the full Laurent variable q to an element of $\mathbb{R}[[\tau]]$ is asserted: the pointwise sheet has a square-root term at $\tau = 0$. The complete Schur base-change proof carries these diagram isomorphisms through F without changing any tableau labels.

Let $F = S_\lambda$, or $F = S_\lambda \circ S_\mu$, be the precise integral Schur functor constructed above. Write $d = |\lambda|$ in the first case and $d = |\lambda| + |\mu|$ in the second. Its tableau weight multiplicities are the previously constructed $m_F(k)$. Functoriality and the full diagonal formula give

$$\begin{aligned} F(D_\tau) &= \text{diag}_{k,j}(\tau^k B(\tau)^k), \\ F(H_\tau) &= \text{diag}_{k,j}(h_\tau^k), \quad F(V_\tau) = \text{diag}_{k,j}((-2\tau)^k), \\ 1 &\leq j \leq m_F(k). \end{aligned} \quad (156)$$

For a single tableau, k is its number of fourth symbols. In the nested case, k is the sum of those counts over the inner tableaux appearing in the outer tableau, with multiplicity. Substitution of the fourth diagonal entry therefore gives its k th power; the other three diagonal entries are one. This also proves the displayed composition entry by entry.

Over $\mathbb{R}[[\tau]]$, B is a unit because its constant term is -42 . An inverse is obtained recursively: if $B = \sum_{i=0}^4 b_i \tau^i$, set $a_0 = -1/42$ and $a_n = -b_0^{-1} \sum_{i=1}^{\min(4,n)} b_i a_{n-i}$. Thus the exact image in the (k, j) target coordinate is $\tau^k \mathbb{R}[[\tau]]$, its kernel is zero, and its cokernel is $\mathbb{R}[[\tau]]/(\tau^k)$. The latter is zero when $k = 0$; for $k > 0$ its basis over $\mathbb{R}$ is $1, \tau, \dots, \tau^{k-1}$, as follows by separating the first k coefficients of a series. Consequently the cokernel length and determinant order both equal $\sum_k k m_F(k)$. Specialization at $\tau = 0$ has rank $m_F(0)$ and kernel and cokernel dimension $\sum_{k>0} m_F(k)$. These are statements over the specified real series ring. The integer factors 21^k and residual characteristics in the earlier integral calculation are retained there; they have not been replaced by assertions about valuations of prime numbers along a real path.

For each $\tau \in I$, identify source and target tableau labels by their explicitly matching based-vector-space map and give the source the Hermitian metric for which that basis is orthonormal. The form defined by $F(D_\tau)$ then has coefficients $\tau^k B(\tau)^k$. Since $B(\tau) < 0$ on I , the even-weight and odd-weight coordinate spaces are respectively positive and negative definite. They are orthogonal and exhaust the space. A larger positive subspace would intersect the negative coordinate space by the dimension formula, and conversely for a larger negative subspace. Thus throughout I the exact inertia is

$$\left(\sum_{k \text{ even}} m_F(k), \sum_{k \text{ odd}} m_F(k), 0 \right).$$

The factor $F(H_\tau)$ is positive definite in the matched metric, so the same indices hold for $F(V_\tau)$.

For the original shape $\nu = (7, 5, 3)$, the completely evaluated profile is

$$(m_\nu(0), \dots, m_\nu(7)) = (27, 81, 162, 270, 300, 252, 126, 42).$$

Its sum is 1260 and its weighted sum is 4725. It follows that

$$\det S_\nu(D_\tau) = \tau^{4725} B(\tau)^{4725}, \quad \text{inertia} = (615, 645, 0), \quad \text{rank}_{\tau=0} = 27. \quad (157)$$

The specialized kernel and cokernel have dimension 1233; the series cokernel has length 4725. The endpoint entries extend to $\tau = 0$, although the conic connection does not extend across its excluded branch locus. No connection value at that branch point is used here.

There is a second exact composition that tests what happens to growth. Use the actual scalar functions of (154): fix $0 < h < 1/100$, $A = 1/2 + h$, $e_0 > 0$, and bounded real ε , and set $g(\tau) = \tau^{-A}(e_0 + \tau^{2h}\varepsilon(\tau))$. Define the explicit new linear map

$$L_\tau = g(\tau)D_\tau$$

between the same two based endpoint spaces. This definition attaches the stated scalar germ to their map; it does not identify an endpoint vector with a fluid velocity or a quantum state. Each tableau monomial has total degree d , including every inner and outer box in the nested case, whence

$$F(L_\tau)_{k,j} = g(\tau)^d C(q)^k = \tau^{k-dA} B(\tau)^k (e_0 + \tau^{2h}\varepsilon(\tau))^d. \quad (158)$$

This proves homogeneity without a division by a factorial. For fixed k, d , the binomial formula and boundedness give $(e_0 + \tau^{2h}\varepsilon)^d = e_0^d + O(\tau^{2h})$. Also $B(\tau)^k = (-42)^k + O(\tau)$ for $k > 0$, and the formula is exactly one for $k = 0$. Since $2h < 1$, multiplication yields

$$F(L_\tau)_{k,j} = (-42)^k e_0^d \tau^{k-dA} + O(\tau^{k-dA+2h}).$$

The leading constant is nonzero. Hence the absolute value tends to zero when $k > dA$, tends to the absolute leading constant when $k = dA$, and diverges when $k < dA$. This classification proves the outcome for the defined map, rather than assuming that scalar multiplication and a degree- d functor have the same effect on an exponent.

In particular $d = 15$ and $0 \leq k \leq 7$ for $\nu = (7, 5, 3)$, so $k - 15A \leq -1/2 - 15h < 0$. Every one of its 1260 diagonal entries diverges in absolute value, with 615 positive and 645 negative entries for sufficiently small τ . Its determinant is exactly

$$\det S_\nu(L_\tau) = \tau^{4725-18900A} B(\tau)^{4725} (e_0 + \tau^{2h} \varepsilon(\tau))^{18900},$$

with leading exponent $-4725 - 18900h$. In contrast, the already proved single scalar product $C(q)g(\tau)$ tends to zero with exponent $1/2 - h$. The displayed functorial computation explains this difference: the degree-fifteen scalar contributes exponent $-15A$, and the largest available fourth-coordinate weight contributes only 7.

All multiplicities here are the unchanged nonnegative tableau counts. The rational source comparison at $q = 1$ is the explicit map already proved in Section 28; this path calculation does not create a generic quantum-group intertwiner or an unrestricted complexity lower bound. It supplies the exact Schur composition, signs, and growth exponents on the specified original endpoint instead.

31 The retained conic and the interacting material-tensor state map

The local source `FABEL_TENSOR_TRANSFER.md`, equations (97)–(105), supplies the full material tensor and finite Wilson box. The source paths and hashes accompany this fragment. The NS input is the explicit conic coordinates and scalar polynomial proved in Section 29. The published manuscript’s global blowup theorem retains its stated independent-verification status; it is not used as a theorem in this calculation.

31.1 Original Hamiltonian and complex six-state map

Fix an integer $L \geq 2$, spacing $a_{\text{YM}} > 0$, and coupling $g > 0$. Here a_{YM} is the source’s spacing a , not the arithmetic coefficient $a = 2s$. Vertices are $\{-L, \dots, L\}^3$; include every positive nearest-neighbour edge $\mathbf{E}_L$ and every elementary face $\mathbf{P}_L$ whose vertices are present, including boundaries. On $\mathcal{Q}_L = \text{SU}(2)^{\mathbf{E}_L}$ use product Haar probability λ . With $T_a = -i\sigma_a/2$ and $X_{e,a}F = \frac{d}{du}F(\dots, e^{uT_a}U_e, \dots)\Big|_{u=0}$, retain

$$\begin{aligned} E_e &= -\sum_{a=1}^3 X_{e,a}^2, & W_p &= \text{tr}(U_i(n)U_j(n + \mathbf{e}_i)U_i(n + \mathbf{e}_j)^{-1}U_j(n)^{-1}), \\ H &= \kappa \sum_e E_e + \sum_p V_p, & \kappa &= 2g^2/a_{\text{YM}}, & V_p &= b_{\text{YM}}(2 - W_p), & b_{\text{YM}} &= 1/(2g^2a_{\text{YM}}). \end{aligned}$$

The gauge group is $\mathcal{G} = \text{SU}(2)^{\{-L, \dots, L\}^3}$, acting by $(h \cdot U)_e = h_{s(e)}^{-1} U_e h_{t(e)}$. Its Hilbert space is $\mathcal{H}_L = L^2(\mathcal{Q}_L, \lambda)$ and its domain is $H^2(\mathcal{Q}_L)$. In particular its original constant $2b_{\text{YM}}|\mathbf{P}_L|$ is retained. Minimizing the closed elliptic form, using compact Sobolev embedding, gives a ground vector. Taking its modulus does not increase the form; elliptic regularity and the strong maximum principle make the ground vector smooth and strictly positive. Denote it by ψ , with $H\psi = E_0\psi$ and $\|\psi\| = 1$. Integration by parts gives, for smooth u ,

$$\langle \psi u, (H - E_0)\psi u \rangle = \kappa \sum_{e,a} \int \psi^2 |X_{e,a}u|^2 d\lambda.$$

This vanishes precisely for constant u , by connectedness, so $\ker(H - E_0) = \mathbb{C}\psi$. Gauge invariance of H and uniqueness of its positive unit ground vector prove gauge invariance of ψ . Write $\mathcal{A} = H - E_0$.

For complex edge weights define on the common smooth domain

$$f_p = \frac{1}{4} \sum_{e \in \partial p} f_e, \quad D_f = \kappa \sum_e f_e E_e + \sum_p f_p V_p, \quad \Xi_f = D_f \psi - \langle \psi, D_f \psi \rangle \psi.$$

Inner products are conjugate linear in the first argument. Thus Ξ_f is complex linear in f , smooth, gauge invariant, and perpendicular to ψ . It belongs to every power domain of H . For complex f this uses $D_f = D_{\text{Re } f} + i D_{\text{Im } f}$ on smooth functions; no self-adjointness of this complex weighted operator is asserted. The physical H and its adjoint remain unchanged. On this smooth domain the ground equation gives $\mathcal{A}\Xi_f = [H, D_f]\psi$ by direct subtraction.

Use the original dimensionless midpoint $m(e) = n + \mathbf{e}_i/2$. The exact complexification of the source's real map is

$$\mathfrak{M} : \text{Sym}_3(\mathbb{C}) \longrightarrow \mathcal{V}_L := C^\infty(\mathcal{Q}_L)^{\mathcal{G}} \cap \psi^\perp, \quad \mathfrak{M}(S) = \Xi_{m^\top S m}.$$

Physical midpoints $a_{\text{YM}} m(e)$ multiply this map by a_{YM}^2 and its raw spectral measures by a_{YM}^4 . For $v \in \mathcal{V}_L$ put $\nu_v(B) = \langle v, 1_B(\mathcal{A})v \rangle$ on $(0, \infty)$. Set $u_i = \Xi_{m_i^2}$, $v_{ij} = \Xi_{2m_i m_j}$ and

$$\nu_0 = \nu_{u_1+u_2+u_3}, \quad \nu_d = \nu_{u_1-u_2}, \quad \nu_o = \nu_{v_{12}}.$$

For every complex symmetric S , the full measure is

$$\nu_{\mathfrak{M}(S)} = \frac{|\text{tr } S|^2}{9} \nu_0 + \frac{1}{2} \sum_i |S_{ii} - \text{tr } S/3|^2 \nu_d + \sum_{i < j} |S_{ij}|^2 \nu_o. \quad (159)$$

To prove this with conjugation retained, signed coordinate permutations act on links by their endpoint permutation and by inversion when an edge reverses. Haar measure and the Casimir are preserved, and the $\text{SU}(2)$ face trace is unchanged by reversal. Their unitaries commute with H and its spectral projections and fix ψ . Reflections kill every mixed u, v spectral Gram entry and every different-pair v, v entry; permutations equate the remaining v, v entries. The u, u Gram matrix has constant diagonal d and constant off-diagonal h ; a transposition and Hermitian symmetry give $h = \bar{h}$. Its quadratic form for complex diagonal coefficients s_i , with mean $s_{\text{av}} = (s_1 + s_2 + s_3)/3$, is

$$(d - h) \sum_i |s_i - s_{\text{av}}|^2 + (d + 2h) |s_1 + s_2 + s_3|^2 / 3.$$

Since $\nu_0 = 3d + 6h$ and $\nu_d = 2(d - h)$, this proves (159) on every Borel set. Integration against $1, E, E^2, e^{-uE}$, with $u \geq 0$, gives norms, excitation forms, squared current norms, and time correlations. No real square has been continued through a Hilbert adjoint.

The kernel of this six-state map is also exact. Decompose $\text{Sym}_3(\mathbb{C}) = \mathcal{S}_0 \oplus \mathcal{S}_d \oplus \mathcal{S}_o$ into scalar, traceless diagonal, and off-diagonal matrices, of dimensions 1, 2, 3. Set $d_j = \nu_j((0, \infty))$. Since the state is perpendicular to the entire zero eigenspace, its norm squared is the total mass in (159). Each of the three coefficient forms is strictly positive on every nonzero matrix in its corresponding channel. Nonnegativity of the three terms therefore proves

$$\ker \mathfrak{M} = \bigoplus_{j: d_j=0} \mathcal{S}_j, \quad \text{rank } \mathfrak{M} = \mathbf{1}_{d_0>0} + 2\mathbf{1}_{d_d>0} + 3\mathbf{1}_{d_o>0}.$$

This evaluates all possible kernel directions without asserting an unknown seed mass positive.

31.2 The complete original tensor on the common conic

Write z_F for the Fabel parameter $z = \sqrt{1 - 8\tau_F} > 0$, distinct from the NS axial coordinate. The unchanged polynomial is

$$\begin{aligned} F_1(x, y, w) &= (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), \\ F_2(x, y, w) &= y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), \\ F_3(x, y, w) &= 2x - 3x^2 y - x^3 w. \end{aligned}$$

With $J = DF$, $x_0 = (1, -3/2, 13/2)$ and $\gamma(z_F) = (z_F^{-1}, -3z_F/2, 13z_F^2/2)$, differentiation gives $\det J = -2$ and

$$J_0 = \begin{pmatrix} -9/16 & -3/8 & -1/8 \\ 3/8 & 25/4 & 3/4 \\ -17/2 & -3 & -1 \end{pmatrix}, \quad J_\gamma = \begin{pmatrix} -9z_F^3/16 & -3z_F/8 & -1/8 \\ 3z_F^2/8 & 25/4 & 3/(4z_F) \\ -17/2 & -3/z_F^2 & -1/z_F^3 \end{pmatrix}.$$

Put $\tau_F = (1 - z_F^2)/8$ and $B_{\tau_F} = I + 6\tau_F E_{23}$. The full tensor is

$$\begin{aligned} C_F(z_F) &= J_0^{-1} B_{\tau_F}^{-1} J_\gamma, \\ &= \begin{pmatrix} (17 - 9z_F^3)/8 & (3 - 3z_F^3)/(4z_F^2) & (1 - z_F^3)/(4z_F^3) \\ (51 - 24z_F^2 - 27z_F^3)/16 & (9 + 8z_F^2 - 9z_F^3)/(8z_F^2) & (3 - 3z_F^3)/(8z_F^3) \\ (-153 + 36z_F^2 + 117z_F^3)/8 & (-27 - 12z_F^2 + 39z_F^3)/(4z_F^2) & (-9 + 13z_F^3)/(4z_F^3) \end{pmatrix}. \end{aligned} \quad (160)$$

Every entry follows by multiplying the displayed matrices, using $B_{\tau_F}^{-1} = I - 6\tau_F E_{23}$. Its inverse is $J_\gamma^{-1} B_{\tau_F} J_0$, and its determinant is one. Put

$$K_F = C_F C_F^\top, \quad v_* = (1/4, 3/8, -9/4)^\top, \quad S_* = v_* v_*^\top = \frac{1}{64} \begin{pmatrix} 4 & 6 & -36 \\ 6 & 9 & -54 \\ -36 & -54 & 324 \end{pmatrix}.$$

The transpose here is not an adjoint. For real $z_F \neq 0$ this K_F is positive definite since C_F is invertible. For complex z_F only complex symmetry is asserted. Each of the nine entries in (160) gives the convergent Laurent limits

$$z_F^3 C_F \longrightarrow v_* e_3^\top, \quad z_F^6 K_F \longrightarrow S_*. \quad (161)$$

Substitution with $xy = -3/2$ gives exactly

$$F(\gamma(z_F)) = (-z_F^2/4, 0, 0), \quad r = -z_F/2, \quad a = -r^2 = -z_F^2/4, \quad s = -z_F^2/8.$$

Indeed $F_1 = (-13 + 9)z_F^2/16$, $F_2 = (-12 + 39 - 27)z_F/8$ and $F_3 = (4 + 9 - 13)/(2z_F)$. This recovers the original inverse chart $(x, y, w) = (-1/(2r), 3r, 26r^2)$; the separate boundary point $(0, 0, a)$ remains in the fibre. Choose the analytic germ $p(z_F) = \sqrt{4 + z_F^2/4}$ with $p(0) = 2$. The exact conic maps are

$$q(z_F) = \frac{p(z_F) - z_F/2}{2}, \quad q(z_F)^{-1} = \frac{p(z_F) + z_F/2}{2}. \quad (162)$$

Their product is one, their difference is $r = -z_F/2$, and their sum is p . Thus $r^2 = -2s$, $p^2 = 4 - 2s$ and $b = c = 0$ retain the earlier coordinate values. The connection is used only where $rp \neq 0$, in a sufficiently small punctured neighbourhood of zero. All factors of the original GCT coefficient remain:

$$\begin{aligned} C(q) &= q^{10} - q^4 - q^{-4} + q^{-10} \\ &= m(n^3 - 5n^2 + 6n - 1)(n - 1), \quad m = z_F^2/4 = -2s, \quad n = m + 4. \end{aligned} \quad (163)$$

For direct verification, multiplying the finite Laurent sums gives $C = (q - q^{-1})^2 [7]_q [3]_q$, $[3]_q = p^2 - 1$ and $[7]_q = p^6 - 5p^4 + 6p^2 - 1$. In particular $z_F^{-2} C(q(z_F)) \rightarrow 21/4$. For real z_F near zero, $q > 0$ and $m > 0$, so $C(q) > 0$. The NS growth conic instead has $s = \tau = 1 - t > 0$ and $r = i\sqrt{2\tau}$; there is no real z_F with this value of s , because $s = -z_F^2/8 \leq 0$ for real z_F .

The strongest direct continuation on this conic is

$$z_F = -i\sqrt{8\tau}, \quad r = i\sqrt{2\tau}, \quad p = \sqrt{4 - 2\tau}, \quad q = \frac{1}{2}(\sqrt{4 - 2\tau} + i\sqrt{2\tau}), \quad 0 < \tau \ll 1. \quad (164)$$

It is exactly the NS conic point. Its Fabel time is $\tau_F = 1/8 + \tau$, so it is not the original real material trajectory before $\tau_F = 1/8$. On this path,

$$C(q) = \tau B(\tau), \quad B(\tau) = -42 + 196\tau - 280\tau^2 + 160\tau^3 - 32\tau^4.$$

This is the full factored polynomial $-2\tau(3 - 2\tau)(7 - 28\tau + 28\tau^2 - 8\tau^3)$. It is negative for sufficiently small positive τ , and $B(0) = -42$.

31.3 State and spectral limits with the adjoint unchanged

Define the actual physical state $\Psi_\tau = \mathfrak{M}(K_F(-i\sqrt{8\tau}))$. The physical Hamiltonian and its real regulator have not been continued. Since $(-i)^6 = -1$ and $8^3 = 512$, (161) gives

$$\tau^3 \Psi_\tau \longrightarrow -\mathfrak{M}(S_*)/512, \quad \tau^2 C(q) \Psi_\tau \longrightarrow (21/256) \mathfrak{M}(S_*). \quad (165)$$

The second limit multiplies the first by $B(\tau)$. Equivalently $z_F^4 = 64\tau^2$ and $z_F^4 C(q) K_F \rightarrow (21/4) S_*$ prove it directly. Convergence holds in every Sobolev norm and after applying $\mathcal{A}$, since the states are combinations of six fixed smooth states with converging scalar coefficients.

The real matrix S_* has trace $337/64$. Its three mean-subtracted diagonal entries are $-325/192$, $-310/192$, and $635/192$. Its off-diagonal entries are $6/64$, $-36/64$, and $-54/64$. Consequently (159) gives

$$\nu_* := \nu_{\mathfrak{M}(S_*)} = \frac{113569}{36864} \nu_0 + \frac{100825}{12288} \nu_d + \frac{531}{512} \nu_o.$$

It follows that

$$\tau^6 \nu_{\Psi_\tau} \longrightarrow \frac{1}{262144} \nu_*, \quad \tau^4 \nu_{C(q) \Psi_\tau} \longrightarrow \frac{441}{65536} \nu_*. \quad (166)$$

These are total-variation limits: in (159) bound the variation of the difference by the sum of the three coefficient errors times the finite masses of the seed measures. Using instead their finite first or second moments proves convergence of those moments. The identities include zero seed states without assuming an unknown moment nonzero or small.

More strongly, at every fixed τ with $C(q) \neq 0$,

$$\nu_{C(q) \Psi_\tau} = |C(q)|^2 \nu_{\Psi_\tau}.$$

This follows from linearity of each spectral projection and the fixed Hilbert adjoint. Thus spectral supports agree, and the energy-to-norm quotients agree whenever the states are nonzero. The negative scalar changes a phase; it scales every positive spectral measure by a positive scalar. Since $\mathcal{A} \geq 0$, none of these states has negative excitation form. This proves an obstruction for this specific proposed sign-to-energy map while retaining its exact state and measure correspondences.

31.4 Arithmetic differentiation and the full quantum source

Let $\mathcal{R} = \mathbb{C}\{z_F\}[z_F^{-1}]$, the ring of convergent meromorphic germs at zero. Equation (162) defines homomorphisms $A = \mathbb{Z}[q, q^{-1}] \rightarrow \mathcal{R}$ and $D = \mathbb{Q}[q, q^{-1}]_{(q-1)} \rightarrow \mathcal{R}$: every denominator nonzero at $q = 1$ becomes a unit analytic germ. This is a germ map, not simultaneous evaluation of every element of D on an entire fixed punctured interval. The arithmetic base map $s \mapsto -z_F^2/8$ factors through its original rank-four conic algebra, with $r = -z_F/2$ and $p = p(z_F)$. In particular the exact common coefficient is

$$C(q) = -2s(7 - 28s + 28s^2 - 8s^3)(3 - 2s).$$

This is the same polynomial acting on the earlier arithmetic operator $\hat{S}$ with s replaced there by $\hat{S}$. The coefficient-ring map asserts neither a representation of that topological operator on $\mathcal{H}_L$ nor membership of these s in its spectrum.

The lifted base derivation on this slice is

$$\nabla_s = -4z_F^{-1} \frac{d}{dz_F}, \quad -\frac{1}{4}\nabla_s^2 = -4z_F^{-2} \frac{d^2}{dz_F^2} + 4z_F^{-3} \frac{d}{dz_F}.$$

Differentiating $s = -z_F^2/8$, $r = -z_F/2$, and $p^2 = 4 - 2s$ gives $\nabla_s s = 1$, $\nabla_s r = -1/r$, $\nabla_s p = -1/p$. Also $q'(z_F) = -q/(2p)$, hence $\nabla_s q = -q/(rp)$. Thus the derivations agree on every generator of the original conic ring. Applying the first-order operator twice gives both terms of the second-order operator, with their signs. Its action on finite-range state germs commutes with the fixed linear map $\mathfrak{M}$, for example

$$-\frac{1}{4}\nabla_s^2 \mathfrak{M}(K_F) = \mathfrak{M}(-4z_F^{-2} K_F'' + 4z_F^{-3} K_F').$$

Under $s = 1 - t$ this retains the scalar $-\partial_t^2/4$. It does not identify that operator with H , or delete radial and axial terms of the full NS heat operator.

Extend the actual honest source lattice by $X_{\mathcal{R}} = \mathcal{R} \otimes_A X_A$. It remains free of rank 1260, with its original four generator matrices, divided powers and both degree-fifteen central characters. The already proved actual source splitting and full tensor-square involution extend through $D \rightarrow \mathcal{R}$, with both idempotents and their exact ranks. Put $\mathcal{V}_{\mathcal{R}} = \mathcal{R} \otimes_{\mathbb{C}} \mathcal{V}_L$; these are algebraic finite-sum tensors, not a Hilbert completion. There is a completely specified external tensor map

$$X_{\mathcal{R}} \otimes_{\mathcal{R}} \text{Sym}_3(\mathcal{R}) \longrightarrow X_{\mathcal{R}} \otimes_{\mathcal{R}} \mathcal{V}_{\mathcal{R}}, \quad x \otimes S \longmapsto x \otimes \mathfrak{M}_{\mathcal{R}}(S). \quad (167)$$

Give both second factors the trivial quantum action. The source coproduct $\Delta(E) = E \otimes K^{-1} + 1 \otimes E$ and $\Delta(F) = F \otimes 1 + K \otimes F$ then acts only on x . For each generator, both compositions in (167) equal its original action on x tensored with $\mathfrak{M}_{\mathcal{R}}(S)$. This proves the intertwiner, its divided-power identities and its central labels; tensor balancing follows from $\mathcal{R}$ -linearity. Its kernel is exactly $X_{\mathcal{R}} \otimes_{\mathbb{C}} \ker \mathfrak{M}$, included by $x \otimes S \mapsto x \otimes (1 \otimes S)$: the complex six-dimensional domain splits as the kernel plus a complementary vector space on which $\mathfrak{M}$ is injective; extending scalars preserves this splitting and its inverse on the image, and $X_{\mathcal{R}}$ is free. Inserting K_F and the actual scalar $C(q)$ composes this map with the state and measure calculations above. Replacing $X_{\mathcal{R}}$ by either extended tensor-square idempotent image gives the same proved map. No quantum action on the YM Hilbert factor, identification $X \rightarrow \mathcal{H}_L$, mass gap, canonical-basis construction or complexity separation is inferred.

32 Research provenance, failed constructions, and public workbenches

This manuscript records a human-directed Codex investigation. The human participant proposed the cross-problem directions and specifically suggested looking for negativity at a finite endpoint. Their initial wording included:

“So we’re looking for something that is under the floor, not necessarily at infinity. Maybe, I don’t know.”

This was an exploratory suggestion, not a theorem or a supplied proof. Codex developed the displayed calculations, wrote the proofs and executable checks, and used separate AI agents to review them. Those reviews are independent calculation routes within an AI workflow; they are not external human peer review or a formal verification of every theorem. The complete arguments, domains, constants, and unsuccessful constructions are retained so that readers can examine the mathematical content directly.

The original retained-fibre, conductor, arithmetic-operator, and GCT calculations preceded this task’s introduction of the public Navier–Stokes manuscript on 8 September 2026. On that date, another mathematics task supplied the public manuscript and the pinned source. The 165-page revision was retained; a subsequent source notification reported the 166-page revision downloaded at 19:17 UTC, with a server last-modified header of 19:09:35 GMT. The two hashes, exact source formulas, selected-passage comparison, and attribution are recorded in Section 29. The date and HTTP observations record source introduction and retrieval; they do not certify the external manuscript’s global theorem. The new concentration-coordinate bridge was derived after that introduction from the explicitly displayed source formulas and the previously retained conic.

Several unsuccessful approaches are part of the result. The printed mixed-sign GCT endpoint required a weight correction. The unsaturated integral presentation retains forty integer-2 torsion summands after inverting the specified quantum integer. Ordinary tensor permutations fail to preserve the generic source action. The replacement involution constructed here is equivariant in degree two but has a nonzero degree-three braid defect. The positive signed-sheet cover recovers the original generators while its quantum relations have nonzero defects in the quotient kernel. Each assertion and the exact repairing map or retained obstruction is proved in the corresponding section; none is erased from the provenance because an initial construction failed.

The finite negative coefficient and the negative signature of the specified Schur form are established calculations. Their derivations do not turn plethysm multiplicities into negative integers. The full source-compatible canonical geometric construction and the unrestricted asymptotic complexity obstruction are still unfinished. This preprint presents proved intermediate maps and the concrete places where the attempted extensions stop, without asserting a resolution of a Millennium problem.

The following read-only public workbenches were supplied by the human participant for public audit as places where Codex worked. They expose the broader exploratory projects, rather than serving as substitutes for the complete proofs in this manuscript. Labels follow the supplied identifications; the fourth link was supplied without a project label.

- [S6 Workbench](#).
- [Erdős–Straus Workbench](#).
- [Navier–Stokes Workbench](#).
- [Additional public workbench](#).

The initial DOI edition is intended to make the mathematics and these workbenches available for scrutiny. A later publication of the contributing transcripts, with profanity filtered and the human suggestions preserved in context, is a separately stated future stage. No such transcript publication is represented as already completed here.

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